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Robotics and Instrumentation

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Lecture Notes

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Course Introduction and Learning Outcomes

About the course

A robot is, in essence, a closed loop of *measurement*, *decision* and *actuation*. Its mechanical arm can only be as precise as the sensors that report where the arm is, and its controller can only act on information that has been correctly measured, conditioned and converted. For this reason robotics and instrumentation are taught together in this course: the first half of each unit builds the mechanical and mathematical picture of a robot, and the second half builds the measurement chain that makes the robot observable and controllable.

The course is organised in four units:

Unit I: Fundamentals of robotics, sensors and instrumentation – robot anatomy, configurations, degrees of freedom, work envelopes, sensors and transducers, static and dynamic characteristics of measurement systems, and measurement of temperature, pressure, force and strain.

Unit II: Actuators, robot mechanisms and measurement systems – electrical, pneumatic and hydraulic actuators, robot joints, serial and parallel manipulators, end-effectors, signal conditioning, instrumentation amplifiers, filters, A/D and D/A conversion and data acquisition.

Unit III: Robot kinematics, trajectory planning and control – rotation matrices, homogeneous transformations, Denavit–Hartenberg representation, forward/inverse/velocity kinematics, Jacobian, trajectory planning, dynamics and basic control.

Unit IV: Industrial automation, embedded systems and machine vision – robotic workcells, PLCs, robot programming, microcontrollers and interfacing, PWM and communication, image acquisition and processing, feature extraction and inspection.

Course learning outcomes

On successful completion of the course the student will be able to:

CO1 Describe the anatomy, classification, configurations, degrees of freedom and work envelope of industrial robots, and select a configuration for a given task.

CO2 Explain the working principles of sensors and transducers used in robotics, and quantify

the static and dynamic performance of a measurement system, including errors, uncertainty and loading effects.

CO3 Model and select electrical, pneumatic and hydraulic actuators and design signal-conditioning and data-acquisition chains for robotic systems.

CO4 Apply coordinate transformations, D–H conventions, forward/inverse kinematics, the Jacobian and trajectory planning to serial manipulators.

CO5 Formulate the dynamics of simple manipulators and implement basic joint-space and task-space control in MATLAB/Simulink.

CO6 Integrate PLCs, microcontrollers and machine-vision systems into automated robotic workcells for smart-manufacturing applications.

How to use these notes

- Each unit begins with the physical idea, develops the mathematics step by step, and closes every major topic with solved examples. Work through the examples with pen and paper before reading the solution.
- Boxes headed **Key points** summarise what must be remembered and revised.
- MATLAB programs are complete and can be copied directly into a MATLAB script (R2019b or later; no toolboxes are required unless stated).
- Every unit ends with a summary, a formula sheet and comparison tables, followed by a question bank of exactly 30 ten-mark questions (concepts, theory, derivations, numerical problems, diagrams and applications). Each question names its topic (section) number and ends with a **Refer to** line that points to the exact sections, figures, tables, equations and worked examples (with page numbers) where the answer is developed; a reference index at the end of each question bank lists the page of every question and of its topic. Technical terms are collected in the glossary at the end of the book.
- Algorithms for computational procedures are numbered by unit (Algorithm 1.1, 1.2, . . .) and are accompanied by flowcharts where helpful.
- Numbers in square brackets, e.g. [1], refer to the consolidated bibliography at the end of the notes (IEEE style).

Recommended textbooks

Primary: Craig [1]; Spong, Hutchinson and Vidyasagar [2]; Lynch and Park [3]; Bentley [10]; Morris and Langari [11]. Supplementary: Klafter *et al.* [5]; Schilling [6]; Groover [7]; Nakra and Chaudhry [13]; Bolton [14]; Fraden [15]; Siciliano and Khatib [8].

Unit I: Fundamentals of Robotics, Sensors and Instrumentation

Unit overview

This unit lays the two foundations on which the rest of the course is built. The first half (Sections 1.1 to 1.6) introduces the robot as a mechanical system: what it is made of, how it is classified, how many independent motions it has and what region of space it can reach. The second half (Sections 1.7 to 1.10 and 1.16 to 1.18) introduces the robot as a measured system: how a physical quantity is converted into a usable signal, how good that conversion is, and which sensors are used for position, velocity, acceleration, force, torque, proximity, temperature, pressure and strain. Section 1.19 provides MATLAB exercises and Section 1.20 the chapter-end material.

Syllabus coverage map. Table 1.1 maps every item of the Unit I syllabus to the section in which it is developed; the same section numbers are used as topic references in the question bank (Section 1.21).

Table 1.1: Unit I syllabus items and the sections that cover them.

No.	Syllabus item	Section	No.	Syllabus item	Section
1	Robotics and automation	1.1	14	Measurement systems	1.7
2	Robot anatomy	1.2	15	Static characteristics	1.8
3	Classification of robots	1.3	16	Dynamic characteristics	1.9
4	Robot configurations	1.4	17	Accuracy	1.8
5	Degrees of freedom	1.5	18	Precision	1.8
6	Work envelope, coordinate systems	1.6	19	Sensitivity	1.8
7	Position sensors	1.11	20	Resolution	1.8
8	Displacement measurement	1.11	21	Measurement uncertainty	1.8.9
9	Velocity measurement	1.12	22	Loading effects	1.8.10
10	Acceleration measurement	1.13	23	Temperature measurement	1.16
11	Force measurement (robotic)	1.15	24	Pressure measurement	1.17
12	Torque measurement	1.15, 1.18	25	Force measurement	1.18
13	Proximity measurement	1.14	26	Strain measurement	1.18

Learning objectives. After studying this unit you should be able to:

(a) define a robot and distinguish fixed, programmable and flexible automation;

(b) identify the components of a robot system and name the joints and links of a manipulator;

(c) classify robots by configuration, control, drive, application and mobility;

(d) compute the degrees of freedom of open and closed chains using Grübler's formula;

(e) sketch and compute the work envelope of Cartesian, cylindrical, spherical, SCARA and articulated robots;

(f) explain the generalised measurement system and quantify accuracy, precision, sensitivity, resolution, uncertainty and loading error;

(g) analyse zero-, first- and second-order instruments in the time and frequency domains;

(h) explain the principle, characteristics and applications of the major robotic sensors and of temperature, pressure, force and strain transducers.

1.1 Introduction to Robotics and Automation

1.1.1 Background

The word *robot* comes from the Czech *robota* (forced labour) and was introduced by Karel Čapek in his 1920 play *R.U.R. (Rossum's Universal Robots)*. The engineering history is more recent. George Devol filed a patent for a "programmed article transfer" device in 1954, and together with Joseph Engelberger founded Unimation, whose *Unimate* arm was installed at a General Motors die-casting plant in 1961 to unload hot castings [7, 8]. Since then the industrial robot has evolved from a hydraulic, point-to-point machine into an electrically driven, sensor-rich, networked device that welds car bodies, assembles electronics, palletises goods and, increasingly, works alongside people as a *collaborative robot (cobot)*.

1.1.2 Definitions

Robot. ISO 8373 defines an *industrial robot* as an "automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications" [16]. Three properties in this definition separate a robot from other machines:

(i) **Reprogrammability** – its motions or auxiliary functions can be changed without physical alteration;

(ii) **Multipurpose** – it can be adapted to a different application by reprogramming and/or changing its tooling;

(iii) **Multiple axes** – it has at least three independently controlled axes.

Robotics is the interdisciplinary engineering science of designing, building, sensing, programming and controlling robots. It combines mechanics (kinematics and dynamics), electrical and electronic engineering (actuators, drives, sensors and instrumentation), control engineering and computer science (planning, perception and learning) [2, 3].

Automation is the use of mechanical, electrical and computer-based systems to perform a process with reduced human intervention. A robot is one tool of automation; conveyors, CNC machines, PLC-controlled machines and vision inspection stations are others.

1.1.3 Types of automation

Automation is commonly divided into three classes according to the trade-off between production volume and product variety [7].

Table 1.2: Types of industrial automation.

	Fixed (hard)	**Programmable**	**Flexible**
Principle	Sequence of operations fixed by equipment configuration	Sequence controlled by a program; changed between batches	Program and tooling changed with negligible changeover time
Volume / variety	Very high volume, single product	Medium volume, batches of different products	Medium volume, mix of products produced continuously
Investment	Very high for custom equipment	High for general-purpose equipment	High for system integration
Examples	Transfer lines, bottling lines	CNC machines, **industrial robots**, PLC systems	Flexible manufacturing systems (FMS), robotic workcells

1.1.4 Why robots? Advantages and limitations

Advantages. Robots provide (a) consistent quality and repeatability of the order of ±0.02 mm to ±0.1 mm on modern arms; (b) the ability to work in hazardous, hot, toxic or monotonous environments (welding, painting, foundry, nuclear); (c) high productivity through continuous operation; (d) flexibility, because one arm can be reprogrammed for a new product; and (e) data generation for quality traceability.

Limitations. High initial cost and integration effort; the need for skilled programming and maintenance; limited dexterity and judgement compared with humans in unstructured tasks; safety requirements (guarding, risk assessment); and social concerns about job displacement.

1.1.5 Laws of robotics and robot generations

Isaac Asimov's three "laws of robotics" (1942) are a literary device rather than an engineering standard, but they introduce an idea that is now codified in safety standards: a robot must not harm a human. Practical robot safety is governed by standards such as ISO 10218 (industrial robots) and ISO/TS 15066 (collaborative operation).

Robots are often described in *generations*: (1) playback robots that repeat a taught sequence; (2) sensor-based robots that adapt to measured conditions (vision, force); (3) intelligent robots that plan, learn and reason about uncertain environments [9].

1.1.6 Applications

Typical industrial applications are *material handling* (pick-and-place, palletising, machine loading), *processing* (spot and arc welding, painting, gluing, grinding, deburring), *assembly* (electronics, automotive sub-assemblies) and *inspection* (vision-guided measurement). Non-industrial applications include surgical robots, warehouse and delivery robots, agricultural robots, autonomous vehicles, exoskeletons and space robots.

Key points

- An industrial robot is an automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes [16].
- Robots belong to *programmable* and *flexible* automation; transfer lines are *fixed* automation.
- Safety of industrial and collaborative robots is governed by ISO 10218 and ISO/TS 15066.

1.2 Robot Anatomy and the Robot System

1.2.1 Components of a robot system

A robot is not just an arm. Figure 1.1 shows the complete robot system. Its principal elements are:

1. **Manipulator (arm and body)** – the mechanical structure of links connected by joints, together with transmissions (gears, belts, harmonic drives, ball screws).
2. **End-effector** – the gripper or tool (welding torch, spray gun, screwdriver, vacuum cup) mounted on the wrist flange. It is task-specific and is not normally supplied with the robot.
3. **Actuators and drives** – electric motors (servo, stepper, BLDC), pneumatic or hydraulic cylinders and motors, together with the power amplifiers (servo drives) that energise them (Unit II).
4. **Sensors** – *internal* sensors measure the robot's own state (joint position, velocity, motor current, temperature); *external* sensors measure the environment and the task (vision, force/torque, proximity, tactile).
5. **Controller** – the computer that stores programs, performs kinematic computation and trajectory generation, closes the servo loops, and communicates with other equipment (PLC, vision system).
6. **Teach pendant and programming interface** – a hand-held unit or an off-line programming PC used to teach positions, edit and run programs (Unit IV).

7. **Power supply** – electric power for motors and electronics, compressed air for pneumatic tools and hydraulic power units where hydraulic drives are used.

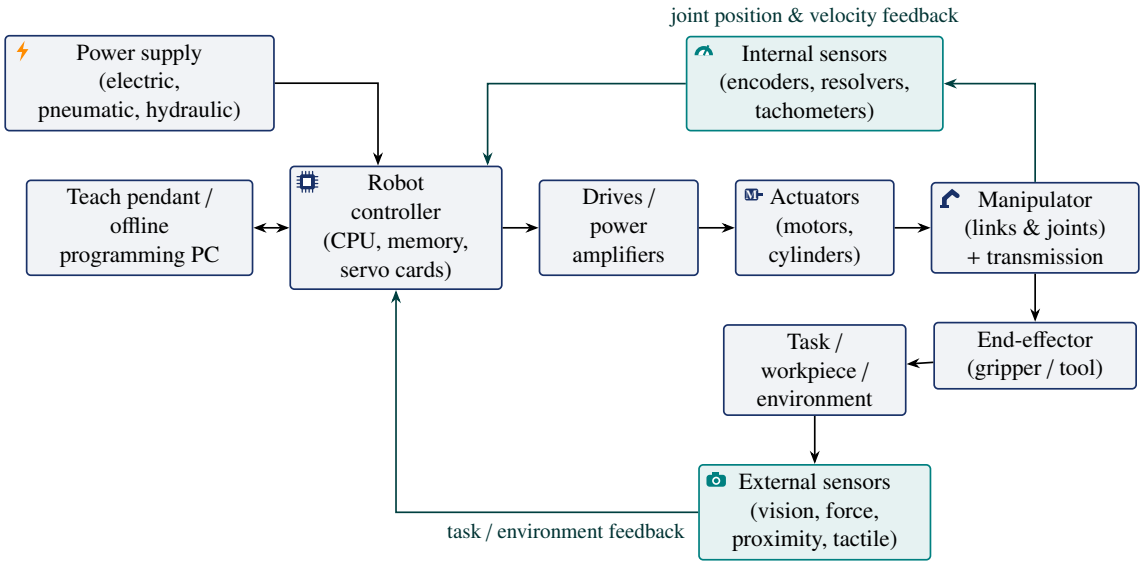


Figure 1.1: Block diagram of a complete robot system. Internal sensors close the joint servo loops; external sensors provide task and environment information to the controller.

1.2.2 Anatomy of the manipulator

The manipulator is a chain of rigid bodies called *links* connected by *joints*. The fixed link attached to the floor is the *base* (link 0); the last link carries the wrist flange to which the end-effector is attached. Figure 1.2 labels a six-axis articulated robot using the anthropomorphic terminology common in industry.

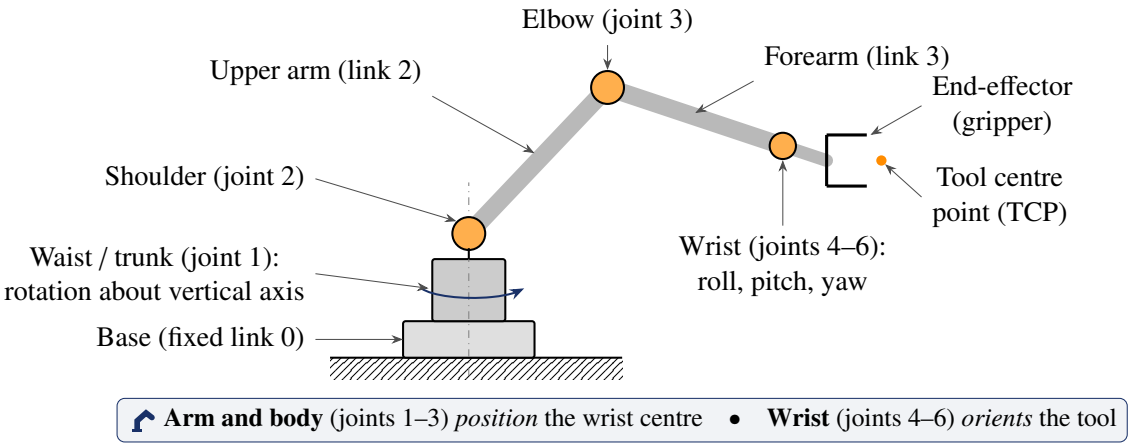


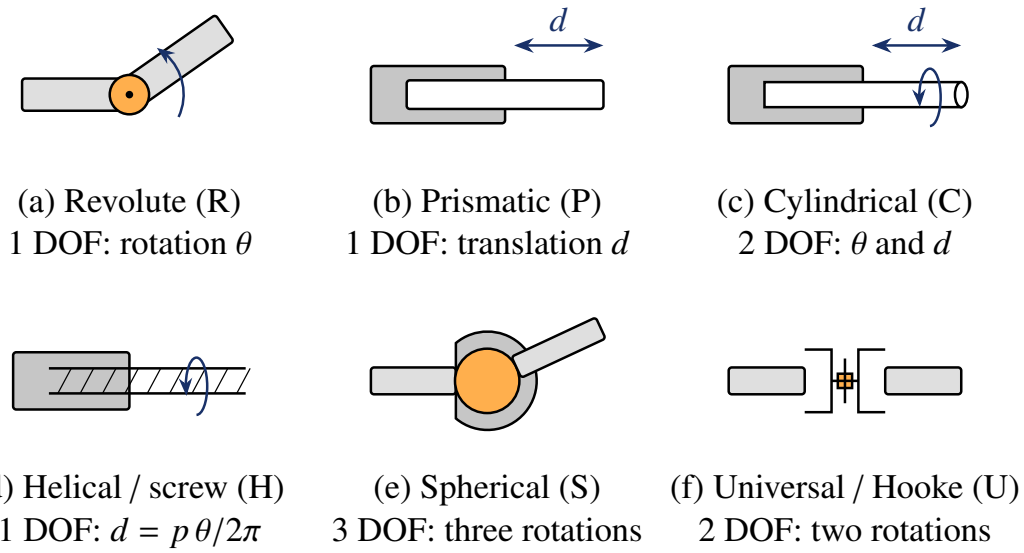
Figure 1.2: Anatomy of a six-axis articulated industrial robot. Joints 1–3 (waist, shoulder, elbow) form the arm and position the wrist; joints 4–6 form the wrist and orient the end-effector.

- **Body and arm** (joints 1–3): their purpose is to *position* the wrist centre anywhere in the work envelope. Three independent motions are needed to reach a point in three-dimensional space.

- **Wrist** (joints 4–6): its purpose is to *orient* the end-effector. The three wrist motions are *roll* (rotation about the arm axis), *pitch* (up–down rotation) and *yaw* (left–right rotation). When the three wrist axes intersect at one point the wrist is called a *spherical wrist*; this greatly simplifies inverse kinematics (Unit III).
- **Tool centre point (TCP)** – the reference point on the end-effector (e.g. the tip of a welding torch or the centre between gripper fingers) whose position and orientation are programmed.

1.2.3 Joints

A joint (kinematic pair) constrains the relative motion of the two links it connects. Figure 1.3 illustrates the joints used in robots and mechanisms, and Table 1.3 lists the degrees of freedom f and constraints c that each joint provides in space (where $f + c = 6$) [3].



(a) Revolute (R) 1 DOF: rotation θ | (b) Prismatic (P) 1 DOF: translation d | (c) Cylindrical (C) 2 DOF: θ and d

(d) Helical / screw (H) 1 DOF: $d = p\,\theta/2\pi$ | (e) Spherical (S) 3 DOF: three rotations | (f) Universal / Hooke (U) 2 DOF: two rotations

Figure 1.3: Common joint types. Industrial manipulators use almost exclusively revolute (R) and prismatic (P) joints; spherical (S) and universal (U) joints appear in parallel robots.

Table 1.3: Degrees of freedom and constraints of common joints (spatial case).

Joint	Symbol	DOF f	Constraints c	Typical use
Revolute (hinge)	R	1	5	Waist, shoulder, elbow of articulated arms
Prismatic (sliding)	P	1	5	Cartesian axes, telescopic arms
Helical (screw)	H	1	5	Ball-screw driven axes
Cylindrical	C	2	4	SCARA quill (rotate + slide)
Universal (Hooke)	U	2	4	Legs of parallel robots
Spherical (ball)	S	3	3	Legs of parallel robots, human shoulder

In industrial literature a different set of letters is also used to describe the arm and body joints of robots [7]: **L** (linear), **O** (orthogonal sliding), **R** (rotational, axis perpendicular to the links), **T** (twisting, axis parallel to the links) and **V** (revolving, input link parallel and output link

perpendicular to the axis). In these notes we use the R/P notation of Table 1.3, which is the convention of modern kinematics texts [1–3].

1.2.4 Links and joint variables

Each joint i has one *joint variable*: the angle θ_i for a revolute joint or the displacement d_i for a prismatic joint. The vector of all joint variables

$$\mathbf{q} = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}^{\mathsf{T}}, \qquad q_i = \begin{cases} \theta_i & \text{revolute joint} \\ d_i & \text{prismatic joint} \end{cases} \tag{1.1}$$

is called the *joint-space* (configuration) vector. Given $\mathbf{q}$, the position and orientation of every link are known; the mapping from $\mathbf{q}$ to the end-effector pose is *forward kinematics*, studied in Unit III.

1.2.5 Robot specifications

Robots are compared using the following specifications, which are defined and tested according to ISO 9283 [17]:

- **Payload** – the maximum mass (including end-effector) that can be carried at rated speed and accuracy, e.g. 6 kg, 210 kg.
- **Reach** – the maximum horizontal distance from the base axis to the wrist centre.
- **Number of axes** – usually 4 (SCARA, palletising), 6 (general purpose) or 7 (redundant, collaborative).
- **Pose repeatability** RP – the closeness of agreement between positions attained when the same programmed pose is repeated; typically ±0.01 mm to ±0.1 mm.
- **Pose accuracy** AP – the deviation between a commanded pose and the mean of the attained poses; typically an order of magnitude worse than repeatability unless the robot is calibrated.
- **Maximum speed** of each axis (°/s) and of the TCP (m/s).
- **Control resolution** – the smallest increment of motion that the controller can command (Section 1.8).

Key points

- Arm and body (first three joints) *position* the wrist; the wrist (last three joints) *orients* the tool.
- A revolute or prismatic joint adds one DOF; a spherical joint adds three.
- Repeatability is usually much better than absolute accuracy; teach-programmed robots rely on repeatability, offline-programmed robots on accuracy.

1.3 Classification of Robots

Robots can be classified in several independent ways, and it is important to realise that these bases are *orthogonal*: a single robot belongs to one class under every basis. A typical welding robot, for example, is an electrically driven, serial, articulated, continuous-path robot that is programmed with a teach pendant. Figure 1.4 presents the six most useful bases as an infographic, and Table 1.4 adds remarks and examples for each class. When a robot is specified for a new application, the engineer effectively walks through this tree: the task geometry suggests a configuration, the payload and cleanliness requirements suggest the drive, the process (welding versus palletising) dictates the type of motion control, and the batch size decides how the robot should be programmed.

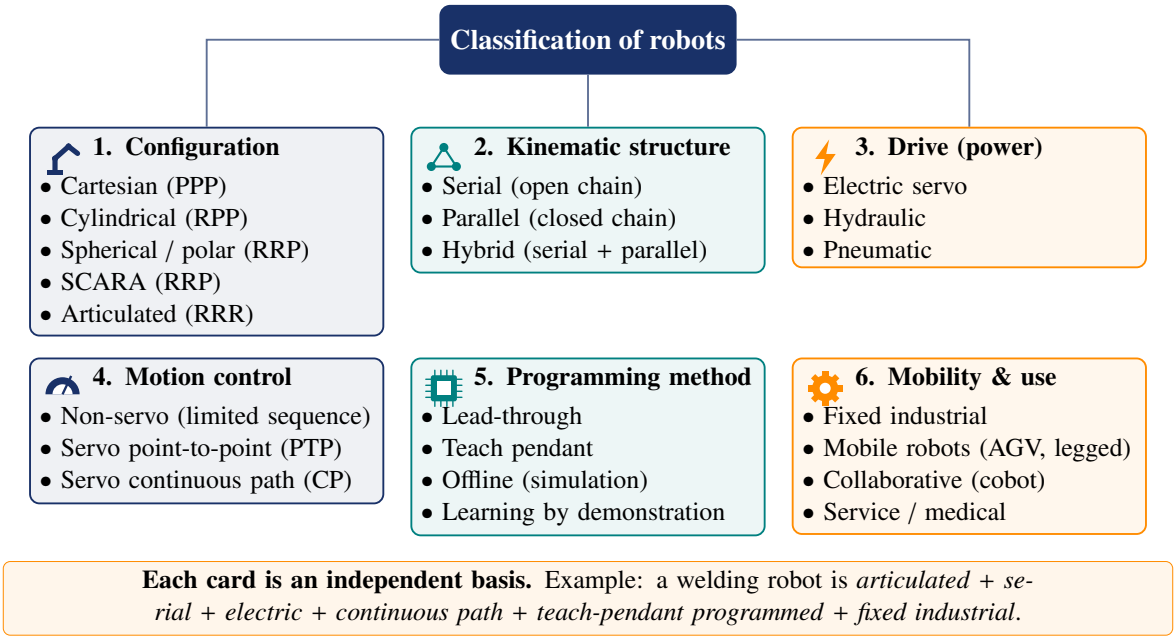


Figure 1.4: Infographic of robot classification. Each column is an independent basis of classification; any real robot takes one entry from every column.

Table 1.4: Classification of robots.

Basis	Classes	Remarks / examples
Physical configuration	Cartesian, cylindrical, spherical (polar), SCARA, articulated, parallel	Determined by the type and order of the first three joints (Section 1.4).
Kinematic structure	Serial (open chain), parallel (closed chain), hybrid	Delta and Stewart platforms are parallel robots.
Drive / power source	Electric, hydraulic, pneumatic	Electric servo drives dominate; hydraulic for very high payload; pneumatic for simple pick-and-place.

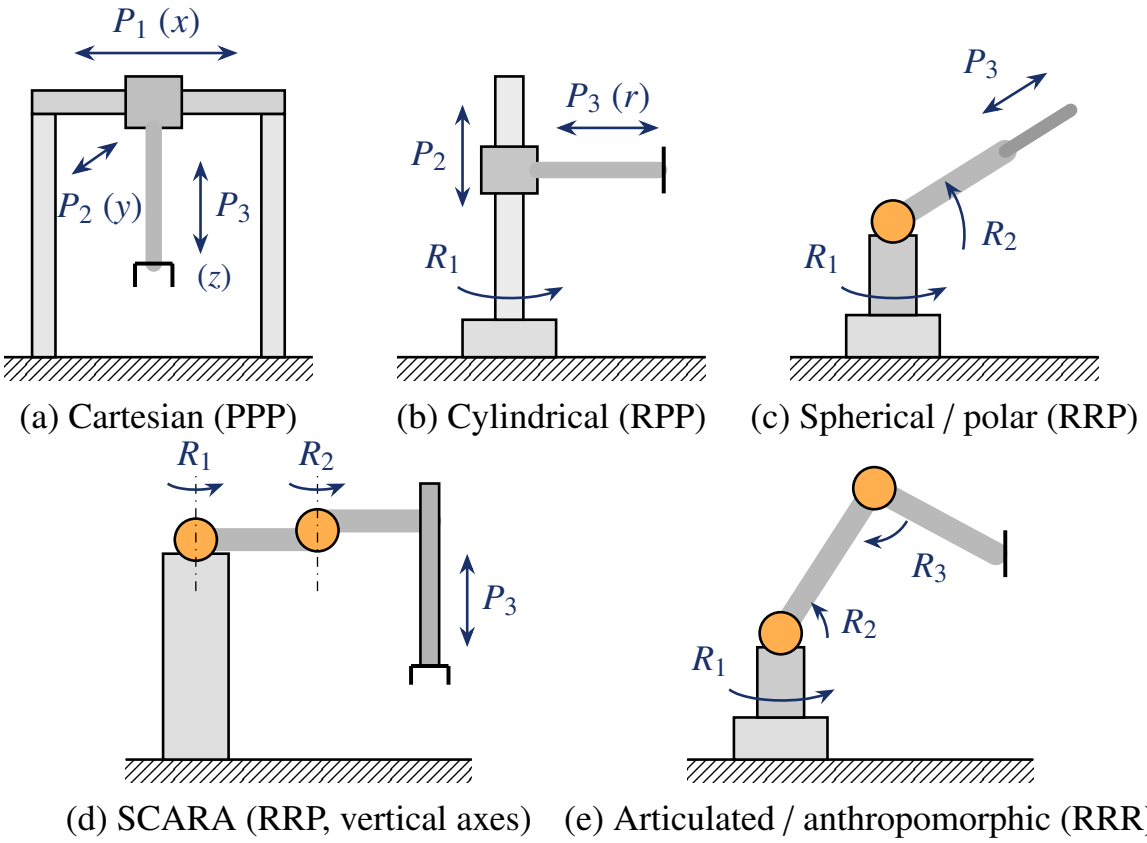
Basis	Classes	Remarks / examples
Control of motion	Non-servo (limited sequence, bang-bang), servo point-to-point (PTP), servo continuous path (CP)	PTP: spot welding, machine loading. CP: arc welding, painting, gluing.
Programming method	Lead-through (walk-through), teach pendant, offline programming, learning by demonstration	See Unit IV.
Intelligence / generation	Playback, sensory-controlled, adaptive / intelligent	Increasing use of sensors and on-board reasoning.
Mobility	Fixed-base manipulators, wheeled and legged mobile robots, aerial and underwater robots, mobile manipulators	Mobile robots require localisation and mapping [9].
Application	Industrial, service, medical, field, space, collaborative	Cobots are designed for safe physical human–robot interaction.

1.3.1 Classification by control: point-to-point versus continuous path

- **Non-servo (limited-sequence) robots** move each axis between mechanical end stops; there is no feedback of position during motion. They are cheap and fast but can only reach the stop positions.
- **Servo point-to-point (PTP) robots** use feedback to reach any programmed point, but the path between points is not controlled precisely.
- **Servo continuous-path (CP) robots** control all axes simultaneously so that the TCP follows a prescribed path (straight line, circular arc) with a specified velocity profile.

1.4 Robot Configurations

The *configuration* of a manipulator is named after the types of its first three (arm and body) joints, because these determine the shape of the work envelope. Figure 1.5 illustrates the five basic serial configurations.



(a) Cartesian (PPP) (b) Cylindrical (RPP) (c) Spherical / polar (RRP)

(d) SCARA (RRP, vertical axes) (e) Articulated / anthropomorphic (RRR)

Figure 1.5: The five basic serial robot configurations. R = revolute joint, P = prismatic joint; arrows indicate the joint motions.

1.4.1 Cartesian (rectangular, gantry) robot – PPP

Three mutually perpendicular prismatic joints move the wrist along x, y and z. If the axes are mounted on an overhead frame the robot is called a *gantry robot*.

- *Kinematics:* the TCP position equals the joint variables directly, $(x, y, z) = (d_1, d_2, d_3)$, so the controller is simple and accuracy is uniform throughout the workspace.
- *Work envelope:* a rectangular box of volume $V = L_x L_y L_z$.
- *Applications:* pick-and-place over large areas, CNC loading, 3-D printing, palletising, laboratory automation.
- *Limitations:* large floor space relative to workspace; cannot reach around obstacles.

1.4.2 Cylindrical robot – RPP

A vertical column rotates about the base (R), a carriage slides vertically along the column (P) and an arm extends radially (P). The TCP position is described naturally in cylindrical coordinates (r, θ, z):

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z. \tag{1.2}$$

Applications: machine loading, assembly, handling at die-casting and moulding machines.

1.4.3 Spherical (polar) robot – RRP

A base rotation θ (R), an elevation (pitch) ϕ (R) and a telescopic radial extension r (P). This was the configuration of the original Unimate. Position in spherical coordinates (elevation ϕ measured from the horizontal plane):

$$x = r\cos\phi\cos\theta, \qquad y = r\cos\phi\sin\theta, \qquad z = r\sin\phi. \tag{1.3}$$

Applications: spot welding, die-casting, forging, machine-tool loading.

1.4.4 SCARA robot – RRP with parallel vertical axes

SCARA stands for *Selective Compliance Assembly Robot Arm* (also *Articulated* Robot Arm). Two revolute joints with vertical, parallel axes move the arm in a horizontal plane and a prismatic quill moves vertically; a fourth revolute joint usually rotates the tool. The arm is *compliant* in the horizontal plane but *rigid* vertically, which is exactly what is needed to insert pegs into holes. The horizontal position is that of a planar two-link arm:

$$x = l_1\cos\theta_1 + l_2\cos(\theta_1+\theta_2), \qquad y = l_1\sin\theta_1 + l_2\sin(\theta_1+\theta_2), \qquad z = z_0 - d_3, \tag{1.4}$$

where l_1 and l_2 are the link lengths, θ_1, θ_2 the joint angles and d_3 the quill stroke. *Applications:* high-speed electronics assembly, packaging, screw driving.

1.4.5 Articulated (anthropomorphic, jointed-arm, revolute) robot – RRR

All three arm joints are revolute: a vertical waist axis and two horizontal axes (shoulder and elbow), mimicking the human arm (Fig. 1.2). This is the most common industrial configuration because it offers the largest work envelope for its footprint and can reach over and around obstacles. *Applications:* arc and spot welding, painting, material handling, machine tending, assembly.

1.4.6 Parallel robots

In a *parallel* (closed-chain) robot the end-effector platform is connected to the base by several independent *legs*. Examples are the *Delta* robot (three legs, high-speed pick-and-place) and the *Stewart–Gough platform* (six extensible UPS legs, flight simulators and machine tools). Parallel robots offer high stiffness, accuracy and speed but a small workspace; they are treated in Unit II.

1.5 Degrees of Freedom

1.5.1 Concept

The *degrees of freedom* (DOF) of a mechanical system is the smallest number of real-valued coordinates needed to represent its configuration [3]. A rigid body moving freely in a plane has

Table 1.5: Comparison of basic robot configurations.

Type	Joints	Work envelope	Advantages	Limitations
Cartesian	PPP	Rectangular box	Simple control, uniform accuracy, high rigidity, easy to program	Large footprint, small workspace-to-size ratio
Cylindrical	RPP	Hollow cylinder	Simple, rigid, good for machine loading	Cannot reach near the column or over obstacles
Spherical	RRP	Portion of a spherical shell	Long horizontal reach	Accuracy varies with radial extension; complex coordinates
SCARA	RRP	Annular cylinder	Fast, compliant in plane, rigid vertically	Limited vertical stroke; planar tasks
Articulated	RRR	Approx. sphere with irregular boundary	Largest reach per footprint, dexterous	Complex kinematics, accuracy varies, singularities
Parallel (Delta)	closed chain	Small dome/cylinder	Very fast, stiff, accurate	Small workspace, complex kinematics

three DOF (x, y, rotation θ); a rigid body in space has six DOF – three translations and three rotations (roll, pitch, yaw). Consequently:

- to place a tool at an arbitrary *position* in space a robot needs at least **3** DOF;
- to place it at an arbitrary *position and orientation* it needs at least **6** DOF;
- a robot with more than 6 DOF in space (e.g. a 7-axis cobot) is *kinematically redundant*: it can reach a given pose in infinitely many ways, which can be used to avoid obstacles and singularities.

1.5.2 Grübler's (Chebychev–Grübler–Kutzbach) formula

Derivation. Consider a mechanism of N links, *including the ground*, connected by J joints. Let m be the DOF of a free rigid body ($m = 3$ for planar, $m = 6$ for spatial mechanisms) and let joint i provide f_i freedoms and therefore $c_i = m - f_i$ constraints.

Step 1: If all links were unconnected, the $N - 1$ moving links would have $m(N - 1)$ DOF (the ground has none).

Step 2: Each joint removes $c_i = m - f_i$ freedoms. Assuming the constraints are independent,

$$\text{DOF} = m(N - 1) - \sum_{i=1}^{J} c_i = m(N - 1) - \sum_{i=1}^{J} (m - f_i).$$

Step 3: Expanding the sum gives **Grübler's formula**:

$$\boxed{\mathrm{DOF} = m\,(N - 1 - J) + \sum_{i=1}^{J} f_i} \tag{1.5}$$

m 3 for planar mechanisms, 6 for spatial mechanisms

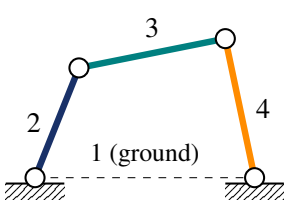
N number of links *including* the ground (fixed base)

J number of joints (each joint connects exactly two links)

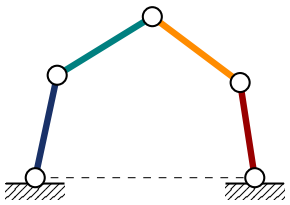
f_i freedoms of joint i (Table 1.3)

For the planar case with only one-DOF joints (R or P) (1.5) reduces to the well-known *Kutzbach criterion* $\mathrm{DOF} = 3(N-1) - 2J_1$, where J_1 is the number of one-DOF joints.

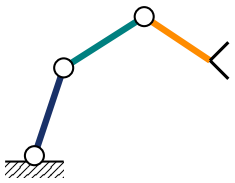
Assumption and caution. The formula assumes that the joint constraints are independent. For mechanisms with special geometry (e.g. parallel links of equal length) some constraints are redundant and the true DOF is larger than the formula predicts. Grübler's formula therefore gives a *lower bound* that is correct for generic geometry.



(a) Four-bar, $N = 4$, $J = 4$ (R): $\mathrm{DOF} = 3(4-1-4)+4 = \mathbf{1}$



(b) Five-bar, $N = 5$, $J = 5$ (R): $\mathrm{DOF} = 3(5-1-5)+5 = \mathbf{2}$



(c) 3R open chain, $N = 4$, $J = 3$: $\mathrm{DOF} = 3(4-1-3)+3 = \mathbf{3}$

Grübler: $\mathrm{DOF} = m(N-1-J) + \sum_{i=1}^{J} f_i, \quad m = 3$ (planar), $m = 6$ (spatial)

Figure 1.6: Planar mechanisms and their degrees of freedom from Grübler's formula with $m = 3$.

Open chains. For a serial (open-chain) robot $J = N - 1$, so the first term vanishes and $\mathrm{DOF} = \sum f_i$: the DOF of a serial robot simply equals the sum of its joint freedoms. This is why a six-joint arm is a 6-DOF robot.

Example 1.1: DOF of a six-axis articulated robot

Given: A spatial serial robot with six revolute joints (Fig. 1.2).

Find: DOF.

Solution: Links including base: $N = 7$. Joints: $J = 6$, each with $f_i = 1$. Spatial mechanism: $m = 6$.

$$\mathrm{DOF} = 6(7-1-6) + \sum_{i=1}^{6} 1 = 0 + 6 = 6.$$

Interpretation: The robot can position *and* orient its tool arbitrarily within its workspace (away from singularities).

Example 1.2: Planar four-bar and five-bar linkages

Given: The planar linkages of Fig. 1.6(a) and (b), all joints revolute.

Solution: Planar: $m = 3$.

- Four-bar: $N = 4$ (including ground), $J = 4$: DOF $= 3(4 - 1 - 4) + 4 = -3 + 4 = 1$.
- Five-bar: $N = 5$, $J = 5$: DOF $= 3(5 - 1 - 5) + 5 = -3 + 5 = 2$.

Interpretation: The four-bar needs one motor (one input crank); the five-bar needs two motors – it is the basis of planar parallel robots used for fast pick-and-place.

Example 1.3: DOF of a Stewart–Gough platform

Given: A spatial parallel robot: a fixed base and a moving platform joined by six legs. Each leg is a UPS chain – a universal joint at the base, a prismatic (actuated) joint in the middle and a spherical joint at the platform – so each leg consists of two links.

Solution:

- Links: ground (1) + platform (1) + 6×2 leg links $= 14 \Rightarrow N = 14$.
- Joints: $6 \times 3 = 18 \Rightarrow J = 18$.
- Joint freedoms: $\sum f_i = 6(2 + 1 + 3) = 36$.

$$\text{DOF} = 6(14 - 1 - 18) + 36 = 6(-5) + 36 = 6.$$

Interpretation: The platform has full six-DOF motion controlled by the six prismatic actuators. If the universal joints were replaced by spherical joints (SPS legs), the formula would give $6(14 - 1 - 18) + 6(3 + 1 + 3) = 12$: each leg would gain an idle spin about its own axis that does not move the platform. Such "internal" freedoms must be recognised when interpreting Grübler's formula [3].

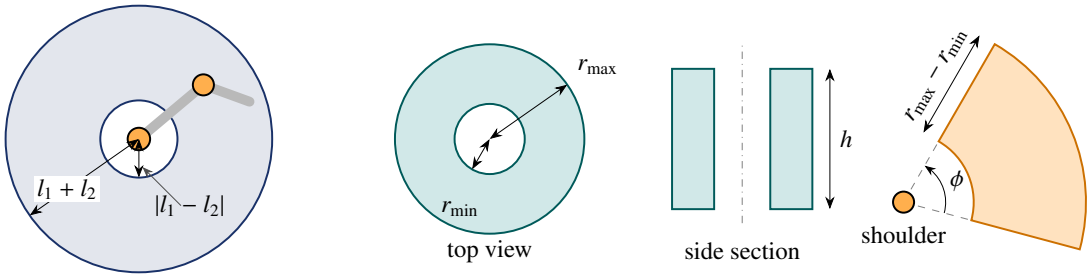
1.6 Work Envelope and Coordinate Systems

1.6.1 Work envelope (workspace)

The *work envelope* or *workspace* is the set of all points that the wrist centre (or TCP) can reach. Two workspaces are distinguished:

- the **reachable workspace** – points reachable with *at least one* orientation;
- the **dexterous workspace** – points reachable with *every* orientation; it is a subset of the reachable workspace.

The shape of the envelope is fixed by the configuration and the joint limits; its size by link lengths and prismatic strokes. Manufacturers publish it as plan and elevation drawings. Figure 1.7 shows the envelopes of three configurations.



(a) Planar 2R arm (top view): annulus, $l_1 > l_2$

(b) Cylindrical robot: hollow cylinder, $V = \pi h\,(r_{\max}^2 - r_{\min}^2)$

(c) Spherical robot: vertical section of a spherical shell

Figure 1.7: Work envelopes: (a) planar 2R arm – annulus between $|l_1 - l_2|$ and $l_1 + l_2$; (b) cylindrical robot – hollow cylinder; (c) spherical robot – part of a spherical shell.

Volume formulas. For a cylindrical robot with radial stroke $r_{\min} \le r \le r_{\max}$, vertical stroke h and base rotation range $\Delta\theta$ (radians), integrating the volume element $r\,dr\,d\theta\,dz$ gives

$$V_{\text{cyl}} = \int_0^{h}\int_0^{\Delta\theta}\int_{r_{\min}}^{r_{\max}} r\,dr\,d\theta\,dz = \frac{\Delta\theta}{2}\left(r_{\max}^2 - r_{\min}^2\right)h. \tag{1.6}$$

For a spherical robot with elevation range $\phi_1 \le \phi \le \phi_2$ (measured from the horizontal) the volume element is $r^2\cos\phi\,dr\,d\phi\,d\theta$, giving

$$V_{\text{sph}} = \frac{\Delta\theta}{3}\left(r_{\max}^3 - r_{\min}^3\right)(\sin\phi_2 - \sin\phi_1)\,. \tag{1.7}$$

For a planar 2R arm with full joint rotation and $l_1 \ge l_2$ the reachable area is an annulus:

$$A_{2R} = \pi\left[(l_1 + l_2)^2 - (l_1 - l_2)^2\right] = 4\pi l_1 l_2. \tag{1.8}$$

Example 1.4: Work volume of a cylindrical robot

Given: Radial stroke from $r_{\min} = 0.25\,\text{m}$ to $r_{\max} = 0.75\,\text{m}$; vertical stroke $h = 0.6\,\text{m}$; base rotation 300°.

Find: Work volume, and the volume if the base could rotate through 360°.

Solution: $\Delta\theta = 300^\circ \times \pi/180^\circ = 5.236\,\text{rad}$. From (1.6):

$$V = \frac{5.236}{2}\,(0.75^2 - 0.25^2)(0.6) = 2.618 \times 0.5 \times 0.6 = 0.785\,\text{m}^3.$$

For $\Delta\theta = 2\pi$: $V = \pi(0.5)(0.6) = 0.942\,\text{m}^3$.

Interpretation: The 60° dead zone (usually behind the robot, where cables and the column

are) removes $1/6$ of the volume.

Example 1.5: Work volume of a spherical robot

Given: $r_{\min} = 0.4\,\text{m}$, $r_{\max} = 1.2\,\text{m}$, base rotation 360°, elevation from $\phi_1 = -30°$ to $\phi_2 = +60°$.

Solution: Using (1.7):

$$\frac{r_{\max}^3 - r_{\min}^3}{3} = \frac{1.728 - 0.064}{3} = 0.5547\,\text{m}^3, \qquad \sin 60° - \sin(-30°) = 0.8660 + 0.5 = 1.3660,$$

$$V = 2\pi \times 0.5547 \times 1.3660 = 4.76\,\text{m}^3.$$

Example 1.6: Reach of a SCARA / planar 2R arm

Given: $l_1 = 0.5\,\text{m}$, $l_2 = 0.3\,\text{m}$, both joints rotate fully.

Solution: Maximum reach $l_1 + l_2 = 0.8\,\text{m}$; minimum reach $|l_1 - l_2| = 0.2\,\text{m}$. Area from (1.8): $A = 4\pi(0.5)(0.3) = 1.885\,\text{m}^2$ (check: $\pi(0.64 - 0.04) = 1.885$).

Interpretation: If $l_1 = l_2$ the inner radius vanishes and the arm can reach its own base axis; equal link lengths therefore maximise the reachable area for a given total length.

1.6.2 Coordinate systems (reference frames)

Robot programs describe positions with respect to coordinate frames. A frame $\{A\}$ is a set of three mutually orthogonal unit vectors $\hat{x}_A, \hat{y}_A, \hat{z}_A$ attached to a point (right-handed, $\hat{x} \times \hat{y} = \hat{z}$). Figure 1.8 shows the frames used by robot controllers [1].

- **Joint coordinate system** – each axis is moved individually; positions are expressed as joint angles/displacements $\mathbf{q}$ (Eq. (1.1)). Used for jogging single axes and for moving out of singularities.
- **World (global) frame {W}** – a fixed Cartesian frame of the cell; used when several robots and machines must share a common reference.
- **Base frame {B}** – attached to the robot base; the robot's own Cartesian x–y–z. Often coincides with {W} for a single robot.
- **Tool frame {T}** – attached to the TCP, usually with z_T (or x_T) along the tool approach direction. Jogging in the tool frame moves the tool along its own axes, e.g. "advance the drill".

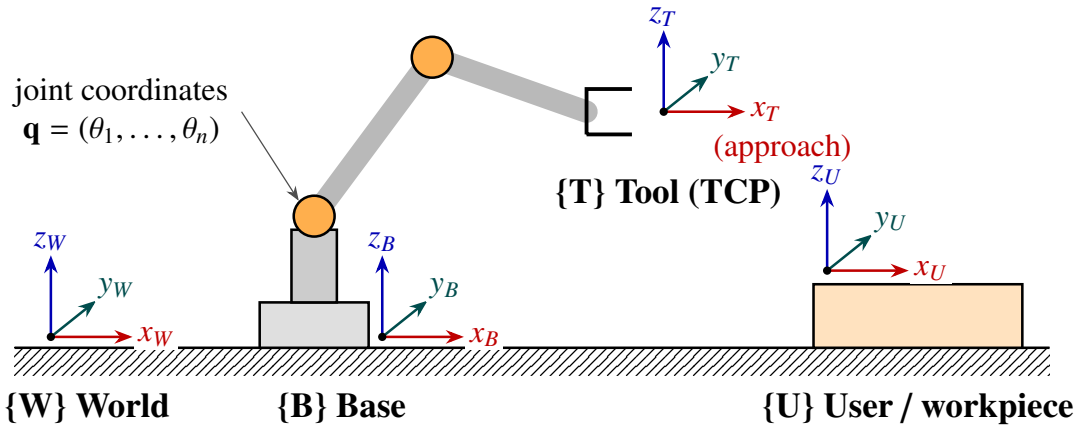


Figure 1.8: Coordinate systems used in robot programming: world {W}, base {B}, tool {T} and user/workpiece {U} frames, plus the joint coordinates $\mathbf{q}$.

- **User (workpiece, fixture) frame {U}** – attached to a fixture or conveyor. If the fixture is moved, only {U} is re-taught and all points defined in it remain valid.

Converting a point from one frame to another is a *coordinate transformation*; rotation matrices and homogeneous transforms that perform these conversions are developed in Unit III.

Key points

- DOF = $m(N - 1 - J) + \sum f_i$; for a serial chain DOF = $\sum f_i$.
- Cylindrical envelope $V = \frac{\Delta\theta}{2}(r_{\max}^2 - r_{\min}^2)h$; spherical $V = \frac{\Delta\theta}{3}(r_{\max}^3 - r_{\min}^3)(\sin\phi_2 - \sin\phi_1)$.
- Teach points in a user frame so that the program survives fixture relocation.

1.7 Measurement Systems

1.7.1 Why measurement matters in robotics

Every closed-loop robot action depends on a measurement: a joint servo compares the encoder reading with the commanded angle; a force-controlled grinding robot compares measured contact force with a set-point; a vision-guided robot locates a part before picking it. Errors in measurement propagate directly into errors of the task. The science of measurement – *metrology* – and its engineering practice – *instrumentation* – therefore form the second pillar of this course [10, 11].

1.7.2 Basic terms

- **Measurand** – the physical quantity to be measured (position, temperature, force, . . .).
- **Sensor** – the element in direct contact with the measurand that responds to it (e.g. a strain-gauge grid, a thermocouple junction).

- **Transducer** – a device that converts energy or a signal from one form to another (e.g. force → strain → resistance change → voltage). In practice the terms sensor and transducer are often used interchangeably; strictly, the sensor is the first stage of a transducer.
- **Active (self-generating) transducer** – produces an output signal without external excitation, taking energy from the measurand: thermocouple, piezoelectric crystal, tachogenerator, photovoltaic cell.
- **Passive (externally powered, modulating) transducer** – requires an external excitation that it modulates: potentiometer, strain gauge, RTD, LVDT, capacitive sensor.
- **Analogue vs digital** – analogue sensors give a continuous output (voltage, current); digital sensors give discrete states or counts (encoders, switches).

1.7.3 The generalised measurement system

Bentley's model [10] describes any measurement system as a chain of four basic elements (Fig. 1.9):

1. **Sensing element** – in contact with the process; its output depends on the measurand (e.g. thermocouple EMF, strain-gauge resistance).
2. **Variable conversion element** – converts the sensing-element output into a more convenient signal, usually a DC voltage, current or frequency (e.g. a Wheatstone bridge).
3. **Signal conditioning / processing element** – amplifies, filters, linearises, compensates and digitises the signal (amplifier, filter, ADC, microcontroller).
4. **Data presentation element** – displays, records or transmits the value to an observer or to a controller (display, data logger, fieldbus).

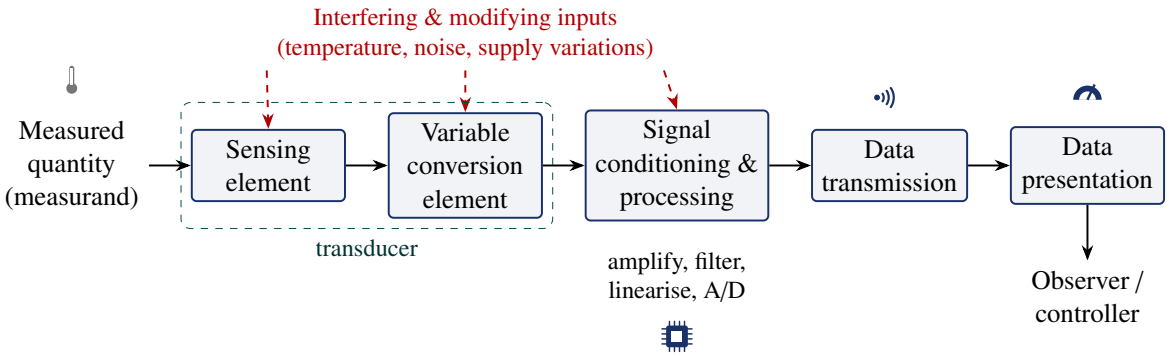


Figure 1.9: Generalised measurement system with interfering and modifying inputs.

Example – robot wrist force sensor. Measurand: contact force. Sensing element: elastic aluminium flexure (force → strain). Variable conversion: bonded strain gauges in Wheatstone bridges (strain → millivolts). Signal conditioning: instrumentation amplifier, low-pass filter, 16-bit ADC, calibration matrix in a microcontroller. Presentation: force/torque values sent to the robot controller over EtherCAT.

Interfering and modifying inputs. Besides the measurand, a real system responds to unwanted inputs. An *interfering input* I_I produces an output by itself (e.g. temperature producing

apparent strain in a strain gauge); a *modifying input* I_M changes the sensitivity to the measurand (e.g. supply-voltage variation changing bridge sensitivity). Good design minimises both by (i) inherent insensitivity, (ii) opposing (compensating) inputs, e.g. dummy gauges, (iii) high-gain feedback, and (iv) filtering.

1.8 Static Characteristics of Instruments

Static characteristics describe the relationship between output and input when the input is constant or changes slowly enough that transients have died out. They are found by *static calibration*: applying known, steady values of the measurand (from a standard of higher accuracy) and recording the outputs.

1.8.1 Range, span and the ideal straight line

- **Input range**: $I_{\min}$ to $I_{\max}$; **output range**: $O_{\min}$ to $O_{\max}$.
- **Span**: $I_{\max} - I_{\min}$ (input span) and $O_{\max} - O_{\min}$ (output span, full-scale output, FSO).
- **Ideal straight line**: the line joining $(I_{\min}, O_{\min})$ and $(I_{\max}, O_{\max})$:

$$O_{\text{ideal}} = KI + a, \qquad K = \frac{O_{\max} - O_{\min}}{I_{\max} - I_{\min}}, \qquad a = O_{\min} - KI_{\min}. \tag{1.9}$$

Example: a pressure transmitter with input range 0 bar to 10 bar and output 4 mA to 20 mA has $K = 1.6\,\text{mA/bar}$ and $a = 4\,\text{mA}$. The *live zero* of 4 mA allows a broken wire (0 mA) to be distinguished from zero pressure.

1.8.2 Accuracy and error

Error is the difference between the measured value O_m and the true value O_t:

$$e = O_m - O_t, \qquad \text{relative error} = \frac{O_m - O_t}{O_t} \times 100\%. \tag{1.10}$$

Accuracy is the closeness of agreement between a measured value and the true value. Since the true value is never known exactly, accuracy is specified as a bound on the error, expressed in one of three ways:

- in units of the measurand, e.g. ±0.5 °C;
- as a percentage of reading (% of rdg);
- as a percentage of full scale (% FS) or span. A % FS specification implies a *constant absolute* error over the range, so the *relative* error grows at low readings. **Rule:** choose an instrument whose range makes normal readings fall in the upper part of the scale.

Example 1.7: Accuracy expressed as percentage of full scale

Given: A temperature indicator with range 0 °C to 200 °C and accuracy ±1% of full scale.

Find: The possible error and relative error when it reads 50 °C, and when it reads 180 °C.

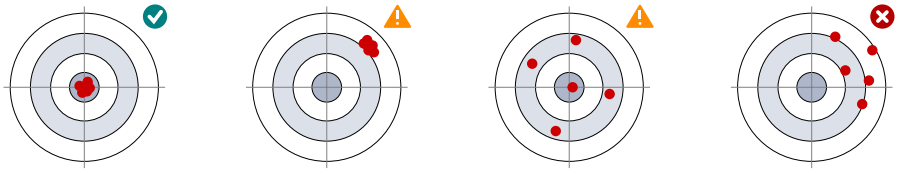
Solution: Absolute error bound = $0.01 \times 200 = \pm 2$ °C everywhere on the scale. At 50 °C: relative error = $\pm 2/50 = \pm 4\%$. At 180 °C: $\pm 2/180 = \pm 1.1\%$.

Interpretation: The same instrument is almost four times less accurate (relatively) near the bottom of its range.

1.8.3 Precision, repeatability and reproducibility

Precision describes the closeness of agreement among repeated measurements of the same quantity – it characterises *random* error, while accuracy also includes *systematic* error (bias). Figure 1.10 illustrates the distinction.

- **Repeatability** – precision under the same conditions (same observer, instrument, location, short time).
- **Reproducibility** – precision under changed conditions (different observers, instruments, laboratories, times).



(a) Accurate and precise

(b) Precise but not accurate (bias)

(c) Accurate on average, not precise

(d) Neither accurate nor precise

accuracy = closeness of the *mean* to the true value (centre); precision = closeness of readings to *each other* (spread)

Figure 1.10: Accuracy versus precision. Precision is the scatter of repeated readings; accuracy is their closeness to the true value (the centre).

Precision is quantified statistically. For n repeated readings $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}, \tag{1.11}$$

where $\bar{x}$ is the sample mean and s the sample standard deviation. The *bias* (systematic error) is $\bar{x} - x_{\text{true}}$.

Example 1.8: Accuracy, bias and precision from repeated readings

Given: A thermometer immersed in a calibration bath at exactly 100.0 °C gives eight readings

(°C): 101.2, 100.8, 101.5, 101.1, 100.9, 101.3, 101.0, 101.2.

Find: Mean, bias, standard deviation and standard uncertainty of the mean.

Solution:

1. Sum = 809.0; mean $\bar{x} = 809.0/8 = 101.125\,^{\circ}\text{C}$.
2. Bias = $101.125 - 100.0 = +1.125\,^{\circ}\text{C}$ (systematic error – the thermometer reads high).
3. Deviations $x_i - \bar{x}$: 0.075, −0.325, 0.375, −0.025, −0.225, 0.175, −0.125, 0.075. Sum of squares = 0.355.
4. $s = \sqrt{0.355/7} = 0.225\,^{\circ}\text{C}$.
5. Standard uncertainty of the mean (Type A, Section 1.8.9): $u = s/\sqrt{n} = 0.225/\sqrt{8} = 0.080\,^{\circ}\text{C}$.

Interpretation: The thermometer is *precise* ($s \approx 0.2\,^{\circ}\text{C}$) but *not accurate* (bias $\approx 1.1\,^{\circ}\text{C}$) – case (b) of Fig. 1.10. The bias can be removed by calibration (subtract 1.125 °C); the scatter cannot.

1.8.4 Sensitivity

Sensitivity is the change in output per unit change in input – the slope of the static characteristic:

$$K = \frac{\Delta O}{\Delta I} \quad \text{(linear)}, \qquad K(I) = \frac{dO}{dI} \quad \text{(non-linear)}. \tag{1.12}$$

Units are output units per input unit, e.g. mV/°C, V/mm, mV/V per unit strain. A high sensitivity is desirable, but it must not be achieved at the expense of range or stability. *Cross-sensitivity* (sensitivity to an interfering input, e.g. $\mu\varepsilon$ per °C) should be low.

Sensitivity drift (span drift) is a change in K caused by environmental conditions, and **zero drift (bias drift)** is a change in the output at zero input (Fig. 1.11(c),(d)). They are quoted, for example, as %FS per °C.

1.8.5 Linearity

Non-linearity is the maximum deviation of the calibration curve from a reference straight line, expressed as a percentage of FSO:

$$\text{Non-linearity} = \frac{\max |O(I) - O_{\text{ref}}(I)|}{O_{\max} - O_{\min}} \times 100\% \ \text{FSO}. \tag{1.13}$$

The reference line may be the end-point line (1.9), the best-fit (least-squares) line, or a line through zero; the specification must state which. The least-squares line is derived in Section 1.8.11.

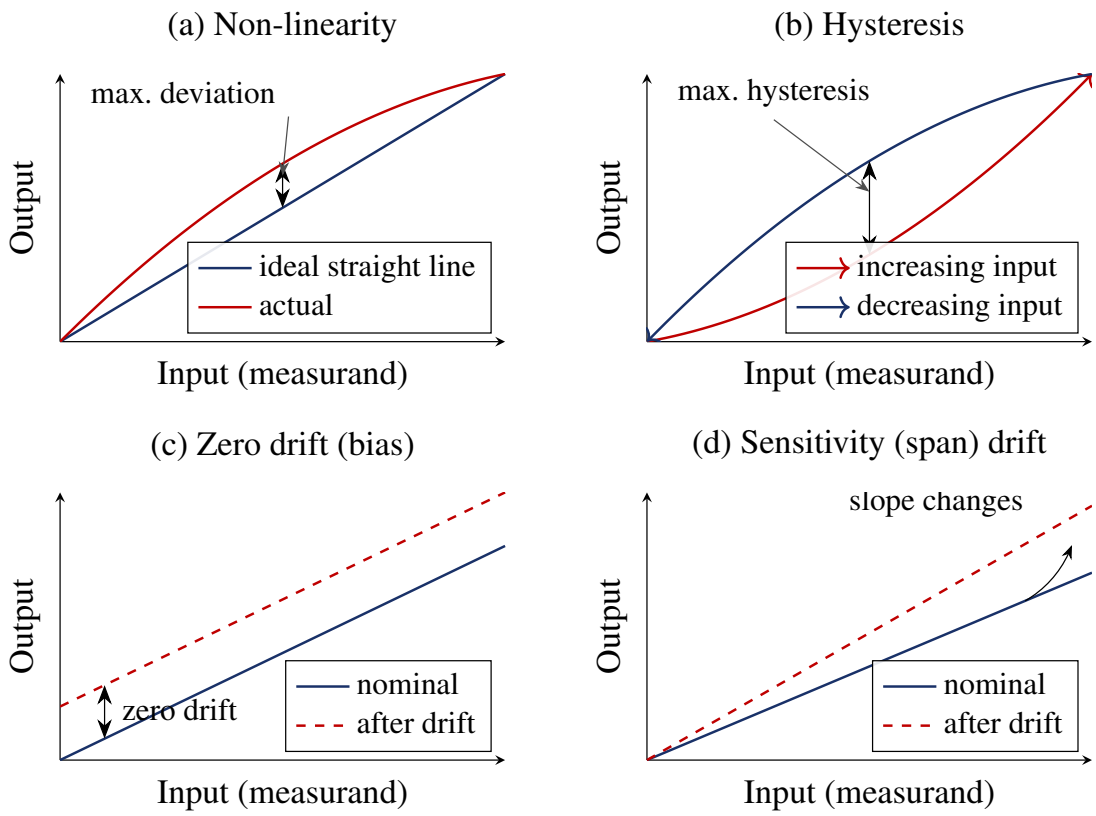


Figure 1.11: Static characteristics: (a) non-linearity, (b) hysteresis, (c) zero drift, (d) sensitivity drift.

1.8.6 Hysteresis, dead band and backlash

Hysteresis is the difference between output readings for the same input when the input is approached from increasing and from decreasing directions (Fig. 1.11(b)). Causes are friction, magnetic hysteresis, elastic hysteresis and backlash in gears:

$$\text{Hysteresis} = \frac{\max |O_{\uparrow}(I) - O_{\downarrow}(I)|}{O_{\max} - O_{\min}} \times 100\% \ \ \text{FSO}. \tag{1.14}$$

Dead band (dead space) is the range of input over which there is no change of output. Gear *backlash* in a robot joint produces a dead band between motor-side encoder and link; this is why precision robots use zero-backlash harmonic or cycloidal reducers.

1.8.7 Resolution and threshold

Resolution is the smallest change in input that produces a detectable change in output. **Threshold** is the smallest input, starting from zero, that produces a detectable output. For digital instruments, resolution equals one count (one least significant bit). For an N-bit converter of full-scale range FSR,

$$\text{Resolution} = \frac{\text{FSR}}{2^{N}}. \tag{1.15}$$

For example, a 12-bit ADC over 10 V resolves $10/4096 = 2.44$ mV (ADCs are treated fully in Unit II). For an incremental encoder the resolution is 360° divided by the number of counts per

revolution (Section 1.11.3).

Do not confuse resolution with accuracy. A digital display with five digits may resolve 0.01 °C while being in error by 1 °C.

1.8.8 Other static characteristics

- **Stability** – ability to retain characteristics over time (long-term drift, % per year).
- **Tolerance** – maximum deviation of a manufactured component from its nominal value (e.g. Pt100 class A tolerance).
- **Environmental effects** – effect of temperature, humidity, vibration, EMI on zero and sensitivity.
- **Input impedance / output impedance** – determine loading effects (Section 1.8.10).

1.8.9 Errors and measurement uncertainty

Classification of errors.

- **Gross errors** – human mistakes in reading, recording or connecting (avoided by care and automation).
- **Systematic errors** – repeatable bias from instrument imperfections (zero offset, wrong calibration), environmental effects, loading and method. They are reduced by calibration and compensation.
- **Random errors** – unpredictable variations from noise, fluctuations and resolution limits. They are reduced by averaging and quantified statistically.

Uncertainty. The *Guide to the Expression of Uncertainty in Measurement* (GUM) [18] defines uncertainty as a parameter characterising the dispersion of values that could reasonably be attributed to the measurand. It is evaluated by two methods:

- **Type A** – statistical analysis of repeated observations: $u_A = s/\sqrt{n}$.
- **Type B** – other information: calibration certificates, manufacturer's specification, resolution. A specification of $\pm a$ with a rectangular distribution gives $u_B = a/\sqrt{3}$; the quantisation error of a digital reading of resolution δ gives $u = \delta/(2\sqrt{3}) = \delta/\sqrt{12}$.

Independent standard uncertainties are combined by the root-sum-square (RSS) rule,

$$u_c = \sqrt{u_1^2 + u_2^2 + \cdots}.$$

The *expanded uncertainty* $U = k\,u_c$ with coverage factor $k = 2$ gives approximately 95% confidence.

Propagation of uncertainty. If a result $y = f(x_1, x_2, \ldots, x_n)$ is calculated from independent

measured quantities with uncertainties $u(x_i)$, a first-order Taylor expansion gives

$$u_c(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u^2(x_i)}. \tag{1.16}$$

Derivation. For small deviations δx_i, $\delta y \approx \sum_i (\partial f/\partial x_i)\,\delta x_i$. Squaring and taking the expectation, cross terms $E[\delta x_i\,\delta x_j]$ vanish for independent quantities, leaving $\sigma_y^2 = \sum_i (\partial f/\partial x_i)^2 \sigma_{x_i}^2$.

Two important special cases follow:

$$y = x_1 \pm x_2: \quad u_c(y) = \sqrt{u^2(x_1) + u^2(x_2)}, \tag{1.17}$$

$$y = c\,x_1^a x_2^b: \quad \frac{u_c(y)}{|y|} = \sqrt{\left(a\frac{u(x_1)}{x_1}\right)^2 + \left(b\frac{u(x_2)}{x_2}\right)^2}. \tag{1.18}$$

The *worst-case* (limiting) error simply adds the magnitudes: $\delta y/y = |a|\,\delta x_1/x_1 + |b|\,\delta x_2/x_2$. RSS is more realistic because independent errors are unlikely to reach their extremes simultaneously.

Algorithm 1.1 collects these rules into a procedure that can be applied to any indirect measurement, such as power computed from voltage and current, or link length computed from two encoder readings.

Algorithm 1.1 Evaluation of combined and expanded measurement uncertainty (GUM)

Require: Model $y = f(x_1, \ldots, x_N)$; repeated readings and/or specifications of each x_i; coverage factor k
Ensure: Estimate y, combined standard uncertainty $u_c(y)$, expanded uncertainty U

```
for each input quantity x_i do
    if repeated readings are available then
        x̄_i ← mean;  u(x_i) ← s/√n                          ▷ Type A
    else
        u(x_i) ← a/√3 for a ±a rectangular limit             ▷ Type B
    end if
    c_i ← ∂f/∂x_i evaluated at the estimates                 ▷ sensitivity coefficient
end for
y ← f(x̄_1, ..., x̄_N)
u_c(y) ← √(Σ_i c_i² u²(x_i))                                 ▷ Eq. (1.16)
U ← k u_c(y)                                                 ▷ k = 2 for ≈ 95%
return y ± U
```

Example 1.9: Uncertainty of electrical power

Given: Motor input power $P = VI$ with $V = 12.0\,\text{V} \pm 0.1\,\text{V}$ and $I = 2.50\,\text{A} \pm 0.03\,\text{A}$ (independent).

Find: P and its uncertainty by RSS and by worst-case.

Solution: $P = 12.0 \times 2.50 = 30.0\,\text{W}$. Relative uncertainties: $0.1/12.0 = 0.833\%$, $0.03/2.50 =$

1.20%.

$$\text{RSS: } \frac{u_P}{P} = \sqrt{0.833^2 + 1.20^2}\,\% = 1.46\% \;\Rightarrow\; u_P = \pm 0.44\,\text{W}.$$

$$\text{Worst case: } 0.833\% + 1.20\% = 2.03\% \;\Rightarrow\; \pm 0.61\,\text{W}.$$

Result: $P = 30.0\,\text{W} \pm 0.44\,\text{W}$ (RSS).

1.8.10 Loading effects

Connecting an instrument to a system extracts energy from it and therefore changes the quantity being measured. This is the *loading effect*. Examples: a thermometer cools the liquid it is dipped in; an accelerometer's mass changes the vibration of a light structure; a voltmeter draws current from a high-impedance circuit.

Electrical loading – derivation. Represent the measured system by its Thévenin equivalent (V_{Th}, R_{Th}) and the instrument by its input resistance R_m (Fig. 1.12). By voltage division,

$$V_m = V_{\text{Th}}\frac{R_m}{R_{\text{Th}} + R_m} \quad\Rightarrow\quad \text{loading error} = \frac{V_m - V_{\text{Th}}}{V_{\text{Th}}} = -\frac{R_{\text{Th}}}{R_{\text{Th}} + R_m}. \tag{1.19}$$

The error tends to zero as $R_m \gg R_{\text{Th}}$: *a voltage-measuring instrument must have an input impedance much higher than the output impedance of the source* (a factor of 100 gives about 1% error). Conversely a current-measuring instrument must have a very low input impedance. In general, the loading error is minimised by *impedance matching* for signal transfer, or by using a buffer (voltage follower) stage.

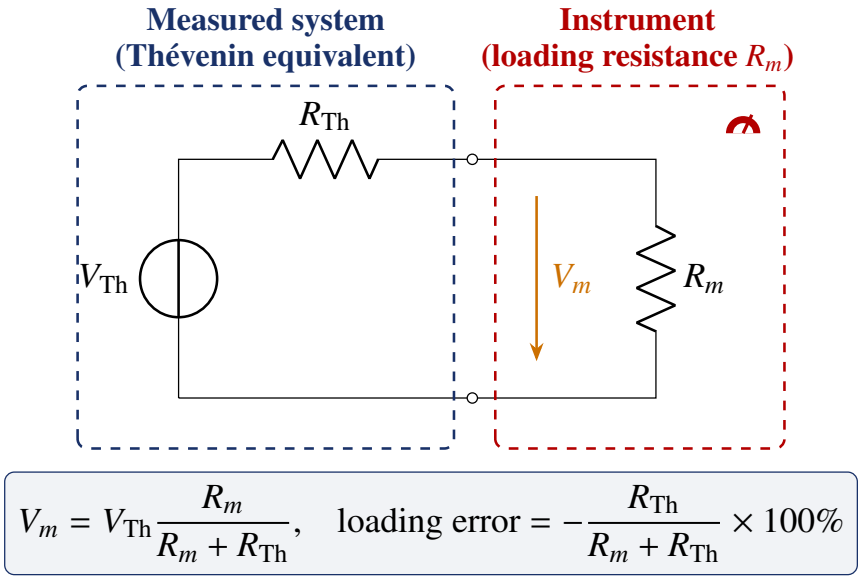


Figure 1.12: Electrical loading: the instrument resistance R_m forms a divider with the source resistance R_{Th}.

Example 1.10: Voltmeter loading

Given: A 12 V supply feeds two 100 kΩ resistors in series. The voltage across the lower resistor is measured with (a) a meter of input resistance 1 MΩ, (b) a DMM of 10 MΩ.

Solution: True voltage $V_{\mathrm{Th}} = 6\,\mathrm{V}$; source resistance $R_{\mathrm{Th}} = 100 \parallel 100 = 50\,\mathrm{k\Omega}$.

(a) $V_m = 6 \times \dfrac{1000}{1050} = 5.714\,\mathrm{V}$; error $= -4.76\%$.

(b) $V_m = 6 \times \dfrac{10000}{10050} = 5.970\,\mathrm{V}$; error $= -0.50\%$.

Interpretation: A tenfold increase in meter resistance reduces the loading error roughly tenfold.

Potentiometer loading is analysed in Section 1.11.1; **mechanical loading** (e.g. accelerometer mass) in Section 1.13.

1.8.11 Static calibration and least-squares fitting

Calibration establishes the relationship between instrument indication and the true value by comparison with a standard at least 3–10 times more accurate. A calibration curve is obtained by fitting a line $y = mx + c$ to n calibration points (x_i, y_i).

Derivation of the least-squares line. Choose m and c to minimise the sum of squared residuals $S(m,c) = \sum_{i=1}^{n} (y_i - mx_i - c)^2$. Setting $\partial S/\partial m = 0$ and $\partial S/\partial c = 0$:

$$\frac{\partial S}{\partial c} = -2\sum (y_i - mx_i - c) = 0 \;\Rightarrow\; \sum y_i = m\sum x_i + nc,$$
$$\frac{\partial S}{\partial m} = -2\sum x_i (y_i - mx_i - c) = 0 \;\Rightarrow\; \sum x_i y_i = m\sum x_i^2 + c\sum x_i.$$

Solving these *normal equations* simultaneously:

$$m = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - \left(\sum x_i\right)^2}, \qquad c = \frac{\sum y_i - m\sum x_i}{n} = \bar{y} - m\bar{x}. \tag{1.20}$$

The slope m is the calibrated sensitivity; c is the zero offset. The quality of fit is described by the coefficient of determination $R^2 = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2$ and by the maximum residual (non-linearity).

Calibration procedure. The complete procedure is summarised as Algorithm 1.2 and as the flowchart of Fig. 1.13. Two points deserve emphasis. First, the inputs should be applied in both increasing and decreasing order, because only then can hysteresis be separated from non-linearity. Second, the result of calibration is used in *inverse* form: in service the instrument delivers y and the controller must compute $x = (y - c)/m$.

Algorithm 1.2 Static calibration by least squares

Require: Reference inputs $x_1, \dots, x_n$ from a standard; measured outputs $y_1, \dots, y_n$; tolerance $\delta_{\max}$ (%FSO)
Ensure: Sensitivity m, offset c, non-linearity N_L, inverse calibration function

$S_x \leftarrow \sum x_i,\ S_y \leftarrow \sum y_i,\ S_{xx} \leftarrow \sum x_i^2,\ S_{xy} \leftarrow \sum x_i y_i$
$m \leftarrow (n S_{xy} - S_x S_y)/(n S_{xx} - S_x^2)$ ▷ slope, Eq. (1.20)
$c \leftarrow (S_y - m S_x)/n$ ▷ intercept
for $i = 1$ **to** n **do**
    $e_i \leftarrow y_i - (m x_i + c)$ ▷ residual
end for
FSO $\leftarrow m\,(x_{\max} - x_{\min});\ N_L \leftarrow 100 \max |e_i|/\text{FSO}$
if $N_L \le \delta_{\max}$ **then**
    return m, c and $x = (y - c)/m$
else
    Adjust the instrument or fit a higher-order polynomial and repeat
end if

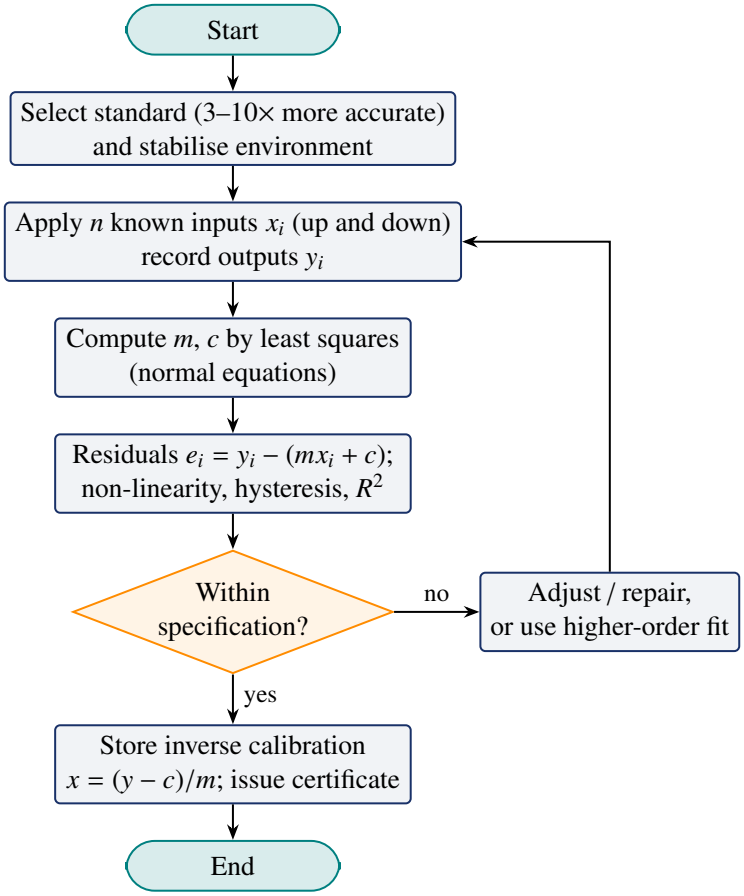


Figure 1.13: Flowchart of the static calibration procedure (Algorithm 1.2).

Example 1.11: Calibration of a displacement sensor

Given: Calibration of a potentiometric displacement sensor against slip gauges:

x (mm)	0	2	4	6	8	10
y (V)	0.05	1.02	2.06	2.95	4.03	5.01

Find: Least-squares sensitivity and offset, and the non-linearity in %FSO.

Solution: $n = 6$, $\sum x = 30$, $\sum y = 15.12$, $\sum x^2 = 220$, $\sum xy = 110.32$.

$$m = \frac{6(110.32) - 30(15.12)}{6(220) - 30^2} = \frac{661.92 - 453.60}{1320 - 900} = \frac{208.32}{420} = 0.4960\ \text{V/mm},$$

$$c = \frac{15.12 - 0.4960 \times 30}{6} = \frac{0.24}{6} = 0.040\ \text{V}.$$

Residuals $y_i - (mx_i + c)$: $+0.010, -0.012, +0.036, -0.066, +0.022, +0.010$ V. Maximum $|-0.066|$ V at $x = 6$ mm. Fitted FSO $= 0.496 \times 10 = 4.96$ V.

$$\text{Non-linearity} = \frac{0.066}{4.96} \times 100 = 1.33\%\ \text{FSO}.$$

Result: $y = 0.496x + 0.040$ (V, x in mm); sensitivity 0.496 V/mm; non-linearity $\approx 1.3\%$ FSO (Fig. 1.14). The inverse relation $x = (y - 0.040)/0.496$ is programmed into the controller to convert voltage to displacement.

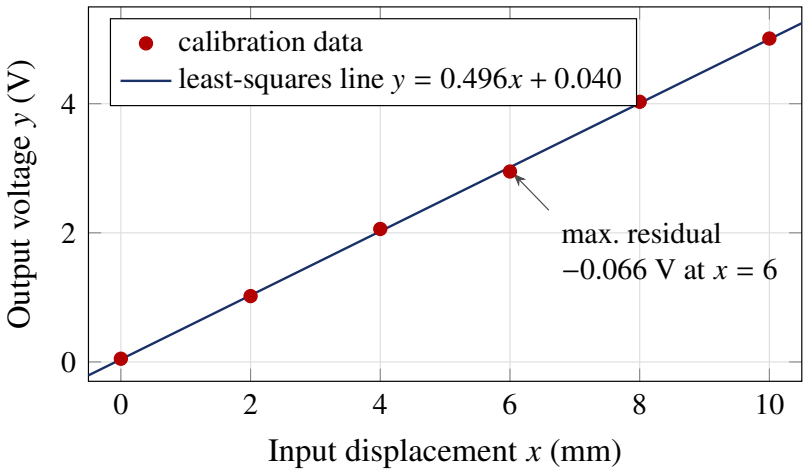


Figure 1.14: Calibration data and least-squares line for Example 1.11.

1.9 Dynamic Characteristics of Instruments

When the measurand changes with time, an instrument does not respond instantaneously because of its mass, thermal capacity, inductance or capacitance. *Dynamic characteristics* de-

scribe this time-dependent behaviour: speed of response, fidelity, lag and dynamic error. They are essential in robotics, where joint velocities, contact forces and vibrations change in milliseconds.

1.9.1 General mathematical model

Most instruments are adequately described by a linear ordinary differential equation with constant coefficients relating output $y(t)$ to input $u(t)$ [12]:

$$a_n \frac{d^n y}{dt^n} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 u. \tag{1.21}$$

The order n classifies the instrument. The standard test inputs are the step, ramp, impulse and sinusoid.

1.9.2 Zero-order instrument

With $n = 0$: $a_0 y = b_0 u \Rightarrow y = Ku$, $K = b_0/a_0$ (static sensitivity). The output follows the input instantly with no distortion. A potentiometer is approximately zero-order (neglecting wiper inertia and inductance).

1.9.3 First-order instrument

With $n = 1$: $a_1 \dot{y} + a_0 y = b_0 u$. Dividing by a_0,

$$\tau \frac{dy}{dt} + y = Ku, \qquad \tau = \frac{a_1}{a_0}\ \text{(time constant, s)}, \qquad K = \frac{b_0}{a_0}. \tag{1.22}$$

Physical example – thermometer. A sensor of mass m, specific heat c_p and surface area A with convection coefficient h is plunged into a fluid at temperature T_f. An energy balance gives $mc_p\, dT/dt = hA(T_f - T)$, i.e. $\tau\, dT/dt + T = T_f$ with $\tau = mc_p/(hA)$. A small, thin sensor in a well-stirred fluid is therefore fast.

Step response – derivation. For a step $u(t) = u_0$ applied at $t = 0$ with $y(0) = 0$, the solution of (1.22) is the sum of the particular solution Ku_0 and the homogeneous solution $Ce^{-t/\tau}$. Applying $y(0) = 0$ gives $C = -Ku_0$:

$$y(t) = Ku_0\left(1 - e^{-t/\tau}\right). \tag{1.23}$$

At $t = \tau$ the output has reached $1 - e^{-1} = 63.2\%$ of its final value; at $t = 4\tau$, 98.2%; at 5τ, 99.3% (Fig. 1.15). The *settling time* to within 2% is 4τ; the *rise time* (10% to 90%) is $\tau \ln 9 \approx 2.2\tau$.

Frequency response. For $u = u_0 \sin \omega t$, substituting $s = j\omega$ into the transfer function $G(s) = K/(\tau s + 1)$ gives the steady-state amplitude ratio and phase:

$$\frac{|Y|}{K|U|} = \frac{1}{\sqrt{1 + (\omega\tau)^2}}, \qquad \phi = -\tan^{-1}(\omega\tau). \tag{1.24}$$

At $\omega = 1/\tau$ (the corner or cut-off frequency) the amplitude ratio is $1/\sqrt{2}$ (-3 dB) and the phase lag is 45°.

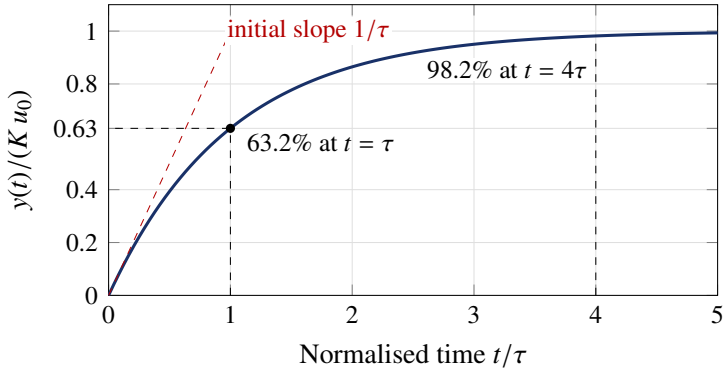


Figure 1.15: Step response of a first-order instrument.

Example 1.12: Thermometer step response

Given: A thermometer with $\tau = 8$ s at 25 °C is suddenly placed in water at 100 °C.

Find: (a) reading after 10 s; (b) time to read 99 °C.

Solution: Step size = 75°C. (a) $T(10) = 25 + 75\left(1 - e^{-10/8}\right) = 25 + 75(1 - 0.2865) = 78.5\,^{\circ}\text{C}$; dynamic error = 21.5 °C. (b) $99 = 25 + 75(1 - e^{-t/8}) \Rightarrow e^{-t/8} = 1/75 \Rightarrow t = 8 \ln 75 = 34.5$ s.

Interpretation: Readings taken before about 4τ–5τ are seriously in error. A robot that uses a temperature reading for process control must allow for this lag.

Example 1.13: Dynamic error with a sinusoidal input

Given: A force sensor with conditioning filter behaves as a first-order system with $\tau = 2$ ms. The force varies sinusoidally at 50 Hz.

Solution: $\omega\tau = 2\pi(50)(0.002) = 0.628$. Amplitude ratio $= 1/\sqrt{1 + 0.628^2} = 0.847$; phase $= -\tan^{-1} 0.628 = -32.1^{\circ}$.

Interpretation: The indicated amplitude is 15.3% low and lags by 32° (1.8 ms). To keep amplitude error below 2%, $\omega\tau \le 0.2$, i.e. $f \le 16$ Hz for this sensor.

1.9.4 Second-order instrument

With $n = 2$: $a_2\ddot{y} + a_1\dot{y} + a_0 y = b_0 u$, written in standard form

$$\frac{1}{\omega_n^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega_n}\frac{dy}{dt} + y = Ku, \qquad \omega_n = \sqrt{\frac{a_0}{a_2}}, \quad \zeta = \frac{a_1}{2\sqrt{a_0 a_2}}, \quad K = \frac{b_0}{a_0}, \tag{1.25}$$

where ω_n is the undamped natural frequency (rad/s) and ζ the damping ratio (dimensionless). **Physical example**: a spring–mass–damper force sensor, $m\ddot{x} + c\dot{x} + kx = F$, gives $\omega_n = \sqrt{k/m}$,

$\zeta = c/(2\sqrt{km})$, $K = 1/k$. Accelerometers, galvanometers, load cells and robot joints with flexible transmissions are second-order.

Step response. The roots of the characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ are $s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$, giving three cases:

- *Under-damped* ($0 < \zeta < 1$): oscillatory response at the damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$:

$$\frac{y(t)}{Ku_0} = 1 - e^{-\zeta\omega_n t}\left[\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right]. \tag{1.26}$$

- *Critically damped* ($\zeta = 1$): $y/(Ku_0) = 1 - (1 + \omega_n t)e^{-\omega_n t}$ – fastest response without overshoot.
- *Over-damped* ($\zeta > 1$): sluggish, non-oscillatory sum of two exponentials.

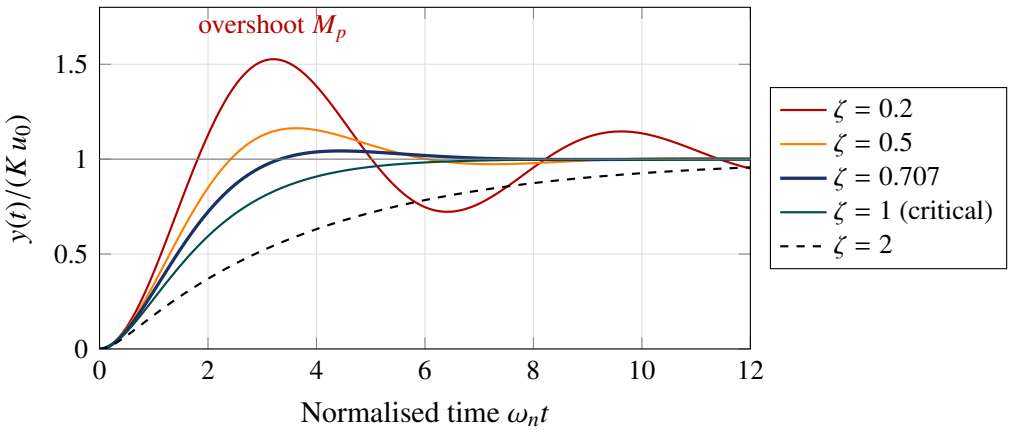


Figure 1.16: Step responses of a second-order instrument for various damping ratios.

Performance indices (under-damped).

$$\text{Peak time } t_p = \frac{\pi}{\omega_d}, \quad \text{Maximum overshoot } M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \quad \text{Settling time (2\%) } t_s \approx \frac{4}{\zeta\omega_n}. \tag{1.27}$$

Instruments are commonly designed with $\zeta \approx 0.6$–0.7: overshoot is small (about 5% at $\zeta = 0.7$), settling is fast, and – as shown next – the amplitude response is flat over the widest band.

Frequency response. With $r = \omega/\omega_n$,

$$\frac{|Y|}{K|U|} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad \phi = -\tan^{-1}\frac{2\zeta r}{1-r^2}. \tag{1.28}$$

For small ζ a resonant peak occurs near $r = 1$. For $\zeta \approx 0.7$ the amplitude ratio stays within a few percent of unity up to $r \approx 0.4$, and the phase is nearly proportional to frequency, so complex waveforms are reproduced without distortion.

Example 1.14: Second-order force sensor

Given: A wrist force sensor has $f_n = 500\,\text{Hz}$ and $\zeta = 0.4$.

Find: Overshoot, peak time and 2% settling time for a step force.

Solution: $\omega_n = 2\pi(500) = 3141.6\,\text{rad/s}$; $\sqrt{1-\zeta^2} = 0.9165$; $\omega_d = 2879.3\,\text{rad/s}$.

$$M_p = \exp\!\left(-\frac{0.4\pi}{0.9165}\right) = e^{-1.371} = 0.254\ (25.4\%),$$

$$t_p = \frac{\pi}{2879.3} = 1.09\,\text{ms}, \qquad t_s = \frac{4}{0.4 \times 3141.6} = 3.18\,\text{ms}.$$

Interpretation: During a collision the indicated force momentarily overshoots by 25%; a force controller using this signal must be tuned accordingly, or the sensor damping increased.

Key points

- First order: $\tau\dot{y} + y = Ku$; 63.2% at $t = \tau$; -3 dB at $\omega = 1/\tau$.
- Second order: ω_n, ζ; $M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$; best flat response at $\zeta \approx 0.7$.
- Fidelity requires the instrument bandwidth to be well above the highest frequency in the measurand.

1.10 Sensors and Transducers for Robotics

1.10.1 Classification of robotic sensors

Robot sensors are grouped by *what* they measure and *where* [7, 8]:

- **Internal (proprioceptive) sensors** measure the robot's own state: joint position (encoders, resolvers, potentiometers), joint velocity (tachogenerators, encoder differentiation), acceleration (IMUs), motor current and joint torque.
- **External (exteroceptive) sensors** measure the environment: *contact* sensors (limit switches, tactile arrays, force/torque sensors) and *non-contact* sensors (proximity, range, vision).

Desirable characteristics of a robotic sensor: adequate accuracy and resolution, wide dynamic range and bandwidth, linearity, low hysteresis, robustness to vibration, temperature and EMI, small size and mass (so as not to load the arm), reliability, ease of interfacing and low cost.

1.11 Position and Displacement Sensors

Table 1.6: Sensors used in robotics, arranged by measurand.

Measurand	Sensors	Typical robotic use
Position / displacement	Potentiometer, LVDT, incremental & absolute encoder, resolver, Hall-effect, capacitive, laser triangulation	Joint angle, gripper opening, part dimension
Velocity	DC/AC tachogenerator, encoder (frequency or period), resolver-to-digital converter	Joint speed loop, conveyor tracking
Acceleration	Piezoelectric, piezoresistive, capacitive MEMS accelerometer, IMU	Vibration monitoring, collision detection, mobile-robot attitude
Force / torque	Strain-gauge load cell, 6-axis F/T sensor, piezoelectric, joint torque sensor, motor current	Assembly, grinding, collaborative safety
Proximity / range	Inductive, capacitive, magnetic (reed, Hall), photoelectric, ultrasonic, LiDAR	Part presence, obstacle avoidance, end-of-travel
Touch / tactile	Microswitches, tactile arrays (resistive, capacitive)	Grasp detection, slip sensing

1.11.1 Resistive potentiometer

Construction and principle. A resistive track (wire-wound, cermet or conductive plastic) of total resistance R_p carries an excitation voltage V_s. A wiper attached to the moving member taps off a fraction x of the track ($0 \le x \le 1$), linear for translational and angular for rotary pots (Fig. 1.17(a)). With no load, the output is a simple voltage divider:

$$V_o = x\,V_s, \qquad x = \frac{d}{L}\ \text{(linear) or}\ \frac{\theta}{\theta_{\max}}\ \text{(rotary)}. \tag{1.29}$$

Loading – derivation. If a load R_L (e.g. an ADC input) is connected, the lower section xR_p is in parallel with R_L:

$$R_{\parallel} = \frac{xR_pR_L}{xR_p + R_L}, \qquad V_o = V_s\frac{R_{\parallel}}{(1-x)R_p + R_{\parallel}}.$$

Substituting and dividing numerator and denominator by R_L:

$$\frac{V_o}{V_s} = \frac{x}{1 + \dfrac{R_p}{R_L}\,x(1-x)}. \tag{1.30}$$

The output is no longer linear (Fig. 1.17(b)). The loading error $V_o - xV_s$ is zero at both ends and maximum near $x \approx 2/3$; for $R_L \gg R_p$ the maximum error is approximately $-0.15(R_p/R_L)\,V_s$. The design rule is $R_L \ge 10R_p$, which conflicts with the wish for a high R_p to limit self-heating – a buffer amplifier resolves the conflict.

Resolution. A wire-wound pot of n turns changes in steps of V_s/n; conductive-plastic pots have essentially infinite (stepless) resolution.

Advantages: simple, cheap, absolute output, no electronics needed. **Limitations:** wear and finite life, contact noise, friction loading, limited speed. Used in low-cost and educational robots, servo motors for hobby robots, and for gripper position.

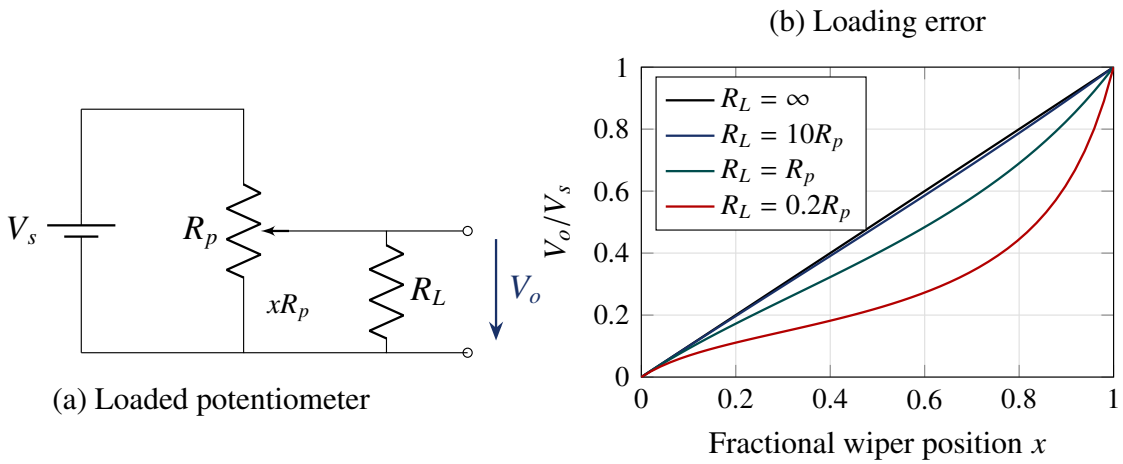


Figure 1.17: (a) Potentiometer with load R_L; (b) effect of loading on the output characteristic.

Example 1.15: Loading of a linear potentiometer

Given: A 100 mm stroke linear potentiometer, $R_p = 10\,\text{k}\Omega$, $V_s = 10\,\text{V}$, connected to a recorder with $R_L = 50\,\text{k}\Omega$. Wiper at 60 mm.

Find: Ideal and actual output, error in %FS.

Solution: $x = 0.6$, $R_p/R_L = 0.2$. Ideal: $V_o = 0.6 \times 10 = 6.000\,\text{V}$.

$$V_o = \frac{10 \times 0.6}{1 + 0.2 \times 0.6 \times 0.4} = \frac{6}{1.048} = 5.725\,\text{V}.$$

Error $= 5.725 - 6.000 = -0.275\,\text{V} = -2.75\%$ FS, corresponding to a displacement error of -2.75 mm. (Numerically, the maximum error, at $x = 0.67$, is $-0.284\,\text{V}$.)

Interpretation: To bring the error below 0.5% FS a buffer or $R_L \geq 300\,\text{k}\Omega$ is required.

1.11.2 Linear variable differential transformer (LVDT)

Construction. An LVDT (Fig. 1.18(a)) has one primary winding P and two identical secondary windings S_1, S_2 placed symmetrically on a hollow former, and a movable soft-iron core attached to the measured member by a non-magnetic push rod. The primary is excited by an AC voltage, typically 1 V to 10 V rms at 1 kHz to 10 kHz.

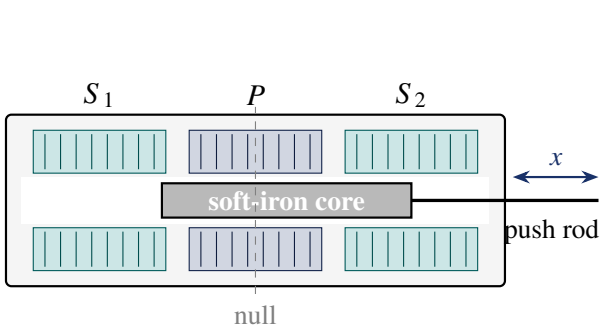
Principle. The core couples flux from the primary into the secondaries. The secondaries are connected in *series opposition*, so the output is the difference of their induced EMFs:

$$V_o = E_{S1} - E_{S2}. \tag{1.31}$$

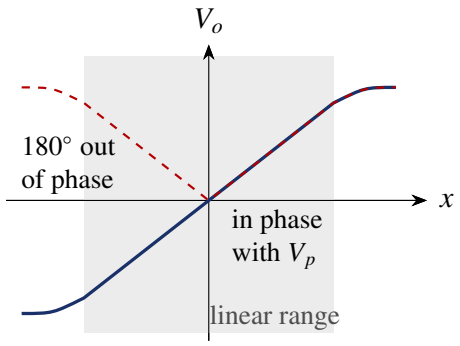
- At the null (central) position, mutual inductances are equal, $E_{S1} = E_{S2}$ and $V_o = 0$ (ideally; a small residual null voltage exists in practice).

- When the core moves towards S_1, E_{S1} increases and E_{S2} decreases; V_o is in phase with the primary voltage.
- When the core moves towards S_2, V_o increases in magnitude but is 180° out of phase.

Within the linear range the rms output magnitude is proportional to displacement, $|V_o| = K|x|$, and the *phase* gives the direction. A *phase-sensitive demodulator* (synchronous rectifier) followed by a low-pass filter produces a signed DC voltage proportional to x (Fig. 1.18(b)).



(a) Construction: primary P and two series-opposed secondaries (core displaced towards S_2)



phase-sensitive (signed) output
rms magnitude $|V_o|$

(b) Output characteristic, $V_o = K\,x$

Figure 1.18: LVDT: (a) construction; (b) output magnitude (dashed) and phase-sensitive (signed) output.

Characteristics. Frictionless (no contact between core and coils), hence infinite resolution, very long life and high repeatability; ranges from ±0.25 mm to ±250 mm; linearity typically 0.25% FS; robust in harsh environments. **Limitations:** need for AC excitation and demodulation electronics; sensitivity to stray magnetic fields; dynamic response limited to roughly one-tenth of the excitation frequency. Rotary variable differential transformers (RVDTs) apply the same principle to angles.

Example 1.16: LVDT measurement

Given: An LVDT with demodulated sensitivity 0.25 V/mm gives an output of −1.60 V.

Solution: $x = V_o/K = -1.60/0.25 = -6.4$ mm, i.e. 6.4 mm from null towards S_2 (negative sign from the 180° phase). If the output were read by a meter with resolution 1 mV, the displacement resolution of the system would be $0.001/0.25 = 4\,\mu$m – the LVDT itself is not the limiting factor.

1.11.3 Optical encoders

Encoders are the dominant position sensors in electric robot joints because their output is inherently digital, accurate and immune to drift.

1.11.3.1 Incremental encoder

Construction. A disc with N equally spaced transparent slots (or a reflective pattern) rotates between a light source (LED) and photodetectors (Fig. 1.19(a),(b)). Two tracks, A and B, are offset by a quarter of a slot pitch, so their square-wave outputs are in *quadrature* (90° electrical phase difference). A third track, Z (index), gives one pulse per revolution as a reference.

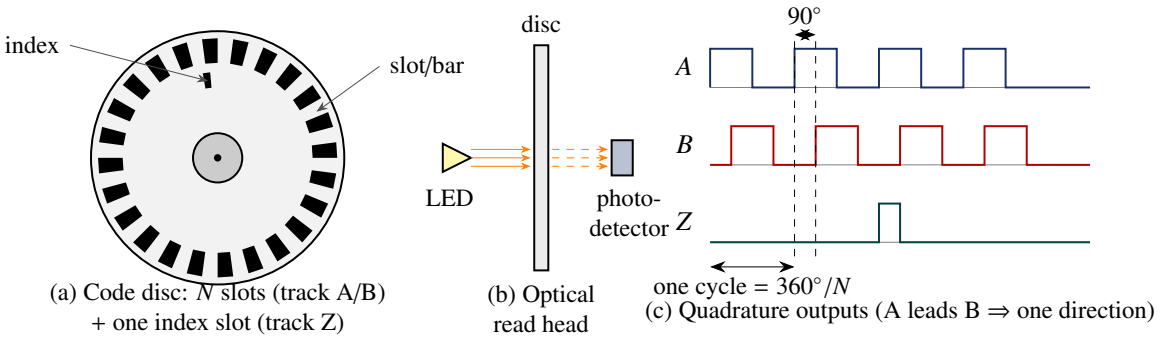


Figure 1.19: Incremental optical encoder: (a) code disc, (b) read head, (c) quadrature outputs A, B and index Z.

Direction sensing. If A leads B the shaft rotates in one direction; if B leads A, in the other. A quadrature decoder counts up or down accordingly.

Resolution and quadrature multiplication. Counting every rising and falling edge of both A and B gives four counts per slot ("×4 decoding"):

$$\Delta\theta = \frac{360^\circ}{4N} \quad \text{(rad: } 2\pi/4N\text{)}. \tag{1.32}$$

If the encoder is on the motor side of a gear reducer of ratio n_g, the joint resolution improves by n_g, and the resulting resolution at the end of a link of length L is $L\,\Delta\theta/n_g$ (radians).

Velocity from an encoder. Two methods are used:

- *Frequency (M) method* – count Δn pulses in a fixed sample time T_s: $\omega = 2\pi\Delta n/(4NT_s)$. Good at high speed.
- *Period (T) method* – measure the time between successive edges with a high-frequency clock. Good at low speed.

Limitation. An incremental encoder gives only *relative* position: on power-up the count is unknown and the robot must perform a homing (referencing) routine using the index pulse or a limit switch. Lost counts due to noise accumulate as errors.

Example 1.17: Encoder resolution and velocity

Given: A 1024-line incremental encoder with ×4 decoding is mounted on a joint motor driving the joint through a 100:1 harmonic drive. The link is 0.8 m long. In a sampling interval of 10 ms the counter advances by 2048 counts.

Find: (a) motor-shaft resolution; (b) joint resolution; (c) linear resolution at the link tip; (d) motor speed.

Solution:

(a) Counts per revolution = $4 \times 1024 = 4096$; $\Delta\theta_m = 360^\circ/4096 = 0.0879^\circ$.

(b) $\Delta\theta_j = 0.0879^\circ/100 = 0.000879^\circ = 1.534 \times 10^{-5}$ rad.

(c) Tip resolution = $0.8 \times 1.534 \times 10^{-5} = 1.23 \times 10^{-5}$ m $\approx 12.3\,\mu$m.

(d) $2048/4096 = 0.5$ rev in 10 ms $\Rightarrow$ 50 rev/s = 3000 rpm (314.2 rad/s).

Interpretation: Gear reduction makes encoder resolution a non-issue at the joint; robot accuracy is instead limited by gear transmission error, link deflection and calibration.

1.11.3.2 Absolute encoder

An absolute encoder has n concentric tracks, each read by its own detector, so that every angular sector has a unique n-bit code. Position is known immediately at power-up, with resolution $360^\circ/2^n$. *Multi-turn* absolute encoders add gear-coupled discs or battery-backed counters to count revolutions – standard on modern industrial robots, which therefore need no homing.

Gray code. With natural binary, several bits change at once at some transitions (e.g. 0111 → 1000); slight misalignment of detectors can produce large spurious readings. Gray (reflected binary) code changes *only one bit* between adjacent positions, eliminating this ambiguity. Conversion rules for n bits (b_{n-1} = most significant):

$$\text{Binary} \rightarrow \text{Gray:} \quad g_{n-1} = b_{n-1}, \qquad g_i = b_{i+1} \oplus b_i, \tag{1.33}$$

$$\text{Gray} \rightarrow \text{Binary:} \quad b_{n-1} = g_{n-1}, \qquad b_i = b_{i+1} \oplus g_i, \tag{1.34}$$

where $\oplus$ denotes exclusive-OR.

Table 1.7: 3-bit binary and Gray codes for an 8-sector absolute encoder (45° per sector).

Sector	Angle	Binary	Gray	Sector	Angle	Binary	Gray
0	0°–45°	000	000	4	180°–225°	100	110
1	45°–90°	001	001	5	225°–270°	101	111
2	90°–135°	010	011	6	270°–315°	110	101
3	135°–180°	011	010	7	315°–360°	111	100

Example 1.18: Absolute encoder design and Gray-code conversion

(a) How many tracks are needed for a resolution better than 0.01°?
$2^n \geq 360/0.01 = 36\,000 \Rightarrow n \geq \log_2 36\,000 = 15.14 \Rightarrow n = 16$ bits, giving $360/65\,536 = 0.0055^\circ$.

(b) Convert binary 1011 to Gray: $g_3 = 1$; $g_2 = 1 \oplus 0 = 1$; $g_1 = 0 \oplus 1 = 1$; $g_0 = 1 \oplus 1 = 0 \Rightarrow$ Gray 1110.

(c) Convert Gray 1110 back: $b_3 = 1$; $b_2 = 1 \oplus 1 = 0$; $b_1 = 0 \oplus 1 = 1$; $b_0 = 1 \oplus 0 = 1 \Rightarrow 1011$ ✓.

1.11.4 Resolvers

A resolver is a rotary transformer: a rotor winding excited at $V_r \sin \omega t$ induces in two stator windings at 90° the voltages $V_{\sin} = kV_r \sin\theta \sin \omega t$ and $V_{\cos} = kV_r \cos\theta \sin \omega t$. The angle is recovered as $\theta = \operatorname{atan2}(V_{\sin}, V_{\cos})$, usually by a resolver-to-digital converter (RDC) that also outputs velocity. Resolvers have no optics or electronics in the motor, so they tolerate high temperature, shock and contamination; they are used in EV traction motors and heavy-duty robot axes.

1.11.5 Hall-effect and capacitive position sensors

Hall effect. When a current I flows through a thin conductor or semiconductor of thickness t in a magnetic flux density B perpendicular to it, the Lorentz force deflects charge carriers and produces a transverse voltage

$$V_H = \frac{IB}{nqt} = R_H \frac{IB}{t}, \tag{1.35}$$

where n is the carrier density, q the electron charge and $R_H = 1/(nq)$ the Hall coefficient. Hall sensors measure the position of a magnet: as linear sensors (gripper position), as switches (end-of-travel, BLDC commutation, Unit II) and as magnetic rotary encoders.

Capacitive sensors. A parallel-plate capacitor has

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}, \tag{1.36}$$

with $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, relative permittivity ε_r, overlap area A and gap d. Displacement can change d (non-linear, high sensitivity $dC/dd = -C/d$), A (linear), or ε_r (level and proximity sensing). Capacitive sensors achieve nanometre resolution in precision metrology and are used as touch and proximity sensors on cobot skins.

1.12 Velocity Sensors

1.12.1 DC tachogenerator

A tachogenerator is a small permanent-magnet DC generator coupled to the shaft. Faraday's law gives a generated EMF proportional to speed:

$$V_t = K_t \, \omega, \tag{1.37}$$

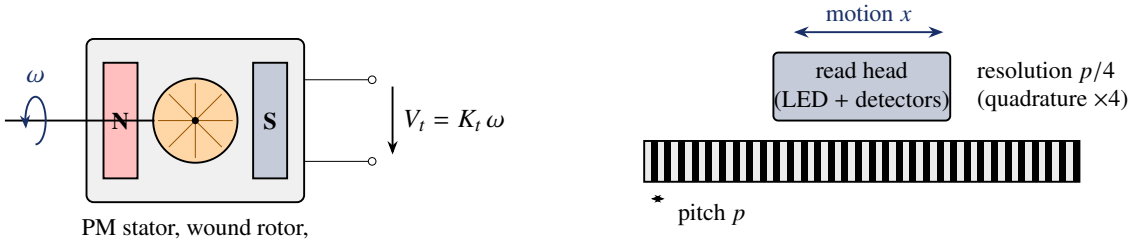
where K_t is the tachometer constant (V/(rad/s), or commonly V per 1000 rpm). Polarity reverses with direction. Ripple from commutation must be filtered. AC tachogenerators (drag-cup or permanent-magnet) give a voltage whose amplitude (and, for PM types, frequency) is proportional to speed.

Example 1.19: Tachogenerator

Given: $K_t = 7$ V per 1000 rpm; output 10.5 V.

Solution: $N = 10.5/7 \times 1000 = 1500$ rpm; $\omega = 1500 \times 2\pi/60 = 157.1$ rad/s.

Construction. Figure 1.20(a) shows the arrangement: a permanent-magnet stator provides a constant field, the rotor carries the armature winding coupled to the joint shaft, and a commutator with brushes rectifies the induced EMF. Because the field is constant, the EMF depends only on speed. Practical limitations are commutator ripple (a few percent of the output), brush wear, and the additional inertia added to the joint; for these reasons tachogenerators have largely been replaced by encoder-based velocity estimation in modern robots, although they remain common in older servo drives and in teaching laboratories.



(a) DC tachogenerator (b) Linear optical encoder (scale + read head)

Figure 1.20: (a) DC tachogenerator coupled to a shaft; (b) linear optical encoder used for prismatic axes and for velocity measurement of linear motion.

Linear velocity and linear encoders. For prismatic axes (gantry and Cartesian robots, linear motors) a *linear encoder* – a glass or steel scale with a grating of pitch p and a read head (Fig. 1.20(b)) – gives position with resolution $p/4$ after quadrature decoding, and linear velocity is obtained exactly as for rotary encoders, $v = \Delta n\,(p/4)/T_s$. Scales with 20 µm pitch and electronic interpolation achieve sub-micrometre resolution.

1.12.2 Velocity from position

In modern robots velocity is usually estimated from the encoder (Section 1.11.3) or the resolver (RDC tracking loop). Simple backward differencing $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T_s$ amplifies quantisation noise; it is normally followed by low-pass filtering or replaced by an observer.

1.13 Acceleration Sensors

1.13.1 Seismic (spring–mass–damper) principle

Every accelerometer is a seismic instrument (Fig. 1.21(a)): a mass m is supported in a case by a spring of stiffness k and damping c. The case moves with the base displacement $x_b(t)$. Let $z = x_m - x_b$ be the relative displacement of mass and case.

Derivation. Newton's law for the mass: $m\ddot{x}_m = -kz - c\dot{z}$. Substituting $x_m = z + x_b$:

$$m\ddot{z} + c\dot{z} + kz = -m\ddot{x}_b \quad \Longrightarrow \quad \ddot{z} + 2\zeta\omega_n\dot{z} + \omega_n^2 z = -a_b(t), \tag{1.38}$$

with $\omega_n = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{km})$. For base accelerations whose frequency is well below ω_n, the derivative terms are negligible and

$$z \approx -\frac{a_b}{\omega_n^2} \qquad (\omega \ll \omega_n) : \tag{1.39}$$

the relative displacement is proportional to acceleration, with sensitivity $1/\omega_n^2$. The frequency response follows (1.28) (Fig. 1.21(b)): with $\zeta \approx 0.7$ the response is flat to within about 1% up to $r \approx 0.4$. A high ω_n gives wide bandwidth but low sensitivity – the fundamental trade-off of accelerometer design. (Conversely, if $\omega \gg \omega_n$ the mass stays still and $z \approx -x_b$: the instrument becomes a seismometer measuring displacement.)

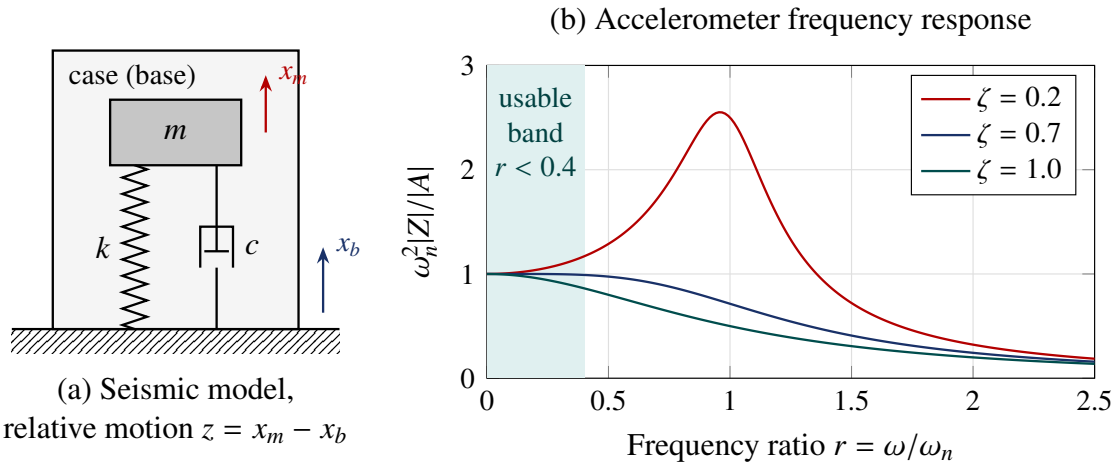


Figure 1.21: (a) Seismic accelerometer model; (b) normalised amplitude response showing the usable band below resonance.

1.13.2 Practical accelerometers

- **Piezoelectric** – the mass compresses a piezoelectric crystal (quartz or PZT) which generates charge $Q = d\,F = d\,m\,a$, where d is the piezoelectric charge coefficient (pC/N). Very high ω_n (tens of kHz), wide bandwidth; cannot measure static (DC) acceleration because charge leaks away; needs a charge amplifier.
- **Piezoresistive** – strain gauges on the spring element; measure down to DC; used for crash and shock.

- **Capacitive MEMS** – a micromachined silicon proof mass between fixed electrodes forms a differential capacitor; measures DC (gravity, tilt) and dynamic acceleration; low cost, integrated with gyroscopes in *inertial measurement units* (IMUs) used in mobile robots, drones and humanoids.

Example 1.20: Accelerometer design

Given: A seismic accelerometer with proof mass 10 g must have $f_n = 5$ kHz.

Find: Spring stiffness and relative deflection for an acceleration of $10g$ (98.1 m/s^2).

Solution: $\omega_n = 2\pi(5000) = 31\,416$ rad/s; $k = m\omega_n^2 = 0.01 \times 31416^2 = 9.87 \times 10^6$ N/m. $z = a/\omega_n^2 = 98.1/(9.87 \times 10^8) = 9.94 \times 10^{-8}$ m ≈ 0.1 µm.

Interpretation: The displacement is tiny, which is why piezoelectric or capacitive transduction (not a pointer) is required. With $\zeta = 0.7$ the usable range is up to about $0.4 f_n = 2$ kHz; with $\zeta \approx 0$ (typical of undamped piezo types) the response rises by about 4% already at $0.2 f_n$, so such sensors are used only up to roughly $0.2 f_n$.

1.14 Proximity and Range Sensors

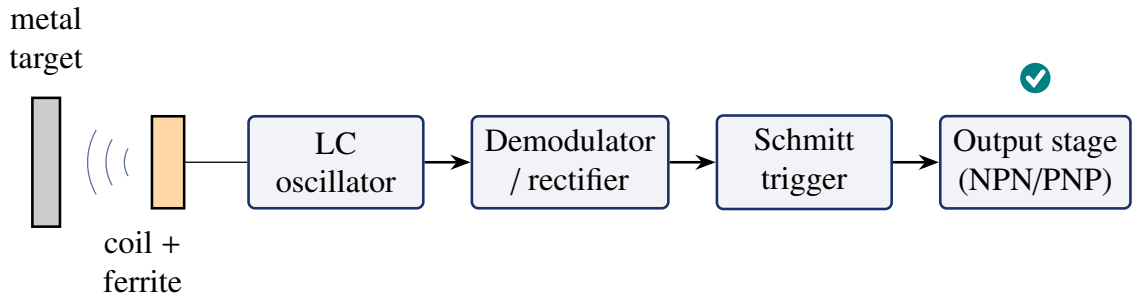
Proximity sensors detect the presence of an object within a defined distance without contact, typically giving a switched (ON/OFF) output; range sensors give the distance itself.

1.14.1 Inductive proximity sensor

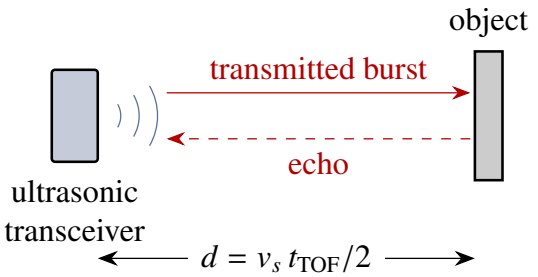
An LC oscillator drives a coil wound on a ferrite pot core that projects an alternating magnetic field from the sensing face (Fig. 1.22(a)). A *metallic* target entering the field carries eddy currents, which absorb energy and damp the oscillation. The demodulated amplitude falls below a Schmitt-trigger threshold and the output stage switches. Only metals are detected; the rated sensing distance S_n (typically 1 mm to 40 mm) applies to mild steel, and is reduced by a correction factor for non-ferrous metals such as aluminium, brass and copper (manufacturer data). Inductive sensors are robust and immune to dirt and oil – the standard sensor for part presence and end-of-travel on robotic cells. Hysteresis between switch-on and switch-off distances prevents chatter.

1.14.2 Capacitive proximity sensor

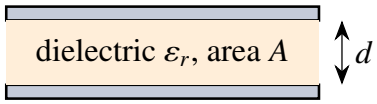
The sensing face is one plate of a capacitor; the object forms the other plate or changes the dielectric (Eq. (1.36)). Capacitive sensors detect metals *and* non-metals – plastics, wood, glass, liquids, human bodies – and are used for level detection and for human proximity on collaborative robots. Their sensitivity is adjusted by a potentiometer to suppress unwanted targets.



(a) Inductive proximity sensor: eddy currents in the target damp the oscillator



(b) Ultrasonic time-of-flight ranging



(c) Capacitive: $C = \varepsilon_0 \varepsilon_r A/d$

Figure 1.22: (a) Inductive proximity sensor; (b) ultrasonic time-of-flight ranging; (c) capacitive sensing principle.

1.14.3 Magnetic sensors

Reed switches (two ferromagnetic reeds in a glass tube) and *Hall switches* detect a permanent magnet, e.g. the piston magnet of a pneumatic cylinder to confirm gripper open/closed.

1.14.4 Photoelectric (optical) sensors

An emitter (LED or laser) and a receiver are arranged in three modes: *through-beam* (emitter and receiver facing each other; longest range, most reliable), *retro-reflective* (beam returned by a reflector) and *diffuse* (light scattered by the object itself). Laser *triangulation* sensors give distance from the position of the reflected spot on a detector array; *time-of-flight* and *LiDAR* sensors measure distance from the travel time of light and are used for navigation and safety scanners.

1.14.5 Ultrasonic sensors

A piezoelectric transducer emits a short burst at 40 kHz to 400 kHz; the echo from the target is received after the time of flight t_{TOF} (Fig. 1.22(b)). Since the sound travels to the target and back,

$$d = \frac{v_s\, t_{\mathrm{TOF}}}{2}, \qquad v_s \approx 331.3 + 0.606\, T \quad [\mathrm{m/s},\ T \text{ in } ^\circ\mathrm{C}]. \tag{1.40}$$

Ultrasonic sensors detect transparent and dark objects that defeat optical sensors, but they have a blind zone close to the transducer, a wide beam, and are affected by temperature and soft

absorbing surfaces.

Example 1.21: Ultrasonic ranging and temperature error

Given: An echo returns after $t_{\mathrm{TOF}} = 5.83\,\mathrm{ms}$. The controller assumes $v_s = 343\,\mathrm{m/s}$ (20 °C), but the cell is at 35 °C.

Solution: Computed distance $= 343 \times 5.83 \times 10^{-3}/2 = 1.000\,\mathrm{m}$. Actual speed $= 331.3 + 0.606 \times 35 = 352.5\,\mathrm{m/s}$; true distance $= 352.5 \times 5.83 \times 10^{-3}/2 = 1.028\,\mathrm{m}$.

Interpretation: The uncompensated reading is 2.7% low; ultrasonic rangers therefore include a temperature sensor for compensation.

Example 1.22: Capacitive displacement sensor

Given: Air-gap sensor, plate area $A = 4\,\mathrm{cm}^2$, gap $d = 1\,\mathrm{mm}$.

Solution: $C = \varepsilon_0 A/d = 8.854 \times 10^{-12} \times 4 \times 10^{-4}/10^{-3} = 3.54\,\mathrm{pF}$. Sensitivity $dC/dd = -C/d = -3.54\,\mathrm{pF/mm}$, i.e. a 1 µm change in gap changes C by about 3.5 fF – measurable with modern capacitance-to-digital converters.

Table 1.8: Comparison of proximity sensors.

Type	Detects	Typical range	Advantages	Limitations
Inductive	Metals only	1 mm to 40 mm	Robust, immune to dirt, long life	Short range; reduced for non-ferrous metals
Capacitive	All materials	1 mm to 30 mm	Detects plastics, liquids, people	Affected by humidity, dust; needs adjustment
Magnetic (reed / Hall)	Magnets	up to 60 mm	Through non-magnetic walls; cheap	Needs a magnet on target
Photoelectric	Most objects	mm to tens of m	Long range, fast	Affected by dust, colour, transparency
Ultrasonic	Most objects incl. transparent	30 mm to several m	Colour-independent, gives distance	Blind zone, temperature dependent, slow

1.15 Force and Torque Sensing in Robots

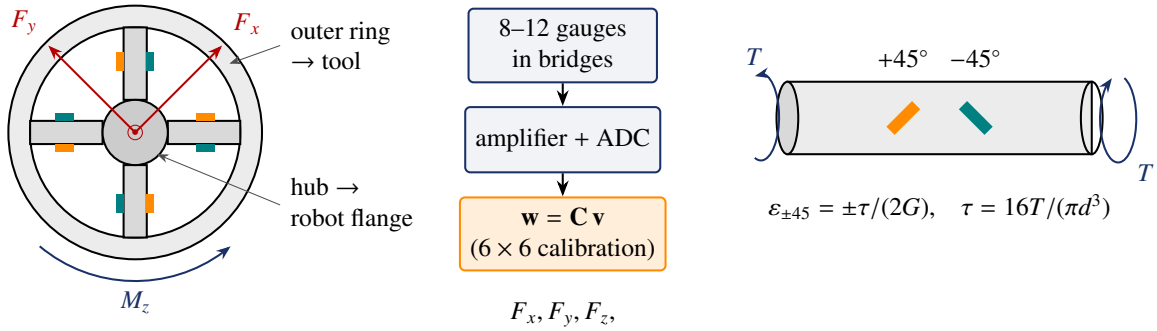
Force and torque sensing allows a robot to control contact: inserting a peg without jamming, grinding with constant pressure, polishing, or detecting a collision with a person. Three approaches are used [8]:

1. **Wrist force/torque (F/T) sensors** – mounted between the flange and the end-effector. An elastic structure (typically a Maltese-cross or hub-and-spoke flexure) carries strain gauges arranged so that six bridge outputs $\mathbf{v}$ are combined through a 6×6 calibration matrix $\mathbf{C}$ to give the wrench

$$\begin{bmatrix} F_x & F_y & F_z & M_x & M_y & M_z \end{bmatrix}^{\mathsf{T}} = \mathbf{C}\,\mathbf{v}. \tag{1.41}$$

2. **Joint torque sensors** – strain-gauged elements in each joint (e.g. in the flexspline of a harmonic drive); used by torque-controlled collaborative arms.
3. **Motor-current estimation** – since motor torque $T = K_T I$ (Unit II), joint torque can be estimated from current without extra sensors, after compensating friction and inertia; cheaper but less accurate.

Figure 1.23 illustrates the two most important hardware arrangements. In the six-axis sensor of part (a), the hub is bolted to the robot flange and the outer ring to the tool; any force or moment transmitted through the sensor bends the four spokes, and gauges on the side faces of each spoke respond to different combinations of the six wrench components. Because each gauge responds to several components (cross-coupling), the outputs must be decoupled by the calibration matrix $\mathbf{C}$ of (1.41), which is obtained by applying known loads in each axis (part (b)). Part (c) shows a joint torque sensor: gauges at ±45° on a shaft respond to the principal strains of torsion (derived in Section 1.18) while rejecting bending and axial loads.



(a) Six-axis F/T sensor: four instrumented spokes

(c) Torque shaft: full bridge of ±45° gauges

Figure 1.23: Force and torque measurement systems in robots: (a) six-axis wrist F/T sensor with instrumented spokes; (b) its signal-processing chain; (c) strain-gauge joint torque shaft.

Complementary *tactile* sensing uses arrays of pressure-sensitive cells on gripper fingers to detect grasp and slip. The underlying measurement of force, torque and strain by strain gauges and load cells is developed in Section 1.18.

Key points

- Loaded pot: $V_o/V_s = x/[1 + (R_p/R_L)x(1 - x)]$; keep $R_L \geq 10R_p$.
- LVDT: magnitude $\propto |x|$, phase gives direction; frictionless, infinite resolution.
- Encoder: $\Delta\theta = 360^\circ/(4N)$ with quadrature; absolute encoders use Gray code.
- Accelerometer: $z \approx -a/\omega_n^2$ for $\omega \ll \omega_n$; use up to ≈ 0.2–$0.4\,\omega_n$.
- Ultrasonic: $d = v_s t/2$; inductive senses metals only; capacitive senses all materials.

1.16 Temperature Measurement

1.16.1 Importance in robotics

Temperature is measured in robotic systems to protect motors and drives (winding and IGBT temperature), to compensate other sensors (strain gauges, ultrasonic rangers, encoders), to monitor battery packs of mobile robots, and to control processes performed by robots (welding, moulding, painting ovens). The SI unit is the kelvin (K); $T[\mathrm{K}] = T[^\circ\mathrm{C}] + 273.15$.

1.16.2 Resistance temperature detectors (RTDs)

Principle. The resistance of a pure metal increases with temperature because lattice vibrations scatter conduction electrons. Platinum is preferred for its stability, wide range and well-defined characteristic. Over a narrow range the relationship is approximately linear:

$$R_T \approx R_0\,(1 + \alpha T)\,, \tag{1.42}$$

where R_0 is the resistance at 0 °C and α the mean temperature coefficient between 0 and 100 °C ($\alpha = 0.003\,85/^\circ\mathrm{C}$ for industrial platinum). The standard characteristic of IEC 60751 [19] is the *Callendar–Van Dusen equation*:

$$R_T = \begin{cases} R_0\left(1 + AT + BT^2\right), & 0\,^\circ\mathrm{C} \leq T \leq 850\,^\circ\mathrm{C} \\ R_0\left[1 + AT + BT^2 + C\,(T - 100)\,T^3\right], & -200\,^\circ\mathrm{C} \leq T < 0\,^\circ\mathrm{C} \end{cases} \tag{1.43}$$

with $A = 3.9083 \times 10^{-3}/^\circ\mathrm{C}$, $B = -5.775 \times 10^{-7}/^\circ\mathrm{C}^2$, $C = -4.183 \times 10^{-12}/^\circ\mathrm{C}^4$. A *Pt100* has $R_0 = 100\,\Omega$, a Pt1000 $1000\,\Omega$ (Fig. 1.26(a)).

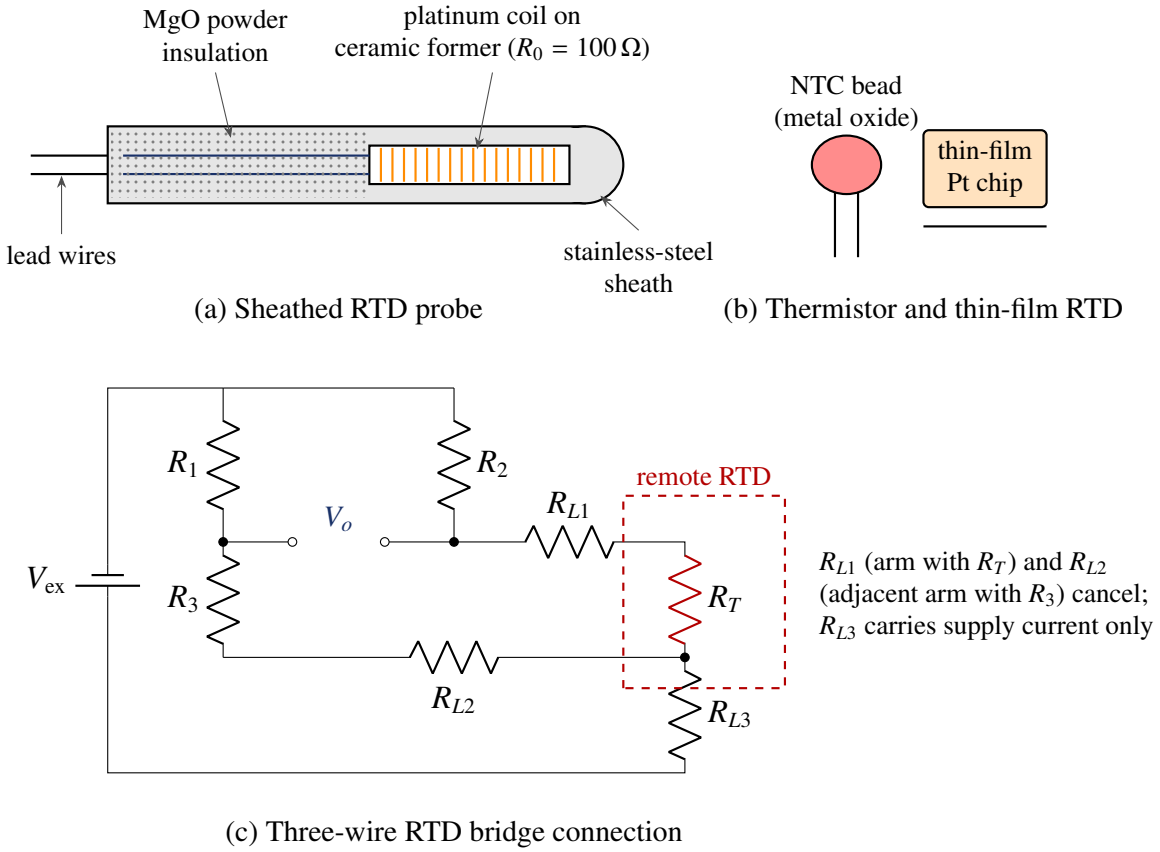
Construction. Figure 1.24 shows typical constructions. Wire-wound (platinum wire on a ceramic former) or thin-film (platinum deposited on a ceramic substrate, laser-trimmed); sheathed in stainless steel for industrial use. Tolerance classes (IEC 60751): class A $\pm(0.15 + 0.002|T|)\,^\circ\mathrm{C}$; class B $\pm(0.3 + 0.005|T|)\,^\circ\mathrm{C}$.

Measurement considerations.

- *Lead-wire resistance* adds directly to R_T ($0.39\,\Omega$ is 1°C for a Pt100). Three-wire con-

nections in a bridge cancel equal lead resistances; four-wire (Kelvin) connections with a constant-current source eliminate them.

- *Self-heating*: the measuring current I dissipates $P = I^2 R_T$, raising the element temperature by P/δ, where δ is the dissipation constant (mW/°C). Currents of about 1 mA are typical.



(a) Sheathed RTD probe

(b) Thermistor and thin-film RTD

(c) Three-wire RTD bridge connection

Figure 1.24: Temperature sensor construction: (a) sheathed wire-wound RTD probe; (b) NTC thermistor bead and thin-film platinum chip; (c) three-wire bridge connection that cancels lead resistance.

The three-wire circuit of Fig. 1.24(c) works because of the bridge rule derived later in Eq. (1.52): lead R_{L1} is in series with the RTD while lead R_{L2} is in series with R_3 in the *adjacent* arm. Equal lead resistances therefore change both arms equally and their effects cancel at balance, while the third lead R_{L3} carries only supply current and does not enter the bridge arms at all.

Example 1.23: Pt100 RTD

Given: A Pt100 at 150 °C; measuring current 1 mA; dissipation constant 2 mW/°C.

Find: (a) R_T by Callendar–Van Dusen; (b) temperature indicated if the linear model with $\alpha = 0.00385$ were used; (c) self-heating error.

Solution:

(a) $R_T = 100\left[1 + 3.9083\times10^{-3}(150) - 5.775\times10^{-7}(150)^2\right] = 100(1 + 0.58625 - 0.01299) = 157.33\,\Omega$.

(b) $T = (157.33/100 - 1)/0.00385 = 148.9\,^\circ\text{C}$ – a linearisation error of $-1.1\,^\circ\text{C}$.

(c) $P = (10^{-3})^2 \times 157.33 = 0.157\,\text{mW}$; $\Delta T = 0.157/2 = 0.079\,^\circ\text{C}$.

Interpretation: For accuracy better than about 1 °C above 100 °C the quadratic CVD equation must be used (in software or a look-up table).

1.16.3 Thermistors

Thermistors are semiconductor (sintered metal-oxide) resistors with a large temperature coefficient. *NTC* thermistors decrease in resistance with temperature; *PTC* types increase sharply above a switching temperature and are used as self-resetting protection. The NTC characteristic is approximated by the β equation:

$$R_T = R_{25}\exp\!\left[\beta\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right], \qquad T \text{ in kelvin},\ T_{25} = 298.15\,\text{K}, \tag{1.44}$$

and more accurately by the *Steinhart–Hart equation* $1/T = a + b\ln R + c(\ln R)^3$. The temperature coefficient $\alpha = (1/R)(dR/dT) = -\beta/T^2$ is about $-4.4\%/^\circ\text{C}$ at 25 °C for $\beta = 3950\,\text{K}$ – more than ten times that of platinum (Fig. 1.26(b)). Figure 1.26(a) and (b) compare the two resistive sensors directly: the RTD curve is almost a straight line with a slope of about $0.39\,\Omega/^\circ\text{C}$, whereas the thermistor resistance falls by more than an order of magnitude over 100 °C and must be plotted on a logarithmic axis. Thermistors are cheap, small, fast and very sensitive, but highly non-linear and limited to about −50 °C to 150 °C. They are embedded in motor windings, battery packs and power electronics.

Example 1.24: NTC thermistor

Given: NTC with $R_{25} = 10\,\text{k}\Omega$, $\beta = 3950\,\text{K}$.

Find: (a) R at 60 °C; (b) temperature when $R = 4.5\,\text{k}\Omega$.

Solution: (a) $R = 10\exp[3950(1/333.15 - 1/298.15)] = 10\exp(-1.3919) = 2.49\,\text{k}\Omega$.
(b) Inverting (1.44): $\dfrac{1}{T} = \dfrac{1}{298.15} + \dfrac{\ln(4.5/10)}{3950} = 3.3540\times10^{-3} - 2.0215\times10^{-4} = 3.1519\times10^{-3}$, so $T = 317.27\,\text{K} = 44.1\,^\circ\text{C}$.

1.16.4 Thermocouples

Principle (Seebeck effect). When two dissimilar metals A and B are joined to form a circuit and the two junctions are at different temperatures T_1 and T_2, an EMF is produced (Fig. 1.25):

$$E \approx S_{AB}\,(T_1 - T_2), \tag{1.45}$$

where S_{AB} is the Seebeck coefficient of the pair (µV/°C), itself a function of temperature. A thermocouple therefore measures a temperature *difference*; the reference (cold) junction temperature must be known.

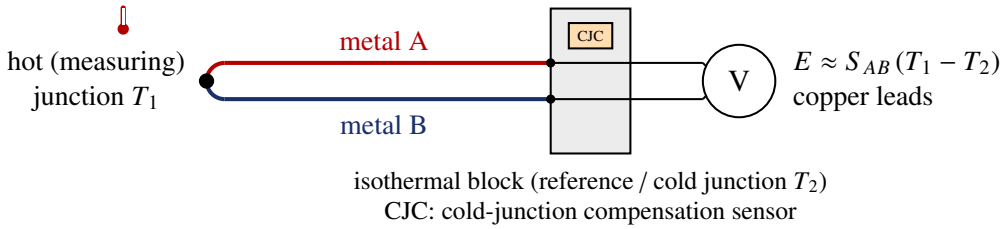


Figure 1.25: Thermocouple circuit with isothermal reference block and cold-junction compensation (CJC).

Laws of thermocouples.

1. *Law of homogeneous circuits:* temperatures along a homogeneous wire do not affect the EMF – only junction temperatures matter.
2. *Law of intermediate metals:* a third metal (e.g. copper leads of a meter) inserted into the circuit does not change the EMF if both its junctions are at the same temperature. This is why the meter can be connected through an isothermal block.
3. *Law of intermediate temperatures:* $E(T_1, T_3) = E(T_1, T_2) + E(T_2, T_3)$. Since reference tables give $E(T, 0)$, the true temperature is found from

$$E(T, 0\,^\circ\mathrm{C}) = E_{\text{measured}}(T, T_{\text{ref}}) + E(T_{\text{ref}}, 0\,^\circ\mathrm{C}). \tag{1.46}$$

Cold-junction compensation (CJC) implements (1.46) electronically: a thermistor, RTD or IC sensor measures the terminal-block temperature and the corresponding EMF is added to the measured EMF before conversion to temperature.

Example 1.25: Type K thermocouple with cold-junction compensation

Given: A type K thermocouple reads 7.138 mV with its reference junction at 25 °C. From type K reference tables: $E(25\,^\circ\mathrm{C}, 0) = 1.000$ mV; $E(200\,^\circ\mathrm{C}, 0) = 8.138$ mV.

Solution: By (1.46): $E(T, 0) = 7.138 + 1.000 = 8.138\ \text{mV} \Rightarrow T = 200\,^\circ\mathrm{C}$.

Common mistake: Converting 7.138 mV directly gives about 175 °C, and adding 25 °C then

Table 1.9: Common thermocouple types (IEC 60584 [20]; approximate values).

Type	Materials (+ / −)	Typical range	Sensitivity	Remarks
J	Iron / Constantan	−40 °C to 750 °C	≈ 52 µV/°C	Reducing atmospheres
K	Chromel / Alumel	−200 °C to 1250 °C	≈ 41 µV/°C	General purpose
T	Copper / Constantan	−200 °C to 350 °C	≈ 43 µV/°C	Low temperature, cryogenics
E	Chromel / Constantan	−200 °C to 900 °C	≈ 68 µV/°C	Highest output
S, R	Pt-Rh / Pt	0 °C to 1600 °C	≈ 10 µV/°C	High temperature, standards
B	Pt-30%Rh / Pt-6%Rh	600 °C to 1700 °C	≈ 8 µV/°C	Very high temperature

gives 200 °C only because type K is nearly linear here; for non-linear regions EMFs, not temperatures, must be added.

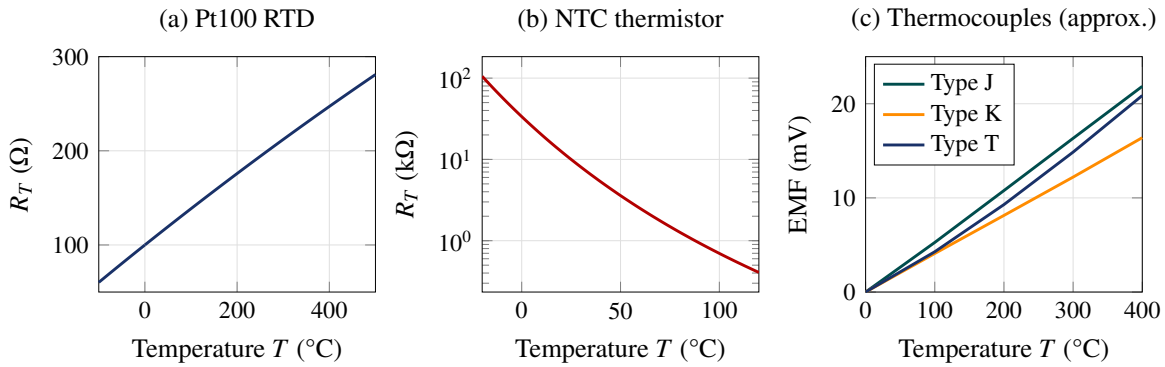


Figure 1.26: Temperature sensor characteristics: (a) Pt100 RTD (Callendar–Van Dusen); (b) NTC thermistor, $R_{25} = 10\,\text{k}\Omega$, $\beta = 3950\,\text{K}$ (log scale); (c) EMF of J, K and T thermocouples with reference at 0 °C (tabulated points).

1.16.5 Other temperature sensors

- **IC (semiconductor) sensors** – exploit the temperature dependence of the base–emitter voltage of transistors; output 10 mV/°C (voltage) or 1 µA/K (current), or a digital value over I²C/SPI. Linear and easy to interface, −55 °C to 150 °C.
- **Radiation (infrared) pyrometers** – non-contact measurement from emitted thermal radiation (Stefan–Boltzmann law); used for moving or very hot objects (weld seams, forging).
- **Bimetallic strips and filled-system thermometers** – mechanical, used as thermostats.

1.17 Pressure Measurement

1.17.1 Definitions

Pressure is force per unit area, $p = F/A$, with SI unit the pascal ($1\,\text{Pa} = 1\,\text{N/m}^2$); $1\,\text{bar} = 10^5\,\text{Pa}$; $1\,\text{atm} = 101.325\,\text{kPa}$. Three references are used:

Table 1.10: Comparison of contact temperature sensors.

Property	RTD (Pt100)	NTC thermistor	Thermocouple	IC sensor
Range	−200 °C to 850 °C	−50 °C to 150 °C	−200 °C to 1700 °C	−55 °C to 150 °C
Sensitivity	≈ 0.39 Ω/°C	Very high, ≈ −4%/°C	Low, 10 µV/°C to 70 µV/°C	10 mV/°C
Linearity	Good	Very poor	Moderate	Excellent
Accuracy / stability	Best	Good (narrow range)	Moderate	Moderate
Excitation	Current source	Current/divider	None (self-generating)	Supply
Response	Moderate	Fast	Fast (fine wire)	Moderate
Cost	High	Low	Low	Low

- **Absolute pressure** – relative to perfect vacuum;
- **Gauge pressure** – relative to local atmospheric pressure, $p_g = p_{\text{abs}} - p_{\text{atm}}$ (negative gauge pressure is *vacuum*);
- **Differential pressure** – difference between two process pressures (flow and level measurement).

In robotics, pressure is measured in pneumatic actuators and grippers, vacuum suction cups (to confirm a part has been picked), hydraulic drives, and soft robots.

1.17.2 Manometers

In a U-tube manometer (Fig. 1.27(c)) the pressure difference is balanced by a liquid column. Hydrostatic equilibrium gives

$$p_1 - p_2 = \rho g h, \tag{1.47}$$

where ρ is the manometric-fluid density (corrected for the density of the fluid above it if significant), g the gravitational acceleration and h the height difference. Manometers are primary standards for low pressures.

Example 1.26: U-tube manometer

Given: Mercury ($\rho = 13\,546\,\text{kg/m}^3$) manometer, $h = 150\,\text{mm}$, air above the mercury.

Solution: $\Delta p = 13546 \times 9.81 \times 0.150 = 19\,933\,\text{Pa} \approx 19.9\,\text{kPa}$ (0.199 bar).

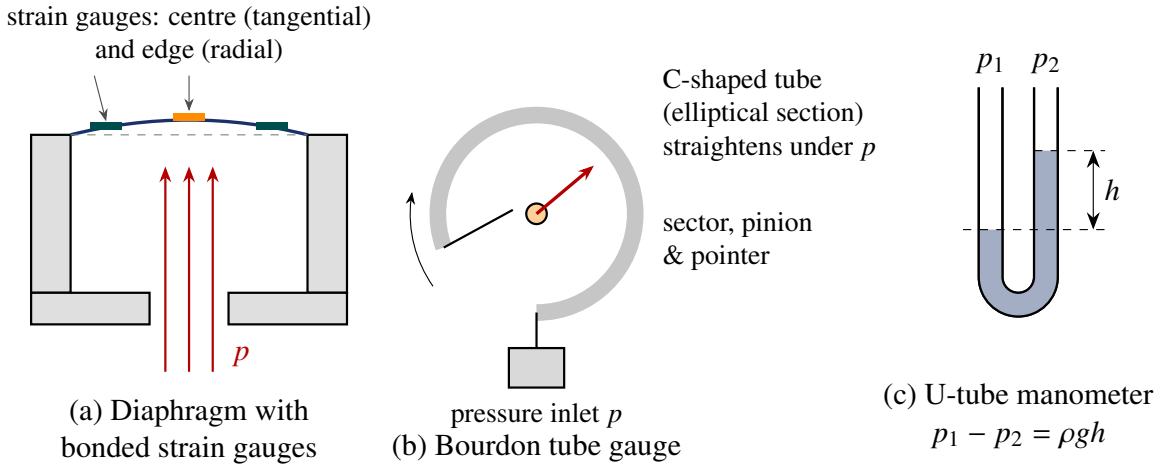
1.17.3 Elastic pressure sensors

Elastic elements convert pressure into displacement or strain, which is then sensed (Fig. 1.27).

- **Bourdon tube** – a C-shaped, spiral or helical tube of elliptical cross-section, closed at the

free end. Pressure tends to make the cross-section circular, which straightens the tube; the tip displacement is amplified by a linkage, sector and pinion to a pointer, or sensed by a potentiometer or LVDT. Range up to several hundred MPa; robust; the standard gauge on pneumatic and hydraulic systems.

- **Bellows** – a corrugated metal cylinder that extends with pressure; large displacement at low pressure.
- **Diaphragm** – a thin circular plate clamped at its edge, deflecting under differential pressure. Deflection or surface strain is measured by strain gauges, capacitance or an LVDT.



(a) Diaphragm with bonded strain gauges

(b) Bourdon tube gauge

(c) U-tube manometer $p_1 - p_2 = \rho g h$

Figure 1.27: Pressure sensors: (a) diaphragm with bonded strain gauges; (b) C-type Bourdon tube gauge; (c) U-tube manometer.

Diaphragm equations. For a clamped circular diaphragm of radius a, thickness t, Young's modulus E and Poisson's ratio ν, small-deflection plate theory gives the centre deflection [12]

$$y_c = \frac{3p\,(1-\nu^2)\,a^4}{16\,E\,t^3}, \tag{1.48}$$

and the radial surface strain is $+3pa^2(1-\nu^2)/(8Et^2)$ at the centre and twice that magnitude, in compression, at the edge. Gauges at the centre and near the edge therefore experience strains of opposite sign and are placed in adjacent bridge arms. Linearity requires y_c to be small compared with t (typically $y_c < t/4$).

Example 1.27: Steel diaphragm pressure sensor

Given: $a = 10\,\text{mm}$, $t = 0.5\,\text{mm}$, $E = 200\,\text{GPa}$, $\nu = 0.3$, $p = 1\,\text{MPa}$.

Solution:

$$y_c = \frac{3(10^6)(1-0.09)(0.01)^4}{16(200\times10^9)(0.0005)^3} = \frac{0.0273}{400} = 6.83\times10^{-5}\,\text{m} = 68.3\,\mu\text{m}.$$

Centre strain $= 3(10^6)(0.01)^2(0.91)/[8(200\times10^9)(0.0005)^2] = 6.83\times10^{-4}$ ($683\,\mu\varepsilon$); edge strain

$\approx -1365\,\mu\varepsilon$.

Check: $y_c/t = 0.137 < 0.25$, so the small-deflection (linear) assumption holds.

1.17.4 Electronic pressure transducers

- **Piezoresistive (silicon) sensors** – four resistors diffused into a micromachined silicon diaphragm form a full bridge; high sensitivity, small size, low cost, internal temperature compensation; the most widely used type (pneumatic and vacuum switches, robot grippers, automotive MAP sensors).
- **Capacitive sensors** – the diaphragm forms one plate of a capacitor; very sensitive at low pressure and stable.
- **Piezoelectric sensors** – generate charge proportional to pressure change; for dynamic pressures only (combustion, blast, hydraulic pulsation).

Example 1.28: Vacuum gripper holding force

Given: A robot vacuum cup of effective diameter 40 mm; the pressure sensor in the vacuum line reads −60 kPa (gauge) after pick-up. Safety factor 2.

Find: Theoretical holding force and the maximum mass that may be lifted vertically.

Solution: $A = \pi(0.020)^2 = 1.257 \times 10^{-3}\,\mathrm{m}^2$; $F = \Delta p\,A = 60\,000 \times 1.257 \times 10^{-3} = 75.4\,\mathrm{N}$. Allowable mass $= 75.4/(2 \times 9.81) = 3.84\,\mathrm{kg}$.

Interpretation: If the measured vacuum rises towards zero during transport (leak, porous part), the controller stops the robot before the part drops – a typical use of a pressure switch in a robotic cell.

1.18 Force, Torque and Strain Measurement

1.18.1 Methods of force measurement

Force can be measured by (i) balancing it against a known force (beam balance); (ii) measuring the acceleration of a known mass ($F = ma$); (iii) measuring the deformation of an elastic element – by far the most common in engineering. The elastic element (column, beam, ring, shear web) is designed so that the strain at the gauge location is proportional to the force; strain gauges convert the strain into a resistance change.

1.18.2 Stress and strain

For a bar of length L under axial force F on cross-section A:

$$\sigma = \frac{F}{A}, \qquad \varepsilon = \frac{\Delta L}{L}, \qquad \sigma = E\varepsilon \ \text{(Hooke's law)}, \qquad \varepsilon_{\text{lat}} = -\nu\,\varepsilon, \tag{1.49}$$

where E is Young's modulus and ν Poisson's ratio. Strain is dimensionless and is expressed in microstrain ($1\,\mu\varepsilon = 10^{-6}$).

1.18.3 The bonded resistance strain gauge

Construction. A metal-foil grid (typically constantan, Cu–Ni) photo-etched on a thin polyimide backing is bonded to the surface with adhesive (Fig. 1.28). The long grid lines are aligned with the strain direction; the end loops are wide to minimise transverse sensitivity. Nominal resistances are 120 Ω, 350 Ω and 1000 Ω.

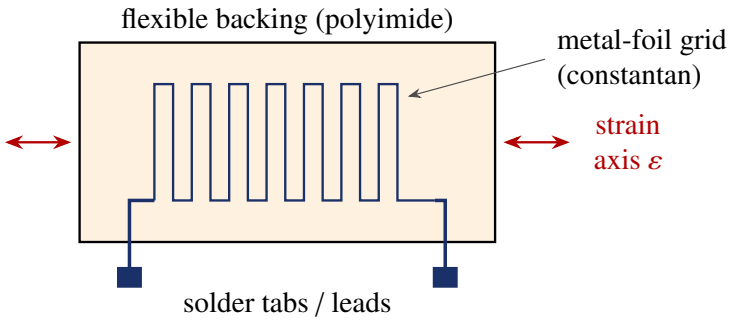


Figure 1.28: Bonded metal-foil strain gauge.

Gauge factor – derivation. The resistance of a conductor is $R = \rho L/A$. Taking logarithms and differentiating,

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dL}{L} - \frac{dA}{A}.$$

For a conductor of circular section, $A = \pi D^2/4$, so $dA/A = 2\,dD/D = 2(-\nu\,\varepsilon)$. With $dL/L = \varepsilon$:

$$\frac{dR}{R} = \varepsilon(1 + 2\nu) + \frac{d\rho}{\rho}.$$

The **gauge factor** is defined as the fractional resistance change per unit strain:

$$G_F = \frac{\Delta R/R}{\varepsilon} = 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon}. \tag{1.50}$$

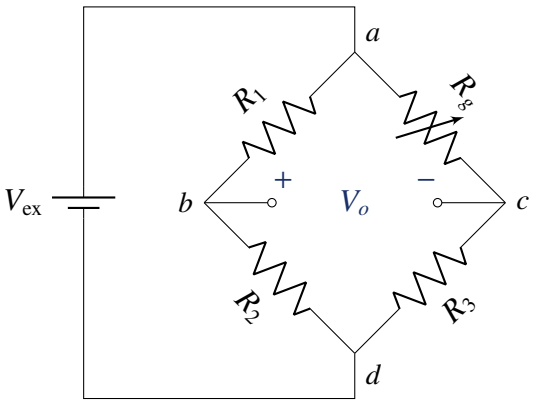
For metals the geometric term $1+2\nu \approx 1.6$ dominates and $G_F \approx 2$; for semiconductor gauges the *piezoresistive* term $(\Delta\rho/\rho)/\varepsilon$ dominates, giving G_F of 50–200 but strong temperature sensitivity and non-linearity.

Temperature effects. A temperature change produces an apparent strain from (i) the temperature coefficient of resistance of the grid and (ii) differential thermal expansion between the gauge and the specimen. Remedies: self-temperature-compensated (STC) gauges matched to

the specimen material, and bridge compensation with a dummy or second active gauge in an adjacent arm.

1.18.4 Wheatstone bridge

Strain-gauge resistance changes are tiny (Example 1.29: 0.126 Ω in 120 Ω), so they are measured with a Wheatstone bridge (Fig. 1.29).



$$V_o = V_{ex}\left(\frac{R_2}{R_1+R_2} - \frac{R_3}{R_g+R_3}\right)$$

R_g = active strain gauge

Figure 1.29: Wheatstone bridge with a strain gauge R_g in one arm (quarter bridge). The output is $V_o = V_b - V_c$.

Derivation. The two half-bridges are voltage dividers across V_{ex}:

$$V_b = V_{ex}\frac{R_2}{R_1+R_2}, \qquad V_c = V_{ex}\frac{R_3}{R_g+R_3}, \qquad V_o = V_b - V_c = V_{ex}\left[\frac{R_2}{R_1+R_2} - \frac{R_3}{R_g+R_3}\right].$$

The bridge is *balanced* ($V_o = 0$) when $R_1R_3 = R_2R_g$. With all four arms equal to R and the gauge changing to $R + \Delta R$:

$$V_o = V_{ex}\left[\frac{1}{2} - \frac{R}{2R+\Delta R}\right] = V_{ex}\frac{\Delta R}{2(2R+\Delta R)} \approx \frac{V_{ex}}{4}\frac{\Delta R}{R} = \frac{V_{ex}\,G_F\,\varepsilon}{4}. \tag{1.51}$$

The small non-linearity from ΔR in the denominator is negligible for metal gauges.

General linearised bridge equation. If every arm changes by a small ΔR_i, linearising about balance gives, for this labelling,

$$\frac{V_o}{V_{ex}} \approx \frac{1}{4}\left(\frac{\Delta R_g}{R} - \frac{\Delta R_3}{R} + \frac{\Delta R_2}{R} - \frac{\Delta R_1}{R}\right). \tag{1.52}$$

Rule: changes in *adjacent* arms (sharing a node, e.g. R_g and R_3) subtract; changes in *opposite* arms (R_g and R_2) add. Consequences:

- **Temperature compensation:** a dummy gauge (unstrained, same temperature) in the adjacent arm R_3 cancels apparent thermal strain.
- **Half bridge:** gauges with equal and opposite strain ($+\varepsilon$ and $-\varepsilon$, e.g. top and bottom of a beam) in adjacent arms add: $V_o \approx V_{ex}G_F\varepsilon/2$, and temperature effects cancel.

- **Full bridge:** four active gauges (two $+\varepsilon$ in opposite arms, two $-\varepsilon$ in the other pair): $V_o \approx V_{\text{ex}} G_F \varepsilon$ – four times the quarter-bridge output, temperature compensated, and linear.

Example 1.29: Quarter-, half- and full-bridge outputs

Given: $120\,\Omega$ gauges, $G_F = 2.1$, strain $500\,\mu\varepsilon$, $V_{\text{ex}} = 5\,\text{V}$.

Solution: $\Delta R = G_F \varepsilon R = 2.1 \times 500 \times 10^{-6} \times 120 = 0.126\,\Omega$.

- Quarter bridge (exact): $V_o = 5 \times 0.126/[2(240.126)] = 1.312\,\text{mV}$; approximation $5 \times 2.1 \times 500 \times 10^{-6}/4 = 1.3125\,\text{mV}$.
- Half bridge: $2.625\,\text{mV}$; full bridge: $5.25\,\text{mV}$.

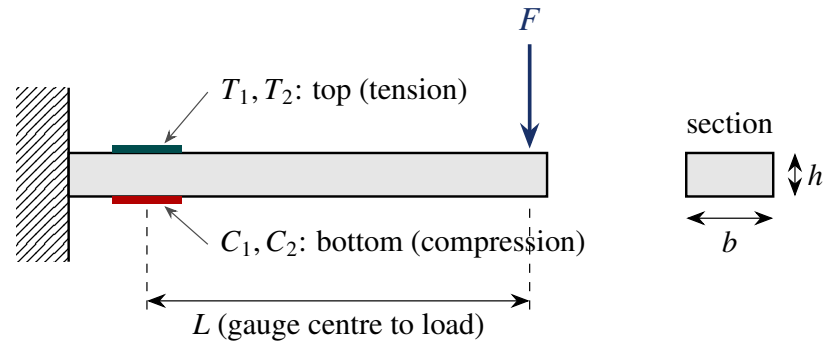
Interpretation: Millivolt outputs require an instrumentation amplifier with gain of the order of 1000 before the ADC (Unit II).

1.18.5 Load cells

A *load cell* is a force transducer consisting of an elastic element with bonded strain gauges in a full bridge. Common elastic elements are the column (compression), S-beam, shear beam, ring and cantilever (bending) beam. For the cantilever of Fig. 1.30, the bending moment at the gauge section is $M = FL$ and the surface stress is $\sigma = Mc/I$ with $c = h/2$ and $I = bh^3/12$:

$$\varepsilon = \frac{\sigma}{E} = \frac{6FL}{E\,b\,h^2}. \tag{1.53}$$

The top surface is in tension ($+\varepsilon$) and the bottom in compression ($-\varepsilon$), so the four gauges form a full bridge with $V_o \approx V_{\text{ex}} G_F \varepsilon$. Load-cell sensitivity is quoted as rated output in mV/V at full load (typically 1 mV/V to 3 mV/V).



surface strain $\varepsilon = \dfrac{6FL}{E\,b\,h^2}$; full bridge: $V_o = V_{\text{ex}}\,GF\,\varepsilon$

Figure 1.30: Cantilever-beam load cell with tension gauges on the top surface and compression gauges on the bottom surface.

Example 1.30: Cantilever load cell for a gripper finger

Given: Aluminium beam ($E = 70\,\text{GPa}$), $b = 20\,\text{mm}$, $h = 5\,\text{mm}$, $L = 100\,\text{mm}$; full bridge, $G_F = 2.0$, $V_{\text{ex}} = 10\,\text{V}$; force $50\,\text{N}$.

Solution:

$$\varepsilon = \frac{6(50)(0.1)}{70\times10^{9}\times0.02\times(0.005)^{2}} = \frac{30}{35\,000} = 8.57\times10^{-4}\ (857\,\mu\varepsilon).$$

$V_o = 10\times2.0\times8.57\times10^{-4} = 17.1\,\text{mV}$; sensitivity $= 17.1/50 = 0.343\,\text{mV/N}$, i.e. rated output $1.71\,\text{mV/V}$ at $50\,\text{N}$.

Check: Stress $= E\varepsilon = 60\,\text{MPa}$, well below the yield strength of structural aluminium alloys, so the beam remains elastic.

1.18.6 Torque measurement

Strain-gauge torque transducer. A shaft of diameter d transmitting torque T has a maximum surface shear stress $\tau = 16T/(\pi d^3)$. Pure shear produces principal strains of $\pm\tau/(2G)$ along directions at $\pm45^\circ$ to the shaft axis, where $G = E/[2(1+\nu)]$ is the shear modulus. Four gauges at $\pm45^\circ$ form a full bridge that responds to torque but cancels bending and axial strains:

$$\varepsilon_{45} = \frac{\tau}{2G} = \frac{8T}{\pi d^{3}G}, \qquad V_o \approx V_{\text{ex}} G_F \varepsilon_{45}. \tag{1.54}$$

Signals from rotating shafts are transmitted by slip rings or, in modern sensors, by contactless telemetry. Alternative principles are *twist-angle* measurement between two encoders (used in series-elastic actuators) and *reaction torque* measurement on the motor housing. Mechanical power follows as $P = T\omega$.

Example 1.31: Joint torque sensor

Given: Steel shaft $d = 20\,\text{mm}$, $G = 80\,\text{GPa}$, $T = 50\,\text{N m}$; full bridge of $\pm45^\circ$ gauges, $G_F = 2.0$, $V_{\text{ex}} = 10\,\text{V}$.

Solution: $\tau = 16(50)/[\pi(0.02)^3] = 31.83\,\text{MPa}$; $\varepsilon_{45} = 31.83\times10^{6}/(2\times80\times10^{9}) = 1.99\times10^{-4}$ $(199\,\mu\varepsilon)$; $V_o = 10\times2.0\times1.99\times10^{-4} = 3.98\,\text{mV}$.

Key points

- $G_F = (\Delta R/R)/\varepsilon \approx 2$ for metal foil gauges.
- Quarter bridge $V_o \approx V_{\text{ex}} G_F \varepsilon/4$; half $\times 2$; full $\times 4$.
- Adjacent arms subtract, opposite arms add – use this for temperature compensation and to reject unwanted loads.
- Cantilever $\varepsilon = 6FL/(Ebh^2)$; shaft torque $\varepsilon_{45} = 8T/(\pi d^3 G)$.

1.19 MATLAB and Simulink Implementation

The programs in this section reproduce the calculations and figures of the unit so that students can vary parameters and observe their effect. They use only core MATLAB functions (no toolboxes) and were verified in GNU Octave (where `histogram` must be replaced by `hist` and `compose` by a manual legend). Each program prints the numerical results of the corresponding worked example so that the code can be checked against the notes.

1.19.1 Program 1 – Static calibration by least squares

Algorithm. (1) Enter calibration data (x_i, y_i); (2) compute m and c from the normal equations (1.20) (and cross-check with `polyfit`); (3) compute residuals, non-linearity (%FSO) and R^2; (4) plot data, fitted line and residuals; (5) form the inverse calibration $x = (y - c)/m$. **Expected output:** $m = 0.4960$ V/mm, $c = 0.0400$ V, non-linearity 1.33% FSO, $R^2 = 0.9996$ (Example 1.11).

Program 1.1: Least-squares calibration of a displacement sensor.

```
% Program 1: Static calibration of a displacement sensor (Example: calibration)
clear; clc; close all;
x = [0 2 4 6 8 10]; % input displacement (mm), reference standard
y = [0.05 1.02 2.06 2.95 4.03 5.01];% sensor output (V)
n = numel(x);

% --- Least-squares slope and intercept from the normal equations
m = (n*sum(x.*y) - sum(x)*sum(y)) / (n*sum(x.^2) - sum(x)^2);
c = (sum(y) - m*sum(x)) / n;
p = polyfit(x, y, 1); % cross-check with built-in fit
yfit = m*x + c;
res = y - yfit; % residuals (V)
FSO = m*(max(x) - min(x)); % fitted full-scale output (V)
nonlin = max(abs(res))/FSO*100; % non-linearity (% FSO)
R2 = 1 - sum(res.^2)/sum((y - mean(y)).^2);

fprintf('Sensitivity m = %.4f V/mm (polyfit %.4f)\n', m, p(1));
fprintf('Offset c = %.4f V\n', c);
```

```
[rmax, imax] = max(abs(res));
fprintf('Max residual = %.3f V at x = %g mm\n', rmax, x(imax));
fprintf('Non-linearity = %.2f %% FSO, R^2 = %.4f\n', nonlin, R2);

% --- Inverse calibration: convert a reading back to displacement
yreading = 3.50;
fprintf('Reading %.2f V -> x = %.3f mm\n', yreading, (yreading - c)/m);

% --- Plots
subplot(2,1,1);
plot(x, y, 'ro', 'MarkerFaceColor', 'r'); hold on; grid on;
xx = linspace(0, 10, 100); plot(xx, m*xx + c, 'b-', 'LineWidth', 1.5);
xlabel('Displacement x (mm)'); ylabel('Output y (V)');
legend('Calibration data', 'Least-squares line', 'Location', 'northwest');
title('Static calibration curve');
subplot(2,1,2);
stem(x, res*1000, 'filled'); grid on;
xlabel('Displacement x (mm)'); ylabel('Residual (mV)'); title('Residuals');
```

1.19.2 Program 2 – Measurement error and uncertainty analysis

Algorithm. (1) Compute mean, bias, standard deviation and Type A uncertainty from repeated readings (Example 1.8); (2) propagate uncertainties analytically for $P = VI$ (Example 1.9); (3) verify by Monte Carlo simulation with 10^5 random samples, treating the stated tolerances as standard uncertainties. **Expected output:** mean 101.125 °C, bias 1.125 °C, $s = 0.225$ °C, $u_A = 0.080$ °C; $u_P \approx 0.44$ W both analytically and by Monte Carlo.

Program 1.2: Error statistics and uncertainty propagation (analytical and Monte Carlo).

```
% Program 2: Measurement errors and uncertainty
clear; clc; close all;
% ---- Part A: repeated readings of a 100.0 degC calibration bath
r = [101.2 100.8 101.5 101.1 100.9 101.3 101.0 101.2];
Ttrue = 100.0;
xbar = mean(r); s = std(r); % std uses (n-1) -> sample standard deviation
uA = s/sqrt(numel(r));
fprintf('Mean = %.3f degC, bias = %+.3f degC\n', xbar, xbar - Ttrue);
fprintf('s = %.3f degC, Type A uncertainty of mean = %.3f degC\n', s, uA);

% ---- Part B: propagation for P = V*I
V = 12.0; uV = 0.10; % volts
I = 2.50; uI = 0.03; % amperes
P = V*I;
uP_rss = P*sqrt((uV/V)^2 + (uI/I)^2);
uP_worst = P*(uV/V + uI/I);
fprintf('P = %.2f W, RSS u = %.3f W, worst case = %.3f W\n', P, uP_rss, uP_worst);

% ---- Part C: Monte Carlo verification (normal distributions)
```

```
N = 1e5;
Vs = V + uV*randn(N,1);
Is = I + uI*randn(N,1);
Ps = Vs.*Is;
fprintf('Monte Carlo: mean P = %.3f W, std = %.3f W\n', mean(Ps), std(Ps));
histogram(Ps, 60); grid on; % (in Octave use hist(Ps,60))
xlabel('Power P (W)'); ylabel('Count'); title('Monte Carlo distribution of P = VI');
```

1.19.3 Program 3 – Static and dynamic response of instruments

Algorithm. (1) Evaluate the first-order step response (1.23) for the thermometer of Example 1.12; (2) evaluate the second-order step response (1.26) for several ζ and compute M_p, t_p, t_s for Example 1.14; (3) plot the first- and second-order amplitude ratios (1.24), (1.28). With the Control System Toolbox the same responses can be obtained with `tf`, `step` and `bode`. **Expected output:** $T(10) = 78.51\,^{\circ}\mathrm{C}$, time to $99\,^{\circ}\mathrm{C} = 34.54\,\mathrm{s}$; $M_p = 25.4\%$, $t_p = 1.091\,\mathrm{ms}$, $t_s = 3.183\,\mathrm{ms}$.

Program 1.3: First- and second-order instrument responses.

```
% Program 3: Dynamic response of first- and second-order instruments
clear; clc; close all;
% ---------------- First-order: thermometer ----------------
tau = 8; T0 = 25; Tf = 100; % s, degC, degC
t = linspace(0, 40, 400);
T = T0 + (Tf - T0)*(1 - exp(-t/tau));
fprintf('T(10 s) = %.2f degC\n', T0 + (Tf-T0)*(1 - exp(-10/tau)));
fprintf('Time to reach 99 degC = %.2f s\n', tau*log((Tf-T0)/(Tf-99)));
figure; plot(t, T, 'LineWidth', 1.5); grid on;
xlabel('Time (s)'); ylabel('Indicated temperature (degC)');
title('First-order step response, \tau = 8 s');

% ---------------- Second-order: force sensor --------------
fn = 500; wn = 2*pi*fn; zeta = [0.2 0.4 0.707 1.0 2.0];
t = linspace(0, 6e-3, 1500);
figure; hold on; grid on;
for z = zeta
    if z < 1
        wd = wn*sqrt(1 - z^2);
        y = 1 - exp(-z*wn*t).*(cos(wd*t) + z/sqrt(1-z^2)*sin(wd*t));
    elseif z == 1
        y = 1 - (1 + wn*t).*exp(-wn*t);
    else
        s1 = -wn*(z - sqrt(z^2-1)); s2 = -wn*(z + sqrt(z^2-1));
        y = 1 + (s2*exp(s1*t) - s1*exp(s2*t))/(s1 - s2);
    end
    plot(t*1e3, y, 'LineWidth', 1.2);
end
xlabel('Time (ms)'); ylabel('y / (K u_0)'); title('Second-order step responses');
```

```
legend(compose('\\zeta = %.3g', zeta), 'Location', 'southeast');
z = 0.4; wd = wn*sqrt(1 - z^2);
fprintf('zeta = 0.4: Mp = %.1f %%, tp = %.3f ms, ts(2%%) = %.3f ms\n', ...
        100*exp(-z*pi/sqrt(1-z^2)), 1e3*pi/wd, 1e3*4/(z*wn));

% ---------------- Frequency response ----------------------
r = linspace(0, 3, 600);
figure; hold on; grid on;
for z = [0.2 0.7 1.0]
    plot(r, 1./sqrt((1 - r.^2).^2 + (2*z*r).^2), 'LineWidth', 1.2);
end
plot(r, 1./sqrt(1 + r.^2), 'k--'); % first order with r = omega*tau
xlabel('\omega/\omega_n (or \omega\tau)'); ylabel('Amplitude ratio');
legend('2nd, \zeta=0.2', '2nd, \zeta=0.7', '2nd, \zeta=1', '1st order');
axis([0 3 0 3]);
```

1.19.4 Program 4 – Sensor characteristic curves

Algorithm. Plot (a) the Pt100 resistance by Callendar–Van Dusen and compare with the linear model; (b) the NTC characteristic by the β equation and its inverse; (c) the loaded-potentiometer characteristic (1.30) and its maximum loading error. **Expected output:** $R_{150} = 157.33\,\Omega$, linear-model temperature 148.90 °C; NTC $R_{60} = 2.49\,\mathrm{k}\Omega$, $T(4.5\,\mathrm{k}\Omega) = 44.12\,^{\circ}\mathrm{C}$; pot output 5.725 V at $x = 0.6$, maximum error -0.284 V at $x \approx 0.67$.

Program 1.4: Characteristic curves of RTD, thermistor and loaded potentiometer.

```
% Program 4: Sensor characteristic curves
clear; clc; close all;
% ---- (a) Pt100 RTD, IEC 60751 Callendar-Van Dusen
A = 3.9083e-3; B = -5.775e-7; C = -4.183e-12; R0 = 100;
cvd = @(T) R0*(1 + A*T + B*T.^2 + C*(T - 100).*T.^3.*(T < 0));
T = -100:1:500;
R = cvd(T);
Rlin = R0*(1 + 0.00385*T);
R150 = cvd(150);
fprintf('Pt100: R(150) = %.3f ohm; linear-model T = %.2f degC\n', ...
        R150, (R150/R0 - 1)/0.00385);
subplot(1,3,1); plot(T, R, T, Rlin, '--'); grid on;
xlabel('T (degC)'); ylabel('R (\Omega)'); legend('CVD', 'linear'); title('Pt100');

% ---- (b) NTC thermistor (beta model)
R25 = 10e3; beta = 3950; T25 = 298.15;
Tc = -20:1:120;
Rn = R25*exp(beta*(1./(Tc + 273.15) - 1/T25));
fprintf('NTC: R(60 degC) = %.3f kohm\n', R25*exp(beta*(1/333.15 - 1/T25))/1e3);
Rm = 4.5e3; Tm = 1/(1/T25 + log(Rm/R25)/beta) - 273.15;
fprintf('NTC: R = 4.5 kohm -> T = %.2f degC\n', Tm);
subplot(1,3,2); semilogy(Tc, Rn/1e3); grid on;
```

```
xlabel('T (degC)'); ylabel('R (k\Omega)'); title('NTC thermistor');

% ---- (c) Loaded potentiometer
Vs = 10; x = linspace(0, 1, 1001);
subplot(1,3,3); hold on; grid on;
for k = [0 0.1 1 5] % k = Rp/RL
    plot(x, Vs*x./(1 + k*x.*(1 - x)));
end
xlabel('x'); ylabel('V_o (V)'); title('Loaded potentiometer');
legend('R_L=\infty', 'R_L=10R_p', 'R_L=R_p', 'R_L=0.2R_p', 'Location', 'northwest');
k = 0.2; % Rp = 10 kohm, RL = 50 kohm
Vo = @(x) Vs*x./(1 + k*x.*(1 - x));
err = Vo(x) - Vs*x; [emin, i] = min(err);
fprintf('Pot: Vo(0.6) = %.3f V; max error %.3f V at x = %.3f\n', Vo(0.6), emin, x(i));
```

1.19.5 Program 5 – Robot workspace and degrees of freedom

Algorithm. (1) A function implements Grübler's formula (1.5); (2) the reachable workspace of a planar 2R arm is generated by sampling the joint space and applying the forward kinematics (1.4); (3) the workspace volumes of Examples 1.4 and 1.5 are computed. **Expected output:** DOF = 6, 1, 2, 6; $A_{2R} = 1.885\,\mathrm{m}^2$; $V_{\mathrm{cyl}} = 0.785\,\mathrm{m}^3$; $V_{\mathrm{sph}} = 4.761\,\mathrm{m}^3$.

Program 1.5: Grübler's formula and the workspace of a planar 2R arm.

```
% Program 5: Degrees of freedom and robot workspace
clear; clc; close all;
grubler = @(m, N, f) m*(N - 1 - numel(f)) + sum(f); % f = vector of joint freedoms
fprintf('6R arm DOF = %d\n', grubler(6, 7, ones(1,6)));
fprintf('Four-bar DOF = %d\n', grubler(3, 4, ones(1,4)));
fprintf('Five-bar DOF = %d\n', grubler(3, 5, ones(1,5)));
fprintf('Stewart UPS DOF = %d\n', grubler(6, 14, repmat([2 1 3], 1, 6)));

% ---- Planar 2R workspace by joint-space sampling
l1 = 0.5; l2 = 0.3;
th1 = 2*pi*rand(1, 20000); th2 = 2*pi*rand(1, 20000);
X = l1*cos(th1) + l2*cos(th1 + th2);
Y = l1*sin(th1) + l2*sin(th1 + th2);
figure; plot(X, Y, '.', 'MarkerSize', 2); axis equal; grid on;
xlabel('x (m)'); ylabel('y (m)'); title('Reachable workspace of a 2R arm');
fprintf('2R annulus area = %.3f m^2\n', pi*((l1 + l2)^2 - (l1 - l2)^2));

% ---- Workspace volumes
Vcyl = deg2rad(300)/2*(0.75^2 - 0.25^2)*0.6;
Vsph = 2*pi/3*(1.2^3 - 0.4^3)*(sind(60) - sind(-30));
fprintf('Cylindrical robot V = %.3f m^3, spherical robot V = %.3f m^3\n', Vcyl, Vsph);
```

1.19.6 Program 6 – Quadrature encoder decoding and velocity estimation

Algorithm. (1) Simulate a joint angle profile; (2) generate the A and B channels of an N-line encoder; (3) decode with a ×4 state-transition table; (4) estimate velocity by the frequency (M) method and compare with the true velocity. **Expected output:** decoded angle tracks the true angle within one count (0.088°); velocity estimate shows quantisation steps of $2\pi/(4NT_s)$.

Program 1.6: Simulation of a quadrature encoder with ×4 decoding.

```
% Program 6: Incremental encoder - quadrature generation, x4 decoding, speed
clear; clc; close all;
N = 1024; % lines per revolution
fs = 200e3; t = 0:1/fs:0.2; % fine time base for the waveforms (s)
theta = 2*pi*(2*t.^2 - 4*t.^3); % true shaft angle (rad), forward then back
q = floor(4*N*theta/(2*pi)); % angle in quarter-cycles (integer)
s = mod(q, 4); % quadrature state 0,1,2,3
A = (s == 1) | (s == 2); % channel A
B = (s == 2) | (s == 3); % channel B (lags A by 90 deg when forward)
% ---- Decoder: rebuild state from (A,B), count +1/-1 on each transition
sHat = 2*B + xor(A, B); % (0,0)->0 (1,0)->1 (1,1)->2 (0,1)->3
d = mod(diff(sHat) + 1, 4) - 1; % +1 forward, -1 reverse, 0 no change
count = [0 cumsum(d)];
thetaHat = count*2*pi/(4*N);
fprintf('Max decoding error = %.4f deg (one count = %.4f deg)\n', ...
        max(abs(rad2deg(thetaHat - theta))), 360/(4*N));

% ---- Velocity by the M-method with Ts = 1 ms
Ts = 1e-3; k = round(Ts*fs);
idx = 1:k:numel(t);
wHat = diff(count(idx))*2*pi/(4*N*Ts); % rad/s
wTrue = 2*pi*(4*t(idx(2:end)) - 12*t(idx(2:end)).^2);
plot(t(idx(2:end)), wTrue, 'k', t(idx(2:end)), wHat, 'r.'); grid on;
xlabel('Time (s)'); ylabel('\omega (rad/s)'); legend('true', 'M-method estimate');
title('Encoder velocity estimation');
```

1.19.7 Simulink models

The following Simulink exercises reinforce the dynamic-characteristics topics. They require only the standard Simulink library.

S1. First-order thermometer. Blocks: `Step` (initial value 25, final value 100, step time 0) → `Transfer Fcn` with numerator `[1]` and denominator `[8 1]` → `Scope`. Because the transfer function starts from zero, add a `Constant` of 25 to the output and use a step from 0 to 75, or set the block's initial state. Measure the time to 63.2% and compare with $\tau = 8$ s.

S2. Second-order force sensor. `Step` → `Transfer Fcn` with numerator `[wn^2]` and denominator `[1 2*zeta*wn wn^2]` (define `wn = 2*pi*500`, `zeta = 0.4` in the workspace) → `Scope`. Use solver `ode45` with maximum step 1×10^{-5} s. Read the overshoot and peak time and compare with Example 1.14. Repeat for $\zeta = 0.707$.

S3. Accelerometer under sinusoidal vibration. Model (1.38) with two `Integrator` blocks in a feedback loop: input $-a_b(t)$ from a `Sine Wave` block, gains $2\zeta\omega_n$ and ω_n^2. Plot $-\omega_n^2 z$ against a_b for $f/f_n = 0.1,\ 0.4,\ 1,\ 2$ and observe amplitude error and phase lag.

S4. Strain-gauge bridge with noise. Generate strain $\varepsilon(t)$ (`Signal Builder`), compute $V_o = V_{ex} G_F \varepsilon / 4$ with `Gain` blocks, add `Band-Limited White Noise`, amplify (gain 1000) and filter with a first-order low-pass `Transfer Fcn 1/(tau_f s + 1)`. Study the trade-off between noise and response time as τ_f changes.

Practical simulation exercises.

(1) Modify Program 1 to fit a second-order polynomial and compare its residuals with the linear fit. When is the higher-order fit justified?

(2) In Program 2, replace the normal distributions by uniform distributions of half-width equal to the tolerances and compare the Monte Carlo standard deviation with the RSS estimate using $a/\sqrt{3}$.

(3) In Program 3, find (numerically) the damping ratio that keeps the amplitude ratio within $\pm 5\%$ up to the highest possible r.

(4) In Program 6, add random noise to channel A and observe the effect on the decoded count. Why do industrial encoders use differential (RS-422) line drivers?

1.20 Unit Summary and Review

1.20.1 Unit summary

1. A robot is an automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes; it belongs to programmable and flexible automation.
2. A robot system comprises manipulator, end-effector, actuators and drives, internal and external sensors, controller, teach pendant and power supply. The arm (first three joints) positions the wrist; the wrist (last three joints) orients the tool.
3. The five basic serial configurations – Cartesian (PPP), cylindrical (RPP), spherical (RRP), SCARA (RRP with vertical axes) and articulated (RRR) – produce box, cylinder, spherical-shell, annular-cylinder and quasi-spherical work envelopes.
4. Degrees of freedom follow from Grübler's formula $m(N-1-J)+\sum f_i$; six DOF are needed for arbitrary position and orientation in space.
5. Robots are programmed in joint, world, base, tool and user coordinate systems.

6. A measurement system consists of sensing, variable conversion, signal processing and data presentation elements, and is affected by interfering and modifying inputs.
7. Static characteristics – accuracy, precision, sensitivity, linearity, hysteresis, resolution, threshold, drift – are obtained by calibration; uncertainty is evaluated by Type A and Type B methods and combined by RSS; loading effects are minimised by impedance considerations.
8. Dynamic characteristics are described by zero-, first- (τ) and second-order (ω_n, ζ) models; fidelity requires adequate bandwidth.
9. Position is sensed by potentiometers, LVDTs, encoders (incremental with quadrature, absolute with Gray code), resolvers, Hall and capacitive sensors; velocity by tachogenerators and encoders; acceleration by seismic (piezoelectric, MEMS) accelerometers; proximity by inductive, capacitive, magnetic, optical and ultrasonic sensors; force and torque by strain-gauge load cells, F/T sensors and joint torque sensors.
10. Temperature is measured with RTDs (Callendar–Van Dusen), thermistors (β equation), thermocouples (Seebeck effect, laws, cold-junction compensation) and IC sensors; pressure with manometers, Bourdon tubes, diaphragms and piezoresistive sensors; strain with bonded gauges in Wheatstone bridges.

1.20.2 Important concepts

- Repeatability versus accuracy of robots; precision versus accuracy of instruments.
- Reachable versus dexterous workspace; redundancy.
- Systematic versus random error; calibration removes bias but not scatter.
- Active versus passive transducers; internal versus external sensors.
- Bandwidth–sensitivity trade-off in seismic sensors.
- Bridge arithmetic: adjacent arms subtract, opposite arms add.

1.20.3 Formula sheet

Table 1.11: Unit I formula sheet.

Quantity	Formula	Ref.
Grübler's formula	$\mathrm{DOF} = m(N-1-J) + \sum f_i$	(1.5)
Cylindrical work volume	$V = \frac{\Delta\theta}{2}(r_{\max}^2 - r_{\min}^2)h$	(1.6)
Spherical work volume	$V = \frac{\Delta\theta}{3}(r_{\max}^3 - r_{\min}^3)(\sin\phi_2 - \sin\phi_1)$	(1.7)
Planar 2R reach	$\lvert l_1 - l_2\rvert \le r \le l_1 + l_2$	(1.8)
Sensitivity	$K = \Delta O / \Delta I$	(1.12)
Mean, standard deviation	$\bar{x} = \frac{1}{n}\sum x_i,\ s = \sqrt{\frac{1}{n-1}\sum (x_i - \bar{x})^2}$	(1.11)

Quantity	Formula	Ref.
Uncertainty propagation	$u_c^2 = \sum (\partial f/\partial x_i)^2 u^2(x_i)$	(1.16)
Digital resolution	$\mathrm{FSR}/2^N$	(1.15)
Loading error	$-R_{\mathrm{Th}}/(R_{\mathrm{Th}} + R_m)$	(1.19)
Least-squares slope	$m = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$	(1.20)
First-order step	$y = Ku_0(1 - e^{-t/\tau})$	(1.23)
First-order frequency	$1/\sqrt{1 + (\omega\tau)^2}$, $\phi = -\tan^{-1}\omega\tau$	(1.24)
Second-order indices	$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$, $t_p = \pi/\omega_d$, $t_s = 4/(\zeta\omega_n)$	(1.27)
Second-order frequency	$1/\sqrt{(1 - r^2)^2 + (2\zeta r)^2}$	(1.28)
Loaded potentiometer	$V_o/V_s = x/[1 + (R_p/R_L)x(1 - x)]$	(1.30)
Encoder resolution	$360°/(4N)$ (incremental), $360°/2^n$ (absolute)	(1.32)
Hall voltage	$V_H = IB/(nqt)$	(1.35)
Capacitance	$C = \varepsilon_0\varepsilon_r A/d$	(1.36)
Tachogenerator	$V_t = K_t\omega$	(1.37)
Seismic accelerometer	$z \approx -a/\omega_n^2$, $\omega_n = \sqrt{k/m}$	(1.39)
Ultrasonic range	$d = v_s t/2$, $v_s \approx 331.3 + 0.606T$	(1.40)
RTD (CVD, $T \geq 0$)	$R_T = R_0(1 + AT + BT^2)$	(1.43)
NTC thermistor	$R_T = R_{25}\exp[\beta(1/T - 1/T_{25})]$	(1.44)
Thermocouple	$E(T, 0) = E(T, T_r) + E(T_r, 0)$	(1.46)
Manometer	$\Delta p = \rho g h$	(1.47)
Diaphragm deflection	$y_c = 3p(1 - \nu^2)a^4/(16Et^3)$	(1.48)
Gauge factor	$G_F = (\Delta R/R)/\varepsilon$	(1.50)
Quarter bridge	$V_o \approx V_{\mathrm{ex}} G_F \varepsilon/4$	(1.51)
Cantilever strain	$\varepsilon = 6FL/(Ebh^2)$	(1.53)
Shaft torque strain	$\varepsilon_{45} = 8T/(\pi d^3 G)$	(1.54)

1.20.4 Comparison tables

Key terminology of this unit is collected in the consolidated glossary at the end of the book (Table 4.9). Comparison tables are provided within the unit: types of automation (Table 1.2), joints (Table 1.3), robot classification (Table 1.4), configurations (Table 1.5), sensors by measurand (Table 1.6), proximity sensors (Table 1.8), thermocouples (Table 1.9) and temperature sensors (Table 1.10). Two further summaries follow.

Table 1.12: Comparison of position sensors.

Sensor	Output	Contact	Strengths	Weaknesses
Potentiometer	Analogue, absolute	Wiper	Cheap, simple	Wear, noise, loading
LVDT	AC, absolute	None	Infinite resolution, rugged	Needs AC excitation, demodulator
Incremental encoder	Digital pulses, relative	None	High resolution, direction	Homing needed, count loss
Absolute encoder	Digital code, absolute	None	Position at power-up	More tracks, cost
Resolver	AC, absolute	None	Very rugged, high temperature	Needs RDC electronics
Hall sensor	Analogue or switch	None	Small, cheap, magnetic	Temperature sensitive, short range

Table 1.13: Static versus dynamic characteristics.

Static characteristics	Dynamic characteristics
Apply to constant or slowly varying inputs	Apply to time-varying inputs
Accuracy, precision, sensitivity, linearity, hysteresis, resolution, threshold, drift	Speed of response, time constant, natural frequency, damping, fidelity, lag, dynamic error, bandwidth
Determined by static calibration	Determined by step, ramp or frequency-response tests
Described by algebraic relations $O = f(I)$	Described by differential equations / transfer functions

1.21 Unit I Question Bank

This question bank contains exactly **30 questions of 10 marks each** (300 marks). Each question states the section (topic) of these notes on which it is based. The questions are distributed over the following categories:

Table 1.14: Distribution of Unit I questions.

Category	Questions	Number
Fundamental concepts and definitions	Q1–Q5	5
Theoretical explanations	Q6–Q12	7
Mathematical derivations	Q13–Q17	5
Numerical problems	Q18–Q23	6
Diagrams and working principles	Q24–Q26	3
Applications and analytical questions	Q27–Q30	4
Total		**30**

Question-to-notes reference index

For every question the table gives the topic (section) number, the page of the question and the page on which the topic begins in these notes. The **Refer to** line under each question also lists the specific figures, tables, equations and worked examples to study.

Table 1.15: Unit I question-to-notes reference index (automatic page references).

Q	Topic	Q. page	Notes page	Q	Topic	Q. page	Notes page
Q1	1.1	69	4	Q16	1.18	72	54
Q2	1.2	69	6	Q17	1.11.1	72	35
Q3	1.3	70	10	Q18	1.6	72	16
Q4	1.8	70	21	Q19	1.8.9	72	25
Q5	1.10	70	34	Q20	1.8.11	72	28
Q6	1.4	70	11	Q21	1.9	72	30
Q7	1.6	70	16	Q22	1.16	73	47
Q8	1.7	70	19	Q23	1.18	73	54
Q9	1.11.3	71	37	Q24	1.11	73	34
Q10	1.14	71	43	Q25	1.15	73	45
Q11	1.16	71	47	Q26	1.12	73	40
Q12	1.17	71	51	Q27	1.4	74	11
Q13	1.5	71	13	Q28	1.8.10	74	27
Q14	1.9	71	30	Q29	1.17	74	51
Q15	1.13	71	41	Q30	1.16	74	47

A. Fundamental concepts and definitions

Q1 **Topic 1.1** – Introduction to Robotics and Automation **[10 marks]**

Define an industrial robot according to ISO 8373 and explain the three features that distinguish a robot from other automatic machines. Define automation and compare fixed, programmable and flexible automation with respect to principle, production volume, product variety, investment and examples. Justify why industrial robots are classed as programmable/flexible automation.

☞ **Refer to:** Section 1.1 (p. 4); Section 1.5 (p. 13); Table 1.2 (p. 5)

Q2 **Topic 1.2** – Robot Anatomy and the Robot System **[10 marks]**

With a neat block diagram, explain the components of a complete robot system. Distinguish clearly between internal and external sensors, and explain the roles of the arm/body and the wrist of a six-axis manipulator. Define the tool centre point.

☞ **Refer to:** Section 1.2 (p. 6); Figure 1.1 (p. 7); Figure 1.2 (p. 7)

Q3 Topic 1.3 – Classification of Robots **[10 marks]**

Classify robots on the basis of (a) configuration, (b) kinematic structure, (c) drive, (d) control of motion, (e) programming method and (f) mobility/application. Explain the difference between non-servo, servo point-to-point and servo continuous-path robots with one application of each.

☞ **Refer to:** Section 1.3 (p. 10); Figure 1.4 (p. 10)

Q4 Topic 1.8 – Static Characteristics: Accuracy, Precision, Sensitivity, Resolution **[10 marks]**

Define and explain, with sketches of the static characteristic where appropriate: accuracy, precision (repeatability and reproducibility), sensitivity, resolution, threshold, linearity, hysteresis, dead band, zero drift and sensitivity drift. Why is it said that "resolution is not accuracy"?

☞ **Refer to:** Section 1.8 (p. 21); Figure 1.11 (p. 24); Figure 1.10 (p. 22); Example 1.7 (p. 21); Example 1.8 (p. 22)

Q5 Topic 1.10 – Sensors and Transducers – Basic Concepts **[10 marks]**

Distinguish between (a) sensor and transducer, (b) active and passive transducers, (c) analogue and digital sensors, and (d) internal (proprioceptive) and external (exteroceptive) robot sensors, giving two examples of each. List the desirable characteristics of a sensor for robotic use.

☞ **Refer to:** Section 1.10 (p. 34); Figure 1.9 (p. 20)

B. Theoretical explanations

Q6 Topic 1.4 – Robot Configurations **[10 marks]**

Sketch and explain the Cartesian, cylindrical, spherical, SCARA and articulated robot configurations. For each, state the joint sequence, the shape of the work envelope, the position equations (where simple), two advantages, one limitation and typical applications.

☞ **Refer to:** Section 1.4 (p. 11); Figure 1.5 (p. 12)

Q7 Topic 1.6 – Work Envelope and Coordinate Systems **[10 marks]**

Explain the terms work envelope, reachable workspace and dexterous workspace. Describe the joint, world, base, tool and user coordinate systems used in robot programming with a neat sketch, and explain the practical advantage of teaching points in a user (workpiece) frame.

☞ **Refer to:** Section 1.6 (p. 16); Figure 1.7 (p. 17); Figure 1.8 (p. 19); Example 1.4 (p. 17); Example 1.5 (p. 18)

Q8 Topic 1.7 – Measurement Systems **[10 marks]**

With a neat block diagram, explain the generalised measurement system and the function of each element using a robot wrist force sensor as the example. Explain interfering and modifying inputs and the methods used to reduce their effects.

☞ **Refer to:** Section 1.7 (p. 19); Figure 1.9 (p. 20)

Q9 **Topic 1.11.3** – Optical Encoders for Position Measurement **[10 marks]**

Explain the construction and working of incremental and absolute optical encoders. Explain how direction is detected using quadrature signals, how ×4 decoding improves resolution, and why absolute encoders use Gray code. Convert binary 0110 to Gray code and Gray 1101 to binary.

☞ **Refer to:** Section 1.11.3 (p. 37); Figure 1.19 (p. 38); Equation (1.33) (p. 39); Equation (1.34) (p. 39); Example 1.17 (p. 38); Example 1.18 (p. 39)

Q10 **Topic 1.14** – Proximity Measurement **[10 marks]**

Explain the working principles of inductive, capacitive, magnetic, photoelectric and ultrasonic proximity sensors. Compare them in a table with respect to detectable materials, range, advantages and limitations, and state where each would be used in a robotic workcell.

☞ **Refer to:** Section 1.14 (p. 43); Figure 1.22 (p. 44); Example 1.21 (p. 45); Example 1.22 (p. 45)

Q11 **Topic 1.16** – Temperature Measurement – Thermocouples **[10 marks]**

Explain the Seebeck effect and the working of a thermocouple. State and explain the laws of homogeneous circuits, intermediate metals and intermediate temperatures. Explain why cold-junction compensation is necessary and how it is implemented.

☞ **Refer to:** Section 1.16 (p. 47); Figure 1.25 (p. 50); Example 1.23 (p. 48); Example 1.24 (p. 49)

Q12 **Topic 1.17** – Pressure Measurement **[10 marks]**

Define absolute, gauge and differential pressure. Explain, with neat sketches, pressure measurement using (a) the U-tube manometer, (b) the Bourdon tube, (c) a diaphragm with strain gauges and (d) a piezoresistive silicon sensor. State where each is used.

☞ **Refer to:** Section 1.17 (p. 51); Figure 1.27 (p. 53); Example 1.26 (p. 52); Example 1.27 (p. 53)

C. Mathematical derivations

Q13 **Topic 1.5** – Degrees of Freedom **[10 marks]**

Derive Grübler's formula for the degrees of freedom of a mechanism. Show that the DOF of a serial chain equals the sum of its joint freedoms. Apply the formula to (a) a planar four-bar linkage, (b) a planar slider–crank mechanism and (c) a 3-RPS spatial parallel robot.

☞ **Refer to:** Section 1.5 (p. 13); Example 1.1 (p. 15); Example 1.2 (p. 16)

Q14 **Topic 1.9** – Dynamic Characteristics – First-Order Instrument **[10 marks]**

Derive the differential equation of a first-order instrument using a thermometer as the example, and obtain its response to a step input. Explain the significance of the time constant. Derive the amplitude ratio and phase for a sinusoidal input and interpret them.

☞ **Refer to:** Section 1.9 (p. 30); Figure 1.15 (p. 32); Example 1.12 (p. 32); Example 1.13 (p. 32)

Q15 **Topic 1.13** – Acceleration Measurement **[10 marks]**

Derive the equation of motion of a seismic instrument and show under what conditions it acts as (a) an accelerometer and (b) a displacement pick-up (seismometer). Derive its frequency response and explain the choice of damping ratio.

☞ **Refer to:** Section 1.13 (p. 41); Figure 1.21 (p. 42); Example 1.20 (p. 43)

Q16 Topic 1.18 – Strain Measurement **[10 marks]**

Derive the expression for the gauge factor of a resistance strain gauge. Derive the output voltage of a Wheatstone bridge with one active gauge, and hence state the outputs of half and full bridges. Explain how a dummy gauge provides temperature compensation.

☞ **Refer to:** Section 1.18 (p. 54); Figure 1.29 (p. 56); Example 1.29 (p. 57); Example 1.30 (p. 57)

Q17 Topic 1.11.1 – Displacement Measurement – Potentiometer **[10 marks]**

Derive the output voltage of a loaded potentiometric displacement sensor and show that the error is zero at both ends of the travel. A 2 kΩ linear potentiometer supplied at 12 V is connected to a load of 10 kΩ. Find the output and the error (%FS) when the wiper is at 75% of its travel, and state the maximum error.

☞ **Refer to:** Section 1.11.1 (p. 35); Figure 1.17 (p. 36); Example 1.15 (p. 36); Example 1.16 (p. 37)

D. Numerical problems

Q18 Topic 1.6 – Work Envelope Calculations **[10 marks]**

(a) A cylindrical robot has a radial stroke from 0.2 m to 0.9 m, a vertical stroke of 0.75 m and a base rotation of 270°. Derive the expression for its work volume and calculate it. (b) A spherical robot has r from 0.5 m to 1.4 m, full base rotation and elevation from −20° to +70°. Calculate its work volume.

☞ **Refer to:** Section 1.6 (p. 16); Example 1.4 (p. 17); Example 1.5 (p. 18)

Q19 Topic 1.8.9 – Accuracy, Precision and Uncertainty **[10 marks]**

(a) A force sensor loaded with a 50.0 N reference weight gives ten readings (N): 50.6, 50.2, 50.9, 50.4, 50.5, 50.3, 50.8, 50.6, 50.4, 50.5. Find the mean, bias, sample standard deviation and Type A standard uncertainty of the mean. (b) The power dissipated in a resistor is found from $P = I^2R$ with $I = 3.00\,\text{A} \pm 0.02\,\text{A}$ and $R = 8.0\,\Omega \pm 0.1\,\Omega$. Find P, its combined (RSS) uncertainty, the expanded uncertainty ($k = 2$) and the worst-case error.

☞ **Refer to:** Section 1.8.9 (p. 25); Equation (1.18) (p. 26); Example 1.9 (p. 26)

Q20 Topic 1.8.11 – Static Calibration **[10 marks]**

A temperature transmitter is calibrated with the following results:

T (°C)	0	25	50	75	100
Output (mA)	4.10	8.05	11.96	16.08	19.98

Determine the least-squares calibration line, the sensitivity, the zero offset and the non-linearity (%FSO). What temperature corresponds to an output of 13.50 mA?

☞ **Refer to:** Section 1.8.11 (p. 28); Equation (1.20) (p. 28); Example 1.11 (p. 28)

Q21 Topic 1.9 – Dynamic Response Calculations **[10 marks]**

(a) A thermometer with $\tau = 12$ s, initially at 20 °C, is plunged into a bath at 80 °C. Find the reading after 15 s and the time to read 79 °C. (b) A force transducer has $f_n = 200$ Hz and $\zeta = 0.6$. Find the maximum overshoot, peak time and 2% settling time for a step input, and the amplitude error and phase lag for a sinusoidal force at 80 Hz.

☞ **Refer to:** Section 1.9 (p. 30); Example 1.12 (p. 32); Example 1.13 (p. 32)

Q22 Topic 1.16 – Temperature Measurement Calculations **[10 marks]**

(a) Find the resistance of a Pt100 at 250 °C using the Callendar–Van Dusen equation and the error that would result from the linear model with $\alpha = 0.003\,85/^\circ\text{C}$. (b) An NTC thermistor has $R_{25} = 10\,\text{k}\Omega$ and $\beta = 3435\,\text{K}$. Find its resistance at 0 °C and the temperature at which $R = 3.0\,\text{k}\Omega$. (c) A type K thermocouple reads 11.006 mV with its reference junction at 30 °C. Given $E_K(30, 0) = 1.203\,\text{mV}$ and $E_K(300, 0) = 12.209\,\text{mV}$, find the measured temperature.

☞ **Refer to:** Section 1.16 (p. 47); Example 1.23 (p. 48); Example 1.24 (p. 49)

Q23 Topic 1.18 – Force and Torque Measurement Calculations **[10 marks]**

(a) A steel cantilever load cell ($E = 200\,\text{GPa}$, $b = 30\,\text{mm}$, $h = 8\,\text{mm}$) carries 200 N at 120 mm from the gauge centre. Four gauges ($G_F = 2.1$) form a full bridge excited at 10 V. Find the surface strain, stress, bridge output and rated output in mV/V. (b) A 25 mm diameter steel shaft ($G = 79.3\,\text{GPa}$) transmits 120 N m. Find the strain in gauges mounted at ±45° and the output of a full bridge with $G_F = 2.0$ and $V_{\text{ex}} = 5\,\text{V}$.

☞ **Refer to:** Section 1.18 (p. 54); Example 1.29 (p. 57); Example 1.30 (p. 57)

E. Diagrams and working principles

Q24 Topic 1.11 – Displacement Measurement – LVDT **[10 marks]**

With a neat sketch, explain the construction and working of an LVDT. Draw its output characteristic and explain the need for phase-sensitive demodulation. An LVDT has a demodulated sensitivity of 0.4 V/mm; interpret outputs of +1.2 V and −0.6 V. List its advantages and limitations.

☞ **Refer to:** Section 1.11 (p. 34); Figure 1.18 (p. 37); Example 1.15 (p. 36); Example 1.16 (p. 37)

Q25 Topic 1.15 – Force and Torque Measurement in Robots **[10 marks]**

With neat diagrams, explain how forces and torques are measured in robots using (a) a six-axis wrist force/torque sensor, (b) a strain-gauge joint torque sensor and (c) motor-current estimation. Explain the role of the calibration matrix and compare the three methods.

☞ **Refer to:** Section 1.15 (p. 45); Figure 1.23 (p. 46)

Q26 Topic 1.12 – Velocity Measurement **[10 marks]**

With neat diagrams, explain velocity measurement using (a) a DC tachogenerator and (b) an incremental encoder by the frequency (M) and period (T) methods. A tachogenerator has $K_t = 3\,\text{V}$ per 1000 rpm; find the speed when it gives 4.5 V. A 2000-line encoder with ×4 decoding records 400 counts in 5 ms; find the speed and the speed resolution of the M method. With a 10 MHz timer, 2500 clock counts are recorded between successive edges; find the speed.

☞ **Refer to:** Section 1.12 (p. 40); Figure 1.20 (p. 41); Example 1.19 (p. 41)

F. Applications and analytical questions

Q27 Topic 1.4 – Selection of Robot Configuration **[10 marks]**

Select, with justification based on work envelope, accuracy, speed and cost, a suitable robot configuration and type of motion control for: (a) insertion of components on printed-circuit boards, (b) palletising 25 kg bags at the end of a packing line, (c) arc welding of a car exhaust assembly, (d) loading and unloading a horizontal injection-moulding machine, (e) high-speed picking of biscuits from a conveyor.

☞ **Refer to:** Section 1.4 (p. 11)

Q28 Topic 1.8.10 – Loading Effects **[10 marks]**

Explain electrical, mechanical and thermal loading effects with examples. A sensor with Thévenin output 2.0 V and source resistance 10 kΩ is connected to an ADC of input resistance 100 kΩ; find the reading and the loading error and suggest a remedy. A 20 g accelerometer is attached to a structure whose vibrating mode has an effective mass of 200 g and natural frequency 500 Hz; estimate the frequency actually measured.

☞ **Refer to:** Section 1.8.10 (p. 27); Equation (1.19) (p. 27); Example 1.10 (p. 27)

Q29 Topic 1.17 – Pressure and Proximity Sensing in a Gripper Application **[10 marks]**

A robot uses four vacuum cups of 30 mm effective diameter at a vacuum of −70 kPa (gauge) to lift cardboard cartons vertically with an upward acceleration of 3 m/s^2. Using a safety factor of 2, find the maximum carton mass. Propose a sensing scheme (sensor types and logic) to confirm part pick-up, detect loss of vacuum during transport and detect the carton on the infeed conveyor; justify each sensor choice.

☞ **Refer to:** Section 1.17 (p. 51); Example 1.26 (p. 52); Example 1.27 (p. 53)

Q30 Topic 1.16 – Selection of Temperature Sensors for Robot Joints **[10 marks]**

Propose and justify a temperature-monitoring scheme for the joint motors and drives of a six-axis robot. Compare RTDs, thermistors, thermocouples and IC sensors for this purpose. If an NTC thermistor ($R_{25} = 10\,\text{k}\Omega$, $\beta = 3950\,\text{K}$) is used at the bottom of a divider with a 10 kΩ resistor from 5 V, find the divider output at 25 °C and 100 °C and comment on the result.

☞ **Refer to:** Section 1.16 (p. 47); Example 1.23 (p. 48); Example 1.24 (p. 49)

Unit II: Actuators, Robot Mechanisms and Measurement Systems

Unit overview

If sensors are the robot's senses (Unit I), actuators are its muscles and the measurement chain is its nervous system. This unit first develops the physics and mathematical models of the actuators that move robot joints and grippers – electric motors of four kinds, pneumatic and hydraulic cylinders, solenoids and linear actuators (Sections 2.1 to 2.8). It then shows how actuators are combined with transmissions into joints, mechanisms, serial and parallel manipulators and end-effectors (Sections 2.9 to 2.11). The second half follows the signal from a sensor to the robot computer: measurement of electrical and physical quantities, signal conditioning, instrumentation amplifiers, filters, shielding and grounding, analogue-to-digital and digital-to-analogue conversion, and complete data acquisition systems (Sections 2.12 to 2.20).

Syllabus coverage map. Table 2.1 maps every syllabus item to its section; these section numbers are the topic numbers used in the question bank.

Table 2.1: Unit II syllabus items and the sections that cover them.

No.	Syllabus item	Section	No.	Syllabus item	Section
1	Electrical actuators	2.1	14	End-effectors	2.11
2	Pneumatic actuators	2.2	15	Grippers	2.11
3	Hydraulic actuators	2.3	16	Electrical quantities	2.12
4	DC motors	2.4	17	Physical quantities	2.13
5	Servo motors	2.5	18	Signal conditioning	2.14
6	Stepper motors	2.6	19	Instrumentation amplifiers	2.15
7	BLDC motors	2.7	20	Filters	2.16
8	Solenoids	2.8	21	Shielding	2.17
9	Linear actuators	2.8	22	Grounding	2.17
10	Robot joints	2.9	23	A/D conversion	2.18
11	Robot mechanisms	2.9	24	D/A conversion	2.19
12	Serial manipulators	2.10	25	Data acquisition systems	2.20
13	Parallel manipulators	2.10			

2.1 Actuators for Robotics: Electrical, Pneumatic and Hydraulic

2.1.1 Definition and role

An *actuator* is a device that converts a controlled input of energy – electrical, pneumatic or hydraulic – into mechanical motion (force and displacement, or torque and rotation). In a robot each joint is driven by one actuator through a transmission, and the end-effector usually has its own small actuator. The choice of actuator technology fixes the robot's payload, speed, precision, cleanliness, safety and cost, so it is one of the earliest and most consequential design decisions [7, 14].

2.1.2 The three actuator families

Figure 2.1 compares the three families visually.

Electrical actuators convert electrical energy into motion by the Lorentz force on a current-carrying conductor in a magnetic field ($\mathbf{F} = I\,\mathbf{L} \times \mathbf{B}$) or by magnetic reluctance forces. They include brushed DC motors, servo motors, stepper motors, brushless DC (BLDC) and AC synchronous motors, solenoids and electric linear actuators. Their decisive advantage is controllability: current, and therefore torque, can be regulated in microseconds by power electronics, and position is measured precisely by encoders. Their weakness is low torque density, which is overcome by high-ratio gearing (harmonic drives, cycloidal reducers). Today nearly all industrial and collaborative robots are electrically driven.

Pneumatic actuators use compressed air, typically at 4 bar to 8 bar gauge, acting on a piston or vane. Air is cheap, clean, readily available in factories, and tolerant of overload (a stalled cylinder simply holds its force). However, air is compressible, so precise intermediate positioning is difficult; pneumatic actuators are normally driven end-to-end against stops. They dominate grippers, vacuum handling and simple pick-and-place units.

Hydraulic actuators use an almost incompressible oil at 70 bar to 350 bar. Because force equals pressure times area, a compact cylinder can produce tens of kilonewtons; hydraulic systems have the highest power-to-weight ratio of the three and are stiff and responsive when controlled by servo-valves. Their drawbacks are leaks, fire risk, noise, maintenance, and the need for a pump, tank and filtration. They are used in very heavy robots (forging manipulators, construction and legged field robots).

The engineering implication of Table 2.2 is that there is no universally best actuator. A typical industrial cell therefore mixes technologies: electric servo motors for the robot joints, a pneumatic gripper or vacuum cups at the wrist, and perhaps a hydraulic press elsewhere in the line.

2.2 Pneumatic Actuators

2.2.1 Construction and working

A pneumatic system (Fig. 2.2) consists of a compressor that raises atmospheric air to the supply pressure p_s, an FRL unit (filter to remove water and particles, regulator to set the working

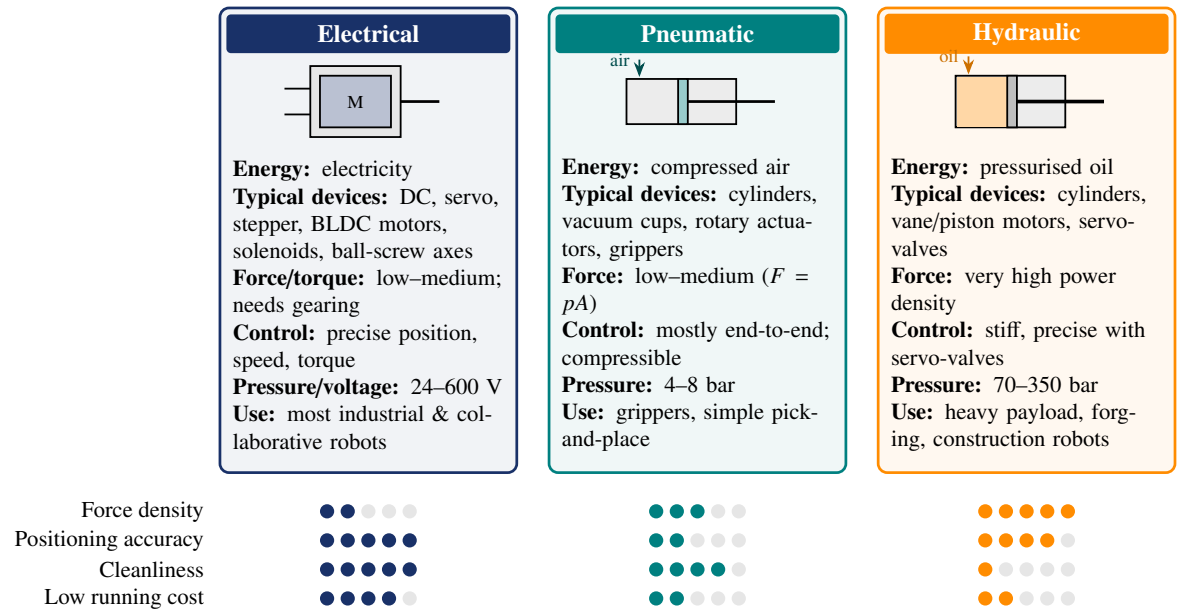


Figure 2.1: Comparison of electrical, pneumatic and hydraulic actuators. Dots show relative ratings on a 1–5 scale (more dots = better).

Table 2.2: Comparison of actuator technologies for robots.

Property	Electrical	Pneumatic	Hydraulic
Energy medium	Electric current	Compressed air, 4–8 bar	Oil, 70–350 bar
Force/power density	Low–medium (needs gearing)	Low–medium	Very high
Positioning	Excellent (servo)	Poor (compressible); end stops	Good with servo-valves
Speed of response	Fast	Fast but spongy	Fast, stiff
Cleanliness	Clean	Clean (exhaust air)	Risk of oil leaks
Infrastructure	Power supply, drives	Compressor, FRL, valves	Pump, tank, filters, cooling
Typical robot use	Joints of most industrial robots and cobots	Grippers, vacuum cups, simple axes	Heavy payload, forging, mobile hydraulics

pressure, lubricator), directional control valves, and actuators. The most common actuator is the *double-acting cylinder*: a piston of area A slides in a barrel; air admitted to port A pushes the piston out (extend stroke) while port B is exhausted, and vice versa. A *single-acting* cylinder is pressurised on one side only and returned by a spring. *Rotary actuators* (vane or rack-and-pinion) produce limited rotation for wrist flips and swivel units.

The *5/2 directional control valve* has five ports (P supply, A and B to the cylinder, two exhausts) and two positions; a solenoid shifts the spool against a spring, so one digital output of a PLC or robot controller decides the direction of motion.

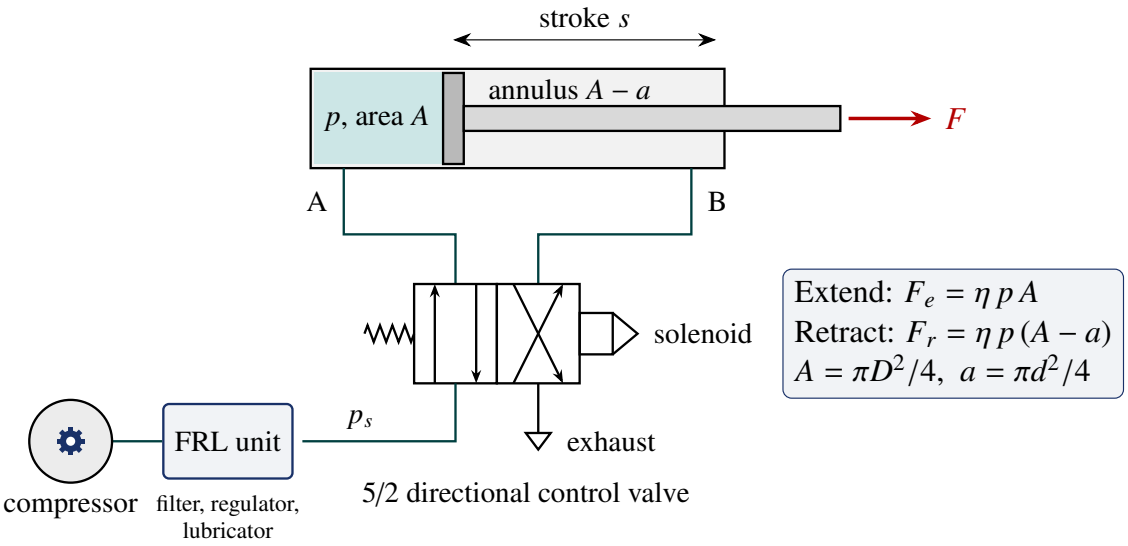


Figure 2.2: Electro-pneumatic circuit: compressor, FRL unit, solenoid-operated 5/2 valve and double-acting cylinder.

2.2.2 Force, speed and air consumption

Force – derivation. Pressure is force per unit area, so the force on the piston is pA. During retraction the rod occupies area a, leaving the annulus $A - a$. Friction in seals is accounted for by an efficiency η (typically 0.85–0.95):

$$F_{\text{ext}} = \eta\, p\, A = \eta\, p\, \frac{\pi D^2}{4}, \qquad F_{\text{ret}} = \eta\, p\,(A - a) = \eta\, p\, \frac{\pi (D^2 - d^2)}{4}, \tag{2.1}$$

where p is the gauge pressure (Pa), D the bore and d the rod diameter (m). The retract force is always smaller.

Air consumption. Each extend–retract cycle fills a volume $(A + (A - a))s$ at absolute pressure $p + p_{\text{atm}}$. Expressed as free (atmospheric) air by Boyle's law $p_1 V_1 = p_2 V_2$ (isothermal):

$$V_{\text{free}} = (2A - a)\, s\, \frac{p + p_{\text{atm}}}{p_{\text{atm}}}. \tag{2.2}$$

Speed is set by flow-control valves restricting the exhaust (meter-out), and cushions at the end of stroke decelerate the piston.

Example 2.1: Pneumatic cylinder sizing (Topic 2.2)

Given: Double-acting cylinder, bore $D = 50\,\text{mm}$, rod $d = 20\,\text{mm}$, $p = 6\,\text{bar}$ gauge, $\eta = 0.9$, stroke 200 mm.

Find: Extend and retract forces; free-air consumption per cycle.

Solution: $A = \pi(0.05)^2/4 = 1.963 \times 10^{-3}\,\text{m}^2$; $a = \pi(0.02)^2/4 = 3.14 \times 10^{-4}\,\text{m}^2$.

$$F_{\text{ext}} = 0.9 \times 6 \times 10^5 \times 1.963 \times 10^{-3} = 1060\,\text{N}, \qquad F_{\text{ret}} = 0.9 \times 6 \times 10^5 \times 1.649 \times 10^{-3} = 891\,\text{N}.$$

$V_{\text{free}} = (2 \times 1.963 - 0.314) \times 10^{-3} \times 0.2 \times (7.013/1.013) = 5.0 \times 10^{-3}\,\text{m}^3 = 5.0\,\text{L}$ per cycle.

Interpretation: At 10 cycles per minute the cylinder consumes 50 L/min of free air; summing such figures over all actuators sizes the compressor.

Advantages: simple, cheap, clean, fast, overload-safe, explosion-proof. **Limitations:** poor intermediate positioning, noise, low efficiency (compressing air wastes energy), limited force.

2.3 Hydraulic Actuators

2.3.1 Principle and circuit

Hydraulics rests on *Pascal's law*: pressure applied to a confined incompressible fluid is transmitted undiminished in all directions. A pump driven by an electric motor delivers flow Q (m^3/s); a relief valve limits the maximum system pressure; a directional or servo/proportional valve meters oil to the cylinder or hydraulic motor; oil returns to the tank through a filter (Fig. 2.3).

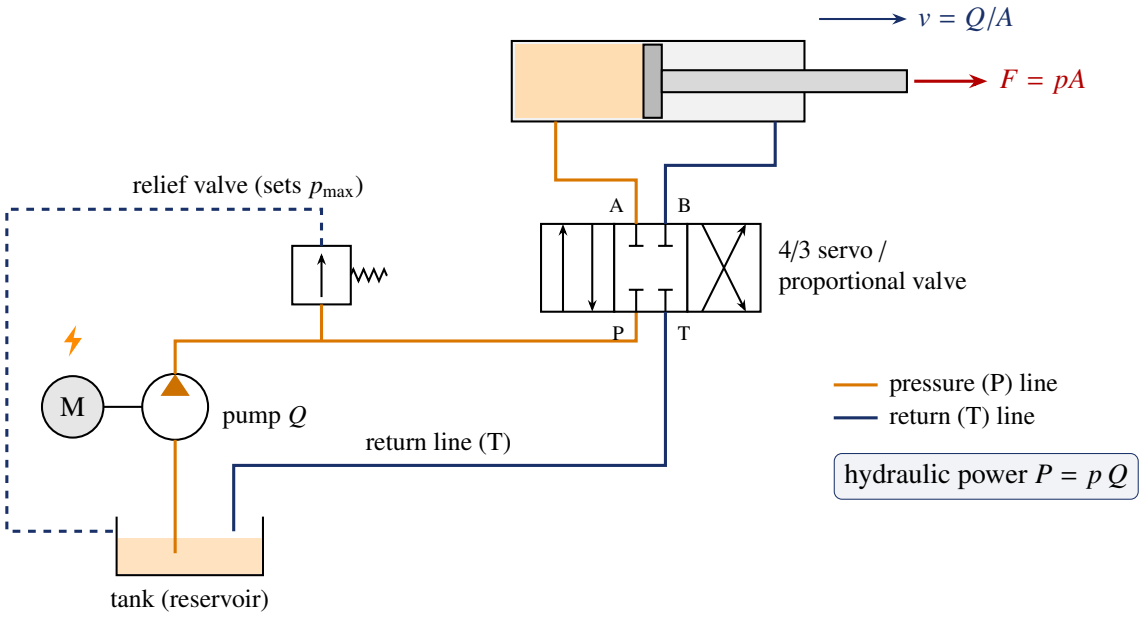


Figure 2.3: Basic hydraulic circuit: motor-driven pump, relief valve, 4/3 directional (or servo) valve, cylinder and tank.

2.3.2 Governing equations

$$F = p\,A \qquad \text{(cylinder force, N)} \tag{2.3}$$

$$v = \frac{Q}{A} \qquad \text{(piston speed, m/s; continuity)} \tag{2.4}$$

$$P = p\,Q = F\,v \qquad \text{(hydraulic power, W)} \tag{2.5}$$

Equation (2.4) follows from conservation of volume for an incompressible fluid: the volume entering the cylinder per second, Q, equals the swept volume per second, Av. Multiplying (2.3) by (2.4) gives (2.5): hydraulic power is pressure times flow. For a hydraulic motor of displacement D_m (m^3/rad) the torque is $T = pD_m$ and the speed $\omega = Q/D_m$.

Example 2.2: Hydraulic cylinder (Topic 2.3)

Given: Bore 80 mm, pressure 10 MPa (100 bar), pump flow 20 L/min.

Solution: $A = \pi(0.08)^2/4 = 5.027 \times 10^{-3}\,\text{m}^2$. $F = 10^7 \times 5.027 \times 10^{-3} = 50.3\,\text{kN}$. $Q = 20 \times 10^{-3}/60 = 3.333 \times 10^{-4}\,\text{m}^3/\text{s}$; $v = 3.333 \times 10^{-4}/5.027 \times 10^{-3} = 0.066\,\text{m/s}$. $P = 10^7 \times 3.333 \times 10^{-4} = 3.33\,\text{kW}$.

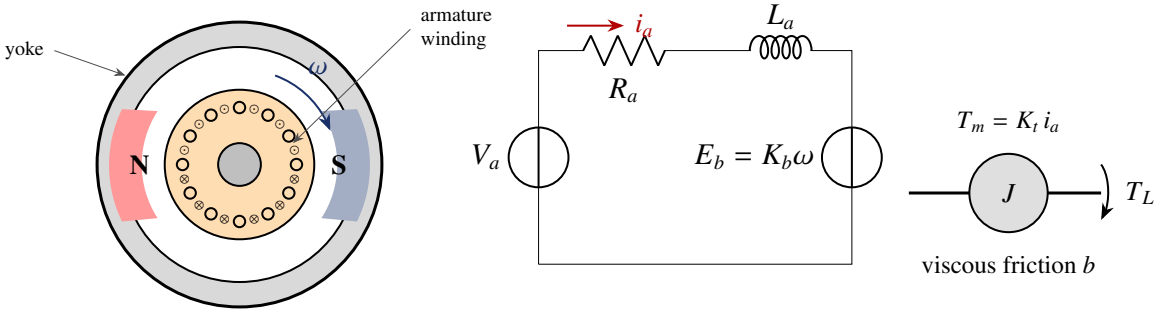
Interpretation: A pneumatic cylinder of the same bore at 6 bar would give only about 3 kN; this 17-fold force advantage explains hydraulics' role in heavy robots.

Advantages: very high force and power density, stiffness, self-lubrication, smooth low-speed control. **Limitations:** leakage, contamination sensitivity, heat, noise, fire hazard, bulky power unit.

2.4 DC Motors

2.4.1 Construction

A permanent-magnet (PM) brushed DC motor (Fig. 2.4(a)) has a stationary *stator* – a steel yoke carrying permanent-magnet poles that produce a radial magnetic field – and a rotating *armature* with windings in slots. The *commutator*, a segmented copper ring on the shaft, and carbon *brushes* reverse the current in each armature coil as it passes from one pole to the next, so the torque on the rotor always acts in the same direction.



(a) Construction: field poles (N, S), yoke, armature conductors, shaft

(b) Equivalent circuit and mechanical model

Figure 2.4: DC motor: (a) cross-section; (b) armature equivalent circuit coupled to the mechanical load.

2.4.2 Mathematical model – derivation

Torque equation. A conductor of length l carrying current i in flux density B experiences force Bil; summed over all conductors at radius r, the torque is proportional to the armature current:

$$T_m = K_t\, i_a, \tag{2.6}$$

where K_t (N m/A) is the torque constant.

Back-EMF equation. The same conductors moving through the field generate an EMF (Faraday's law) proportional to speed:

$$E_b = K_b\, \omega, \tag{2.7}$$

where K_b (V s/rad) is the back-EMF constant. Energy conservation – the electrical power $E_b i_a$ converted equals the mechanical power $T_m \omega$ – shows that $K_t = K_b$ in consistent SI units.

Electrical equation. Kirchhoff's voltage law around the armature circuit of Fig. 2.4(b):

$$v_a = R_a i_a + L_a \frac{di_a}{dt} + K_b \omega. \tag{2.8}$$

Mechanical equation. Newton's law for the rotor and load of inertia J (kg m^2) with viscous friction b (N m s/rad) and load torque T_L:

$$J \frac{d\omega}{dt} + b\,\omega = K_t i_a - T_L. \tag{2.9}$$

Transfer function. Taking Laplace transforms with zero initial conditions and $T_L = 0$, and eliminating $I_a(s)$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(Js + b)(L_a s + R_a) + K_t K_b} \approx \frac{1/K_b}{(\tau_m s + 1)(\tau_e s + 1)}, \tag{2.10}$$

with electrical time constant $\tau_e = L_a/R_a$ and mechanical time constant $\tau_m = JR_a/(K_t K_b)$ (approximation valid for $b \approx 0$ and $\tau_e \ll \tau_m$).

2.4.3 Steady-state characteristics

In steady state ($di/dt = 0$, $d\omega/dt = 0$), substituting $i_a = T/K_t$ into (2.8):

$$\omega = \frac{V_a}{K_b} - \frac{R_a}{K_t K_b}\, T. \tag{2.11}$$

This straight-line *speed–torque characteristic* (Fig. 2.5(a)) is defined by two points: the *no-load speed* $\omega_0 = V_a/K_b$ and the *stall torque* $T_s = K_t V_a/R_a$. Changing V_a shifts the line parallel to itself, which is why armature-voltage (PWM) control is the standard method of speed control. The output power $P_o = T\omega$ is a parabola with maximum $P_{\max} = T_s \omega_0/4$ at half the stall torque; efficiency is highest at low torque (Fig. 2.5(b)).

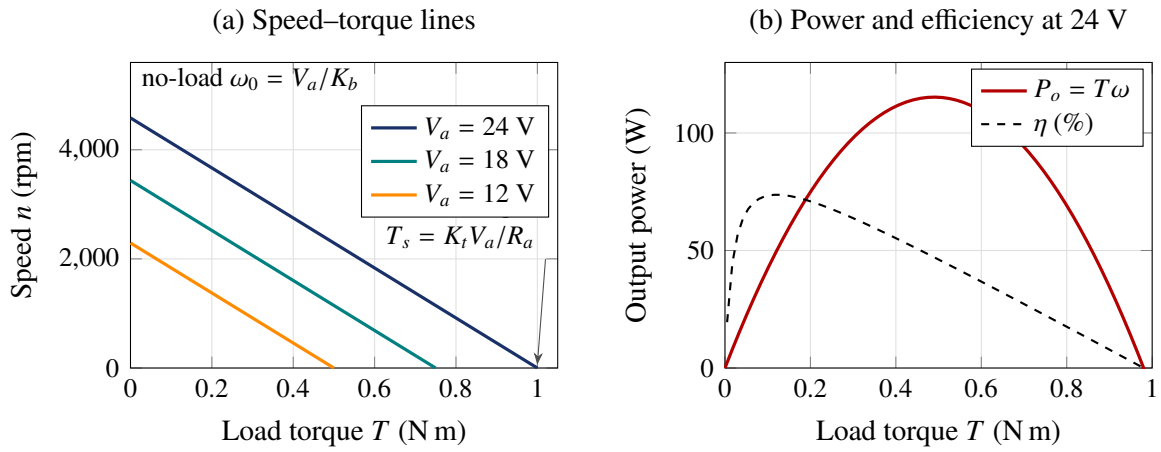


Figure 2.5: DC motor characteristics ($R_a = 1.2\,\Omega$, $K_t = K_b = 0.05\,\text{N m/A}$): (a) speed–torque lines at three voltages; (b) output power and efficiency at 24 V (friction torque 0.02 N m).

Example 2.3: DC motor operating point (Topic 2.4)

Given: $V_a = 24\,\text{V}$, $R_a = 1.2\,\Omega$, $K_t = K_b = 0.05\,\text{N m/A}$, load 0.2 N m; neglect friction.

Find: Current, back-EMF, speed, output and input power, efficiency, stall torque, no-load speed.

Solution:

1. $i_a = T/K_t = 0.2/0.05 = 4\,\text{A}$; $E_b = V_a - i_a R_a = 24 - 4.8 = 19.2\,\text{V}$.
2. $\omega = E_b/K_b = 19.2/0.05 = 384\,\text{rad/s}$ (3667 rpm).
3. $P_o = T\omega = 0.2 \times 384 = 76.8\,\text{W}$; $P_i = V_a i_a = 96\,\text{W}$; $\eta = 80\%$ (the 19.2 W difference is copper loss $i_a^2 R_a$).
4. $T_s = K_t V_a/R_a = 0.05 \times 24/1.2 = 1.0\,\text{N m}$; $\omega_0 = V_a/K_b = 480\,\text{rad/s}$ (4584 rpm); $P_{\max} = 1.0 \times 480/4 = 120\,\text{W}$.

Interpretation: The motor runs at 20% of stall torque, a good efficient operating point; continuous operation near stall would overheat the winding.

Example 2.4: Time constants of a DC motor (Topic 2.4)

Given: Same motor, $J = 2 \times 10^{-5}\,\text{kg m}^2$, $L_a = 0.5\,\text{mH}$.

Solution: $\tau_e = L_a/R_a = 0.5 \times 10^{-3}/1.2 = 0.42\,\text{ms}$; $\tau_m = JR_a/(K_t K_b) = 2 \times 10^{-5} \times 1.2/0.0025 = 9.6\,\text{ms}$.

Interpretation: Since $\tau_e \ll \tau_m$, current responds about 20 times faster than speed; this separation justifies the cascade current–velocity–position control of Section 2.5.

Speed control by PWM and the H-bridge. A DC motor is driven by an H-bridge of four transistors switched by a pulse-width-modulated (PWM) signal at 10 kHz to 40 kHz. The motor's inductance smooths the current, so the effective armature voltage is the average $V_a = D\,V_{\mathrm{dc}}$, where D is the duty cycle; reversing the diagonal pair of transistors reverses the direction. PWM is treated in detail in Unit IV.

Advantages: simple linear control, high starting torque, cheap drives. **Limitations:** brush and commutator wear, sparking (EMI, not for explosive atmospheres), heat generated in the rotor is hard to remove.

2.5 Servo Motors and Servo Control

2.5.1 What a servo motor is

A *servo motor* is not a different motor type but a motor – DC, BLDC or AC synchronous – combined with a feedback sensor and a controller so that it follows a commanded position, speed or torque precisely. Industrial robot joints use *AC servo motors* (permanent-magnet synchronous motors) with absolute encoders; small robots and models use *RC servos*, which contain a DC motor, gearbox, potentiometer and control circuit in one housing.

2.5.2 Cascade control architecture

Figure 2.6(a) shows the standard industrial architecture. Three nested loops run at increasing rates:

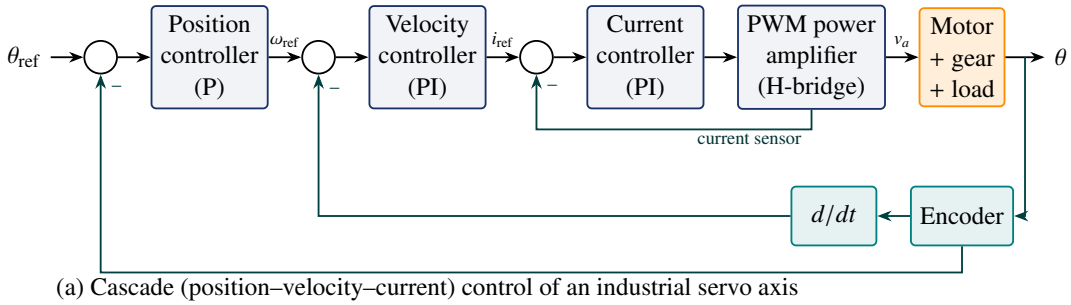
1. the **current (torque) loop** (10 kHz to 20 kHz) regulates armature current, hence torque, using a PI controller and current sensors;
2. the **velocity loop** (2 kHz to 8 kHz) compares commanded and measured speed (from the differentiated encoder signal) and outputs a current reference;
3. the **position loop** (0.5 kHz to 2 kHz) compares commanded and measured position and outputs a velocity reference.

Each inner loop is several times faster than the loop around it – exactly the time-scale separation shown in Example 2.4 – so each outer loop sees its inner loop as a nearly ideal actuator, and each loop can be tuned independently.

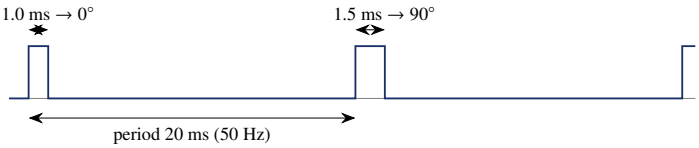
RC servo command. The RC servo of Fig. 2.6(b) receives a pulse every 20 ms; the pulse width t_p from 1.0 ms to 2.0 ms maps linearly to an angle of 0° to 180°:

$$\theta = 180^\circ \,\frac{t_p - 1.0\,\mathrm{ms}}{1.0\,\mathrm{ms}}. \tag{2.12}$$

For example $t_p = 1.25$ ms commands 45° (duty cycle $1.25/20 = 6.25\%$).



(a) Cascade (position–velocity–current) control of an industrial servo axis



(b) Command signal of a hobby (RC) servo: pulse width sets the angle (1.0–2.0 ms → 0°–180°)

Figure 2.6: Servo motor control: (a) cascade position–velocity–current loops of an industrial servo drive; (b) pulse-width command of an RC servo.

2.5.3 Position servo model and PID control

With the velocity and current loops closed, or for a simple DC motor whose electrical time constant is negligible, the motor from voltage to angle behaves as $\Theta(s)/V(s) = K/[s(\tau_m s + 1)]$. Closing a proportional position loop with gain K_p gives the second-order closed-loop system

$$\frac{\Theta(s)}{\Theta_{\text{ref}}(s)} = \frac{K K_p/\tau_m}{s^2 + s/\tau_m + K K_p/\tau_m}, \qquad \omega_n = \sqrt{\frac{K K_p}{\tau_m}}, \quad \zeta = \frac{1}{2\sqrt{K K_p \tau_m}}. \tag{2.13}$$

Increasing K_p speeds the response but reduces damping – the classic trade-off already met for instruments in Unit I. A PID controller

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \frac{de}{dt} \tag{2.14}$$

adds derivative action for damping and integral action to remove steady-state error caused by friction or gravity. Algorithm 2.1 gives the discrete implementation used in drive firmware.

2.6 Stepper Motors

2.6.1 Construction and types

A stepper motor rotates in discrete angular steps, one step for each input pulse, so its position can be controlled *open loop* by counting pulses. Three types exist:

- **Permanent-magnet (PM)** – magnetised rotor; large steps (7.5°–15°); cheap.
- **Variable-reluctance (VR)** – toothed soft-iron rotor that aligns with energised stator poles to minimise reluctance; no detent torque.

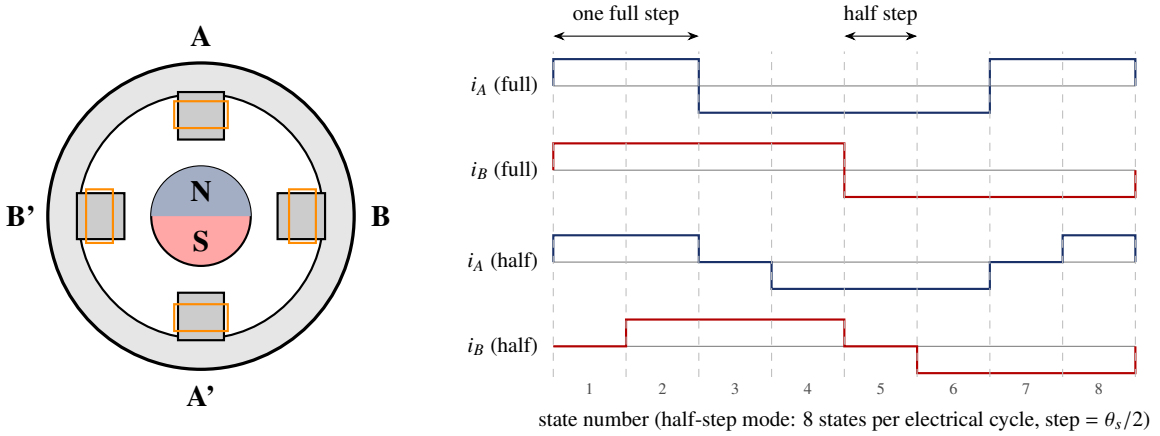
Algorithm 2.1 Discrete PID speed/position control of a motor

Require: Gains K_p, K_i, K_d; sample time T_s; output limits $u_{\min}, u_{\max}$
Ensure: PWM duty cycle (or voltage) applied to the motor every T_s

$I \leftarrow 0$, $e_{\text{prev}} \leftarrow 0$
loop ▷ executed by a timer interrupt every T_s
  read reference r_k and measured output y_k (encoder)
  $e_k \leftarrow r_k - y_k$
  $I \leftarrow I + K_i T_s e_k$ ▷ integral term
  $D \leftarrow K_d(e_k - e_{\text{prev}})/T_s$ ▷ derivative term
  $u \leftarrow K_p e_k + I + D$
  if $u > u_{\max}$ **or** $u < u_{\min}$ **then**
    saturate u; $I \leftarrow I - K_i T_s e_k$ ▷ anti-windup
  end if
  write duty cycle $D_{\text{PWM}} = u/V_{\text{dc}}$ to the PWM timer; $e_{\text{prev}} \leftarrow e_k$
end loop

- **Hybrid** – toothed rotor with an axial magnet; fine steps (typically 1.8°, 200 steps/rev) and high torque; the standard type in 3-D printers, CNC and small robots.

Figure 2.7(a) shows a simple two-phase PM stepper: energising phase A pulls the rotor's south pole to pole A; switching to phase B rotates it 90°, and so on.



(a) Two-phase PM stepper: phases A–A′ and B–B′ (90° step)

(b) Phase currents: full step (two phases on) and half step

Figure 2.7: Stepper motor: (a) two-phase PM stepper; (b) phase-current sequences for full-step and half-step drive.

2.6.2 Step angle and speed

For a motor with m phases and N_r rotor teeth (or pole pairs),

$$\theta_s = \frac{360^\circ}{m\,N_r}\ \text{(VR)}, \qquad \theta_s = \frac{360^\circ}{2\,m\,N_r}\ \text{(hybrid)}, \qquad S = \frac{360^\circ}{\theta_s}\ \text{steps/rev}. \tag{2.15}$$

A two-phase hybrid with $N_r = 50$ gives $\theta_s = 1.8°$; a three-phase VR motor with four rotor teeth gives 30°. With pulse frequency f (steps/s) the speed is

$$n = \frac{60\,f}{S} \ \text{rpm}. \tag{2.16}$$

2.6.3 Drive modes

- **Wave drive** – one phase on at a time; lowest torque.
- **Full step (two-phase on)** – both phases always energised; 41% more torque than wave drive.
- **Half step** – alternates one-phase and two-phase states; doubles resolution (8 states per electrical cycle, Fig. 2.7(b)).
- **Microstepping** – phase currents follow sine and cosine levels, dividing each full step into 4–256 microsteps for smooth motion.

The full-step sequence for a bipolar two-phase motor (A+B+, A−B+, A−B−, A+B−) is generated by Algorithm 2.2; reversing the table order reverses rotation.

Algorithm 2.2 Stepper motor sequencing (full or half step)

Require: Sequence table $\mathcal{S}$ (4 states full-step or 8 states half-step); number of steps n; direction $d = \pm 1$; step rate f
Ensure: Phase-coil states applied to the H-bridge driver

```
k ← 0
for j = 1 to n do
    k ← (k + d) mod |S|
    output coil states S[k] to the driver      ▷ e.g. full step: {(+,+),(−,+),(−,−),(+,−)}
    wait 1/f                                   ▷ ramp f up and down to avoid lost steps
end for
```

Torque characteristics and lost steps. The *holding torque* is the maximum static torque with the windings energised. As step rate increases, winding inductance limits current and torque falls; above the *pull-in* curve the motor cannot start without ramping, and above the *pull-out* curve it loses synchronism ("misses steps"). Because open-loop control cannot detect lost steps, critical applications add an encoder ("closed-loop stepper").

Example 2.5: Stepper-driven lead screw (Topic 2.6)

Given: 1.8° hybrid stepper driving a lead screw of pitch 5 mm; full-step pulse rate 1 kHz; option of 1/16 microstepping.

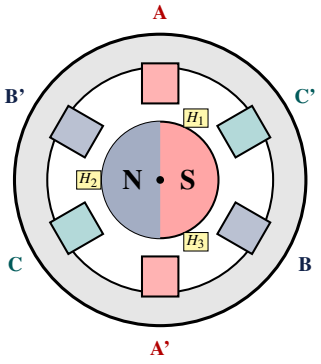
Solution: $S = 360/1.8 = 200$ steps/rev. Speed $n = 60 \times 1000/200 = 300$ rpm. Linear resolution $= 5/200 = 0.025$ mm per step; linear speed $= 5 \times 300/60 = 25$ mm/s. With 1/16 microstepping: 3200 microsteps/rev, nominal resolution 1.56 µm (actual accuracy is limited by

microstep non-linearity, typically a few percent of a full step).

2.7 Brushless DC (BLDC) Motors

2.7.1 Construction and electronic commutation

A BLDC motor is an "inside-out" DC motor (Fig. 2.8(a)): the permanent magnets are on the rotor and the three-phase windings on the stator, where heat is easily removed. The mechanical commutator is replaced by *electronic commutation*: three Hall sensors spaced 120° (electrical) detect the rotor position, and a three-phase inverter energises the appropriate pair of phases. In *six-step (trapezoidal) commutation* two phases conduct at any time and the conducting pair changes every 60° electrical, keeping the stator field about 90° ahead of the rotor field for maximum torque.



(a) Three-phase BLDC: PM rotor, stator windings A, B, C and Hall sensors

(b) Back-EMF, Hall signals and six-step phase current

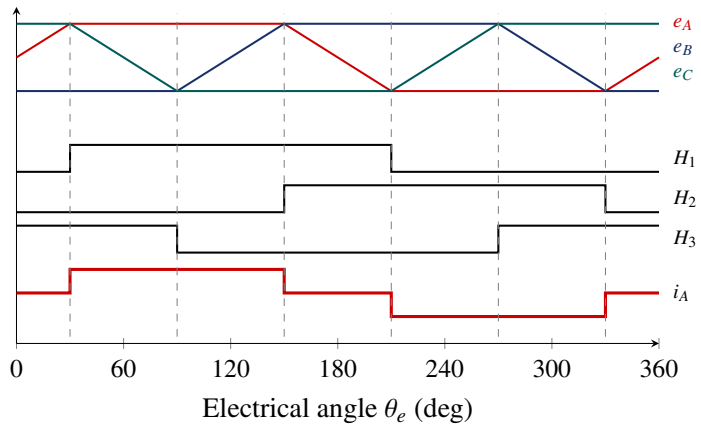


Figure 2.8: BLDC motor: (a) construction with Hall sensors; (b) trapezoidal back-EMFs, Hall signals and phase-A current under six-step commutation (commutation instants dashed).

Electrical model. For each phase $x \in \{a, b, c\}$,

$$v_x = R\, i_x + L\frac{di_x}{dt} + e_x, \qquad e_x = K_e\, \omega\, f_x(\theta_e), \tag{2.17}$$

where $f_x(\theta_e)$ is the normalised trapezoidal shape of Fig. 2.8(b) and $\theta_e = p\,\theta_m$ is the electrical angle for a motor with p pole pairs. The electromagnetic torque is $T = (e_a i_a + e_b i_b + e_c i_c)/\omega$; with two phases conducting a constant current I during the flat part of their EMFs, $T = K_t I$, so in terms of control the BLDC behaves like a brushed DC motor.

Sinusoidal (field-oriented) control. Industrial robot servo motors are *PMSMs* with sinusoidal back-EMF and are driven with sinusoidal currents using field-oriented control (FOC) and an encoder, which eliminates the torque ripple of six-step commutation.

Example 2.6: BLDC motor (Topic 2.7)

Given: 8-pole ($p = 4$) BLDC, $K_t = 0.1\,\mathrm{N\,m/A}$, 10 A, 3000 rpm.

Solution: Torque $T = K_t I = 1.0\,\mathrm{N\,m}$. $\omega = 3000 \times 2\pi/60 = 314.2\,\mathrm{rad/s}$; mechanical power $= T\omega = 314\,\mathrm{W}$. Electrical frequency $f_e = p\,n/60 = 4 \times 50 = 200\,\mathrm{Hz}$; commutation events $= 6p = 24$ per revolution, i.e. every 0.83 ms. Back-EMF (in SI, $K_e = K_t$) $= 0.1 \times 314.2 = 31.4\,\mathrm{V}$.

Interpretation: The drive must supply at least $31.4 + IR$ volts; commutation timing errors of even 0.1 ms cause noticeable torque ripple at this speed.

Table 2.3: Comparison of robot motors.

Property	Brushed DC	Servo (AC PMSM)	Stepper	BLDC
Commutation	Mechanical	Electronic (FOC)	Electronic (sequence)	Electronic (Hall/six-step)
Feedback	Optional	Always (encoder)	Usually none	Hall sensors
Control	Voltage/PWM	Cascade loops	Open-loop pulses	Six-step or FOC
Torque at speed	Good	Excellent	Falls rapidly	Excellent
Efficiency	70–85%	90–95%	Low (current at standstill)	85–92%
Maintenance	Brushes wear	None	None	None
Cost	Low	High	Low	Medium
Robot use	Toys, small grippers	Industrial robot joints	3-D printers, small Cartesian	Mobile robots, drones, cobots

Table 2.3 summarises the trade-offs: steppers win on simplicity for low-speed open-loop positioning, brushed DC on cost, BLDC on efficiency and speed, and the AC servo on dynamic precision – which is why it drives virtually every industrial robot joint.

2.8 Solenoids and Linear Actuators

2.8.1 Solenoids

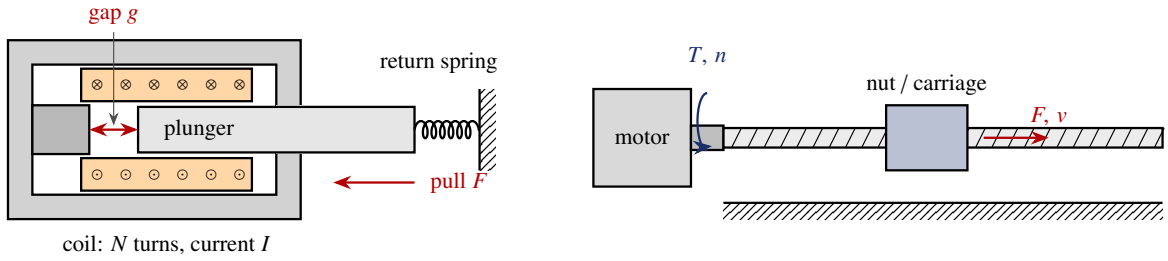
A solenoid (Fig. 2.9(a)) is a coil wound around a movable ferromagnetic plunger. Current in the coil magnetises the plunger, which is pulled into the coil to reduce the air gap and hence the magnetic reluctance; a spring returns it when the current stops. Solenoids operate valves, locks, brakes and simple grippers.

Force – derivation. For an air gap g of cross-section A, the reluctance is dominated by the gap, so the flux density is $B = \mu_0 NI/g$. The magnetic energy stored in the gap is $W = \frac{B^2}{2\mu_0} Ag$. With

constant current, the force is $F = \partial W'/\partial g$ in magnitude, giving

$$F = \frac{(NI)^2 \mu_0 A}{2g^2}, \tag{2.18}$$

where N is the number of turns, I the current (A), $\mu_0 = 4\pi \times 10^{-7}$ H/m. The force rises steeply as the gap closes ($\propto 1/g^2$), so solenoids have short strokes and strong holding force at the end of travel.



(a) Solenoid: $F = \dfrac{(NI)^2 \mu_0 A}{2g^2}$ (b) Ball-screw linear actuator: $F = \dfrac{2\pi\eta T}{p}$, $v = p\,n$

Figure 2.9: Linear actuators: (a) solenoid with plunger and return spring; (b) motor-driven ball-screw actuator.

Example 2.7: Solenoid force (Topic 2.8)

Given: $N = 1000$, $I = 0.5$ A, $A = 1$ cm^2, $g = 2$ mm.

Solution: $F = \dfrac{(500)^2 (4\pi \times 10^{-7})(10^{-4})}{2(0.002)^2} = \dfrac{3.14 \times 10^{-5}}{8 \times 10^{-6}} = 3.9$ N. At $g = 1$ mm the force quadruples to 15.7 N (ideal, ignoring iron saturation).

2.8.2 Electric linear actuators

Rotary motors are converted to linear motion by *lead screws*, *ball screws*, *belts* and *rack and pinion*; *linear motors* produce linear force directly. For a screw of lead (pitch) p (m/rev), driven at torque T and speed n (rev/s) with efficiency η, energy balance over one revolution ($2\pi T\eta = F p$) gives

$$F = \frac{2\pi\eta T}{p}, \qquad v = p\,n. \tag{2.19}$$

Ball screws ($\eta \approx 0.9$) use recirculating balls for low friction and backlash; lead screws ($\eta \approx 0.3$–0.5) are cheaper and often self-locking.

Example 2.8: Ball-screw axis (Topic 2.8)

Given: $T = 0.5$ N m, $p = 10$ mm, $\eta = 0.9$, $n = 1500$ rpm.

Solution: $F = 2\pi(0.9)(0.5)/0.01 = 283\,\text{N}$; $v = 0.01 \times 1500/60 = 0.25\,\text{m/s}$.

2.9 Robot Joints, Transmissions and Mechanisms

2.9.1 Robot joints

Unit I introduced the kinematic joint types (Table 1.3). Physically, an industrial robot joint is an assembly: a servo motor with brake and encoder, a speed reducer, bearings that carry the link loads, and cabling. *Revolute joints* (R) are used for the waist, shoulder, elbow and wrist of articulated arms because rotary motors and bearings are compact and sealable; *prismatic joints* (P) are built from ball screws, belts, racks or linear motors on linear guides and are used in Cartesian, cylindrical and SCARA axes.

2.9.2 Transmissions and gear reduction

Electric motors are fast (2000 rpm to 6000 rpm) and weak, whereas robot joints are slow and must deliver large torques, so a reducer of ratio n is inserted between them. For an ideal reducer

$$\omega_L = \frac{\omega_m}{n}, \qquad T_L = n\,\eta\,T_m, \tag{2.20}$$

where η is the gear efficiency.

Reflected inertia – derivation. The kinetic energy of the load, $\tfrac{1}{2}J_L\omega_L^2 = \tfrac{1}{2}(J_L/n^2)\omega_m^2$, appears at the motor shaft as an inertia J_L/n^2. The total inertia seen by the motor is therefore

$$J_{\text{eq}} = J_m + \frac{J_L}{n^2}. \tag{2.21}$$

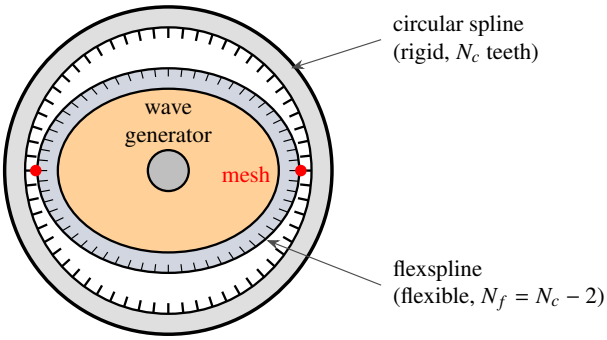
This has a profound consequence: with $n \approx 100$ the motor sees the arm's inertia reduced by a factor of 10^4, so variations of arm inertia with configuration hardly affect the joint servo. Maximising load acceleration for a given motor torque gives the optimal ratio $n_{\text{opt}} = \sqrt{J_L/J_m}$ (inertia matching).

Harmonic (strain-wave) drive. The most common robot reducer (Fig. 2.10(a)) has three parts: an elliptical *wave generator* on the motor shaft, a thin flexible cup called the *flexspline* with N_f external teeth, and a rigid *circular spline* with $N_c = N_f + 2$ internal teeth. The wave generator deforms the flexspline so it meshes with the circular spline at two diametrically opposite zones. Each revolution of the wave generator advances the flexspline by $N_c - N_f = 2$ teeth relative to the circular spline, giving (circular spline fixed, flexspline output, direction reversed)

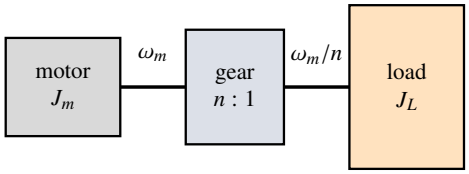
$$n = \frac{N_f}{N_c - N_f}. \tag{2.22}$$

Harmonic drives offer ratios of 50–160 in a single stage, near-zero backlash, compactness and a hollow shaft for cables; their disadvantages are torsional compliance and limited life of the

flexspline. *Cycloidal (RV) reducers* are used in the heavy base axes of large robots for their stiffness and shock capacity; *planetary gears* and *timing belts* are used in lighter axes.



(a) Harmonic (strain-wave) drive: $n = N_f/(N_c - N_f)$, zero backlash



$$J_{\text{eq}} = J_m + \frac{J_L}{n^2}, \quad T_L^{(m)} = \frac{T_L}{n\eta}$$

(b) Reflected inertia and torque

Figure 2.10: (a) Harmonic drive (wave generator, flexspline and circular spline); (b) inertia and torque reflected through a reducer.

Example 2.9: Harmonic drive and reflected inertia (Topic 2.9)

Given: Flexspline 200 teeth, circular spline 202 teeth; motor inertia $J_m = 1 \times 10^{-4}\,\text{kg m}^2$; arm inertia about the joint $J_L = 2\,\text{kg m}^2$.

Solution: $n = 200/(202 - 200) = 100$. Reflected inertia $J_L/n^2 = 2/10^4 = 2 \times 10^{-4}\,\text{kg m}^2$; $J_{\text{eq}} = 3 \times 10^{-4}\,\text{kg m}^2$. Optimum ratio $\sqrt{2/10^{-4}} = 141$.

Interpretation: With $n = 100$ the load contributes two-thirds of the inertia seen by the motor – close to the inertia-matched condition, giving good acceleration.

2.9.3 Robot mechanisms

A *mechanism* is an assembly of links and joints that transforms an input motion into a desired output motion. Robots use several classical mechanisms:

- **Four-bar and parallelogram linkages** – in palletising robots a parallelogram keeps the wrist flange horizontal and allows the elbow motor to be mounted near the base, reducing moving mass.
- **Slider–crank** – converts rotation into reciprocating translation (gripper jaws, presses).
- **Cam and follower** – prescribed motion profiles in fixed automation and grippers.
- **Rack and pinion, screws, belts** – rotary-to-linear conversion for prismatic axes.
- **Wedge and toggle mechanisms** – force amplification in grippers (Section 2.11).

Their mobility follows Grübler's formula (Eq. (1.5)).

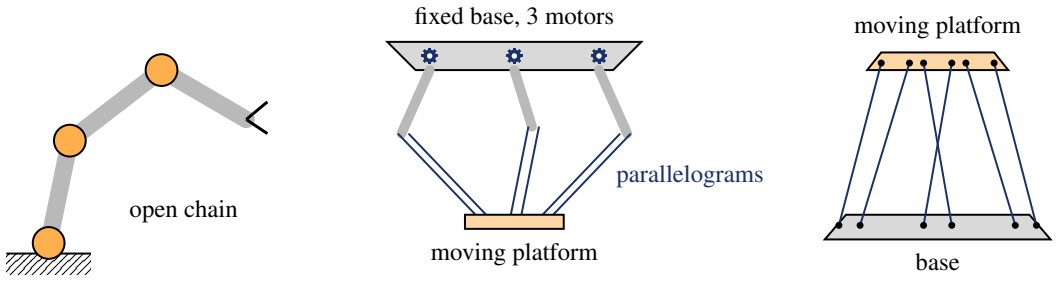
2.10 Serial and Parallel Manipulators

2.10.1 Serial manipulators

In a *serial* (open-chain) manipulator the links form a single chain from base to end-effector, each joint carrying all the links after it (Fig. 2.11(a)). Serial arms have large workspaces relative to their size and simple forward kinematics (Unit III), and are the form of almost all industrial robots. Their disadvantages follow from the chain structure: each actuator must carry the mass of the following motors and links, so payload-to-weight ratio is low (typically 1:10 to 1:20); link and joint deflections and errors *add* along the chain; and stiffness is limited.

2.10.2 Parallel manipulators

In a *parallel* (closed-chain) manipulator the moving platform is connected to the base by several independent *legs*, each typically with one actuated joint (Fig. 2.11(b),(c)). The **Delta robot** has three rotary actuators on the fixed base driving light parallelogram arms; the parallelograms keep the platform parallel to the base, giving three translational DOF and accelerations above $10g$ for picking. The **Stewart–Gough platform** has six extensible legs and six DOF; it is used in flight and driving simulators, machining heads and precision positioners. Because the load is shared among the legs and the motors are fixed to the base, parallel robots are stiff, fast and accurate; but their workspace is small, their inverse kinematics is easy while forward kinematics is difficult (the opposite of serial arms), and they have more complex singularities.



(a) Serial (open chain): large workspace; errors accumulate link by link

(b) Parallel: Delta robot (3-DOF translation); light moving mass

(c) Stewart–Gough platform: 6 UPS legs, 6 DOF; high stiffness

Figure 2.11: (a) Serial manipulator; (b) Delta parallel robot; (c) Stewart–Gough platform.

2.11 End-Effectors and Grippers

2.11.1 End-effectors

The *end-effector* is the device attached to the wrist flange that performs the task. It is either a **tool** – spot-welding gun, arc-welding torch, spray gun, spindle for drilling or deburring, screwdriver, glue dispenser, laser head – or a **gripper** for grasping objects. *Tool changers* allow a robot to exchange end-effectors automatically, and *compliance devices* such as the remote-centre-compliance (RCC) unit allow small lateral and angular misalignments during insertion

Table 2.4: Serial versus parallel manipulators.

Feature	Serial	Parallel
Kinematic structure	Open chain	Closed chains (legs)
Workspace	Large	Small
Stiffness, accuracy	Lower; errors accumulate	Higher; errors averaged
Payload / own weight	Low	High
Speed / acceleration	Moderate	Very high (Delta)
Forward kinematics	Easy (closed form)	Difficult (numerical)
Inverse kinematics	Difficult (multiple solutions)	Easy (leg by leg)
Examples	6-axis articulated, SCARA	Delta, Stewart platform, hexapod

(Unit IV).

2.11.2 Types of grippers

Figure 2.12 shows the three most common types.

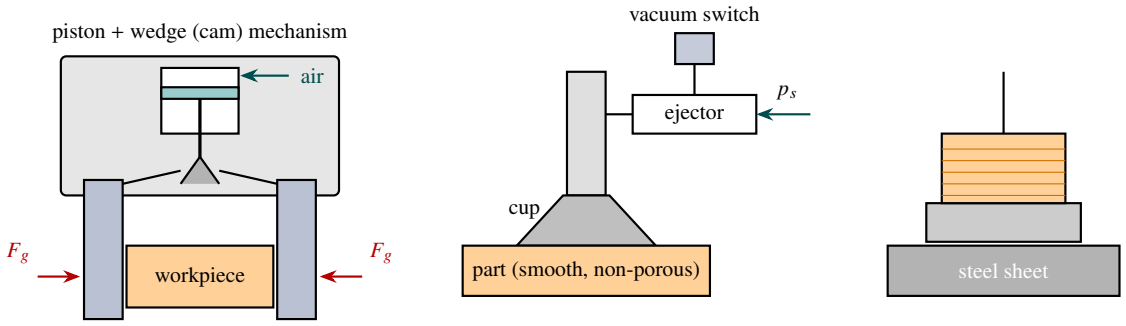
- **Mechanical grippers** – two or three fingers driven by a pneumatic piston or electric motor through a linkage, wedge, rack-and-pinion or screw. *Parallel-jaw* grippers keep the jaws parallel; *angular* grippers pivot. Grasping is by *friction* (flat jaws) or *encompassing/form closure* (contoured jaws that enclose the part).
- **Vacuum grippers** – suction cups evacuated by a venturi ejector or vacuum pump; ideal for flat, smooth, non-porous parts (sheet metal, glass, cartons). Holding force $F = n\,\Delta p\,A$ (Unit I).
- **Magnetic grippers** – electromagnets or switchable permanent magnets for ferrous parts; fast, tolerant of holes, but residual magnetism and multiple-sheet pick-up must be managed.
- **Others** – adhesive, needle grippers for textiles, soft pneumatic grippers for food, and multi-fingered dexterous hands.

2.11.3 Gripping force – derivation

For a friction grip with n_f jaws, each pressing with normal force F_g and coefficient of friction μ, the total friction force must support the weight and inertial force of the part. For vertical motion with upward acceleration a and safety factor S:

$$n_f\,\mu\,F_g \;\geq\; S\,m\,(g+a) \quad\Rightarrow\quad F_g \;\geq\; \frac{S\,m\,(g+a)}{n_f\,\mu}. \tag{2.23}$$

For an encompassing (form-closure) grip the part is retained by geometry and the required force is much smaller.



(a) Pneumatic parallel-jaw gripper

(b) Vacuum gripper: $F = n\,\Delta p\,A$

(c) Electromagnetic gripper (ferrous parts only)

Figure 2.12: Grippers: (a) pneumatic parallel-jaw gripper with wedge mechanism; (b) vacuum gripper with venturi ejector and vacuum switch; (c) electromagnetic gripper.

Example 2.10: Gripper force and cylinder (Topic 2.11)

Given: Part mass 2 kg; two-finger friction grip, $\mu = 0.3$; vertical acceleration 5 m/s^2; $S = 2$. Gripper cylinder bore 32 mm at 6 bar.

Solution: $F_g \geq 2 \times 2 \times (9.81 + 5)/(2 \times 0.3) = 98.7$ N per finger. Cylinder force $= 6 \times 10^5 \times \pi(0.032)^2/4 = 483$ N; even after a wedge mechanism with an output-to-input ratio of 0.5 and losses, each finger receives well over 98.7 N.

Interpretation: Friction grips need forces several times the part weight; rubber jaw pads (higher μ) or form-fitting jaws reduce the requirement and protect the part.

Table 2.5: Comparison of gripper technologies.

	Mechanical	Vacuum	Magnetic
Parts handled	Most rigid parts	Flat, smooth, non-porous	Ferrous only
Holding force	High, controllable	Limited by area and vacuum	High
Positioning of part	Centres the part	Little centring	Little centring
Speed	Medium	Fast	Fast
Limitations	Part-specific jaws	Leakage, porous parts	Residual magnetism, heat

2.12 Measurement of Electrical Quantities

2.12.1 Why electrical measurements matter in robotics

A robot controller continuously measures electrical quantities: motor phase currents for torque control, DC-bus voltage for protection, power consumption for energy management, and the

voltages produced by every sensor. The principles below underlie both bench instruments and the embedded measurement circuits inside servo drives [10, 11].

2.12.2 Voltage and current

Voltage is measured by a high-input-impedance voltmeter or ADC connected in parallel; the loading analysis of Unit I (Eq. (1.19)) requires $R_m \gg R_{\text{Th}}$. High voltages are scaled by resistive dividers $v_o = v_i R_2/(R_1 + R_2)$, often with isolation amplifiers for safety.

Current is measured by one of three methods:

- *Shunt resistor* – a small, precise resistance R_{sh} in series; $v = IR_{sh}$ is amplified by a differential amplifier. Cheap and accurate, but dissipates I^2R_{sh} and is not isolated.
- *Hall-effect current sensor* – the conductor's magnetic field, concentrated by a ferrite core, acts on a Hall element (Eq. (1.35)); provides galvanic isolation and measures DC and AC. Closed-loop (compensated) versions achieve high linearity. Standard in servo drives.
- *Current transformer* – for AC only; secondary current $I_s = I_p/N$.

Example 2.11: Shunt design (Topic 2.12)

Given: A shunt must give 50 mV at 100 A.

Solution: $R_{sh} = 0.05/100 = 0.5\,\text{m}\Omega$; dissipation at full current = $100^2 \times 0.5 \times 10^{-3} = 5\,\text{W}$. An amplifier of gain 50 scales 50 mV to 2.5 V for a 0–3.3 V ADC.

2.12.3 Power and energy

For DC, $P = VI$. For sinusoidal AC, with rms values and phase angle ϕ between voltage and current:

$$P = V_{\text{rms}} I_{\text{rms}} \cos\phi \ \text{(W)}, \qquad Q = V_{\text{rms}} I_{\text{rms}} \sin\phi \ \text{(var)}, \qquad S = V_{\text{rms}} I_{\text{rms}} \ \text{(VA)}, \tag{2.24}$$

where $\cos\phi$ is the power factor. Digital power meters sample v and i simultaneously and compute $P = \frac{1}{N}\sum v_k i_k$, which is valid for any waveform. Energy is $E = \int P\,dt$. *Example:* 230 V, 5 A, $\cos\phi = 0.8$ gives $P = 920\,\text{W}$, $S = 1150\,\text{VA}$, $Q = 690\,\text{var}$.

2.12.4 Resistance, impedance and frequency

Medium resistances are measured with the Wheatstone bridge (balance $R_x = R_2R_3/R_1$, Unit I) or by ohmmeters (constant current, measured voltage). Low resistances (shunts, windings, contacts) require the *four-wire (Kelvin)* method: current is injected through one pair of leads and voltage sensed by a separate pair carrying negligible current, so lead resistance is eliminated – the same idea as the four-wire RTD. Inductance and capacitance are measured by AC bridges or LCR meters. Frequency and period are measured by digital counters: counting cycles in a gate

time T_g gives $f = N/T_g$ (resolution $1/T_g$), while counting a reference clock during one period is better at low frequencies – the same M/T trade-off as encoder speed measurement (Unit I).

2.13 Measurement of Physical Quantities

Physical (non-electrical) quantities are measured by converting them to electrical signals with the transducers of Unit I and then conditioning those signals as described in the following sections. Table 2.6 collects the principal methods relevant to robotic and automated systems, including flow and level, which arise in process-handling cells and in hydraulic/pneumatic systems.

Table 2.6: Measurement of physical quantities in robotic and automation systems.

Quantity	Principle / transducer	Governing relation
Displacement, angle	Potentiometer, LVDT, encoder, resolver	$V_o = xV_s$; $\Delta\theta = 360°/4N$
Velocity, speed	Tachogenerator, encoder, inductive pick-up on toothed wheel	$V = K_t\omega$; $n = 60f/z$
Acceleration, vibration	Piezoelectric / MEMS accelerometer	$z = -a/\omega_n^2$
Force, torque, weight	Strain-gauge load cell, torque shaft	$V_o = V_{\text{ex}} G_F \varepsilon$
Pressure	Diaphragm, piezoresistive, Bourdon	$\Delta p = \rho g h$; bridge output
Temperature	RTD, thermistor, thermocouple, IC	CVD; β equation; Seebeck
Flow (liquid/gas)	Orifice/venturi (differential pressure), turbine, electromagnetic, thermal mass	$Q = C_d A_2 \sqrt{2\Delta p/[\rho(1-\beta^4)]}$
Level	Hydrostatic pressure, ultrasonic, capacitive, float	$h = p/(\rho g)$; $d = v_s t/2$
Humidity	Capacitive polymer sensor	C varies with RH

Speed by inductive pick-up. A toothed steel wheel with z teeth passing an inductive or Hall sensor produces pulses at $f = zn/60$; hence $n = 60f/z$. A 60-tooth wheel giving 1200 Hz indicates 1200 rpm. **Flow by differential pressure.** Bernoulli's equation and continuity applied between the pipe and the throat of an orifice or venturi give the relation in Table 2.6, where β is the ratio of throat to pipe diameter and C_d the discharge coefficient; flow is proportional to $\sqrt{\Delta p}$, so a differential-pressure transmitter followed by a square-root extractor gives a linear flow signal.

2.14 Signal Conditioning

2.14.1 Purpose

Raw sensor outputs are rarely suitable for direct digitisation: a strain-gauge bridge gives a few millivolts riding on a common-mode voltage of several volts; a thermocouple gives microvolts per degree; an RTD needs excitation; signals are contaminated by noise and may be at dangerous potentials. *Signal conditioning* is the set of operations that converts the sensor output into

a clean, correctly scaled signal suited to the ADC (Fig. 2.13):

1. **Excitation** of passive sensors (constant voltage or current);
2. **Conversion** of resistance, capacitance or inductance change into voltage (bridges, dividers, oscillators);
3. **Amplification** to match the ADC range (instrumentation amplifiers, Section 2.15);
4. **Filtering** to remove noise and prevent aliasing (Section 2.16);
5. **Isolation** for safety and to break ground loops (Section 2.17);
6. **Linearisation and compensation** (cold-junction compensation, look-up tables);
7. **Transmission** over long distances as a 4 mA to 20 mA current loop or digitally.

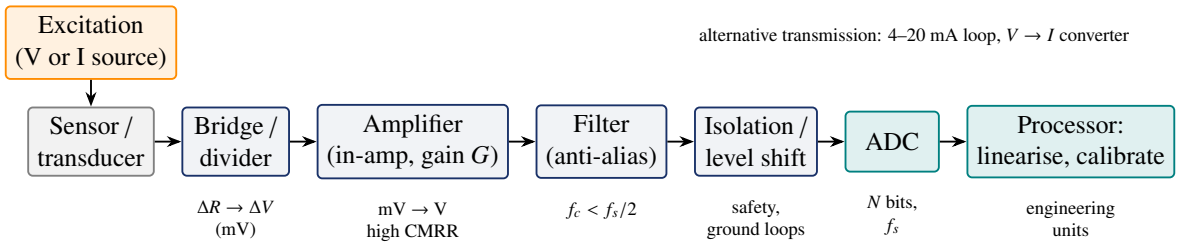


Figure 2.13: Signal-conditioning chain from sensor to processor.

2.14.2 Operational-amplifier building blocks

Most conditioning circuits use operational amplifiers. With the ideal op-amp rules – (i) no current flows into the inputs, (ii) with negative feedback the output adjusts so that $v_+ = v_-$ – the three basic circuits of Fig. 2.14 are analysed as follows.

Inverting amplifier. $v_+ = 0 \Rightarrow v_- = 0$ (virtual earth). The input current v_i/R_1 must flow through R_f: $v_i/R_1 = -v_o/R_f$, so

$$v_o = -\frac{R_f}{R_1} v_i. \tag{2.25}$$

Non-inverting amplifier. $v_- = v_i$ and $v_- = v_o R_1/(R_1 + R_f)$ (divider), so

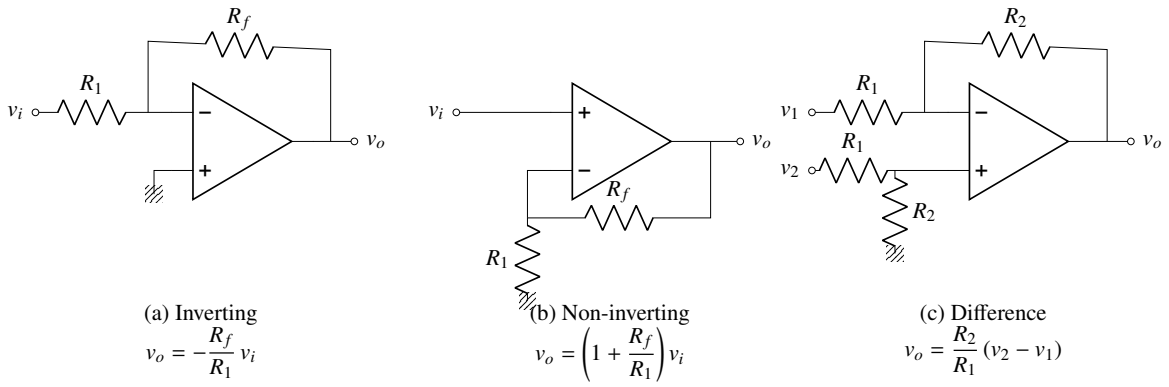
$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i, \tag{2.26}$$

with very high input impedance – the buffer used to cure loading (Unit I) is the special case $R_f = 0$.

Difference amplifier. By superposition, with equal resistor ratios,

$$v_o = \frac{R_2}{R_1} (v_2 - v_1). \tag{2.27}$$

It amplifies the difference and rejects the common-mode voltage, but its input impedance is only R_1 and its common-mode rejection depends on resistor matching – limitations that lead to the instrumentation amplifier.



(a) Inverting $v_o = -\dfrac{R_f}{R_1} v_i$

(b) Non-inverting $v_o = \left(1 + \dfrac{R_f}{R_1}\right) v_i$

(c) Difference $v_o = \dfrac{R_2}{R_1} (v_2 - v_1)$

Figure 2.14: Op-amp building blocks: (a) inverting, (b) non-inverting, (c) difference amplifier.

The 4–20 mA current loop. A transmitter converts the conditioned signal x (0–100% of span) into $I = 4 + 16x$ mA. Current is unaffected by wire resistance and noise pick-up is low; the live zero (4 mA) detects broken wires and powers two-wire transmitters. A 250 Ω resistor converts the loop current to 1 V to 5 V at the receiver.

2.15 Instrumentation Amplifiers

2.15.1 Requirements

An instrumentation amplifier (in-amp) amplifies a small differential signal $v_d = v_2 - v_1$ while rejecting a large common-mode voltage $v_{cm} = (v_1 + v_2)/2$. It must have (i) very high, balanced input impedance (no loading of the bridge), (ii) high and accurate gain set by one resistor, (iii) a high *common-mode rejection ratio*

$$\mathrm{CMRR} = \frac{A_d}{A_{cm}}, \qquad \mathrm{CMRR_{dB}} = 20 \log_{10} \frac{A_d}{A_{cm}}, \tag{2.28}$$

typically 90 dB to 120 dB, and (iv) low offset, drift and noise.

2.15.2 Three-op-amp instrumentation amplifier – derivation

Figure 2.15 shows the classic circuit. The first stage consists of two non-inverting buffers A1 and A2 sharing the gain resistor R_G.

Step 1: By the virtual-short rule, the inverting inputs of A1 and A2 are at v_1 and v_2, so the voltage across R_G is $v_2 - v_1$ and the current through it is $i = (v_2 - v_1)/R_G$.

Step 2: No current enters the op-amp inputs, so the same current flows through both resistors R. The voltage across the chain $R + R_G + R$ is therefore $v_{o2} - v_{o1} = i(2R + R_G) = (v_2 - v_1)(1 + 2R/R_G)$.

Step 3: The common-mode voltage appears equally at both first-stage outputs and is *not* amplified by the first stage (gain 1 for v_{cm}).

Step 4: The difference stage A3 (Eq. (2.27)) amplifies $v_{o2} - v_{o1}$ by R_3/R_2 and rejects the remaining common mode.

$$\boxed{v_o = \left(1 + \frac{2R}{R_G}\right)\frac{R_3}{R_2}(v_2 - v_1)} \tag{2.29}$$

Because the differential gain is raised in the first stage while the common-mode gain is unity there, the overall CMRR is improved by the first-stage gain. Integrated in-amps (laser-trimmed resistors) need only the external R_G.

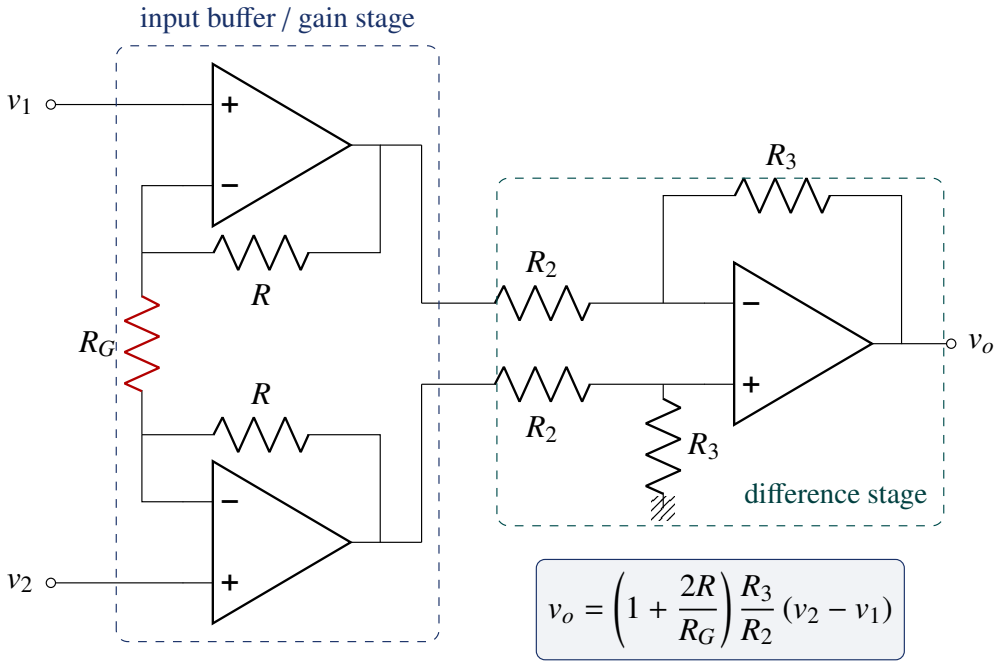


Figure 2.15: Three-op-amp instrumentation amplifier with gain resistor R_G.

Example 2.12: In-amp for a strain-gauge bridge (Topic 2.15)

Given: In-amp with $R = 25\,\text{k}\Omega$, $R_3/R_2 = 1$; full-bridge output 5 mV full scale; bridge common-mode voltage 2.5 V; CMRR 100 dB. Required output 5 V.

Solution: Required gain $G = 5/0.005 = 1000 = 1 + 2R/R_G \Rightarrow R_G = 2\times25\,000/999 = 50.05\,\Omega$. CMRR $= 10^5 \Rightarrow A_{cm} = A_d/\text{CMRR} = 1000/10^5 = 0.01$. Common-mode error at output $= 0.01 \times 2.5 = 25\,\text{mV}$, equivalent to 25 µV at the input, i.e. 0.5% of full scale.

Interpretation: Even 100 dB CMRR leaves a 0.5% error, which is constant and can be calibrated out; a simple difference amplifier built with 1% resistors (worst-case CMRR only about 34 dB at unity gain) would be useless here.

2.16 Filters

2.16.1 Purpose and classification

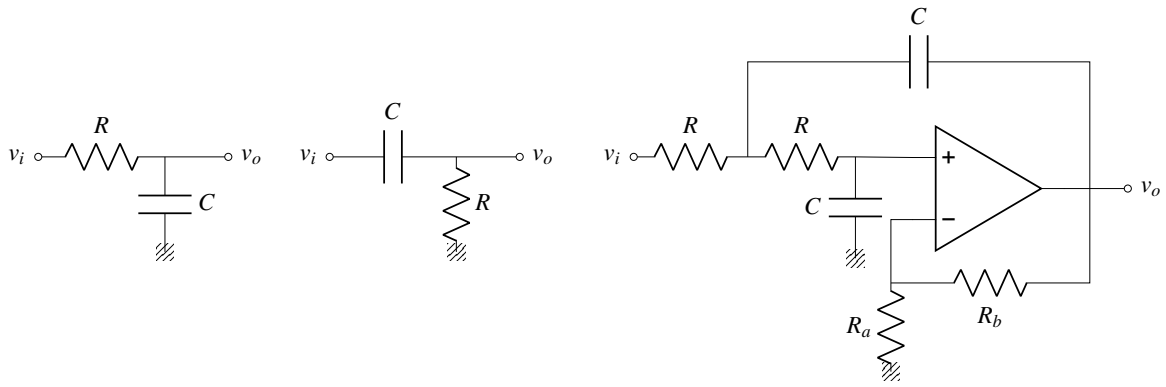
Filters pass desired frequency components and attenuate others. In instrumentation they remove noise (mains hum, PWM ripple, vibration) and, crucially, band-limit signals before sampling (anti-aliasing, Section 2.18). Filters are *low-pass*, *high-pass*, *band-pass* or *band-stop (notch)*; *passive* (R, L, C only) or *active* (with op-amps, providing gain and buffering); *analogue* or *digital*.

2.16.2 First-order RC filters – derivation

For the RC low-pass of Fig. 2.16(a), the capacitor and resistor form an impedance divider:

$$H(s) = \frac{V_o}{V_i} = \frac{1/(sC)}{R + 1/(sC)} = \frac{1}{1 + sRC}, \qquad |H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \qquad f_c = \frac{1}{2\pi RC}. \tag{2.30}$$

At f_c the gain is -3 dB; above it the gain falls at -20 dB/decade. Interchanging R and C gives the high-pass $H(s) = sRC/(1 + sRC)$. A high-pass followed by a low-pass gives a band-pass (Fig. 2.17(a)).



(a) RC low-pass $H(s) = \dfrac{1}{1 + sRC}$

(b) RC high-pass $H(s) = \dfrac{sRC}{1 + sRC}$

(c) Sallen–Key 2nd-order low-pass $\omega_0 = 1/RC$, $K = 1 + R_b/R_a = 3 - 1/Q$

Figure 2.16: Filter circuits: (a) RC low-pass, (b) RC high-pass, (c) Sallen–Key second-order active low-pass.

2.16.3 Higher-order and active filters

A single RC stage attenuates slowly. The **Butterworth** filter of order n has the maximally flat magnitude

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^{2n}}}, \tag{2.31}$$

with roll-off $-20n$ dB/decade (Fig. 2.17(b)). Second-order sections are built with the **Sallen–Key** circuit (Fig. 2.16(c)); with equal R and C, $\omega_0 = 1/(RC)$ and the quality factor is set by the gain $K = 1 + R_b/R_a = 3 - 1/Q$. A Butterworth section ($Q = 0.707$) needs $K = 1.586$. Higher orders cascade such sections. Other approximations trade flatness for steeper roll-off (Chebyshev) or linear phase (Bessel).

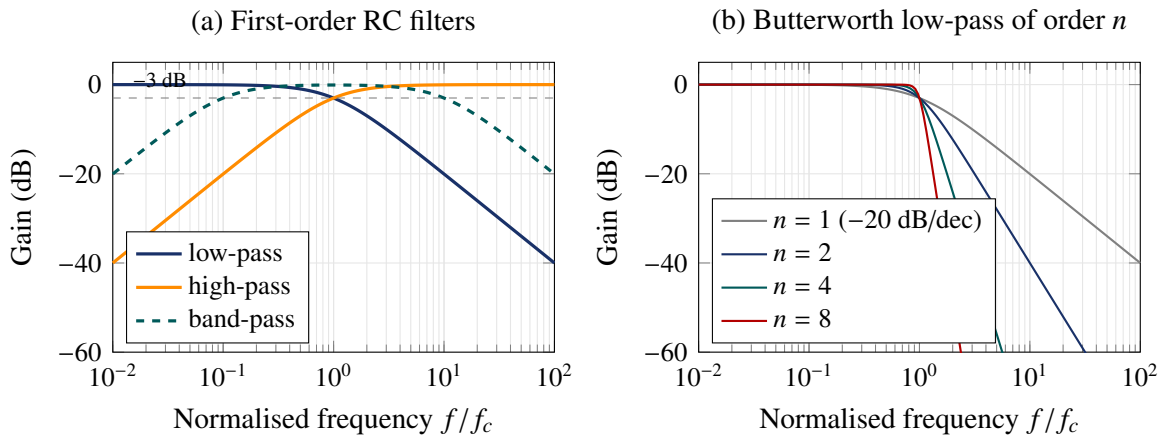


Figure 2.17: Filter magnitude responses: (a) first-order low-, high- and band-pass; (b) Butterworth low-pass of increasing order.

Example 2.13: Low-pass filter design (Topic 2.16)

Given: RC low-pass with $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$; compare with a fourth-order Butterworth with $f_c = 100\,\text{Hz}$.

Solution: $f_c = 1/(2\pi \times 10^4 \times 10^{-7}) = 159\,\text{Hz}$. At 1 kHz: $|H| = 1/\sqrt{1 + (1000/159)^2} = 0.157$ (-16.1 dB). Butterworth $n = 4$ at 500 Hz ($\omega/\omega_c = 5$): $|H| = 1/\sqrt{1 + 5^8} = 0.0016$ (-55.9 dB).

Interpretation: The fourth-order filter attenuates a signal five times above cut-off by a factor of 625, whereas the first-order stage attenuates a signal six times above cut-off by a factor of only 6.4 – the reason anti-aliasing filters are of high order.

Digital filters. After digitisation, filters are implemented in software. A *moving average* $y_k = \frac{1}{M}\sum_{i=0}^{M-1} x_{k-i}$ (FIR) suppresses random noise; the first-order IIR filter $y_k = \alpha x_k + (1-\alpha)y_{k-1}$, with $\alpha = T_s/(\tau + T_s)$, is the discrete equivalent of the RC low-pass.

Table 2.7: Analogue versus digital filters.

	Analogue	**Digital**
Implementation	R, L, C, op-amps	Software / DSP / FPGA
Anti-aliasing	Essential (before ADC)	Cannot prevent aliasing
Flexibility	Fixed by components	Reprogrammable, adaptive
Precision & drift	Component tolerance, temperature drift	Exact, no drift
Linear phase	Difficult	Easy (symmetric FIR)
Bandwidth	Up to GHz	Limited by sampling rate

2.17 Shielding and Grounding

2.17.1 Noise coupling mechanisms

A robot cell is electrically hostile: servo drives switch hundreds of volts at tens of kilohertz, welding equipment draws kiloamperes, and motor cables run beside millivolt sensor cables. Noise couples into measurement circuits by four mechanisms:

1. **Capacitive (electric-field) coupling** through stray capacitance C_s from a source with changing voltage: injected current $i = C_s\, dv/dt$.
2. **Inductive (magnetic-field) coupling** into the loop area A of the signal circuit: induced EMF $e = -d\Phi/dt = -A\, dB/dt$.
3. **Conductive coupling** through shared conductors, especially ground connections (ground loops).
4. **Radiated coupling** of high-frequency electromagnetic waves.

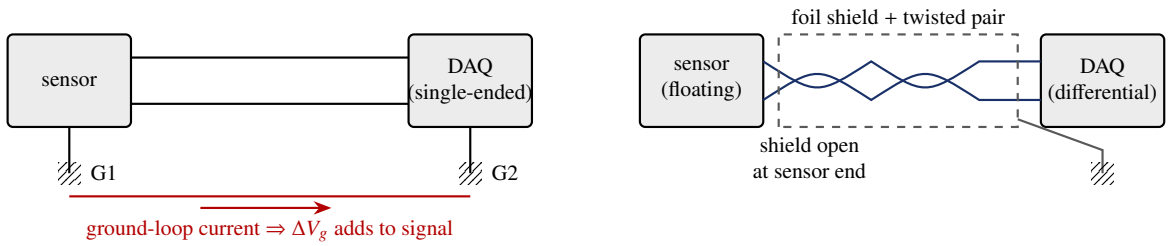
2.17.2 Shielding

- **Electrostatic shield** – a conducting foil or braid around the signal wires, connected to ground, intercepts capacitive coupling and diverts the current to ground.
- **Twisted pairs** – twisting the two signal wires makes successive loops cancel induced EMFs and minimises loop area; this is the most effective defence against magnetic coupling at low frequencies.
- **Magnetic shielding** – high-permeability (mu-metal) enclosures for low-frequency fields; conductive enclosures for high frequencies.
- **Separation** – route power and signal cables in separate trays and cross them at right angles.

2.17.3 Grounding

A *ground loop* forms when a signal circuit is grounded at two points that are at different potentials (Fig. 2.18(a)). Current flowing in the loop develops a voltage ΔV_g that appears in series with the signal. Remedies:

- **Single-point (star) grounding** – connect each circuit's ground to one reference point.
- **Ground the cable shield at one end only** (normally the receiving instrument) for low-frequency signals, so that no current flows in the shield (Fig. 2.18(b)).
- **Differential inputs and in-amps** – the ground-potential difference becomes a common-mode voltage that is rejected (Eq. (2.28)).
- **Isolation** – isolation amplifiers, optocouplers or digital isolators break the galvanic path entirely.
- Separate *power ground*, *signal ground* and *protective earth*, joined at one point.



(a) Wrong: two grounds form a loop

(b) Right: shielded twisted pair, shield grounded at one end only, differential input

Figure 2.18: Grounding and shielding: (a) a ground loop created by grounding both ends; (b) shielded twisted pair with the shield grounded at one end and a differential input.

2.18 Analogue-to-Digital Conversion

2.18.1 Sampling and the Nyquist criterion

An ADC converts a continuous signal into a sequence of numbers by two operations: *sampling* in time at frequency $f_s = 1/T_s$, and *quantisation* in amplitude (Fig. 2.19). The **Nyquist–Shannon sampling theorem** states that a band-limited signal containing no frequencies above $f_{\max}$ is completely represented by its samples if

$$f_s > 2 f_{\max}. \tag{2.32}$$

If this condition is violated, a component at frequency f appears after sampling at the *alias* frequency $|f - k f_s|$ (nearest integer k): a 900 Hz signal sampled at 1 kHz masquerades as 100 Hz (Fig. 2.19(c)). Aliasing cannot be removed after sampling; hence an analogue anti-aliasing low-pass filter precedes every ADC, and in practice f_s is chosen 5–10 times $f_{\max}$.

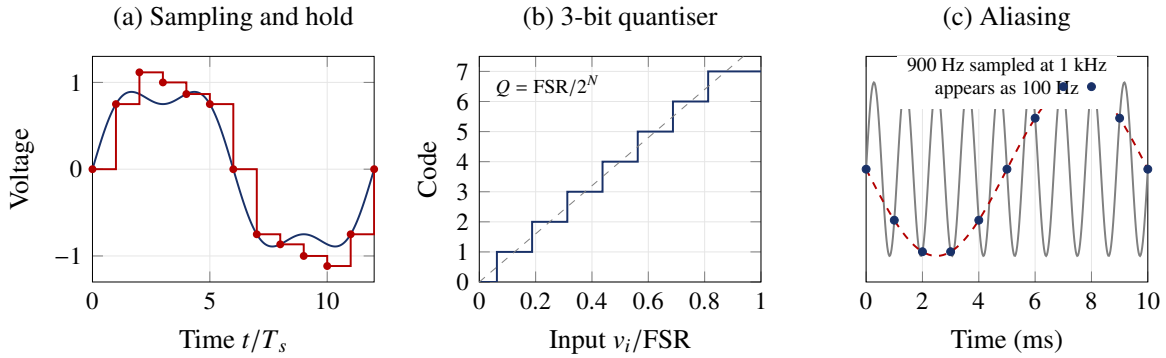


Figure 2.19: Analogue-to-digital conversion: (a) sampling and zero-order hold; (b) 3-bit quantiser characteristic; (c) aliasing of a 900 Hz signal sampled at 1 kHz.

2.18.2 Quantisation, resolution and SNR

An N-bit ADC with full-scale range FSR has 2^N codes and quantisation step (1 LSB)

$$Q = \frac{\mathrm{FSR}}{2^N}. \tag{2.33}$$

The digital output for input v_i (unipolar) is $D = \mathrm{round}(v_i/Q)$, so the *quantisation error* lies within $\pm Q/2$. Treating it as uniformly distributed, its rms value is $Q/\sqrt{12}$. For a full-scale sine wave of rms value $\mathrm{FSR}/(2\sqrt{2})$, the ideal signal-to-quantisation-noise ratio is

$$\mathrm{SNR} = 20\log_{10}\frac{\mathrm{FSR}/(2\sqrt{2})}{Q/\sqrt{12}} = 20\log_{10}\left(2^N\sqrt{1.5}\right) = 6.02N + 1.76\ \mathrm{dB}. \tag{2.34}$$

Example 2.14: ADC resolution for a thermocouple (Topic 2.18)

Given: 12-bit ADC, 0 V to 5 V; type K thermocouple ($\approx 41\ \mu\mathrm{V}/^\circ\mathrm{C}$) amplified by 100.

Solution: $Q = 5/4096 = 1.22\ \mathrm{mV}$; maximum quantisation error $\pm 0.61\ \mathrm{mV}$; ideal SNR = $6.02(12)+1.76 = 74.0\ \mathrm{dB}$ (16-bit: 98.1 dB). Amplified sensitivity = $4.1\ \mathrm{mV}/^\circ\mathrm{C}$, so temperature resolution = $1.22/4.1 = 0.30\ ^\circ\mathrm{C}$.

Interpretation: To resolve 0.1 °C either the gain must be tripled (reducing the range) or a 14-bit or better converter used.

2.18.3 ADC architectures

Successive approximation (SAR). A sample-and-hold freezes the input; an internal N-bit DAC, a comparator and the SAR logic perform a binary search (Fig. 2.20(a)). Starting with the MSB, each bit is set to 1, the DAC output compared with the input, and the bit kept if $v_{\mathrm{in}} \geq v_{\mathrm{DAC}}$ (Algorithm 2.3). Conversion takes N clock cycles. SAR ADCs (8–18 bits, up to a few MS/s) are the standard converters in microcontrollers and DAQ cards.

Algorithm 2.3 Successive-approximation A/D conversion

Require: Held input v_{in}; reference V_{ref}; resolution N bits
Ensure: Output code $D = b_{N-1}\ldots b_0$

```
D ← 0
for k = N − 1 downto 0 do
    D ← D + 2^k                          ▷ trial: set bit k
    v_DAC ← V_ref D/2^N
    if v_in < v_DAC then
        D ← D − 2^k                      ▷ clear the bit
    end if
end for
return D
```

Example 2.15: 4-bit SAR conversion (Topic 2.18)

Given: $V_{\mathrm{ref}} = 5\ \mathrm{V}$, $v_{\mathrm{in}} = 3.3\ \mathrm{V}$, $N = 4$ ($Q = 0.3125\ \mathrm{V}$).

Solution:

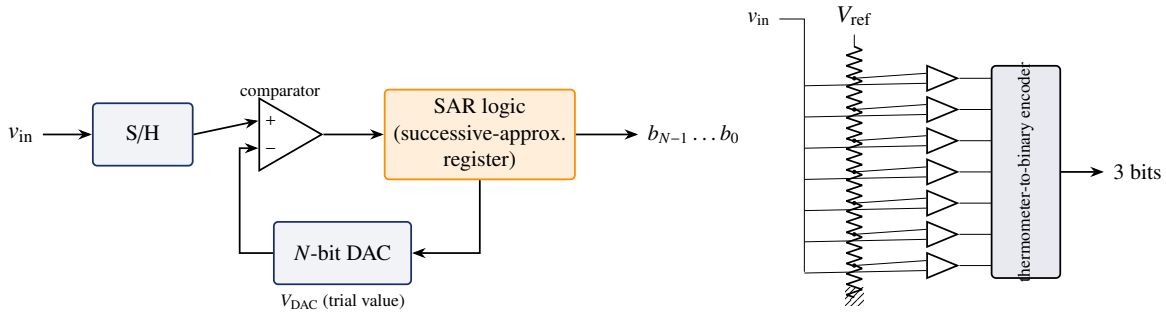
Step	Trial code	v_{DAC} (V)	$v_{in} \geq v_{DAC}$?	Result
1	1000	2.5000	yes	1xxx
2	1100	3.7500	no	10xx
3	1010	3.1250	yes	101x
4	1011	3.4375	no	1010

Output $1010_2 = 10$, representing 3.125 V; error 0.175 V $< Q$ (truncating converter).

Flash (parallel) ADC. A resistor ladder generates $2^N - 1$ reference levels, each compared simultaneously with the input by its own comparator; an encoder converts the "thermometer" code to binary (Fig. 2.20(b)). Conversion occurs in one clock – the fastest architecture (GS/s, used in oscilloscopes and high-speed vision) – but comparator count doubles with each bit, limiting resolution to about 8 bits.

Dual-slope (integrating) ADC. The input is integrated for a fixed time T_1, then a reference of opposite polarity is integrated until the output returns to zero, taking T_2. Since the integrator charge balances, $v_{in}T_1 = V_{ref}T_2$, so $v_{in} = V_{ref}T_2/T_1$, independent of the integrator R and C. Choosing T_1 equal to a multiple of the mains period rejects 50 Hz hum. Slow but very accurate: used in digital multimeters and panel meters.

Sigma-delta (ΣΔ) ADC. A 1-bit modulator oversamples the input at a very high rate; noise shaping pushes quantisation noise to high frequencies, which a digital decimation filter removes, yielding 16–24 bits at modest bandwidth. Ideal for load cells, thermocouples and audio.



(a) Successive-approximation (SAR) ADC: one bit decided per clock, N clocks per conversion

(b) 3-bit flash ADC: $2^N - 1 = 7$ comparators

Figure 2.20: ADC architectures: (a) successive approximation; (b) 3-bit flash converter.

2.19 Digital-to-Analogue Conversion

2.19.1 Principle

A DAC converts an N-bit code D into a voltage (or current) proportional to it:

$$v_o = V_{ref}\frac{D}{2^N} = V_{ref}\left(\frac{b_{N-1}}{2} + \frac{b_{N-2}}{4} + \cdots + \frac{b_0}{2^N}\right), \tag{2.35}$$

Table 2.8: Comparison of ADC architectures.

Type	Resolution	Speed	Strengths	Typical use
Flash	6–8 bit	up to GS/s	Fastest, one-step	Oscilloscopes, video
SAR	8–18 bit	kS/s–MS/s	Balanced, low power	Microcontrollers, DAQ, motor control
Dual-slope	12–20 bit	1–100 S/s	Accurate, rejects mains	DMMs, panel meters
Sigma-delta	16–24 bit	S/s–kS/s	Highest resolution	Load cells, thermocouples, audio

with resolution $V_{\text{ref}}/2^N$. DACs produce analogue command signals (e.g. ±10 V speed commands to older servo drives, proportional-valve set-points) and are the heart of SAR ADCs.

2.19.2 Architectures

Binary-weighted resistor DAC. Each bit switches a resistor $R, 2R, 4R, \ldots, 2^{N-1}R$ into the virtual-earth node of an inverting summer. It is simple but needs a wide range of precisely matched resistors, which is impractical beyond about 8 bits.

R–2R ladder DAC. Only two resistor values are used (Fig. 2.21). *Derivation:* looking into any node of the ladder towards the terminated end, the resistance is $2R \parallel 2R = R$; adding the series R gives $2R$ again. Hence at every node the current splits equally, and the leg currents are $V_{\text{ref}}/(2R)$, $V_{\text{ref}}/(4R)$, … – binary-weighted. Legs whose bits are 1 feed the virtual-earth bus; with $R_f = R$ the output is $v_o = -V_{\text{ref}} D/2^N$. Ease of matching makes R–2R the standard architecture. Other types include current-steering (fast), string (inherently monotonic) and ΣΔ DACs.

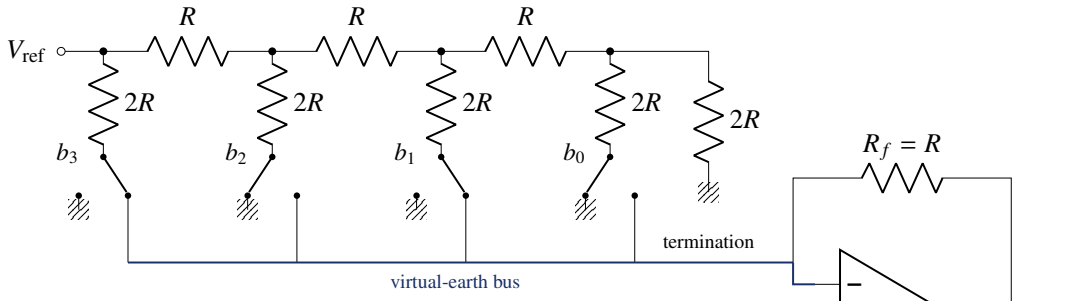


Figure 2.21: 4-bit R–2R ladder DAC with output op-amp.

Example 2.16: R–2R DAC (Topic 2.19)

Given: 8-bit DAC, $V_{\text{ref}} = 5$ V, input 10110100.

Solution: $D = 128 + 32 + 16 + 4 = 180$. $|v_o| = 5 \times 180/256 = 3.516\,\text{V}$; resolution $= 5/256 = 19.5\,\text{mV}$.

DAC specifications: resolution, settling time, monotonicity, differential and integral non-linearity (DNL, INL), offset and gain error. A DAC whose DNL exceeds -1 LSB can be non-monotonic, which is dangerous inside a control loop.

2.20 Data Acquisition Systems

2.20.1 Architecture

A data acquisition (DAQ) system acquires signals from many sensors, digitises them and delivers them to a computer, and often also generates analogue and digital outputs (Fig. 2.22). Its components are:

1. sensors and individual **signal conditioning** (excitation, amplification, filtering, isolation);
2. an analogue **multiplexer** (MUX) that connects one channel at a time to the converter;
3. a **programmable-gain amplifier** (PGA) matching each channel to the ADC range;
4. a **sample-and-hold** (S/H) that freezes the signal during conversion (unnecessary for slowly varying signals; essential for SAR converters);
5. the **ADC**;
6. **timing and control** logic (sample clock, triggering, channel sequencing);
7. a **processor and FIFO memory** buffering data;
8. the **host interface** (USB, Ethernet, PCIe, EtherCAT) and software;
9. **outputs**: DACs, digital I/O, counters/timers.

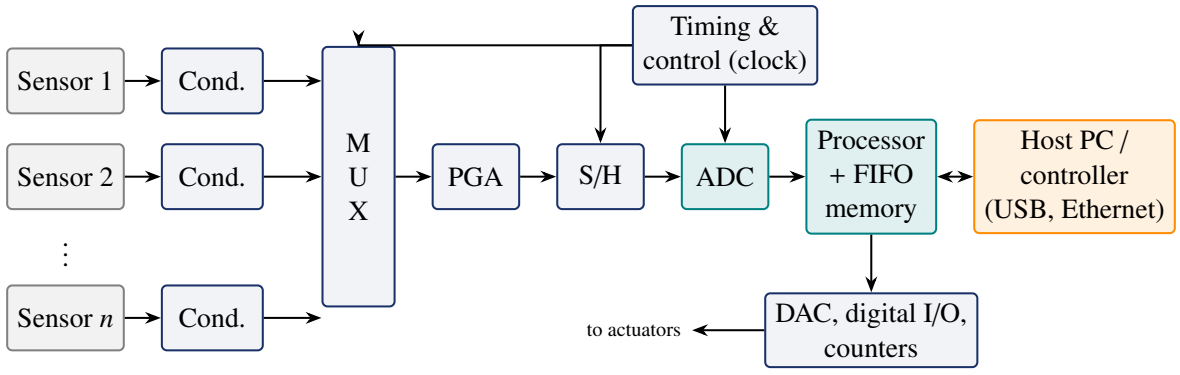


Figure 2.22: Architecture of a multichannel data acquisition system.

2.20.2 Design calculations

With n multiplexed channels each sampled at f_s, the ADC must convert at nf_s; the data rate is

$$R_b = n\, f_s\, N \quad \text{(bit/s)}. \tag{2.36}$$

Multiplexing introduces a *skew* of $1/(nf_s)$ between channels; when phase relationships matter (e.g. three motor phase currents or a six-axis F/T sensor), *simultaneous-sampling* DAQs with one S/H or ADC per channel are used.

Example 2.17: DAQ throughput (Topic 2.20)

Given: 8 channels, 10 kS/s per channel, 16-bit samples.

Solution: Aggregate rate $= 8 \times 10^4 = 80\,\text{kS/s}$; $R_b = 8 \times 10^4 \times 16 = 1.28\,\text{Mbit/s}$ (160 kB/s); inter-channel skew $= 1/(80\,000) = 12.5\,\mu\text{s}$. Each channel's anti-aliasing filter must cut off well below 5 kHz.

Interpretation: USB 2.0 full speed (12 Mbit/s) easily carries this data; for vision or high-speed vibration data, faster buses are needed.

Key points

- DC motor: $v_a = R_a i_a + L_a\, di_a/dt + K_b\omega$, $T = K_t i_a$; speed–torque is a straight line from V/K_b to $K_t V/R_a$.
- Stepper $\theta_s = 360^\circ/(mN_r)$ (VR); speed $n = 60f/S$.
- In-amp gain $(1 + 2R/R_G)R_3/R_2$; CMRR $= A_d/A_{cm}$.
- Nyquist $f_s > 2f_{\max}$; $Q = \text{FSR}/2^N$; SNR $= 6.02N + 1.76$ dB.
- Ground shields at one end; use differential inputs, twisted pairs and isolation.

2.21 MATLAB and Simulink Implementation

The following programs use only core MATLAB (and run in GNU Octave); each prints values that can be checked against the worked examples of this unit.

2.21.1 Program 2.1 – DC motor modelling

Objective. Simulate the armature current and speed of the DC motor of Example 2.3 when 24 V is applied against a 0.2 N m load. **Theory.** Equations (2.8) and (2.9) are integrated numerically by the explicit Euler method with a step of 1 µs, much smaller than $\tau_e = 0.42$ ms. **Expected output:** steady current 4.00 A, speed 384 rad/s (3667 rpm) in agreement with the analytical

solution; $\tau_e = 0.417\,\text{ms}$, $\tau_m = 9.60\,\text{ms}$. The plot shows the current spike at start (limited only by R_a and L_a) decaying as back-EMF builds – the reason drives limit starting current.

Program 2.1: DC motor model: step response and steady state.

```
% Program 2.1: DC motor model -- step response and steady state
clear; clc; close all;
Ra = 1.2; La = 0.5e-3; Kt = 0.05; Kb = 0.05; % ohm, H, N.m/A, V.s/rad
J = 2e-5; b = 0; % kg.m^2, N.m.s/rad
Va = 24; TL = 0.2; % V, N.m (load applied from t = 0)
dt = 1e-6; t = 0:dt:0.08; N = numel(t);
ia = zeros(1,N); w = zeros(1,N);
for k = 1:N-1 % explicit Euler integration
    dia = (Va - Ra*ia(k) - Kb*w(k))/La; % electrical equation
    dw = (Kt*ia(k) - b*w(k) - TL*(w(k)>0 | Kt*ia(k)>TL))/J; % mechanical (static load)
    ia(k+1) = ia(k) + dt*dia;
    w(k+1) = max(w(k) + dt*dw, 0);
end
fprintf('Steady state: i = %.2f A, w = %.1f rad/s (%.0f rpm)\n', ia(end), w(end), w(end)*60/(2*p
fprintf('Analytical : i = %.2f A, w = %.1f rad/s\n', TL/Kt, (Va - TL/Kt*Ra)/Kb);
fprintf('tau_e = %.3f ms, tau_m = %.2f ms\n', 1e3*La/Ra, 1e3*J*Ra/(Kt*Kb));
subplot(2,1,1); plot(t*1e3, w*60/(2*pi)); grid on; ylabel('Speed (rpm)');
title('DC motor step response (24 V, 0.2 N.m load)');
subplot(2,1,2); plot(t*1e3, ia); grid on; xlabel('Time (ms)'); ylabel('Current (A)');
```

2.21.2 Program 2.2 – Stepper motor operation

Objective. Implement Algorithm 2.2 and display the half-step coil sequence. **Expected output:** 200 steps/rev, 300 rpm at 1 kHz; after 16 half steps the shaft has turned 14.4°. Changing `dir` to -1 reverses the sequence and the rotation.

Program 2.2: Stepper motor sequencing (full step and half step).

```
% Program 2.2: Stepper motor sequencing (full step and half step)
clear; clc; close all;
full = [ 1 1; -1 1; -1 -1; 1 -1]; % (A,B) coil polarities
half = [ 1 0; 1 1; 0 1; -1 1; -1 0; -1 -1; 0 -1; 1 -1];
thetaS = 1.8; S = 360/thetaS; % step angle, steps/rev
f = 1000; % pulses per second
fprintf('Steps/rev = %d, speed at %d Hz = %.0f rpm\n', S, f, 60*f/S);
nsteps = 16; dir = +1; k = 0; seq = zeros(nsteps, 2);
for j = 1:nsteps % Algorithm: stepper sequencing
    k = mod(k + dir, size(half,1));
    seq(j,:) = half(k+1,:);
end
ang = (1:nsteps)*thetaS/2; % half step = thetaS/2
fprintf('After %d half steps the shaft has turned %.1f deg\n', nsteps, ang(end));
```

```
subplot(2,1,1); stairs(0:nsteps-1, seq(:,1), 'LineWidth', 1.5); ylim([-1.5 1.5]); grid on; ylabe
title('Half-step drive sequence');
subplot(2,1,2); stairs(0:nsteps-1, seq(:,2), 'r', 'LineWidth', 1.5); ylim([-1.5 1.5]); grid on;
xlabel('Step number'); ylabel('i_B');
```

2.21.3 Program 2.3 – Servo response with PID control

Objective. Simulate a servo joint (DC motor with 50:1 gearbox) following a 90° step under the PID law of Algorithm 2.1, including voltage saturation and anti-windup. **Expected output:** overshoot $\approx 0.9\%$, 2% settling time ≈ 0.28 s, final angle approaching 90° (within 1° at 0.4 s while the integral term removes the residual error). Students should vary K_p, K_i and K_d and observe the trade-offs predicted by Eq. (2.13).

Program 2.3: PID position control of a geared DC motor.

```
% Program 2.3: Servo position control of a DC motor with a PID controller
clear; clc; close all;
Ra = 1.2; Kt = 0.05; Kb = 0.05; J = 2e-5; b = 1e-5; n = 50; % gear ratio n
Jt = J; Vmax = 24;
Kp = 40; Ki = 20; Kd = 0.6; % PID gains (volts per rad of load error)
Ts = 1e-4; t = 0:Ts:0.4; N = numel(t);
thref = (pi/2)*ones(1,N); % 90 deg step command at the load
thm = 0; wm = 0; I = 0; eprev = 0; thL = zeros(1,N); u = zeros(1,N);
for k = 1:N
    thL(k) = thm/n;
    e = thref(k) - thL(k);
    I = I + Ki*Ts*e;
    D = Kd*(e - eprev)/Ts; eprev = e;
    u(k) = Kp*e + I + D;
    if abs(u(k)) > Vmax, I = I - Ki*Ts*e; u(k) = sign(u(k))*Vmax; end % anti-windup
    ia = (u(k) - Kb*wm)/Ra; % electrical time constant neglected
    dw = (Kt*ia - b*wm)/Jt;
    wm = wm + Ts*dw; thm = thm + Ts*wm;
end
yfin = thL(end); Mp = (max(thL) - pi/2)/(pi/2)*100;
ts = t(find(abs(thL - pi/2) > 0.02*pi/2, 1, 'last'));
fprintf('Final angle = %.2f deg, overshoot = %.1f %%, settling (2%%) = %.3f s\n', rad2deg(yfin),
subplot(2,1,1); plot(t, rad2deg(thL), t, rad2deg(thref), '--'); grid on; ylabel('\theta_L (deg)'
title('PID position control of a geared DC motor');
subplot(2,1,2); plot(t, u); grid on; xlabel('Time (s)'); ylabel('Motor voltage (V)');
```

2.21.4 Program 2.4 – PWM generation

Objective. Generate a 20 kHz PWM signal by comparing a duty-cycle reference with a sawtooth carrier – exactly what a microcontroller timer does in hardware – and verify that its mean equals DV_{dc}. **Expected output:** mean voltage 15.6 V for $D = 0.65$, $V_{\mathrm{dc}} = 24$ V.

Program 2.4: PWM generation by carrier comparison.

```
% Program 2.4: PWM generation by carrier comparison and average voltage
clear; clc; close all;
Vdc = 24; fpwm = 20e3; D = 0.65; % supply, PWM frequency, duty cycle
fs = 400*fpwm; t = 0:1/fs:3/fpwm; % three PWM periods
carrier = mod(t*fpwm, 1); % sawtooth 0..1
pwm = Vdc*(D > carrier); % comparator output
fprintf('Duty = %.2f, mean voltage = %.2f V (theory %.2f V)\n', D, mean(pwm(1:end-1)), D*Vdc);
plot(t*1e6, pwm, 'LineWidth', 1.5); hold on;
plot(t*1e6, Vdc*carrier, 'k:', t*1e6, Vdc*D*ones(size(t)), 'r--'); grid on;
xlabel('Time (\mus)'); ylabel('Voltage (V)'); legend('PWM output', 'carrier (scaled)', 'referenc
title('PWM generation by comparing a reference with a sawtooth carrier');
```

2.21.5 Program 2.5 – ADC signal acquisition

Objective. Sample and quantise a near-full-scale sine with a 12-bit, 5 V ADC and verify Eqs. (2.33) and (2.34), and compute an alias frequency. **Expected output:** $Q = 1.221\,\text{mV}$, maximum error 0.61 mV ($Q/2$), rms error 0.350 mV ($\approx Q/\sqrt{12}$), measured SNR 74.1 dB versus theoretical 74.0 dB; a 900 Hz input aliases to 100 Hz. A non-coherent test frequency (47.3 Hz) is used so that the quantisation error is spread uniformly.

Program 2.5: ADC sampling, quantisation error and SNR.

```
% Program 2.5: ADC signal acquisition -- sampling, quantisation and SNR
clear; clc; close all;
FSR = 5; Nbits = 12; Q = FSR/2^Nbits;
f0 = 47.3; fs = 1000; t = 0:1/fs:10-1/fs; % 47.3 Hz sine (non-coherent) sampled at 1 kHz
v = FSR/2 + (FSR/2 - Q)*sin(2*pi*f0*t); % near full-scale unipolar sine
code = round(v/Q); code = min(max(code, 0), 2^Nbits - 1);
vq = code*Q; e = vq - v;
SNRmeas = 20*log10(std(v)/std(e));
fprintf('Q = %.3f mV, max |error| = %.3f mV, rms error = %.3f mV (Q/sqrt(12) = %.3f mV)\n', ...
        1e3*Q, 1e3*max(abs(e)), 1e3*std(e), 1e3*Q/sqrt(12));
fprintf('SNR measured = %.1f dB, theory 6.02N+1.76 = %.1f dB\n', SNRmeas, 6.02*Nbits + 1.76);
fa = abs(900 - round(900/fs)*fs);
fprintf('A 900 Hz input sampled at %d Hz aliases to %d Hz\n', fs, fa);
subplot(2,1,1); stairs(t(1:40), vq(1:40)); hold on; plot(t(1:40), v(1:40), 'r'); grid on;
ylabel('Voltage (V)'); title('Sampled and quantised signal');
subplot(2,1,2); plot(t, 1e3*e); grid on; xlabel('Time (s)'); ylabel('Error (mV)');
```

2.21.6 Program 2.6 – Instrumentation amplifier and filter design

Objective. Compute the in-amp gain resistor and common-mode error of Example 2.12, plot RC and Butterworth responses, and filter a noisy bridge signal with the first-order IIR filter. **Expected output:** $R_G = 50.05\,\Omega$, common-mode error 25.0 mV; RC $f_c = 159.2\,\text{Hz}$,

$|H(1\,\text{kHz})| = 0.157$; the IIR filter ($\tau = 20\,\text{ms}$) reduces the noise standard deviation by about a factor of ten at the cost of a 20 ms time constant in the step response.

Program 2.6: Instrumentation amplifier gain, analogue filter responses and digital IIR filtering.

```
% Program 2.6: Instrumentation amplifier and filter design
clear; clc; close all;
% ---- In-amp gain resistor and common-mode error
R = 25e3; Greq = 1000; RG = 2*R/(Greq - 1);
CMRRdB = 100; Acm = Greq/10^(CMRRdB/20); Vcm = 2.5;
fprintf('R_G = %.2f ohm; common-mode output error = %.1f mV\n', RG, 1e3*Acm*Vcm);
% ---- Analogue filter responses
f = logspace(0, 4, 400); fc = 1/(2*pi*10e3*100e-9);
H1 = 1./sqrt(1 + (f/fc).^2);
semilogx(f, 20*log10(H1), 'LineWidth', 1.5); hold on;
for n = [2 4]
    semilogx(f, -10*log10(1 + (f/100).^(2*n)), 'LineWidth', 1.2);
end
grid on; xlabel('Frequency (Hz)'); ylabel('Gain (dB)'); ylim([-80 5]);
legend(sprintf('RC, f_c = %.0f Hz', fc), 'Butterworth n=2, 100 Hz', 'Butterworth n=4, 100 Hz');
fprintf('RC f_c = %.1f Hz, |H(1 kHz)| = %.3f\n', fc, 1/sqrt(1 + (1000/fc)^2));
% ---- Digital first-order IIR filter on a noisy bridge signal
fs = 1000; t = 0:1/fs:1; x = 5e-3*(t > 0.2) + 0.5e-3*randn(size(t));
tau = 0.02; alpha = (1/fs)/(tau + 1/fs); y = zeros(size(x));
for k = 2:numel(x), y(k) = alpha*x(k) + (1 - alpha)*y(k-1); end
fprintf('Noise std before = %.3f mV, after IIR = %.3f mV\n', 1e3*std(x(t<0.2)), 1e3*std(y(t>0.1
figure; plot(t, 1e3*x, t, 1e3*y, 'LineWidth', 1.2); grid on;
xlabel('Time (s)'); ylabel('Bridge output (mV)'); legend('raw', 'IIR filtered');
```

2.21.7 Simulink models

S2.1 DC motor block diagram. Build Eqs. (2.8) and (2.9) with `Sum`, `Gain` ($1/L_a$, R_a, K_t, K_b, $1/J$) and two `Integrator` blocks; apply a `Step` voltage and a `Step` load torque at 50 ms. Observe the speed droop predicted by Eq. (2.11).

S2.2 Cascade servo. Add PI current, PI velocity and P position loops around S2.1 (as in Fig. 2.6(a)), tuning the inner loop first.

S2.3 PWM H-bridge. Use a `Repeating Sequence` carrier and `Relational Operator` to generate PWM, feed it to the motor model and compare with the average-voltage model.

S2.4 DAQ chain. Cascade `Transfer Fcn` (anti-aliasing Butterworth), `Zero-Order Hold` (sampling) and `Quantizer` (interval Q) blocks; drive with a sine above and below $f_s/2$ to visualise aliasing.

2.22 Unit II Summary and Review

2.22.1 Unit summary

1. Electrical actuators dominate robot joints because of their controllability; pneumatic actuators dominate grippers; hydraulic actuators serve very high payloads ($F = pA$, $v = Q/A$, $P = pQ$).
2. The DC motor is described by $v_a = R_a i_a + L_a\, di_a/dt + K_b\omega$ and $J\dot{\omega} + b\omega = K_t i_a - T_L$; its speed–torque characteristic is linear and speed is controlled by PWM.
3. Servo motors close current, velocity and position loops; RC servos are commanded by 1–2 ms pulses; PID control with anti-windup is the standard algorithm.
4. Stepper motors move in steps $\theta_s = 360^\circ/(mN_r)$ under open-loop control; BLDC motors replace the commutator with Hall sensors and six-step or field-oriented commutation.
5. Solenoids give short-stroke force $F \propto (NI)^2/g^2$; ball screws convert torque to thrust $F = 2\pi\eta T/p$.
6. Gear reducers (harmonic, cycloidal) multiply torque and reduce reflected inertia by n^2; serial manipulators offer workspace, parallel manipulators stiffness and speed.
7. Grippers (mechanical, vacuum, magnetic) are sized from $F_g \geq S\,m(g + a)/(n_f\mu)$.
8. Signal conditioning provides excitation, amplification, filtering, isolation, linearisation and transmission; the in-amp gain is $(1 + 2R/R_G)R_3/R_2$ with high CMRR.
9. Filters remove noise and prevent aliasing; shielding, twisted pairs, single-point grounding and differential inputs control interference.
10. ADCs sample at $f_s > 2f_{\max}$ and quantise with step $Q = \mathrm{FSR}/2^N$; SAR, flash, dual-slope and sigma-delta architectures trade speed for resolution; R–2R DACs output $V_{\mathrm{ref}}D/2^N$; DAQ systems multiplex, condition and digitise many channels.

2.22.2 Formula sheet

Table 2.9: Unit II formula sheet.

Quantity	Formula	Ref.
Pneumatic cylinder force	$F = \eta pA$; $F_r = \eta p(A - a)$	(2.1)
Free-air consumption	$(2A - a)s(p + p_a)/p_a$	(2.2)
Hydraulic force, speed, power	$F = pA$, $v = Q/A$, $P = pQ$	(2.3)–(2.5)
DC motor torque, back-EMF	$T = K_t i_a$, $E_b = K_b\omega$	(2.6), (2.7)
Armature equation	$v_a = R_a i_a + L_a\dot{i}_a + K_b\omega$	(2.8)
Speed–torque line	$\omega = V_a/K_b - R_a T/(K_t K_b)$	(2.11)
Motor time constants	$\tau_e = L_a/R_a$, $\tau_m = JR_a/(K_t K_b)$	(2.10)

Quantity	Formula	Ref.
RC servo angle	$\theta = 180^\circ(t_p - 1)/1$ (ms)	(2.12)
PID law	$u = K_p e + K_i \int e + K_d \dot{e}$	(2.14)
Step angle, speed	$\theta_s = 360^\circ/(mN_r)$; $n = 60f/S$	(2.15), (2.16)
Solenoid force	$F = (NI)^2 \mu_0 A/(2g^2)$	(2.18)
Ball screw	$F = 2\pi\eta T/p$, $v = pn$	(2.19)
Gear, reflected inertia	$T_L = n\eta T_m$; $J_{\text{eq}} = J_m + J_L/n^2$	(2.20), (2.21)
Harmonic drive	$n = N_f/(N_c - N_f)$	(2.22)
Gripping force	$F_g \geq S\,m(g + a)/(n_f \mu)$	(2.23)
AC power	$P = VI\cos\phi$	(2.24)
Op-amp gains	$-R_f/R_1$; $1 + R_f/R_1$; R_2/R_1	(2.25)–(2.27)
In-amp gain	$(1 + 2R/R_G)R_3/R_2$	(2.29)
CMRR	$20\log(A_d/A_{cm})$	(2.28)
RC filter	$f_c = 1/(2\pi RC)$	(2.30)
Butterworth	$\lvert H\rvert = [1 + (\omega/\omega_c)^{2n}]^{-1/2}$	(2.31)
Nyquist	$f_s > 2f_{\max}$	(2.32)
Quantisation step, SNR	$Q = \text{FSR}/2^N$; $6.02N + 1.76$ dB	(2.33), (2.34)
DAC output	$v_o = V_{\text{ref}} D/2^N$	(2.35)
DAQ data rate	$R_b = nf_sN$	(2.36)

2.23 Unit II Question Bank

Exactly **30 questions of 10 marks each** (300 marks). Distribution: concepts Q1–Q5; theory Q6–Q12; derivations Q13–Q17; numerical problems Q18–Q23; diagrams and working principles Q24–Q26; applications Q27–Q30.

Question-to-notes reference index

For every question the table gives the topic (section) number, the page of the question and the page on which the topic begins in these notes. The **Refer to** line under each question also lists the specific figures, tables, equations and worked examples to study.

Table 2.10: Unit II question-to-notes reference index (automatic page references).

Q	Topic	Q. page	Notes page	Q	Topic	Q. page	Notes page
Q1	2.1	115	75	Q16	2.18	117	103
Q2	2.9	115	90	Q17	2.19	117	105
Q3	2.11	115	92	Q18	2.2	117	76

Q4	2.14	115	96	Q19	2.4	118	80
Q5	2.12	116	94	Q20	2.6	118	84
Q6	2.2	116	76	Q21	2.15	118	98
Q7	2.3	116	79	Q22	2.20	118	107
Q8	2.6	116	84	Q23	2.19	118	105
Q9	2.7	116	87	Q24	2.5	118	83
Q10	2.10	116	92	Q25	2.8	118	88
Q11	2.18	116	103	Q26	2.20	119	107
Q12	2.17	117	101	Q27	2.1	119	75
Q13	2.4	117	80	Q28	2.9	119	90
Q14	2.15	117	98	Q29	2.11	119	92
Q15	2.16	117	99	Q30	2.14	119	96

A. Fundamental concepts and definitions

Q1 **Topic 2.1** – Electrical, Pneumatic and Hydraulic Actuators **[10 marks]**

Define an actuator. Classify robot actuators into electrical, pneumatic and hydraulic types, explaining the energy-conversion principle of each, and compare them in a table with respect to force density, positioning accuracy, speed, cleanliness, infrastructure and typical robotic applications.

☞ **Refer to:** Section 2.1 (p. 75); Figure 2.1 (p. 77); Table 2.2 (p. 77)

Q2 **Topic 2.9** – Robot Joints and Mechanisms **[10 marks]**

Describe the physical construction of revolute and prismatic robot joints. Explain why speed reducers are used, and describe harmonic drives, cycloidal reducers, ball screws and parallelogram mechanisms as used in robots.

☞ **Refer to:** Section 2.9 (p. 90); Figure 2.10 (p. 91); Equation (2.20) (p. 90); Equation (2.21) (p. 90); Example 2.9 (p. 91)

Q3 **Topic 2.11** – End-Effectors and Grippers **[10 marks]**

Distinguish between tools and grippers as end-effectors. Classify grippers (mechanical, vacuum, magnetic, others), explaining the working principle, advantages and limitations of each, and distinguish friction grip from encompassing grip.

☞ **Refer to:** Section 2.11 (p. 92); Figure 2.12 (p. 94); Table 2.5 (p. 94); Example 2.10 (p. 93)

Q4 **Topic 2.14** – Signal Conditioning **[10 marks]**

Define signal conditioning and explain each of its functions – excitation, conversion, amplification, filtering, isolation, linearisation and transmission – with a block diagram. Explain the advantages of the 4–20 mA current loop.

☞ **Refer to:** Section 2.14 (p. 96); Figure 2.13 (p. 97)

Q5 Topic 2.12 – Measurement of Electrical Quantities **[10 marks]**

Explain the methods of measuring voltage, current (shunt, Hall-effect sensor, current transformer), AC power and low resistance (four-wire method). State the advantages and limitations of each current-measurement method for use in servo drives.

☞ **Refer to:** Section 2.12 (p. 94); Example 2.11 (p. 95)

B. Theoretical explanations

Q6 Topic 2.2 – Pneumatic Actuators **[10 marks]**

With a neat circuit diagram, explain an electro-pneumatic system comprising compressor, FRL unit, 5/2 solenoid valve and double-acting cylinder. Distinguish single- and double-acting cylinders, and state the advantages and limitations of pneumatics in robotics.

☞ **Refer to:** Section 2.2 (p. 76); Figure 2.2 (p. 78); Example 2.1 (p. 78)

Q7 Topic 2.3 – Hydraulic Actuators **[10 marks]**

Explain Pascal's law and the working of a hydraulic actuation system with pump, relief valve, directional/servo valve, cylinder and tank. Derive the relations for force, speed and power, and discuss where hydraulic robots are used.

☞ **Refer to:** Section 2.3 (p. 79); Figure 2.3 (p. 79); Example 2.2 (p. 80)

Q8 Topic 2.6 – Stepper Motors **[10 marks]**

Explain the construction and working of permanent-magnet, variable-reluctance and hybrid stepper motors. Explain wave, full-step, half-step and microstepping drive modes, and the terms holding torque, pull-in and pull-out torque and lost steps.

☞ **Refer to:** Section 2.6 (p. 84); Figure 2.7 (p. 85); Example 2.5 (p. 86)

Q9 Topic 2.7 – BLDC Motors **[10 marks]**

Explain the construction of a BLDC motor and the principle of electronic commutation using Hall sensors. Draw the back-EMF, Hall-signal and phase-current waveforms for six-step commutation and compare BLDC motors with brushed DC motors.

☞ **Refer to:** Section 2.7 (p. 87); Figure 2.8 (p. 87); Example 2.6 (p. 87)

Q10 Topic 2.10 – Serial and Parallel Manipulators **[10 marks]**

Explain serial and parallel manipulators with neat sketches, describing the Delta robot and the Stewart–Gough platform. Compare serial and parallel manipulators in a table and give applications of each.

☞ **Refer to:** Section 2.10 (p. 92); Figure 2.11 (p. 92); Table 2.4 (p. 93)

Q11 Topic 2.18 – A/D Conversion Architectures **[10 marks]**

Explain with block diagrams the successive-approximation, flash, dual-slope and sigma-delta ADCs. Show the step-by-step conversion of 2.0 V by a 6-bit SAR ADC with $V_{\text{ref}} = 3.3$ V and compare the four architectures.

☞ **Refer to:** Section 2.18 (p. 103); Figure 2.20 (p. 105); Table 2.8 (p. 106); Example 2.14 (p. 104); Example 2.15 (p. 104); Algorithm 2.3 (p. 104)

Q12 **Topic 2.17** – Shielding and Grounding **[10 marks]**

Explain the capacitive, inductive, conductive and radiated noise-coupling mechanisms in a robot cell. Explain ground loops and describe shielding and grounding practices that minimise interference.

☞ **Refer to:** Section 2.17 (p. 101); Figure 2.18 (p. 103)

C. Mathematical derivations

Q13 **Topic 2.4** – DC Motor Model **[10 marks]**

Derive the electrical and mechanical equations of a permanent-magnet DC motor, its transfer function $\Omega(s)/V_a(s)$ and its steady-state speed–torque characteristic. Show that $K_t = K_b$ in SI units and identify the no-load speed and stall torque.

☞ **Refer to:** Section 2.4 (p. 80); Example 2.3 (p. 81); Example 2.4 (p. 82)

Q14 **Topic 2.15** – Instrumentation Amplifier **[10 marks]**

Draw the three-op-amp instrumentation amplifier and derive its output voltage. Explain why the first stage improves CMRR, and define CMRR.

☞ **Refer to:** Section 2.15 (p. 98); Figure 2.15 (p. 99); Example 2.12 (p. 99)

Q15 **Topic 2.16** – Filters **[10 marks]**

Derive the transfer function, magnitude response and cut-off frequency of first-order RC low-pass and high-pass filters. Write the Butterworth magnitude response and explain the role of filter order. Explain the Sallen–Key second-order active filter.

☞ **Refer to:** Section 2.16 (p. 99); Figure 2.17 (p. 101); Figure 2.16 (p. 100); Example 2.13 (p. 100)

Q16 **Topic 2.18** – Sampling and Quantisation **[10 marks]**

State and explain the Nyquist sampling theorem and aliasing. Derive the quantisation step, the rms quantisation error and the ideal SNR $6.02N + 1.76$ dB of an N-bit ADC.

☞ **Refer to:** Section 2.18 (p. 103); Example 2.14 (p. 104); Example 2.15 (p. 104)

Q17 **Topic 2.19** – D/A Conversion **[10 marks]**

Derive the output of a binary-weighted-resistor DAC and of an R–2R ladder DAC. Explain why the R–2R ladder is preferred, and define DAC resolution, settling time, monotonicity, DNL and INL.

☞ **Refer to:** Section 2.19 (p. 105); Figure 2.21 (p. 106); Example 2.16 (p. 106)

D. Numerical problems

Q18 **Topic 2.2** – Pneumatic and Hydraulic Calculations **[10 marks]**

(a) A double-acting pneumatic cylinder has bore 63 mm, rod 20 mm, supply 5 bar and efficiency 0.85. Find the extend and retract forces. (b) A hydraulic cylinder of bore 100 mm must exert 60 kN at 0.05 m/s. Find the pressure, pump flow (L/min) and hydraulic power.

☞ **Refer to:** Section 2.2 (p. 76); Example 2.1 (p. 78)

Q19 Topic 2.4 – DC Motor Calculations **[10 marks]**

A 48 V DC motor has $R_a = 0.8\,\Omega$ and $K_t = K_b = 0.12$ SI units. For a load of 1.5 N m (friction neglected) find the current, back-EMF, speed (rpm), output and input power and efficiency. Also find the stall torque, no-load speed and maximum output power.

☞ **Refer to:** Section 2.4 (p. 80); Example 2.3 (p. 81); Example 2.4 (p. 82)

Q20 Topic 2.6 – Stepper Motor Calculations **[10 marks]**

(a) Find the step angle of a two-phase hybrid stepper with 50 rotor teeth and of a four-phase VR stepper with 6 rotor teeth. (b) The hybrid motor must rotate 90° in 0.5 s; find the pulse rate in full-step and half-step modes. (c) It drives a lead screw of pitch 4 mm with 1/8 microstepping; find the linear resolution.

☞ **Refer to:** Section 2.6 (p. 84); Example 2.5 (p. 86)

Q21 Topic 2.15 – Instrumentation Amplifier Calculations **[10 marks]**

An integrated in-amp has $G = 1 + 49.4\,\text{k}\Omega/R_G$ and CMRR 110 dB. It amplifies the output of a full strain-gauge bridge ($G_F = 2$, $V_{\text{ex}} = 10$ V, strain $400\,\mu\varepsilon$) to 4 V. Find the bridge output, the required gain and R_G, and the output error caused by a common-mode voltage of 5 V.

☞ **Refer to:** Section 2.15 (p. 98); Example 2.12 (p. 99)

Q22 Topic 2.20 – ADC and DAQ Calculations **[10 marks]**

A 16-bit bipolar ADC has range ±10 V. Find its LSB and ideal SNR. Four channels with content up to 2 kHz are sampled at 10 kS/s each through a multiplexer; find the ADC conversion rate, data rate and inter-channel skew, and the alias frequency of a 9 kHz interference.

☞ **Refer to:** Section 2.20 (p. 107); Example 2.17 (p. 108)

Q23 Topic 2.19 – DAC and Filter Calculations **[10 marks]**

(a) A 10-bit R–2R DAC with $V_{\text{ref}} = 10$ V receives code 1010100101. Find the output magnitude and resolution. (b) An anti-aliasing RC filter uses $R = 4.7\,\text{k}\Omega$, $C = 33$ nF. Find f_c and the attenuation at 5 kHz; comment on its adequacy for $f_s = 10\,\text{kS/s}$.

☞ **Refer to:** Section 2.19 (p. 105); Example 2.16 (p. 106)

E. Diagrams and working principles

Q24 Topic 2.5 – Servo Motors **[10 marks]**

With a neat block diagram, explain the cascade position–velocity–current control of an industrial servo motor and the reason for the nesting of loops. Explain the RC servo and its pulse-width command, and find the angle for a 1.75 ms pulse.

☞ **Refer to:** Section 2.5 (p. 83); Figure 2.6 (p. 84); Example 2.4 (p. 82)

Q25 Topic 2.8 – Solenoids and Linear Actuators **[10 marks]**

With neat sketches explain the working of a solenoid and of a ball-screw linear actuator, deriving the force relations. A solenoid has 800 turns, 1 A, gap area 2 cm^2 and gap 3 mm: find the force. A ball screw (lead 5 mm, $\eta = 0.9$) is driven at 0.8 N m and 3000 rpm: find the thrust and speed.

☞ **Refer to:** Section 2.8 (p. 88); Figure 2.9 (p. 89); Example 2.7 (p. 89); Example 2.8 (p. 89)

Q26 **Topic 2.20** – Data Acquisition and Physical Measurements **[10 marks]**

With a neat block diagram, explain each element of a multichannel data acquisition system. Explain how speed and liquid level are measured as physical quantities, and calculate (a) the speed indicated by a 48-tooth wheel producing 2400 Hz, (b) the water level corresponding to a bottom pressure of 24.5 kPa.

☞ **Refer to:** Section 2.20 (p. 107); Figure 2.22 (p. 107); Example 2.17 (p. 108)

F. Applications and analytical questions

Q27 **Topic 2.1** – Actuator Selection **[10 marks]**

Select and justify the actuator (and motor type) for: (a) joints of a six-axis welding robot; (b) a gripper picking cartons; (c) a 3-D printer axis; (d) a forging manipulator handling 2 t; (e) the wheels of a warehouse mobile robot; (f) a door lock in a safety fence.

☞ **Refer to:** Section 2.1 (p. 75)

Q28 **Topic 2.9** – Reducer Selection and Reflected Inertia **[10 marks]**

A harmonic drive has a flexspline of 160 teeth and a circular spline of 162 teeth. The motor runs at 3000 rpm delivering 1.2 N m; drive efficiency is 0.8; the arm inertia about the joint is 1.5 kg m^2. Find the ratio, output speed and torque, and the reflected inertia. Discuss the effect of the reducer on control.

☞ **Refer to:** Section 2.9 (p. 90); Example 2.9 (p. 91)

Q29 **Topic 2.11** – Gripper Design **[10 marks]**

A two-finger friction gripper must lift a 5 kg part vertically with an acceleration of 4 m/s^2; $\mu = 0.4$, safety factor 2. Find the required gripping force per finger. Suggest a gripper type and the sensors needed to confirm a secure grasp, and how the design would change for a cardboard carton and for a steel sheet.

☞ **Refer to:** Section 2.11 (p. 92); Example 2.10 (p. 93)

Q30 **Topic 2.14** – Design of a Measurement Chain **[10 marks]**

Design the signal chain from a 100 kg, 2 mV/V load cell excited at 10 V to a 3.3 V, 16-bit microcontroller ADC sampling at 100 Hz. Specify the amplifier gain, filter, grounding and shielding, and calculate the weight resolution.

☞ **Refer to:** Section 2.14 (p. 96)

Unit III: Robot Kinematics, Trajectory Planning and Control

Unit overview

This unit provides the mathematical machinery of robot motion. We begin with the description of position and orientation by coordinate frames, rotation matrices and homogeneous transformations (Sections 3.1 to 3.3), formalise link geometry with the Denavit–Hartenberg convention (Section 3.4), and use it to solve the forward, inverse and velocity kinematics problems, including the Jacobian, singularities and workspace (Sections 3.5 to 3.9). We then generate smooth motions in joint and Cartesian space (Sections 3.10 to 3.12), relate joint torques to end-effector forces (Section 3.13), derive equations of motion by the Lagrangian method (Sections 3.14 and 3.15) and close the unit with joint-space and task-space control (Sections 3.16 and 3.17). The treatment follows Craig [1], Spong *et al.* [2] and Lynch and Park [3].

Table 3.1: Unit III syllabus items and the sections that cover them.

No.	Syllabus item	Section	No.	Syllabus item	Section
1	Coordinate transformations	3.1	11	Joint-space trajectories	3.10
2	Rotation matrices	3.2	12	Cartesian trajectories	3.11
3	Homogeneous transformation	3.3	13	Velocity profiles	3.12
4	Denavit–Hartenberg representation	3.4	14	Acceleration profiles	3.12
5	Forward kinematics	3.5	15	Robot statics	3.13
6	Inverse kinematics	3.6	16	Robot dynamics	3.14
7	Velocity kinematics	3.7	17	Lagrangian formulation	3.15
8	Jacobian matrix	3.7	18	Joint-space control	3.16
9	Singularities	3.8	19	Task-space control	3.17
10	Workspace	3.9			

3.1 Coordinate Frames and Coordinate Transformations

3.1.1 Position and frames

To describe where a robot's tool is we attach a right-handed Cartesian *coordinate frame* to each rigid body of interest: the base, each link, the tool, the workpiece and the camera (Unit I, Fig. 1.8). A frame $\{A\}$ is defined by an origin and three mutually orthogonal unit vectors $\hat{\mathbf{x}}_A, \hat{\mathbf{y}}_A, \hat{\mathbf{z}}_A$ with $\hat{\mathbf{x}}_A \times \hat{\mathbf{y}}_A = \hat{\mathbf{z}}_A$. The position of a point P is a 3×1 vector whose components are the projections of P on the frame axes; the leading superscript names the frame in which the

vector is expressed:

$$ {}^{A}\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \text{(m)}. \tag{3.1} $$

3.1.2 The coordinate-transformation problem

The same physical point has different coordinates in different frames. The fundamental problem of robot kinematics is to convert coordinates between frames – for example, a camera reports a part's position in the camera frame, but the robot must move to it in its base frame. In general frame $\{B\}$ is both *translated* and *rotated* relative to $\{A\}$ (Fig. 3.1). If ${}^{A}\mathbf{p}_{B,\text{org}}$ locates the origin of $\{B\}$ in $\{A\}$ and ${}^{A}_{B}\mathbf{R}$ describes its orientation, vector addition gives the general transformation

$$ {}^{A}\mathbf{p} = {}^{A}_{B}\mathbf{R}\ {}^{B}\mathbf{p} + {}^{A}\mathbf{p}_{B,\text{org}}. \tag{3.2} $$

Pure translation ($\{B\}$ parallel to $\{A\}$): ${}^{A}_{B}\mathbf{R} = \mathbf{I}$, so ${}^{A}\mathbf{p} = {}^{B}\mathbf{p} + {}^{A}\mathbf{p}_{B,\text{org}}$. **Pure rotation** (common origin): ${}^{A}\mathbf{p} = {}^{A}_{B}\mathbf{R}\ {}^{B}\mathbf{p}$. The rotation matrix is the subject of the next section.

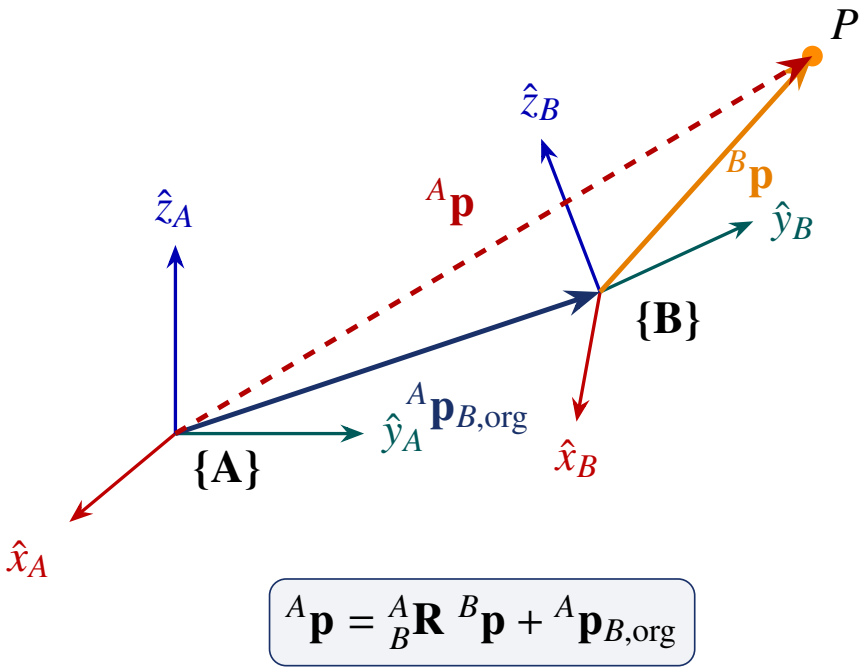


Figure 3.1: General coordinate transformation between frames {A} and {B}: the vector to P in {A} is the rotated vector in {B} plus the position of the origin of {B}.

3.2 Rotation Matrices

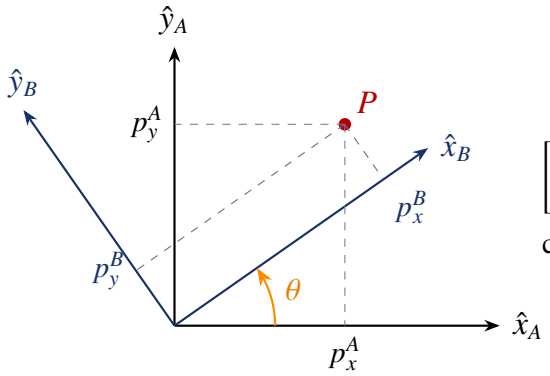
3.2.1 Definition

The orientation of $\{B\}$ relative to $\{A\}$ is described by writing the unit vectors of $\{B\}$ in the coordinates of $\{A\}$ and stacking them as columns:

$$ {}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{\mathbf{x}}_B & {}^{A}\hat{\mathbf{y}}_B & {}^{A}\hat{\mathbf{z}}_B \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{x}}_B \cdot \hat{\mathbf{x}}_A & \hat{\mathbf{y}}_B \cdot \hat{\mathbf{x}}_A & \hat{\mathbf{z}}_B \cdot \hat{\mathbf{x}}_A \\ \hat{\mathbf{x}}_B \cdot \hat{\mathbf{y}}_A & \hat{\mathbf{y}}_B \cdot \hat{\mathbf{y}}_A & \hat{\mathbf{z}}_B \cdot \hat{\mathbf{y}}_A \\ \hat{\mathbf{x}}_B \cdot \hat{\mathbf{z}}_A & \hat{\mathbf{y}}_B \cdot \hat{\mathbf{z}}_A & \hat{\mathbf{z}}_B \cdot \hat{\mathbf{z}}_A \end{bmatrix}. \tag{3.3} $$

Each element is the cosine of the angle between two axes, so ${}^{A}_{B}\mathbf{R}$ is also called the *direction-cosine matrix*.

Why it transforms coordinates – derivation. A point with coordinates ${}^{B}\mathbf{p} = (p_x^B, p_y^B, p_z^B)$ is the vector $p_x^B\hat{\mathbf{x}}_B + p_y^B\hat{\mathbf{y}}_B + p_z^B\hat{\mathbf{z}}_B$. Expressing each unit vector in $\{A\}$ and collecting terms gives ${}^{A}\mathbf{p} = [{}^{A}\hat{\mathbf{x}}_B \; {}^{A}\hat{\mathbf{y}}_B \; {}^{A}\hat{\mathbf{z}}_B]\,{}^{B}\mathbf{p} = {}^{A}_{B}\mathbf{R}\,{}^{B}\mathbf{p}$. Figure 3.2 shows the planar case geometrically: the coordinates in $\{A\}$ are obtained by projecting the rotated components back onto the axes of $\{A\}$.



$$ \begin{bmatrix} p_x^A \\ p_y^A \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} p_x^B \\ p_y^B \end{bmatrix} $$

columns of ${}^{A}_{B}\mathbf{R}$ = unit vectors $\hat{x}_B, \hat{y}_B$ in {A}

Figure 3.2: Geometric interpretation of a planar rotation matrix: projections of P on the axes of {A} and the rotated frame {B}.

3.2.2 Properties

Because the columns are mutually orthogonal unit vectors:

(i) $\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}$ – the matrix is *orthonormal*, so its inverse is simply its transpose: ${}^{B}_{A}\mathbf{R} = {}^{A}_{B}\mathbf{R}^{-1} = {}^{A}_{B}\mathbf{R}^{\mathsf{T}}$;

(ii) $\det \mathbf{R} = +1$ for a right-handed frame;

(iii) only three of the nine elements are independent (six orthonormality constraints) – orientation has three DOF;

(iv) rotation preserves lengths and angles: $\|\mathbf{R}\mathbf{p}\| = \|\mathbf{p}\|$;

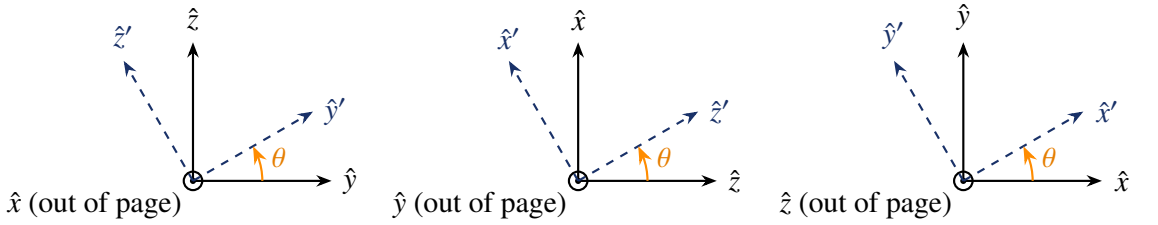
(v) the set of all such matrices forms the special orthogonal group $SO(3)$; products of rotations are rotations, but *matrix multiplication is not commutative* – the order of rotations matters.

3.2.3 Elementary rotations

Rotating $\{B\}$ by angle θ about one axis of $\{A\}$ leaves that axis unchanged and rotates the other two in their plane (Fig. 3.3). Applying (3.3) with $c\theta = \cos\theta$, $s\theta = \sin\theta$:

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}, \quad \mathbf{R}_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}, \quad \mathbf{R}_z(\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{3.4}$$

Note the sign pattern of $\mathbf{R}_y$: the $+s\theta$ appears in the top-right because the cyclic order is $z \to x$ for a rotation about y. Positive angles follow the right-hand rule.



Rotation about $\hat{x}$: $R_x(\theta)$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$

Rotation about $\hat{y}$: $R_y(\theta)$ $\begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$

Rotation about $\hat{z}$: $R_z(\theta)$ $\begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Figure 3.3: Elementary rotations about the x, y and z axes and the corresponding matrices.

Example 3.1: Coordinates after a rotation (Topic 3.2)

Given: $\{B\}$ is rotated 30° about $\hat{\mathbf{z}}_A$; ${}^B\mathbf{p} = (1, 2, 0)$ m.

Solution: ${}^A\mathbf{p} = \mathbf{R}_z(30°)\,{}^B\mathbf{p} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.134 \\ 2.232 \\ 0 \end{bmatrix}$ m.

Check: length $\sqrt{0.134^2 + 2.232^2} = 2.236 = \sqrt{5}$, unchanged as property (iv) requires.

3.2.4 Composition of rotations

If $\{C\}$ is described relative to $\{B\}$ and $\{B\}$ relative to $\{A\}$, the chain rule of subscripts and superscripts gives ${}^A_C\mathbf{R} = {}^A_B\mathbf{R}\,{}^B_C\mathbf{R}$. Two interpretations follow:

- rotations about the axes of the *current* (moving) frame are composed by *post*-multiplication;
- rotations about the axes of the *fixed* frame are composed by *pre*-multiplication.

Example: $\mathbf{R}_z(90°)\mathbf{R}_y(90°) = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$, whereas $\mathbf{R}_y(90°)\mathbf{R}_z(90°) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ – different final orientations, confirming non-commutativity.

3.2.5 Minimal representations of orientation

A rotation matrix uses nine numbers for three DOF; minimal or compact descriptions are used in programming and teach pendants.

- **Roll–pitch–yaw (fixed X–Y–Z, equivalent to current Z–Y–X Euler)**: roll γ about x, pitch β about y, yaw α about z, all about fixed axes:

$$\mathbf{R}_{XYZ}(\gamma,\beta,\alpha) = \mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_x(\gamma) = \begin{bmatrix} c_\alpha c_\beta & c_\alpha s_\beta s_\gamma - s_\alpha c_\gamma & c_\alpha s_\beta c_\gamma + s_\alpha s_\gamma \\ s_\alpha c_\beta & s_\alpha s_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta c_\gamma - c_\alpha s_\gamma \\ -s_\beta & c_\beta s_\gamma & c_\beta c_\gamma \end{bmatrix}. \tag{3.5}$$

  Inverse: $\beta = \operatorname{atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right)$, $\alpha = \operatorname{atan2}(r_{21}, r_{11})$, $\gamma = \operatorname{atan2}(r_{32}, r_{33})$, valid for $\beta \neq \pm 90°$ (representation singularity, "gimbal lock").

- **Z–Y–Z Euler angles** (ϕ, θ, ψ): $\mathbf{R} = \mathbf{R}_z(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\psi)$; natural for spherical wrists.
- **Axis–angle**: rotation by θ about unit axis $\hat{\mathbf{k}}$; Rodrigues' formula $\mathbf{R} = \mathbf{I} + \sin\theta\,[\hat{\mathbf{k}}]_\times + (1 - \cos\theta)[\hat{\mathbf{k}}]_\times^2$.
- **Unit quaternion q** $= (\cos\frac{\theta}{2}, \hat{\mathbf{k}}\sin\frac{\theta}{2})$: four numbers, no singularities, efficient interpolation (used by many robot controllers, e.g. ABB).

Example 3.2: Roll–pitch–yaw (Topic 3.2)

Given: $\gamma = 30°, \beta = 45°, \alpha = 60°$.

Solution: From (3.5):

$$\mathbf{R} = \begin{bmatrix} 0.3536 & -0.5732 & 0.7392 \\ 0.6124 & 0.7392 & 0.2803 \\ -0.7071 & 0.3536 & 0.6124 \end{bmatrix}.$$

Inverse check: $\beta = \operatorname{atan2}(0.7071, \sqrt{0.3536^2 + 0.6124^2}) = 45°$; $\alpha = \operatorname{atan2}(0.6124, 0.3536) = 60°$; $\gamma = \operatorname{atan2}(0.3536, 0.6124) = 30°$.

3.3 Homogeneous Transformations

3.3.1 Definition

Equation (3.2) mixes a multiplication and an addition, which becomes clumsy when many frames are chained. Appending a fourth coordinate of 1 to every position vector (*homogeneous*

Table 3.2: Comparison of orientation representations.

Representation	Numbers	Advantages	Limitations
Rotation matrix	9	Direct composition and transformation; no singularity	Redundant; must stay orthonormal
Euler / RPY angles	3	Minimal, intuitive, used on pendants	Singularities (gimbal lock); order-dependent
Axis–angle	4 (3)	Physical meaning	Ambiguous at $\theta = 0$
Unit quaternion	4	No singularity; smooth interpolation (slerp)	Less intuitive; double cover ($\pm\mathbf{q}$)

coordinates $\tilde{\mathbf{p}} = [\mathbf{p}^\mathsf{T}\ 1]^\mathsf{T}$) allows rotation and translation to be combined in a single 4×4 matrix:

$$ {}^{A}\tilde{\mathbf{p}} = {}^{A}_{B}\mathbf{T}\,{}^{B}\tilde{\mathbf{p}}, \qquad {}^{A}_{B}\mathbf{T} = \begin{bmatrix} {}^{A}_{B}\mathbf{R} & {}^{A}\mathbf{p}_{B,\mathrm{org}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}. \tag{3.6} $$

Multiplying out the block matrix reproduces (3.2) exactly. ${}^{A}_{B}\mathbf{T}$ has a double meaning: it describes frame $\{B\}$ relative to $\{A\}$ (its columns are the axes and origin of $\{B\}$), and it is the operator mapping coordinates from $\{B\}$ to $\{A\}$ (Fig. 3.4).

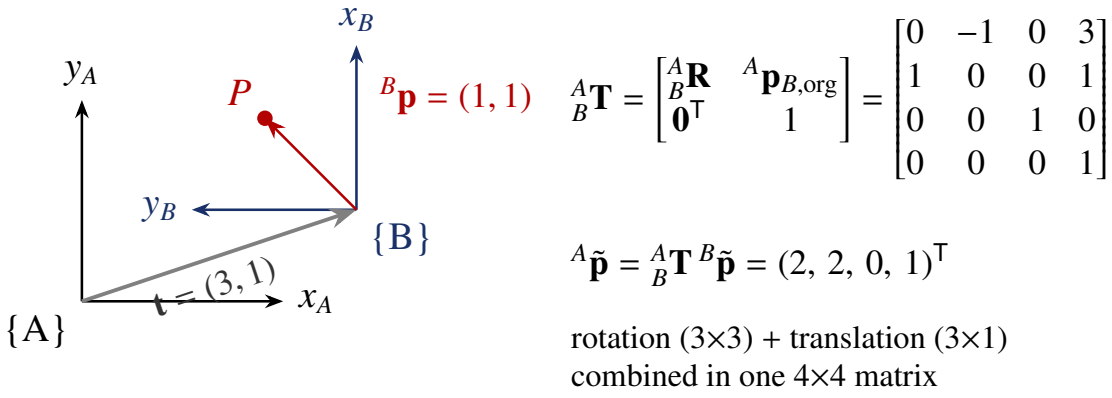


Figure 3.4: Homogeneous transformation combining a 90° rotation and a translation (3, 1) m.

3.3.2 Composition and inverse

Chains of frames compose by matrix multiplication:

$$ {}^{A}_{C}\mathbf{T} = {}^{A}_{B}\mathbf{T}\,{}^{B}_{C}\mathbf{T}, \tag{3.7} $$

which is the key to forward kinematics: ${}^{0}_{n}\mathbf{T} = {}^{0}_{1}\mathbf{T}\,{}^{1}_{2}\mathbf{T}\cdots{}^{n-1}_{n}\mathbf{T}$.

Inverse – derivation. To invert ${}^{A}\mathbf{p} = \mathbf{R}\,{}^{B}\mathbf{p} + \mathbf{t}$, multiply by $\mathbf{R}^\mathsf{T}$: ${}^{B}\mathbf{p} = \mathbf{R}^\mathsf{T}\,{}^{A}\mathbf{p} - \mathbf{R}^\mathsf{T}\mathbf{t}$. Hence

$$ {}^{B}_{A}\mathbf{T} = {}^{A}_{B}\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}, \tag{3.8} $$

which is far cheaper than a general 4×4 inversion and is used continually in robot software.

Example 3.3: Homogeneous transformation (Topic 3.3)

Given: $\{B\}$ rotated 90° about $\hat{\mathbf{z}}_A$ with origin at ${}^{A}\mathbf{p}_{B,\text{org}} = (3, 1, 0)$ m; ${}^{B}\mathbf{p} = (1, 1, 0)$ m.

Solution: ${}^{A}_{B}\mathbf{T} = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; ${}^{A}\tilde{\mathbf{p}} = (0 - 1 + 3,\ 1 + 0 + 1,\ 0,\ 1) = (2, 2, 0, 1)$. Inverse by (3.8): $\mathbf{R}^{\mathsf{T}} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $-\mathbf{R}^{\mathsf{T}}\mathbf{t} = (-1, 3, 0)$; applying it to $(2, 2, 0)$ returns $(1, 1, 0)$ ✓.

Application: a camera frame $\{C\}$ known relative to the robot base, ${}^{0}_{C}\mathbf{T}$ (from hand–eye calibration), converts a detected part position ${}^{C}\mathbf{p}$ to robot coordinates ${}^{0}\mathbf{p} = {}^{0}_{C}\mathbf{T}\,{}^{C}\mathbf{p}$ (Unit IV).

Algorithm for coordinate transformation. Given the transforms of a chain of frames, (1) multiply them in order from the reference frame outward to obtain the composite transform, (2) append 1 to the point, (3) multiply, (4) drop the last component; to go the other way, use (3.8). Program 3.1 (Program 3.1) implements this.

3.4 Denavit–Hartenberg Representation

3.4.1 Motivation

A general homogeneous transform between two link frames needs six parameters. Denavit and Hartenberg (1955) showed that by choosing the frames cleverly, the transform between consecutive links of a serial chain can be described by only **four** parameters, one of which is the joint variable. The DH convention provides a systematic, universally understood way to tabulate a robot's geometry and compute its forward kinematics.

3.4.2 Frame assignment rules (standard DH)

For a manipulator with n joints and links 0 (base) to n:

1. Number the joints 1 to n. Axis $\hat{\mathbf{z}}_{i-1}$ is the axis of joint i (rotation axis for R, translation direction for P).
2. Establish base frame $\{0\}$ with $\hat{\mathbf{z}}_0$ along joint 1; $\hat{\mathbf{x}}_0, \hat{\mathbf{y}}_0$ arbitrary (right-handed).
3. For $i = 1, \ldots, n$: place origin o_i where the *common normal* between $\hat{\mathbf{z}}_{i-1}$ and $\hat{\mathbf{z}}_i$ meets $\hat{\mathbf{z}}_i$ (if the axes intersect, at the intersection; if parallel, anywhere convenient, usually where the normal from o_{i-1} meets $\hat{\mathbf{z}}_i$).
4. Take $\hat{\mathbf{x}}_i$ along the common normal, pointing from $\hat{\mathbf{z}}_{i-1}$ to $\hat{\mathbf{z}}_i$ (if the axes intersect, along $\hat{\mathbf{z}}_{i-1} \times \hat{\mathbf{z}}_i$).
5. Complete $\hat{\mathbf{y}}_i = \hat{\mathbf{z}}_i \times \hat{\mathbf{x}}_i$; place the tool frame $\{n\}$ at the TCP.

3.4.3 The four DH parameters

With frames assigned as above (Fig. 3.5):

- a_i (*link length*) – distance from $\hat{\mathbf{z}}_{i-1}$ to $\hat{\mathbf{z}}_i$ measured along $\hat{\mathbf{x}}_i$;
- α_i (*link twist*) – angle from $\hat{\mathbf{z}}_{i-1}$ to $\hat{\mathbf{z}}_i$ measured about $\hat{\mathbf{x}}_i$;
- d_i (*link offset*) – distance from $\hat{\mathbf{x}}_{i-1}$ to $\hat{\mathbf{x}}_i$ measured along $\hat{\mathbf{z}}_{i-1}$ (*variable* for a prismatic joint);
- θ_i (*joint angle*) – angle from $\hat{\mathbf{x}}_{i-1}$ to $\hat{\mathbf{x}}_i$ measured about $\hat{\mathbf{z}}_{i-1}$ (*variable* for a revolute joint).

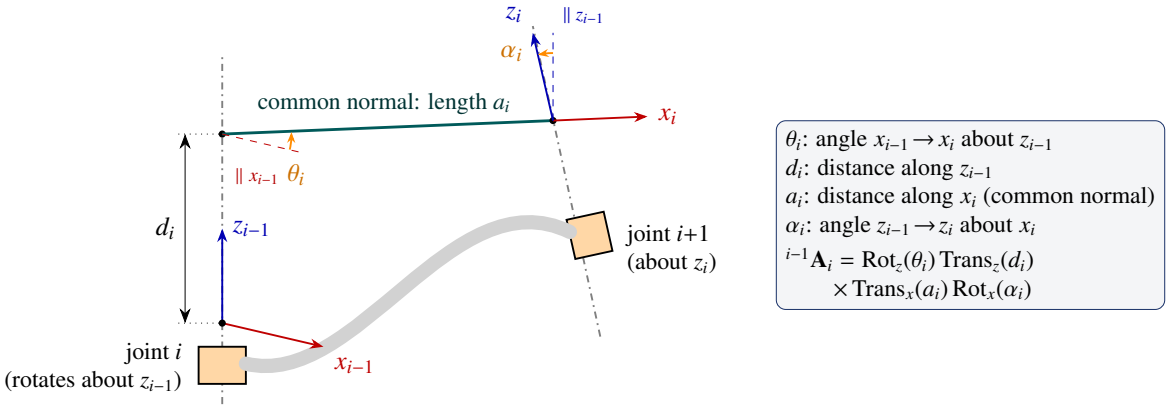


Figure 3.5: Denavit–Hartenberg frame assignment and the four parameters $\theta_i, d_i, a_i, \alpha_i$ relating frame $\{i-1\}$ to frame $\{i\}$.

3.4.4 The DH link transformation – derivation

Frame $\{i\}$ is reached from $\{i-1\}$ by four elementary motions, each about or along an axis of the current frame (hence post-multiplied): rotate θ_i about $\hat{\mathbf{z}}$, translate d_i along $\hat{\mathbf{z}}$, translate a_i along the new $\hat{\mathbf{x}}$, rotate α_i about $\hat{\mathbf{x}}$:

$$
\begin{aligned}
{}^{i-1}\mathbf{A}_i &= \mathrm{Rot}_z(\theta_i)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i) \\
&= \begin{bmatrix}
c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\
s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\
0 & s\alpha_i & c\alpha_i & d_i \\
0 & 0 & 0 & 1
\end{bmatrix}.
\end{aligned}
\tag{3.9}
$$

The forward kinematics is then the product

$$
{}^{0}_{n}\mathbf{T}(\mathbf{q}) = {}^{0}\mathbf{A}_1(q_1)\,{}^{1}\mathbf{A}_2(q_2)\cdots{}^{n-1}\mathbf{A}_n(q_n) = \begin{bmatrix} \mathbf{R}(\mathbf{q}) & \mathbf{p}(\mathbf{q}) \\ \mathbf{0} & 1 \end{bmatrix}. \tag{3.10}
$$

Algorithm 3.1 DH parameter determination and forward kinematics

Require: Robot geometry (joint axes, link dimensions); joint values $\mathbf{q}$
Ensure: DH table and tool pose ${}^{0}_{n}\mathbf{T}$
Assign $\hat{\mathbf{z}}_{i-1}$ along each joint axis i; choose base frame $\{0\}$
for $i = 1$ **to** n **do**
    locate common normal of $\hat{\mathbf{z}}_{i-1}, \hat{\mathbf{z}}_i$; set origin o_i and $\hat{\mathbf{x}}_i$ along it
    read a_i, α_i (along/about $\hat{\mathbf{x}}_i$) and d_i, θ_i (along/about $\hat{\mathbf{z}}_{i-1}$); mark the joint variable
end for
$\mathbf{T} \leftarrow \mathbf{I}_4$
for $i = 1$ **to** n **do**
    $\mathbf{T} \leftarrow \mathbf{T} \cdot {}^{i-1}\mathbf{A}_i(\theta_i, d_i, a_i, \alpha_i)$ ▷ Eq. (3.9)
end for
return DH table, $\mathbf{T}$ (position = column 4, orientation = upper-left 3×3)

Table 3.3: DH parameters of three common manipulators (* = joint variable).

Robot	Link	a_i	α_i	d_i	θ_i
Planar 2R	1	l_1	0	0	θ_1^*
	2	l_2	0	0	θ_2^*
SCARA (RRPR)	1	a_1	0	d_1	θ_1^*
	2	a_2	180°	0	θ_2^*
	3	0	0	d_3^*	0
	4	0	0	d_4	θ_4^*
Articulated (RRR)	1	0	90°	d_1	θ_1^*
	2	a_2	0	0	θ_2^*
	3	a_3	0	0	θ_3^*

3.4.5 DH tables of common robots

Table 3.3 shows how compactly the convention encodes geometry. In the SCARA, the twist of 180° on link 2 flips $\hat{\mathbf{z}}$ downward so that the prismatic variable d_3 moves the quill down. In the articulated arm, the 90° twist on link 1 turns the vertical waist axis into the horizontal shoulder axis.

3.5 Forward Kinematics

3.5.1 Problem statement

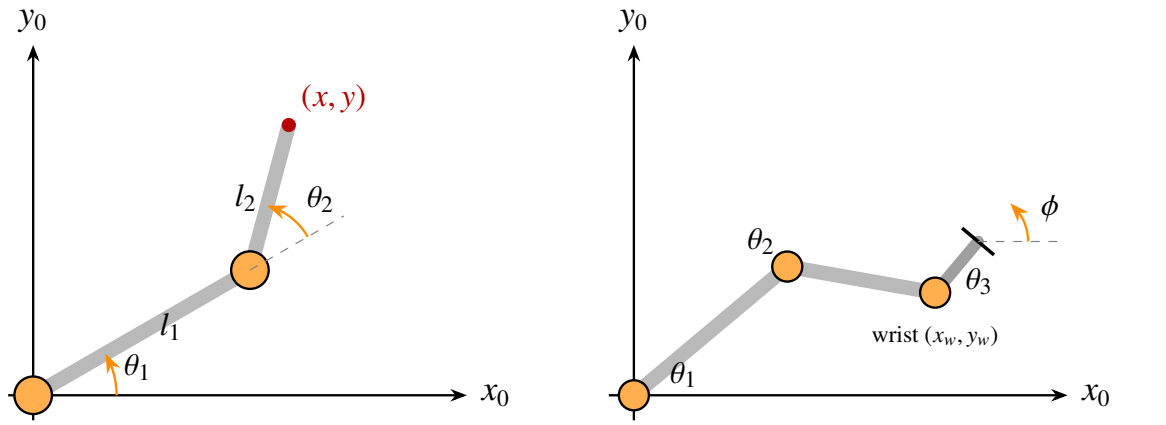
Forward kinematics computes the position and orientation of the end-effector from the joint variables: $\mathbf{x} = \mathbf{f}(\mathbf{q})$. For serial robots the solution is unique and is obtained directly from (3.10). It is needed to display the TCP pose on the teach pendant, to check workspace limits and to close Cartesian control loops.

3.5.2 Planar 2R manipulator

For the arm of Fig. 3.6(a) (DH table in Table 3.3), multiplying ${}^{0}\mathbf{A}_1\,{}^{1}\mathbf{A}_2$ and using $\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 = \cos(\theta_1+\theta_2)$:

$$ {}^{0}_{2}\mathbf{T} = \begin{bmatrix} c_{12} & -s_{12} & 0 & l_1c_1 + l_2c_{12} \\ s_{12} & c_{12} & 0 & l_1s_1 + l_2s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \begin{aligned} x &= l_1c_1 + l_2c_{12} \\ y &= l_1s_1 + l_2s_{12} \\ \phi &= \theta_1 + \theta_2 \end{aligned} \tag{3.11} $$

where $c_1 = \cos\theta_1$, $s_{12} = \sin(\theta_1+\theta_2)$, etc. The same result follows from vector addition of the two link vectors.



(a) Planar 2R arm (b) Planar 3R arm with orientation $\phi = \theta_1 + \theta_2 + \theta_3$

Figure 3.6: (a) Planar 2R manipulator; (b) planar 3R manipulator with tool orientation ϕ.

Example 3.4: Forward kinematics of a 2R arm (Topic 3.5)

Given: $l_1 = 0.5$ m, $l_2 = 0.3$ m, $\theta_1 = 30^\circ$, $\theta_2 = 45^\circ$.

Solution: $x = 0.5\cos 30^\circ + 0.3\cos 75^\circ = 0.4330 + 0.0776 = 0.5107$ m; $y = 0.5\sin 30^\circ + 0.3\sin 75^\circ = 0.2500 + 0.2898 = 0.5398$ m; $\phi = 75^\circ$.

3.5.3 Planar 3R manipulator

Adding a third link l_3 (Fig. 3.6(b)) gives position and orientation in the plane (3 DOF):

$$ x = l_1c_1 + l_2c_{12} + l_3c_{123}, \quad y = l_1s_1 + l_2s_{12} + l_3s_{123}, \quad \phi = \theta_1 + \theta_2 + \theta_3. \tag{3.12} $$

Example: $L = (0.4, 0.3, 0.1)$ m, $\theta = (30^\circ, -45^\circ, 60^\circ)$ gives $x = 0.7069$ m, $y = 0.1931$ m, $\phi = 45^\circ$.

3.5.4 SCARA and articulated robots

SCARA (Table 3.3): $x = a_1c_1 + a_2c_{12}$, $y = a_1s_1 + a_2s_{12}$, $z = d_1 - d_3 - d_4$, tool yaw $= \theta_1 + \theta_2 - \theta_4$. With $a_1 = 0.4$, $a_2 = 0.3\,\text{m}$, $d_1 = 0.4\,\text{m}$, $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$, $d_3 = 0.1\,\text{m}$, $d_4 = 0$: $(x, y, z) = (0.346,\ 0.500,\ 0.300)$ m.

Articulated (RRR): multiplying the three DH matrices gives

$$x = c_1\,(a_2c_2 + a_3c_{23}), \qquad y = s_1\,(a_2c_2 + a_3c_{23}), \qquad z = d_1 + a_2s_2 + a_3s_{23}. \tag{3.13}$$

Example 3.5: Forward kinematics of an articulated arm (Topic 3.5)

Given: $d_1 = 0.4\,\text{m}$, $a_2 = 0.5\,\text{m}$, $a_3 = 0.4\,\text{m}$; $\boldsymbol{\theta} = (30^\circ, 45^\circ, -60^\circ)$.

Solution: Horizontal reach $r = a_2c_2 + a_3c_{23} = 0.5(0.7071) + 0.4\cos(-15^\circ) = 0.3536 + 0.3864 = 0.7400\,\text{m}$. $x = 0.866 \times 0.7400 = 0.6408\,\text{m}$; $y = 0.5 \times 0.7400 = 0.3700\,\text{m}$; $z = 0.4 + 0.5(0.7071) + 0.4\sin(-15^\circ) = 0.4 + 0.3536 - 0.1035 = 0.6500\,\text{m}$. The full matrix product ${}^0_3\mathbf{T}$ (Program 3.2) gives the same last column and the orientation $\mathbf{R} = \begin{bmatrix} 0.837 & 0.224 & 0.5 \\ 0.483 & 0.129 & -0.866 \\ -0.259 & 0.966 & 0 \end{bmatrix}$.

3.6 Inverse Kinematics

3.6.1 Problem statement and difficulties

Inverse kinematics (IK) finds joint variables that place the end-effector at a desired pose: $\mathbf{q} = \mathbf{f}^{-1}(\mathbf{x})$. It is required whenever a task is specified in Cartesian space (welding seam, camera-detected part). IK is harder than FK because the equations are non-linear and trigonometric, and:

- a solution *exists* only if the target is inside the workspace;
- there may be *multiple solutions* (elbow up/down, shoulder left/right, wrist flip – up to eight for a six-axis arm, or infinitely many for a redundant arm);
- closed-form solutions exist only for special geometries (e.g. three consecutive intersecting or parallel axes – Pieper's condition); otherwise numerical methods are used.

3.6.2 Geometric solution of the 2R arm – derivation

Step 1 (elbow angle). Squaring and adding the two equations of (3.11):

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1l_2(c_1c_{12} + s_1s_{12}) = l_1^2 + l_2^2 + 2l_1l_2\cos\theta_2,$$

which is the law of cosines for the triangle formed by the two links and the line to the target (Fig. 3.7):

$$\cos\theta_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2 l_1 l_2} \equiv D, \qquad \theta_2 = \operatorname{atan2}\!\left(\pm\sqrt{1 - D^2},\; D\right). \tag{3.14}$$

A real solution requires $|D| \le 1$ (target in the workspace); the two signs give the two *solution branches*. Using atan2 rather than $\cos^{-1}$ preserves the correct quadrant.

Step 2 (shoulder angle). The target direction is $\gamma = \operatorname{atan2}(y, x)$, and the angle between link 1 and the target line is $\beta = \operatorname{atan2}(l_2 s_2,\; l_1 + l_2 c_2)$, so

$$\theta_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(l_2 \sin\theta_2,\; l_1 + l_2\cos\theta_2). \tag{3.15}$$

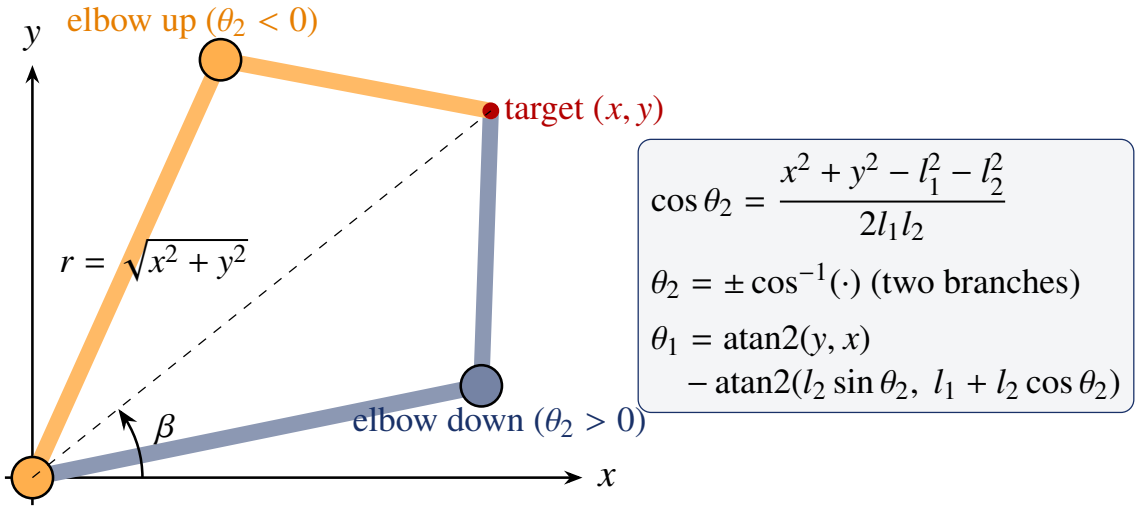


Figure 3.7: Inverse kinematics of the 2R arm: two solution branches (elbow down and elbow up) reach the same target.

Example 3.6: Inverse kinematics of a 2R arm (Topic 3.6)

Given: $l_1 = 0.5$ m, $l_2 = 0.3$ m; target $(0.5, 0.4)$ m.

Solution: $D = (0.25{+}0.16{-}0.25{-}0.09)/(2{\times}0.5{\times}0.3) = 0.07/0.30 = 0.2333$, so $\theta_2 = \pm 76.51°$.

- Branch 1 ($\theta_2 = +76.51°$): $\theta_1 = \operatorname{atan2}(0.4, 0.5) - \operatorname{atan2}(0.2917, 0.5700) = 38.66° - 27.10° = 11.56°$.
- Branch 2 ($\theta_2 = -76.51°$): $\theta_1 = 38.66° + 27.10° = 65.76°$.

Check: FK of $(11.56°, 76.51°)$ gives $x = 0.5\cos 11.56° + 0.3\cos 88.07° = 0.4899 + 0.0101 = 0.500$, $y = 0.1002 + 0.2998 = 0.400$ ✓.

Interpretation: The controller selects the branch closest to the current configuration, or the one avoiding obstacles and joint limits.

3.6.3 Planar 3R with orientation

For a desired pose (x, y, ϕ) of the 3R arm, first find the *wrist point* by stepping back along the tool:

$$x_w = x - l_3 \cos\phi, \qquad y_w = y - l_3 \sin\phi, \tag{3.16}$$

then solve the 2R problem for (θ_1, θ_2) with target (x_w, y_w) and finally $\theta_3 = \phi - \theta_1 - \theta_2$. *Example:* $L = (0.4, 0.3, 0.1)$ m, target $(0.5, 0.3)$ m with $\phi = 0$: $(x_w, y_w) = (0.4, 0.3)$, $D = 0$, so $\theta_2 = \pm 90°$; taking $\theta_2 = -90°$: $\theta_1 = 36.87° + 36.87° = 73.74°$, $\theta_3 = 0 - 73.74° + 90° = 16.26°$.

3.6.4 Articulated arm and kinematic decoupling

For the RRR arm of (3.13): $\theta_1 = \operatorname{atan2}(y, x)$ (or $\theta_1 + \pi$, "shoulder back"); with $r = \sqrt{x^2 + y^2}$ and $s = z - d_1$, the remaining two joints form a planar 2R problem in the (r, s) plane: $\cos\theta_3 = (r^2 + s^2 - a_2^2 - a_3^2)/(2a_2a_3)$, $\theta_2 = \operatorname{atan2}(s, r) - \operatorname{atan2}(a_3 s_3, a_2 + a_3 c_3)$. For the target of Example 3.5 this returns $(30°, 45°, -60°)$ and the second branch $(30°, -7.66°, 60°)$.

For a six-axis robot with a *spherical wrist*, the problem *decouples*: the wrist centre $\mathbf{p}_w = \mathbf{p} - d_6\,\mathbf{R}\hat{\mathbf{z}}$ depends only on θ_1–θ_3 (solved as above), and the wrist angles θ_4–θ_6 are then found as Euler angles of ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^{\mathsf{T}}\mathbf{R}$.

3.6.5 Numerical inverse kinematics

When no closed form exists, IK is solved iteratively by the Newton–Raphson method using the Jacobian of Section 3.7 (Algorithm 3.2). Near singularities, the damped-least-squares update $\Delta\mathbf{q} = \mathbf{J}^{\mathsf{T}}(\mathbf{J}\mathbf{J}^{\mathsf{T}} + \lambda^2\mathbf{I})^{-1}\mathbf{e}$ is used instead of $\mathbf{J}^{-1}\mathbf{e}$.

Algorithm 3.2 Inverse kinematics: analytical 2R solution and numerical Newton–Raphson

Require: Target $\mathbf{x}_d$; link parameters; initial guess $\mathbf{q}_0$; tolerance ε
Ensure: Joint vector $\mathbf{q}$ (or "unreachable")

Analytical (2R):

$D \leftarrow (x^2 + y^2 - l_1^2 - l_2^2)/(2l_1l_2)$
if $|D| > 1$ **then**
  return unreachable
end if
$\theta_2 \leftarrow \operatorname{atan2}(\sigma\sqrt{1 - D^2}, D)$, $\sigma = \pm 1$ (branch)
$\theta_1 \leftarrow \operatorname{atan2}(y, x) - \operatorname{atan2}(l_2 \sin\theta_2, l_1 + l_2\cos\theta_2)$

Numerical (general):

$\mathbf{q} \leftarrow \mathbf{q}_0$
repeat
  $\mathbf{e} \leftarrow \mathbf{x}_d - \mathbf{f}(\mathbf{q})$ ▷ forward kinematics error
  $\mathbf{q} \leftarrow \mathbf{q} + \mathbf{J}^{\dagger}(\mathbf{q})\,\mathbf{e}$ ▷ pseudo-inverse or damped LS
until $\|\mathbf{e}\| < \varepsilon$ or iteration limit
return $\mathbf{q}$

3.7 Velocity Kinematics and the Jacobian

3.7.1 Definition

Differentiating the forward kinematics $\mathbf{x} = \mathbf{f}(\mathbf{q})$ with respect to time by the chain rule gives

$$\dot{\mathbf{x}} = \frac{\partial \mathbf{f}}{\partial \mathbf{q}}\,\dot{\mathbf{q}} = \mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}}, \qquad J_{ij} = \frac{\partial f_i}{\partial q_j}. \tag{3.17}$$

The *Jacobian* $\mathbf{J}(\mathbf{q})$ is a configuration-dependent matrix that maps joint velocities to end-effector velocities (Fig. 3.8). For a spatial robot it is $6 \times n$: three rows for linear velocity $\mathbf{v}$ and three for angular velocity $\boldsymbol{\omega}$. It is the most important matrix in robotics after the transformation matrix: it governs velocity control, singularity analysis, static force transmission and numerical IK.

3.7.2 Jacobian of the 2R arm – derivation

Differentiating (3.11):

$$\dot{x} = -l_1 s_1 \dot{\theta}_1 - l_2 s_{12}(\dot{\theta}_1 + \dot{\theta}_2), \qquad \dot{y} = l_1 c_1 \dot{\theta}_1 + l_2 c_{12}(\dot{\theta}_1 + \dot{\theta}_2),$$

so that

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \underbrace{\begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix}}_{\mathbf{J}(\boldsymbol{\theta})} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \qquad \det \mathbf{J} = l_1 l_2 \sin\theta_2. \tag{3.18}$$

The determinant result follows by expanding: $(-l_1 s_1 - l_2 s_{12}) l_2 c_{12} + l_2 s_{12}(l_1 c_1 + l_2 c_{12}) = l_1 l_2 (s_{12} c_1 - s_1 c_{12}) = l_1 l_2 \sin\theta_2$.

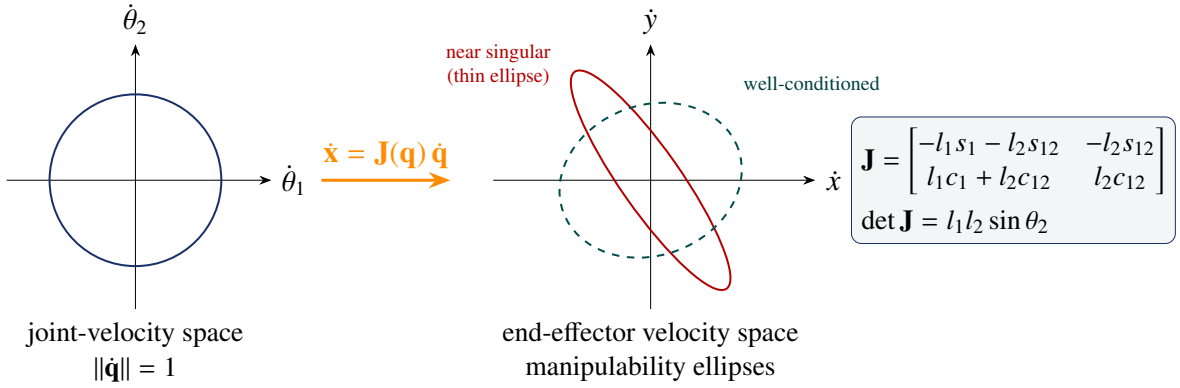


Figure 3.8: The Jacobian maps the unit circle of joint velocities into a manipulability ellipse of end-effector velocities; the ellipse collapses near a singularity.

3.7.3 Geometric Jacobian of a general serial robot

For joint i with axis $\hat{\mathbf{z}}_{i-1}$ through origin $\mathbf{o}_{i-1}$ (from the DH frames), and end-effector origin $\mathbf{o}_n$, column i of the $6 \times n$ Jacobian is

$$\mathbf{J}_i = \begin{cases} \begin{bmatrix} \hat{\mathbf{z}}_{i-1} \times (\mathbf{o}_n - \mathbf{o}_{i-1}) \\ \hat{\mathbf{z}}_{i-1} \end{bmatrix} & \text{revolute} \\ \begin{bmatrix} \hat{\mathbf{z}}_{i-1} \\ \mathbf{0} \end{bmatrix} & \text{prismatic} \end{cases} \tag{3.19}$$

Physical meaning: a revolute joint rotating at unit rate moves the end-effector with velocity $\hat{\mathbf{z}} \times \mathbf{r}$ (rigid-body rotation) and turns it at unit rate about $\hat{\mathbf{z}}$; a prismatic joint translates it along $\hat{\mathbf{z}}$ without rotation. All vectors are taken from the columns of the transforms ${}^{0}_{i-1}\mathbf{T}$, so the Jacobian is computed directly from forward kinematics (Algorithm 3.3).

Algorithm 3.3 Jacobian computation and singularity detection

Require: DH table; joint vector $\mathbf{q}$; threshold $w_{\min}$
Ensure: Jacobian $\mathbf{J}(\mathbf{q})$; singularity flag
$\mathbf{T} \leftarrow \mathbf{I}_4$; store $\hat{\mathbf{z}}_0 = (0, 0, 1)$, $\mathbf{o}_0 = \mathbf{0}$
for $i = 1$ **to** n **do**
  $\mathbf{T} \leftarrow \mathbf{T}\,{}^{i-1}\mathbf{A}_i$; store $\hat{\mathbf{z}}_i$ (column 3) and $\mathbf{o}_i$ (column 4)
end for
for $i = 1$ **to** n **do**
  if joint i revolute **then**
    $\mathbf{J}_i \leftarrow [\hat{\mathbf{z}}_{i-1} \times (\mathbf{o}_n - \mathbf{o}_{i-1});\ \hat{\mathbf{z}}_{i-1}]$
  else
    $\mathbf{J}_i \leftarrow [\hat{\mathbf{z}}_{i-1};\ \mathbf{0}]$
  end if
end for
$w \leftarrow \sqrt{\det(\mathbf{J}\mathbf{J}^{\mathsf{T}})}$ ▷ manipulability measure
singular $\leftarrow (w < w_{\min})$
return $\mathbf{J}$, singular

3.7.4 Inverse velocity

To move the tool with a desired Cartesian velocity (straight-line motion, conveyor tracking), the joint rates are obtained by inverting (3.17):

$$\dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\,\dot{\mathbf{x}} \quad \text{(square } \mathbf{J}\text{)}, \qquad \dot{\mathbf{q}} = \mathbf{J}^{\dagger}\dot{\mathbf{x}} = \mathbf{J}^{\mathsf{T}}(\mathbf{J}\mathbf{J}^{\mathsf{T}})^{-1}\dot{\mathbf{x}} \quad \text{(redundant)}. \tag{3.20}$$

This is *resolved-rate* motion control.

Example 3.7: Jacobian and velocities of a 2R arm (Topic 3.7)

Given: $l_1 = 0.5\,\text{m}$, $l_2 = 0.3\,\text{m}$, $\boldsymbol{\theta} = (30^\circ, 45^\circ)$.

Find: (a) $\mathbf{J}$ and $\det \mathbf{J}$; (b) tool velocity for $\dot{\boldsymbol{\theta}} = (0.5, -0.2)$ rad/s; (c) joint rates for $\dot{\mathbf{x}} = (0.1, 0)$ m/s.

Solution: (a) $\mathbf{J} = \begin{bmatrix} -0.5398 & -0.2898 \\ 0.5107 & 0.0776 \end{bmatrix}$; $\det \mathbf{J} = 0.1061 = 0.5 \times 0.3 \times \sin 45^\circ$ ✓. (b) $\dot{\mathbf{x}} = \mathbf{J}\dot{\boldsymbol{\theta}} = (-0.2119,\ 0.2398)$ m/s. (c) $\dot{\boldsymbol{\theta}} = \mathbf{J}^{-1}\dot{\mathbf{x}} = \dfrac{1}{0.1061}\begin{bmatrix} 0.0776 & 0.2898 \\ -0.5107 & -0.5398 \end{bmatrix}\begin{bmatrix} 0.1 \\ 0 \end{bmatrix} = (0.0732,\ -0.4815)$ rad/s.

Interpretation: Moving the tool horizontally requires the elbow to turn more than six times faster than the shoulder in this configuration.

3.8 Singularities

3.8.1 Definition and consequences

A *singular configuration* is one in which the Jacobian loses rank: for a square Jacobian, $\det \mathbf{J}(\mathbf{q}) = 0$. At a singularity:

- the end-effector cannot move in one or more directions, whatever the joint velocities (a DOF is lost);
- bounded Cartesian velocities near the singularity require *unbounded* joint velocities, since $\mathbf{J}^{-1}$ contains $1/\det \mathbf{J}$ – the robot "flails" or stops with a speed error;
- IK solutions merge (branches meet) or become infinite in number;
- conversely, the robot can resist large forces in some directions with little joint torque (Section 3.13).

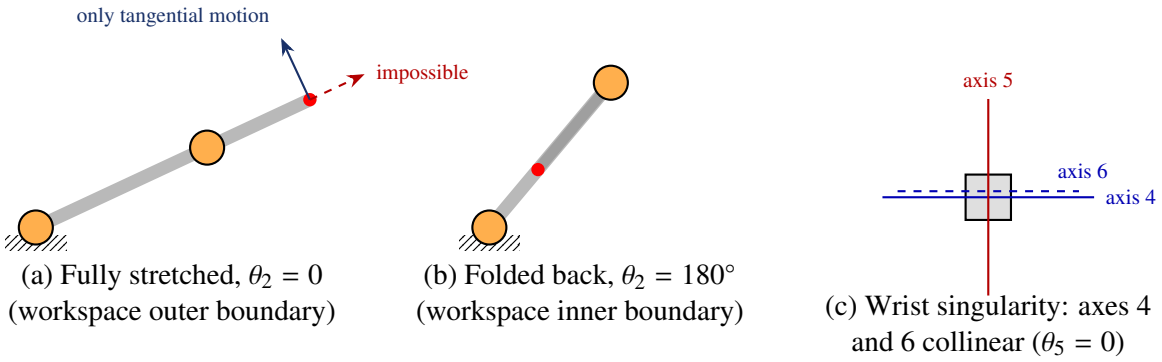
3.8.2 Types

For the 2R arm, $\det \mathbf{J} = l_1 l_2 \sin\theta_2 = 0$ when $\theta_2 = 0$ (fully stretched) or $\theta_2 = 180^\circ$ (folded) (Fig. 3.9(a),(b)). In both, the two links are collinear and the tool can move only perpendicular to them.

- **Boundary singularities** occur on the workspace boundary (arm fully stretched or folded); they are avoided by not driving to the reach limit.
- **Interior singularities** occur inside the workspace – e.g. the *wrist singularity* of a six-axis robot when axes 4 and 6 become collinear ($\theta_5 = 0$, Fig. 3.9(c)), the *shoulder singularity* when the wrist centre lies on axis 1, and the *elbow singularity* when the arm is stretched. They are the dangerous ones because planned straight-line paths may pass through them.

Manipulability $w = \sqrt{\det(\mathbf{J}\mathbf{J}^{\mathsf{T}})}$ (for square $\mathbf{J}$, $w = |\det \mathbf{J}|$) measures distance from singularity and is proportional to the area/volume of the velocity ellipse of Fig. 3.8. For the 2R arm

$w = l_1 l_2 |\sin\theta_2|$, maximum at $\theta_2 = \pm 90°$: the elbow at a right angle is the most dexterous posture.



(a) Fully stretched, $\theta_2 = 0$ (workspace outer boundary)

(b) Folded back, $\theta_2 = 180°$ (workspace inner boundary)

(c) Wrist singularity: axes 4 and 6 collinear ($\theta_5 = 0$)

Figure 3.9: Singular configurations: (a) stretched and (b) folded 2R arm (boundary singularities); (c) wrist singularity of a six-axis robot.

Handling singularities. Controllers detect $w < w_{\min}$ (Algorithm 3.3) and respond by limiting speed, switching to joint interpolation, using damped-least-squares inverses, or re-planning the path; programmers avoid them by choosing wrist orientations that keep θ_5 away from zero.

3.9 Robot Workspace

The workspace concepts of Unit I (Section 1.6) can now be stated precisely. The **reachable workspace** is the set $\{\mathbf{p}(\mathbf{q}) : \mathbf{q} \in \mathcal{Q}\}$ of positions attainable for joint values within their limits $\mathcal{Q}$; the **dexterous workspace** is the subset of positions attainable with every orientation. For a planar 3R arm with full joint rotation (Fig. 3.10), the reachable workspace is the disc $r \le L_1 + L_2 + L_3$ (inner radius $\max(0, |L_1 - L_2| - L_3)$), while the dexterous workspace requires the whole circle of radius L_3 around the wrist to lie inside the 2R annulus:

$$|L_1 - L_2| + L_3 \;\le\; r \;\le\; L_1 + L_2 - L_3. \tag{3.21}$$

For $L = (0.4, 0.3, 0.1)$ m this gives a reachable disc of radius 0.8 m but a dexterous annulus of only 0.2 m to 0.6 m. Workspace boundaries coincide with boundary singularities; joint limits carve further holes. Workspaces are computed numerically by sampling the joint space and applying forward kinematics (Program 3.1) and are essential for cell layout (Unit IV).

3.10 Trajectory Planning in Joint Space

3.10.1 Path versus trajectory

A *path* is the geometric locus of configurations the robot passes through; a *trajectory* is a path with a timing law – position, velocity and acceleration as functions of time. Trajectories must be smooth: discontinuous velocity implies infinite acceleration and torque, and discontinuous acceleration (infinite *jerk*) excites vibration of links and gears. In *joint-space* planning each joint is interpolated directly between its start and goal values (obtained by IK at the end points

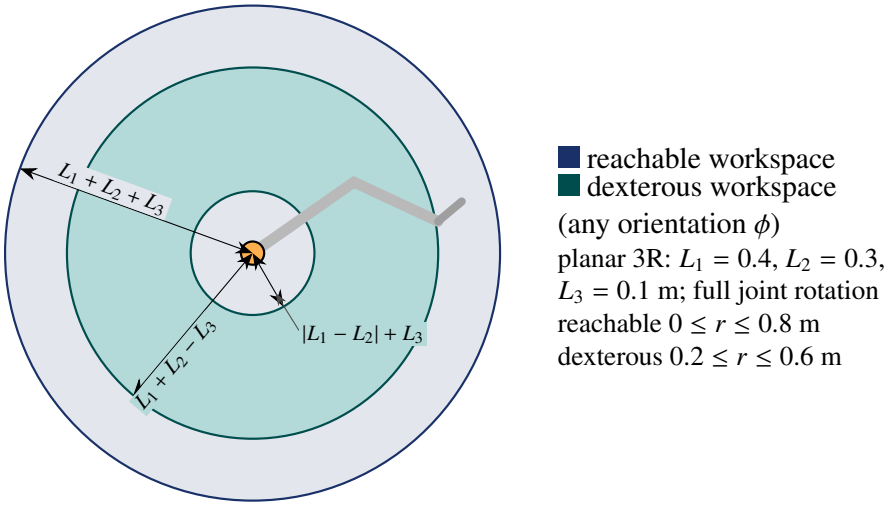


Figure 3.10: Reachable and dexterous workspaces of a planar 3R manipulator.

only); all joints are synchronised to start and finish together.

3.10.2 Cubic polynomial – derivation

Four boundary conditions – $\theta(0) = \theta_0$, $\theta(t_f) = \theta_f$, $\dot\theta(0) = 0$, $\dot\theta(t_f) = 0$ – can be met by a cubic $\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$. Substituting:

$$a_0 = \theta_0, \quad a_1 = 0, \quad \theta_0 + a_2 t_f^2 + a_3 t_f^3 = \theta_f, \quad 2a_2 t_f + 3a_3 t_f^2 = 0.$$

Solving the last two equations simultaneously:

$$a_0 = \theta_0, \quad a_1 = 0, \quad a_2 = \frac{3(\theta_f - \theta_0)}{t_f^2}, \quad a_3 = -\frac{2(\theta_f - \theta_0)}{t_f^3}. \tag{3.22}$$

The velocity $\dot\theta = 2a_2 t + 3a_3 t^2$ is parabolic with maximum $1.5\,\Delta\theta/t_f$ at $t_f/2$; the acceleration $2a_2 + 6a_3 t$ is linear with maximum $6\Delta\theta/t_f^2$ at the ends (Fig. 3.11). With non-zero end velocities (via points), $a_1 = \dot\theta_0$ and a_2, a_3 change accordingly.

3.10.3 Quintic polynomial

The cubic has non-zero acceleration at start and stop, i.e. a step in acceleration (infinite jerk). Imposing also $\ddot\theta(0) = \ddot\theta(t_f) = 0$ needs six conditions and a quintic. With $\tau = t/t_f$ and $\Delta\theta = \theta_f - \theta_0$, the solution is

$$\theta(t) = \theta_0 + \Delta\theta\left(10\tau^3 - 15\tau^4 + 6\tau^5\right), \tag{3.23}$$

with peak velocity $1.875\,\Delta\theta/t_f$ and peak acceleration $5.77\,\Delta\theta/t_f^2$. It is smoother, but its peak velocity is 25% higher than the cubic's for the same duration.

Example 3.8: Cubic and quintic trajectories (Topic 3.10)

Given: A joint moves from $\theta_0 = 10°$ to $\theta_f = 70°$ in $t_f = 3$ s, starting and ending at rest.

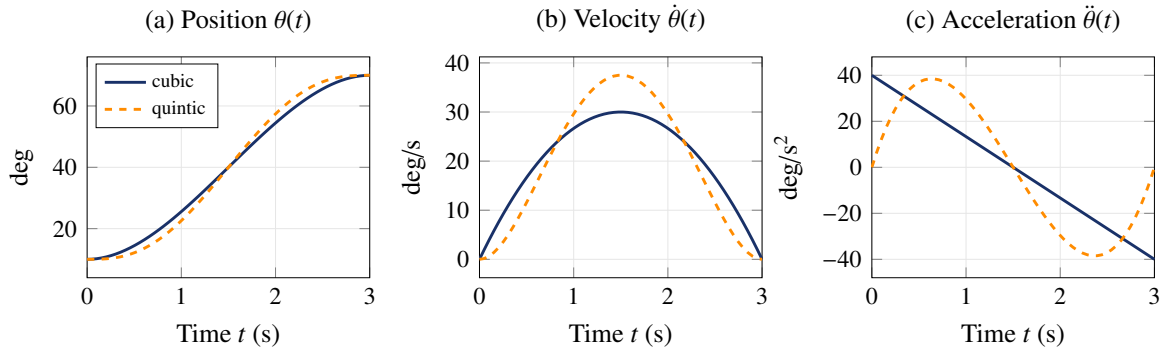


Figure 3.11: Cubic (solid) and quintic (dashed) joint trajectories from 10° to 70° in 3 s: (a) position, (b) velocity, (c) acceleration.

Solution (cubic): $a_2 = 3(60)/9 = 20°/s^2$, $a_3 = -2(60)/27 = -4.444°/s^3$: $\theta(t) = 10 + 20t^2 - 4.444t^3$. At $t = 1.5$ s: $\theta = 10 + 45 - 15 = 40°$ (midpoint), $\dot{\theta}_{\max} = 40(1.5) - 13.33(2.25) = 30°/s$; $\ddot{\theta}_{\max} = 40°/s^2$ at $t = 0$.

Solution (quintic): $\dot{\theta}_{\max} = 1.875 \times 60/3 = 37.5°/s$; $\ddot{\theta}_{\max} = 5.774 \times 60/9 = 38.5°/s^2$, but acceleration starts and ends at zero.

3.10.4 Linear segments with parabolic blends (LSPB)

Industrial controllers usually move at a constant cruise velocity V for most of the motion, with constant-acceleration blends at the ends – a *trapezoidal velocity profile* (Fig. 3.12). By symmetry, with blend time t_b and acceleration $a = V/t_b$, the area under the velocity profile equals the displacement: $V(t_f - t_b) = \theta_f - \theta_0$, giving

$$t_b = \frac{\theta_0 - \theta_f + V t_f}{V}, \qquad a = \frac{V}{t_b}, \qquad \frac{\Delta\theta}{t_f} < V \le \frac{2\Delta\theta}{t_f}. \tag{3.24}$$

The upper limit $V = 2\Delta\theta/t_f$ gives $t_b = t_f/2$ (a triangular, bang–bang profile).

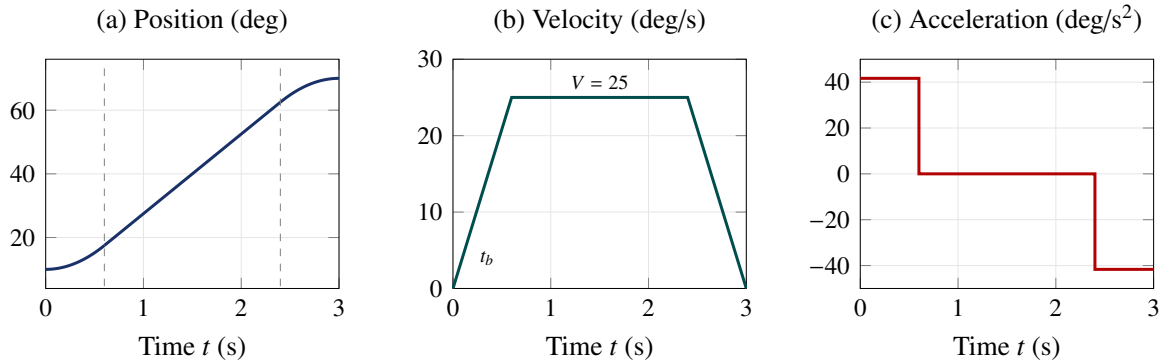


Figure 3.12: LSPB (trapezoidal-velocity) trajectory: (a) position, (b) velocity, (c) acceleration.

Example 3.9: Trapezoidal profile (Topic 3.10)

Given: $\Delta\theta = 60°$ in $t_f = 3\,\text{s}$ with cruise velocity $V = 25°/\text{s}$ (allowed range 20–40°/s).

Solution: $t_b = (-60 + 75)/25 = 0.6\,\text{s}$; $a = 25/0.6 = 41.7°/\text{s}^2$. Angle at end of first blend $= \frac{1}{2}at_b^2 = 7.5°$; cruise segment $25 \times 1.8 = 45°$; total $7.5 + 45 + 7.5 = 60°$ ✓.

Multi-axis synchronisation and via points. For coordinated joint motion the slowest joint (largest $\Delta\theta/V_{\max}$) sets t_f; the other joints are scaled to finish simultaneously. A path through intermediate *via points* is planned as a sequence of segments with continuous velocity (cubic splines) or by blending ("fly-by" zones on industrial controllers).

3.11 Trajectory Planning in Cartesian Space

3.11.1 Principle

Many tasks require the TCP to follow a specified geometric path: a straight weld seam, a circular glue bead, insertion along a hole axis. In *Cartesian-space* planning the path is defined in task coordinates, a timing law $s(t) \in [0, 1]$ (cubic, quintic or trapezoidal, as above) is applied, and IK converts each intermediate pose to joint values at the controller's interpolation rate (typically 1 ms to 4 ms). For a straight line,

$$\mathbf{p}(t) = \mathbf{p}_0 + s(t)\,(\mathbf{p}_f - \mathbf{p}_0), \tag{3.25}$$

and orientation is interpolated by angle–axis or quaternion slerp. A circular arc through three points is interpolated by angle about its centre.

3.11.2 Joint versus Cartesian interpolation

Figure 3.13 compares the two for a 2R arm moving from $(0.6, 0.2)$ to $(0.3, 0.5)$ m. The joint values at start and goal are $(-9.26°, 78.46°)$ and $(28.07°, 90.00°)$. Linear joint interpolation produces a curved tool path whose midpoint $(0.474, 0.381)$ m lies 39 mm from the straight line; Cartesian interpolation keeps the tool exactly on the line by solving IK at every point – at the midpoint $(0.45, 0.35)$ m it requires $(6.17°, 92.87°)$ rather than the joint-interpolated $(9.41°, 84.23°)$. Robot languages provide both: `MOVJ/PTP` (joint) and `MOVL/LIN` (linear), see Unit IV.

3.12 Velocity and Acceleration Profiles

The timing law determines the velocity and acceleration profiles, which must respect the drive limits $|\dot{q}| \le V_{\max}$, $|\ddot{q}| \le a_{\max}$ and, for smoothness, a jerk limit $|\dddot{q}| \le j_{\max}$.

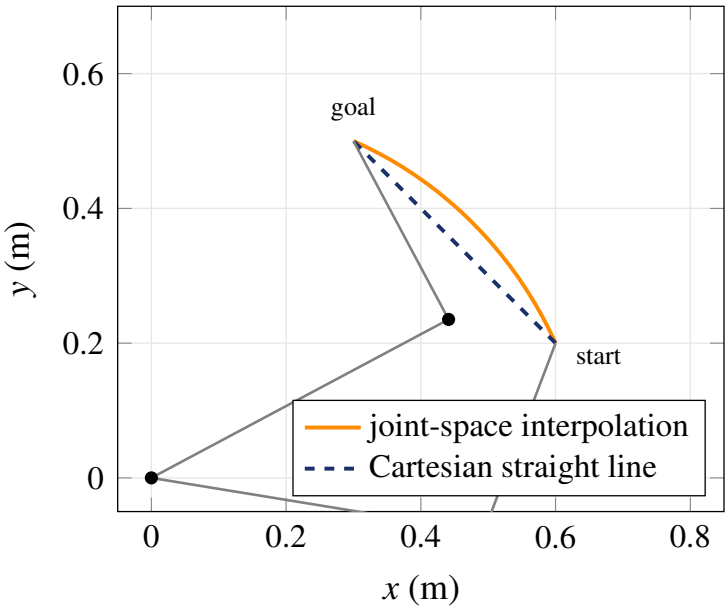


Figure 3.13: Tool paths of a 2R arm for joint-space interpolation (curved) and Cartesian straight-line interpolation, with the start and goal arm configurations.

Table 3.4: Joint-space versus Cartesian-space trajectory planning.

	Joint space	**Cartesian space**
Path of the tool	Not straight; hard to predict	Exactly as specified (line, arc)
Computation	Low: IK only at end points	High: IK at every interpolation step
Singularities	No problem (no Jacobian inverse)	Path may cross singularities ⇒ high joint speeds
Joint limits/branches	Always respected	May become unreachable mid-path
Speed	Fastest, time-optimal per joint	Limited by Cartesian speed and singularities
Typical use	Free-space motion, pick-and-place transfers	Welding, gluing, insertion, conveyor tracking

- **Trapezoidal velocity** (LSPB): constant acceleration, cruise, constant deceleration. Minimum time for given $V_{\max}$ and $a_{\max}$; acceleration is discontinuous (infinite jerk at three instants). If $\Delta\theta < V_{\max}^2/a_{\max}$ the cruise phase vanishes and the profile becomes triangular with peak velocity $\sqrt{a_{\max}\Delta\theta}$ and $t_f = 2\sqrt{\Delta\theta/a_{\max}}$.
- **S-curve (jerk-limited)**: acceleration ramps linearly (seven-segment profile), giving trapezoidal acceleration; used in high-speed robots to reduce vibration and gear wear.
- **Polynomial**: cubic (parabolic velocity, linear acceleration) and quintic (smooth bell-shaped velocity, zero end accelerations) (Fig. 3.11).

The minimum-time trapezoidal move with full cruise phase takes

$$t_f = \frac{\Delta\theta}{V_{\max}} + \frac{V_{\max}}{a_{\max}}, \tag{3.26}$$

Algorithm 3.4 Trajectory generation (joint or Cartesian)

Require: Start/goal (joint values or poses); duration t_f (or $V_{\max}, a_{\max}$); interpolation period T_s
Ensure: Joint set-points $\mathbf{q}_k$ every T_s

```
choose timing law s(t): cubic, quintic or LSPB, s(0) = 0, s(t_f) = 1
for t = 0 to t_f step T_s do
    if joint interpolation then
        q ← q_0 + s(t)(q_f − q_0)
    else
        p ← p_0 + s(t)(p_f − p_0); orientation by slerp
        q ← IK(p, R; branch nearest previous q)
        if q undefined or near singular then
            stop / re-plan
        end if
    end if
    send q to joint servos
end for
```

obtained by adding the cruise time $(\Delta\theta - V_{\max}^2/a_{\max})/V_{\max}$ to the two blend times $V_{\max}/a_{\max}$. *Example:* $\Delta\theta = 90°$, $V_{\max} = 180°/\text{s}$, $a_{\max} = 720°/\text{s}^2$: since $90 > 180^2/720 = 45°$, a cruise exists and $t_f = 0.5 + 0.25 = 0.75$ s.

3.13 Robot Statics

3.13.1 Force–torque relationship – derivation by virtual work

When the robot exerts a force $\mathbf{F}$ (and moment) on its environment – pushing a tool, holding a payload – joint torques $\boldsymbol{\tau}$ are required. At static equilibrium the work done by the joints in a virtual displacement $\delta\mathbf{q}$ equals the work done at the end-effector in the corresponding displacement $\delta\mathbf{x} = \mathbf{J}\,\delta\mathbf{q}$:

$$\boldsymbol{\tau}^{\mathsf{T}}\delta\mathbf{q} = \mathbf{F}^{\mathsf{T}}\delta\mathbf{x} = \mathbf{F}^{\mathsf{T}}\mathbf{J}\,\delta\mathbf{q} \quad \text{for all } \delta\mathbf{q},$$

hence

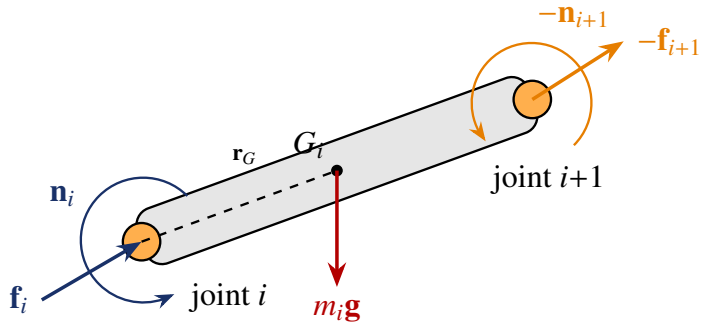
$$\boxed{\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}(\mathbf{q})\,\mathbf{F}} \tag{3.27}$$

The *same* Jacobian that maps velocities forward maps forces backward. Consequences: (i) at a singularity there are force directions requiring no joint torque (the structure carries them); (ii) force capability is described by a force ellipse that is the inverse of the velocity ellipse – directions of high speed are directions of low force. Gravity torques of the links add to (3.27).

Link equilibrium. Alternatively the free-body diagram of each link (Fig. 3.14) gives $\mathbf{f}_i - \mathbf{f}_{i+1} + m_i\mathbf{g} = \mathbf{0}$ and the moment balance; iterating from the tool to the base yields the joint torques $\tau_i = \mathbf{n}_i \cdot \hat{\mathbf{z}}_{i-1}$ (static Newton–Euler recursion).

Example 3.10: Joint torques for a tip force (Topic 3.13)

Given: 2R arm of Example 3.7 at $(30°, 45°)$ pressing down on a surface with $\mathbf{F} = (0, -20)$ N



Static equilibrium of link i:
$\mathbf{f}_i - \mathbf{f}_{i+1} + m_i\mathbf{g} = \mathbf{0}$
$\mathbf{n}_i - \mathbf{n}_{i+1} - \mathbf{r}_{i,i+1} \times \mathbf{f}_{i+1}$
$+ \mathbf{r}_G \times m_i\mathbf{g} = \mathbf{0}$
joint torque $\tau_i = \mathbf{n}_i \cdot \hat{\mathbf{z}}_{i-1}$

Figure 3.14: Free-body diagram of link i showing the joint forces and moments exerted by the neighbouring links and gravity.

(link weights neglected).

Solution: $\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}\mathbf{F} = \begin{bmatrix} -0.5398 & 0.5107 \\ -0.2898 & 0.0776 \end{bmatrix} \begin{bmatrix} 0 \\ -20 \end{bmatrix} = \begin{bmatrix} -10.21 \\ -1.55 \end{bmatrix}$ N m.

Check: τ_1 equals force times the horizontal distance of the tip from joint 1 (20×0.5107), as elementary statics requires.

3.14 Robot Dynamics

3.14.1 Purpose

Kinematics describes motion without regard to forces; *dynamics* relates joint torques to the resulting motion. Dynamic models are needed to size motors and reducers, to simulate robots, and for model-based control (computed torque, gravity and feed-forward compensation). Two formulations are standard: the *Newton–Euler* recursion (forces and moments on each link, efficient for computation) and the *Lagrangian* formulation (energy-based, systematic for derivation by hand).

3.14.2 Structure of the equations of motion

For an n-joint rigid robot both methods yield

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \boldsymbol{\tau}_f(\dot{\mathbf{q}}) = \boldsymbol{\tau}, \tag{3.28}$$

where $\mathbf{M}$ is the $n \times n$ *inertia (mass) matrix*, $\mathbf{C}\dot{\mathbf{q}}$ contains *Coriolis* ($\dot{q}_i\dot{q}_j$) and *centrifugal* ($\dot{q}_i^2$) terms, $\mathbf{G}$ the *gravity* torques and $\boldsymbol{\tau}_f$ friction. Important properties used in control design: $\mathbf{M}$ is symmetric and positive definite; $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew-symmetric (energy conservation); the equations are linear in the inertial parameters (enabling identification and adaptive control). The model is strongly non-linear and coupled: the torque at one joint depends on the motion of all others.



(a) Single link: $\tau = I_O\ddot{\theta} + mgl_c\cos\theta$ (b) Two-link arm with point masses at link ends

Figure 3.15: Dynamic models: (a) single link rotating in a vertical plane; (b) two-link planar arm with point masses at the link ends.

3.15 Lagrangian Formulation

3.15.1 Lagrange's equation

Define the *Lagrangian* $L = K - P$, the difference between total kinetic energy K and potential energy P expressed in generalised coordinates $\mathbf{q}$. The equations of motion are

$$\tau_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i}, \qquad i = 1, \ldots, n, \tag{3.29}$$

where τ_i is the generalised force (torque for revolute, force for prismatic joints). The method requires only scalar energies – no internal constraint forces – and the procedure of Fig. 3.16 applies to any robot. For a manipulator, $K = \tfrac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{M}(\mathbf{q})\dot{\mathbf{q}}$, with $\mathbf{M} = \sum_i \left(m_i \mathbf{J}_{vi}^{\mathsf{T}}\mathbf{J}_{vi} + \mathbf{J}_{\omega i}^{\mathsf{T}}\mathbf{I}_i\mathbf{J}_{\omega i}\right)$ built from the link-centre Jacobians, which links dynamics to the velocity kinematics of Section 3.7.

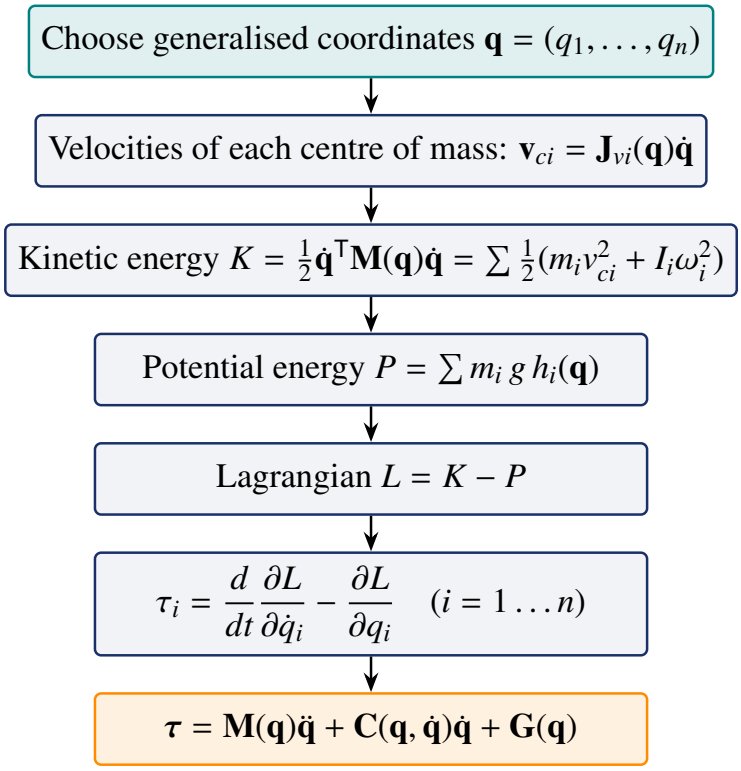


Figure 3.16: Flowchart of the Lagrangian formulation of manipulator dynamics.

3.15.2 Example 1: single link – derivation

A uniform link of mass m and length l, pivoted at one end and rotating in a vertical plane (Fig. 3.15(a)), with θ measured from the horizontal and centre of mass at $l_c = l/2$:

$$K = \tfrac{1}{2} I_O \dot{\theta}^2 \;\; (I_O = m l^2/3), \qquad P = m g l_c \sin\theta, \qquad L = \tfrac{1}{2} I_O \dot{\theta}^2 - m g l_c \sin\theta.$$

$\partial L/\partial \dot{\theta} = I_O \dot{\theta}$, $\frac{d}{dt}(\cdot) = I_O \ddot{\theta}$, $\partial L/\partial \theta = -m g l_c \cos\theta$, hence

$$\tau = I_O \ddot{\theta} + m g l_c \cos\theta. \tag{3.30}$$

Example 3.11: Single-link dynamics (Topic 3.15)

Given: $m = 2\,\text{kg}$, $l = 0.5\,\text{m}$; horizontal position $\theta = 0$.

Solution: $I_O = 2(0.25)/3 = 0.1667\,\text{kg m}^2$. Holding torque $= m g l_c = 2(9.81)(0.25) = 4.905\,\text{N m}$. To accelerate upward at $2\,\text{rad/s}^2$: $\tau = 0.1667(2) + 4.905 = 5.24\,\text{N m}$.

Interpretation: Gravity dominates; the motor must supply the holding torque continuously, which is why gravity compensation (springs or counterweights) is used on large robots.

3.15.3 Example 2: two-link planar arm – derivation

With point masses m_1, m_2 at the ends of links l_1, l_2 (Fig. 3.15(b)) and angles from the horizontal:

- Positions: $\mathbf{p}_1 = (l_1 c_1, l_1 s_1)$, $\mathbf{p}_2 = (l_1 c_1 + l_2 c_{12},\; l_1 s_1 + l_2 s_{12})$.
- Kinetic energy: $K = \tfrac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \tfrac{1}{2} m_2 \left[l_1^2 \dot{\theta}_1^2 + l_2^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + 2 l_1 l_2 c_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \right]$.
- Potential energy: $P = m_1 g l_1 s_1 + m_2 g (l_1 s_1 + l_2 s_{12})$.

Applying (3.29) and collecting terms gives (3.28) with

$$\mathbf{M} = \begin{bmatrix} (m_1 + m_2) l_1^2 + m_2 l_2^2 + 2 m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix},$$

$$\mathbf{C}\dot{\mathbf{q}} = \begin{bmatrix} -m_2 l_1 l_2 s_2 \,(2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ m_2 l_1 l_2 s_2 \,\dot{\theta}_1^2 \end{bmatrix}, \qquad \mathbf{G} = \begin{bmatrix} (m_1 + m_2) g l_1 c_1 + m_2 g l_2 c_{12} \\ m_2 g l_2 c_{12} \end{bmatrix}. \tag{3.31}$$

The term $2\dot{\theta}_1 \dot{\theta}_2$ is Coriolis and $\dot{\theta}_2^2$, $\dot{\theta}_1^2$ centrifugal; $\mathbf{M}$ depends on θ_2 – the arm's inertia about the shoulder is largest when stretched.

Example 3.12: Two-link torques (Topic 3.15)

Given: $m_1 = m_2 = 1\,\text{kg}$, $l_1 = 0.5\,\text{m}$, $l_2 = 0.3\,\text{m}$; $\boldsymbol{\theta} = (0, 90^\circ)$, at rest, required $\ddot{\boldsymbol{\theta}} = (1, 1)\,\text{rad/s}^2$.

Solution: $c_2 = 0$: $\mathbf{M} = \left[\begin{smallmatrix} 0.59 & 0.09 \\ 0.09 & 0.09 \end{smallmatrix}\right]\,\mathrm{kg\,m^2}$; velocities zero so $\mathbf{C}\dot{\mathbf{q}} = \mathbf{0}$; $\mathbf{G} = (2{\times}9.81{\times}0.5{+}0,\ 0) = (9.81, 0)\,\mathrm{N\,m}$ (link 2 vertical: $c_{12} = 0$). $\boldsymbol{\tau} = \mathbf{M}\ddot{\boldsymbol{\theta}} + \mathbf{G} = (0.68 + 9.81,\ 0.18) = (10.49,\ 0.18)\,\mathrm{N\,m}$.

3.16 Joint-Space Control

3.16.1 Independent joint control

Most industrial robots control each joint independently with the cascade servo of Unit II, treating the coupling torques of (3.28) as disturbances. This works because the reducer divides the varying link inertia by n^2 (Eq. (2.21)). Modelling joint i as $J_{\text{eff}}\ddot{q} + B\dot{q} = \tau - \tau_d$ with a PD law $\tau = K_p e + K_d \dot{e}$ (error $e = q_d - q$) gives the closed loop $J_{\text{eff}}\ddot{e} + (B + K_d)\dot{e} + K_p e = \tau_d$, a second-order system with

$$\omega_n = \sqrt{K_p / J_{\text{eff}}}, \qquad \zeta = \frac{B + K_d}{2\sqrt{K_p J_{\text{eff}}}}. \tag{3.32}$$

A constant disturbance such as gravity leaves a steady-state error τ_d / K_p, removed by integral action (PID) or by *gravity compensation*.

PD with gravity compensation. The law

$$\boldsymbol{\tau} = \mathbf{K}_p \mathbf{e} + \mathbf{K}_d \dot{\mathbf{e}} + \hat{\mathbf{G}}(\mathbf{q}) \tag{3.33}$$

(Fig. 3.17(a)) drives the robot to any set-point $\mathbf{q}_d$ with zero steady-state error when $\hat{\mathbf{G}} = \mathbf{G}$; global asymptotic stability is proved with the Lyapunov function $V = \tfrac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{M}\dot{\mathbf{q}} + \tfrac{1}{2}\mathbf{e}^{\mathsf{T}}\mathbf{K}_p\mathbf{e}$ using the skew-symmetry property.

3.16.2 Computed-torque (inverse-dynamics) control

For fast trajectory tracking the non-linear dynamics are cancelled by

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q})\left[\ddot{\mathbf{q}}_d + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e}\right] + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}). \tag{3.34}$$

Substituting into (3.28) and cancelling $\mathbf{M}$ (invertible) gives the decoupled linear error dynamics $\ddot{\mathbf{e}} + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e} = \mathbf{0}$. Choosing diagonal gains $K_p = \omega_n^2$, $K_d = 2\zeta\omega_n$ assigns identical, configuration-independent responses to every joint. *Example:* for $\omega_n = 10\,\mathrm{rad/s}$ and critical damping, $K_p = 100\ \mathrm{s^{-2}}$ and $K_d = 20\ \mathrm{s^{-1}}$.

Example 3.13: PD gains for an independent joint (Topic 3.16)

Given: Effective inertia at the motor $J_{\text{eff}} = 3 \times 10^{-4}\,\mathrm{kg\,m^2}$ (Example 2.9), $B \approx 0$; desired $\omega_n = 50\,\mathrm{rad/s}$, $\zeta = 1$.

Solution: $K_p = J_{\text{eff}}\omega_n^2 = 3\times10^{-4}\times2500 = 0.75\,\text{N m/rad}$; $K_d = 2\zeta\omega_n J_{\text{eff}} = 2\times50\times3\times10^{-4} = 0.03\,\text{N m s/rad}$ (motor side).

Interpretation: 2% settling time $\approx 5.8/\omega_n = 0.12\,\text{s}$ for a critically damped system (approximately; $4/(\zeta\omega_n)$ underestimates for $\zeta = 1$).

Algorithm 3.5 PID-based joint control of a robot

Require: Trajectory $\mathbf{q}_d(t), \dot{\mathbf{q}}_d(t)$; gains $\mathbf{K}_p, \mathbf{K}_i, \mathbf{K}_d$; model $\hat{\mathbf{G}}$; period T_s
Ensure: Joint torque commands
1: $\mathbf{I} \leftarrow \mathbf{0}$
2: **loop** every T_s
3: read $\mathbf{q}, \dot{\mathbf{q}}$ from encoders
4: $\mathbf{e} \leftarrow \mathbf{q}_d - \mathbf{q}$; $\dot{\mathbf{e}} \leftarrow \dot{\mathbf{q}}_d - \dot{\mathbf{q}}$; $\mathbf{I} \leftarrow \mathbf{I} + \mathbf{e}T_s$
5: $\boldsymbol{\tau} \leftarrow \mathbf{K}_p\mathbf{e} + \mathbf{K}_i\mathbf{I} + \mathbf{K}_d\dot{\mathbf{e}} + \hat{\mathbf{G}}(\mathbf{q})$
6: saturate $\boldsymbol{\tau}$; anti-windup; send current references $i = \tau/(nK_t)$ to drives
7: **end loop**

3.17 Task-Space Control

When the task is specified in Cartesian coordinates – and especially when the robot interacts with its environment – it is natural to define the error directly in task space, $\mathbf{e}_x = \mathbf{x}_d - \mathbf{x}$, with $\mathbf{x} = \mathbf{f}(\mathbf{q})$ from forward kinematics.

- **Jacobian-transpose control** (Fig. 3.17(b)): a virtual spring–damper at the end-effector produces a force that is mapped to joint torques by the statics relation (3.27):

$$\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}(\mathbf{q})\left[\mathbf{K}_p\mathbf{e}_x - \mathbf{K}_d\dot{\mathbf{x}}\right] + \hat{\mathbf{G}}(\mathbf{q}). \tag{3.35}$$

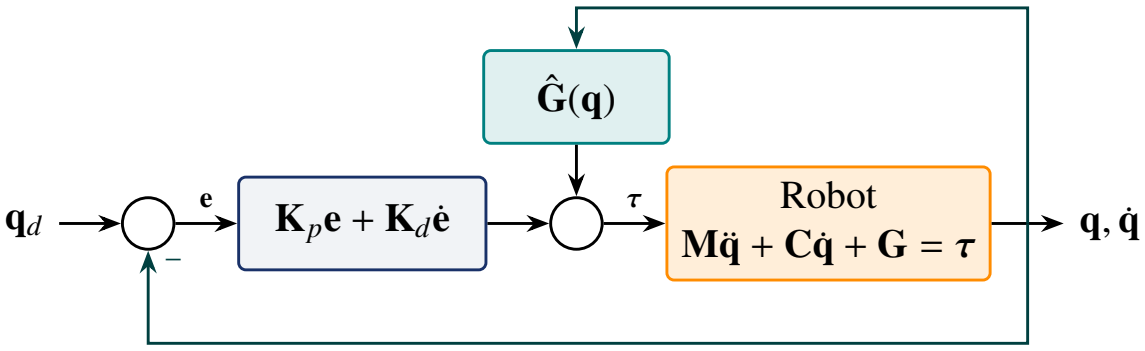
  No inverse kinematics or Jacobian inversion is needed, so it is robust near singularities; the stiffness $\mathbf{K}_p$ directly sets how compliant the tool is – the basis of *impedance control* used in assembly and collaborative robots.

- **Resolved-rate (kinematic) control**: $\dot{\mathbf{q}}_{\text{ref}} = \mathbf{J}^{-1}(\mathbf{q})\,[\dot{\mathbf{x}}_d + \mathbf{K}\mathbf{e}_x]$ sent to joint velocity servos; widely used for teleoperation, visual servoing and conveyor tracking.

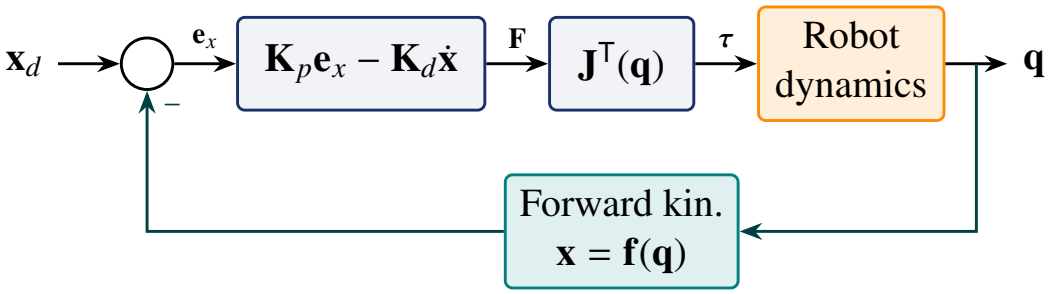
- **Operational-space (inverse-dynamics) control** cancels the dynamics in task coordinates for decoupled Cartesian behaviour.

3.18 MATLAB and Simulink Implementation

The following programs (core MATLAB; tested in GNU Octave) implement the kinematic, trajectory and control algorithms of this unit. Each prints values that reproduce the worked examples. Students with the Robotics System Toolbox or Peter Corke's Robotics Toolbox can cross-check the results with `rigidBodyTree` or `SerialLink` objects.



(a) Joint-space PD control with gravity compensation



(b) Task-space control via the Jacobian transpose (+$\hat{\mathbf{G}}$ added at $\boldsymbol{\tau}$)

Figure 3.17: Robot control architectures: (a) joint-space PD control with gravity compensation; (b) task-space control through the Jacobian transpose.

Table 3.5: Comparison of robot control schemes.

Scheme	Model needed	Strengths	Limitations
Independent joint PID	None (per-joint gains)	Simple, robust, industry standard	Coupling errors at high speed
PD + gravity compensation	$\mathbf{G}(\mathbf{q})$	Zero set-point error, provably stable	Tracking errors in fast motion
Computed torque	Full $\mathbf{M}, \mathbf{C}, \mathbf{G}$	Decoupled, uniform tracking	Sensitive to model errors; heavy computation
Jacobian transpose / impedance	Kinematics (+$\mathbf{G}$)	Natural compliance, no IK	Steady-state error under load unless integral added
Resolved rate	Kinematics	Direct Cartesian velocity	Fails near singularities

3.18.1 Program 3.1 – Coordinate transformations and workspace

Objective. Verify the rotation and homogeneous-transformation results of Examples 3.1 to 3.3 and compute the reachable workspace of a planar 3R arm by random joint sampling. **Expected output:** $(-0.134, 2.232, 0)$; $\det \mathbf{R} = 1$ and $\|\mathbf{R}^{\mathsf{T}}\mathbf{R} - \mathbf{I}\| \approx 10^{-16}$; recovered angles $30^\circ, 45^\circ, 60^\circ$; transformed point $(2, 2, 0)$ and back $(1, 1, 0)$; maximum sampled reach 0.800 m. The scatter plot reproduces the disc of Fig. 3.10.

Program 3.1: Rotation matrices, homogeneous transformations and workspace sampling.

```
% Program 3.1: Rotation matrices, homogeneous transformations and workspace
clear; clc; close all;
Rx = @(t) [1 0 0; 0 cos(t) -sin(t); 0 sin(t) cos(t)];
Ry = @(t) [cos(t) 0 sin(t); 0 1 0; -sin(t) 0 cos(t)];
Rz = @(t) [cos(t) -sin(t) 0; sin(t) cos(t) 0; 0 0 1];
pA = Rz(deg2rad(30))*[1; 2; 0]; % Example: rotation of a point
fprintf('Rz(30)*[1 2 0]'' = [%.3f %.3f %.3f]\n', pA);
R = Rz(deg2rad(60))*Ry(deg2rad(45))*Rx(deg2rad(30)); % roll-pitch-yaw
fprintf('det(R) = %.4f, ||R''R - I|| = %.1e\n', det(R), norm(R'*R - eye(3)));
yaw = atan2(R(2,1), R(1,1));
pitch = atan2(-R(3,1), sqrt(R(1,1)^2 + R(2,1)^2));
roll = atan2(R(3,2), R(3,3));
fprintf('Recovered roll/pitch/yaw = %.1f %.1f %.1f deg\n', rad2deg([roll pitch yaw]));
% ---- Homogeneous transformation and its inverse
T = [Rz(pi/2) [3;1;0]; 0 0 0 1];
p = T*[1;1;0;1];
Tinv = [T(1:3,1:3)' -T(1:3,1:3)'*T(1:3,4); 0 0 0 1];
fprintf('T*[1 1 0 1]'' = [%g %g %g], back = [%g %g %g]\n', p(1:3), Tinv(1:3,:)*p);
% ---- Workspace of a planar 3R arm by joint-space sampling
L = [0.4 0.3 0.1]; N = 30000; q = 2*pi*rand(3, N);
x = L(1)*cos(q(1,:)) + L(2)*cos(q(1,:)+q(2,:)) + L(3)*cos(sum(q,1));
y = L(1)*sin(q(1,:)) + L(2)*sin(q(1,:)+q(2,:)) + L(3)*sin(sum(q,1));
plot(x, y, '.', 'MarkerSize', 1); axis equal; grid on; xlabel('x (m)'); ylabel('y (m)');
title('Reachable workspace of a planar 3R arm');
fprintf('Max reach sampled = %.3f m (theory %.3f m)\n', max(hypot(x,y)), sum(L));
```

3.18.2 Program 3.2 – DH parameters and forward kinematics

Objective. Implement Algorithm 3.1 with the DH matrix (3.9) as an anonymous function and evaluate the articulated arm of Example 3.5 and the SCARA of Table 3.3. **Expected output:** position $(0.6408, 0.3700, 0.6500)$ m with the orientation matrix quoted in Example 3.5; SCARA tool at $(0.3464, 0.5000, 0.3000)$ m.

Program 3.2: DH forward kinematics of articulated and SCARA robots.

```
% Program 3.2: DH parameters and forward kinematics
clear; clc;
```

```
dh = @(th,d,a,al) [cos(th) -sin(th)*cos(al) sin(th)*sin(al) a*cos(th);
                   sin(th) cos(th)*cos(al) -cos(th)*sin(al) a*sin(th);
                   0 sin(al) cos(al) d;
                   0 0 0 1];
% ---- Articulated (RRR) arm: rows = [theta d a alpha]
q = deg2rad([30 45 -60]);
DH = [q(1) 0.4 0 pi/2;
      q(2) 0 0.5 0;
      q(3) 0 0.4 0];
T = eye(4);
for i = 1:size(DH,1) % Algorithm: DH forward kinematics
    T = T*dh(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
end
fprintf('Articulated arm 0T3 =\n'); fprintf('%8.4f %8.4f %8.4f %8.4f\n', T');
% ---- SCARA (RRPR): theta1=30, theta2=60, d3=0.1, theta4=20 deg
S = [deg2rad(30) 0.4 0.4 0; deg2rad(60) 0 0.3 pi; 0 0.1 0 0; deg2rad(20) 0 0 0];
T = eye(4); for i = 1:4, T = T*dh(S(i,1), S(i,2), S(i,3), S(i,4)); end
fprintf('SCARA tool position = [%.4f %.4f %.4f] m\n', T(1:3,4));
```

3.18.3 Program 3.3 – Inverse kinematics

Objective. Compute both IK branches of the 2R arm (Example 3.6) and the 3R solution with orientation, verifying each by forward kinematics. **Expected output:** $(11.56°, 76.51°)$ and $(65.76°, -76.51°)$, both reproducing $(0.5, 0.4)$ m; 3R: $(73.74°, -90°, 16.26°)$. The plot shows the elbow-up and elbow-down postures of Fig. 3.7.

Program 3.3: Inverse kinematics of planar 2R and 3R arms.

```
% Program 3.3: Inverse kinematics of 2R and 3R planar arms (both branches)
clear; clc; close all;
l1 = 0.5; l2 = 0.3; x = 0.5; y = 0.4;
D = (x^2 + y^2 - l1^2 - l2^2)/(2*l1*l2);
if abs(D) > 1, error('Target outside the workspace'); end
for sgn = [1 -1]
    th2 = atan2(sgn*sqrt(1 - D^2), D);
    th1 = atan2(y, x) - atan2(l2*sin(th2), l1 + l2*cos(th2));
    xc = l1*cos(th1) + l2*cos(th1+th2); yc = l1*sin(th1) + l2*sin(th1+th2);
    fprintf('branch %+d: th1 = %7.2f deg, th2 = %7.2f deg -> FK (%.4f, %.4f)\n', ...
            sgn, rad2deg(th1), rad2deg(th2), xc, yc);
    plot([0 l1*cos(th1) xc], [0 l1*sin(th1) yc], 'o-', 'LineWidth', 2); hold on;
end
axis equal; grid on; title('Two IK solutions of a 2R arm'); xlabel('x (m)'); ylabel('y (m)');
% ---- 3R with orientation
L = [0.4 0.3 0.1]; xd = 0.5; yd = 0.3; phi = 0;
xw = xd - L(3)*cos(phi); yw = yd - L(3)*sin(phi);
D = (xw^2 + yw^2 - L(1)^2 - L(2)^2)/(2*L(1)*L(2));
t2 = atan2(-sqrt(1 - D^2), D);
t1 = atan2(yw, xw) - atan2(L(2)*sin(t2), L(1) + L(2)*cos(t2));
```

```
t3 = phi - t1 - t2;
fprintf('3R IK: theta = [%.2f %.2f %.2f] deg\n', rad2deg([t1 t2 t3]));
```

3.18.4 Program 3.4 – Jacobian and singularity analysis

Objective. Evaluate the 2R Jacobian, forward and inverse velocity maps and the static torque of Examples 3.7 and 3.10, then scan the manipulability over the elbow angle. **Expected output:** $\det \mathbf{J} = 0.1061$; $\dot{\mathbf{x}} = (-0.2119, 0.2398)$ m/s; $\dot{\boldsymbol{\theta}} = (0.0732, -0.4815)$ rad/s; $\boldsymbol{\tau} = (-10.213, -1.553)$ N m; singularities found at $\theta_2 = 0°, \pm 180°$, with maximum manipulability at $\pm 90°$.

Program 3.4: Jacobian, velocity mapping, statics and singularity scan.

```
% Program 3.4: Jacobian, velocities, statics and singularity analysis of a 2R arm
clear; clc; close all;
l1 = 0.5; l2 = 0.3;
J = @(t) [-l1*sin(t(1))-l2*sin(t(1)+t(2)), -l2*sin(t(1)+t(2));
           l1*cos(t(1))+l2*cos(t(1)+t(2)), l2*cos(t(1)+t(2))];
th = deg2rad([30 45]); Jt = J(th);
fprintf('det J = %.4f (l1 l2 sin th2 = %.4f)\n', det(Jt), l1*l2*sin(th(2)));
fprintf('xdot for qdot=[0.5 -0.2]: [%.4f %.4f] m/s\n', Jt*[0.5; -0.2]);
fprintf('qdot for xdot=[0.1 0] : [%.4f %.4f] rad/s\n', Jt\[0.1; 0]);
fprintf('tau for F=[0 -20] N : [%.3f %.3f] N.m\n', Jt'*[0; -20]);
% ---- Singularity scan: manipulability versus elbow angle
t2 = linspace(-pi, pi, 721); w = zeros(size(t2));
for k = 1:numel(t2), w(k) = abs(det(J([0 t2(k)]))); end
plot(rad2deg(t2), w, 'LineWidth', 1.5); grid on;
xlabel('\theta_2 (deg)'); ylabel('w = |det J|'); title('Manipulability of a 2R arm');
sing = rad2deg(t2(w < 1e-3));
fprintf('Singular elbow angles found near: %s deg\n', mat2str(unique(round(sing/10)*10)));
```

3.18.5 Program 3.5 – Trajectory planning

Objective. Generate the joint trajectories of Examples 3.8 and 3.9 and the joint motion required for a straight-line Cartesian path with a cubic timing law (Algorithm 3.4). **Expected output:** cubic peak velocity 30 °/s and peak acceleration $\approx 40\,°/s^2$ (numerical differentiation gives 39.5); quintic peak velocity 37.5 °/s; LSPB $t_b = 0.60$ s, $a = 41.7\,°/s^2$; joints at the Cartesian midpoint $(6.17°, 92.87°)$, as in Section 3.11.

Program 3.5: Cubic, quintic, LSPB and Cartesian straight-line trajectories.

```
% Program 3.5: Joint-space (cubic, quintic, LSPB) and Cartesian trajectories
clear; clc; close all;
th0 = 10; thf = 70; tf = 3; t = linspace(0, tf, 301); D = thf - th0;
cub = th0 + 3*D/tf^2*t.^2 - 2*D/tf^3*t.^3;
```

```
s = t/tf; quin = th0 + D*(10*s.^3 - 15*s.^4 + 6*s.^5);
V = 25; tb = (th0 - thf + V*tf)/V; a = V/tb; % LSPB
lspb = zeros(size(t));
for k = 1:numel(t)
    if t(k) < tb, lspb(k) = th0 + 0.5*a*t(k)^2;
    elseif t(k) <= tf - tb, lspb(k) = th0 + 0.5*a*tb^2 + V*(t(k) - tb);
    else, lspb(k) = thf - 0.5*a*(tf - t(k))^2;
    end
end
dt = t(2) - t(1);
fprintf('Cubic : max vel %.1f deg/s, max acc %.1f deg/s^2\n', max(gradient(cub,dt)), max(abs(gra
fprintf('Quintic: max vel %.1f deg/s\n', max(gradient(quin,dt)));
fprintf('LSPB : tb = %.2f s, a = %.1f deg/s^2, final = %.2f deg\n', tb, a, lspb(end));
subplot(1,2,1); plot(t, cub, t, quin, '--', t, lspb, ':', 'LineWidth', 1.4); grid on;
xlabel('t (s)'); ylabel('\theta (deg)'); legend('cubic', 'quintic', 'LSPB', 'Location', 'southea
% ---- Cartesian straight line for a 2R arm with IK at each point
l1 = 0.5; l2 = 0.3; p0 = [0.6 0.2]; pf = [0.3 0.5];
ik = @(p) [atan2(p(2),p(1)) - atan2(l2*sqrt(1-((p*p'-l1^2-l2^2)/(2*l1*l2))^2), ...
           l1 + l2*(p*p'-l1^2-l2^2)/(2*l1*l2)), acos((p*p'-l1^2-l2^2)/(2*l1*l2))];
sq = 3*s.^2 - 2*s.^3; Q = zeros(numel(s), 2);
for k = 1:numel(s), Q(k,:) = ik(p0 + sq(k)*(pf - p0)); end
qm = ik((p0 + pf)/2);
fprintf('Cartesian midpoint joints = [%.2f %.2f] deg\n', rad2deg(qm));
subplot(1,2,2); plot(t, rad2deg(Q), 'LineWidth', 1.4); grid on;
xlabel('t (s)'); ylabel('joint angle (deg)'); legend('\theta_1', '\theta_2'); title('Joints for
```

3.18.6 Program 3.6 – Robot joint control

Objective. Simulate the two-link dynamics (3.31) under the control laws (3.33) and (3.34) for a set-point step to $(60^\circ, 30^\circ)$ with $\omega_n = 10\,\text{rad/s}$, $\zeta = 1$. **Expected output:** torque check $(10.49, 0.18)$ N m as in Example 3.12; both controllers converge with negligible final error; the computed-torque responses of the two joints are identical critically damped curves, whereas the PD-plus-gravity responses show coupling between the joints during the transient.

Program 3.6: Two-link dynamics with PD-plus-gravity and computed-torque control.

```
% Program 3.6: Two-link arm dynamics with PD + gravity and computed-torque control
clear; clc; close all;
m1 = 1; m2 = 1; l1 = 0.5; l2 = 0.3; g = 9.81;
M = @(q) [(m1+m2)*l1^2 + m2*l2^2 + 2*m2*l1*l2*cos(q(2)), m2*l2^2 + m2*l1*l2*cos(q(2));
          m2*l2^2 + m2*l1*l2*cos(q(2)), m2*l2^2];
C = @(q,dq) [-m2*l1*l2*sin(q(2))*(2*dq(1)*dq(2) + dq(2)^2); m2*l1*l2*sin(q(2))*dq(1)^2];
G = @(q) [(m1+m2)*g*l1*cos(q(1)) + m2*g*l2*cos(q(1)+q(2)); m2*g*l2*cos(q(1)+q(2))];
fprintf('Check: tau at q=(0,90deg), qdd=(1,1) = [%.2f %.2f] N.m\n', M([0 pi/2])*[1;1] + G([0 pi/
qd = [pi/3; pi/6]; % set-point (60, 30 deg)
Kp = diag([100 100]); Kd = diag([20 20]); % wn = 10 rad/s, zeta = 1
dt = 1e-3; T = 0:dt:2; names = {'PD+gravity', 'computed torque'};
for law = 1:2
```

```
    q = [0; 0]; dq = [0; 0]; Q = zeros(2, numel(T));
    for k = 1:numel(T)
        e = qd - q; de = -dq;
        if law == 1, tau = M(qd)*(Kp*e + Kd*de) + G(q); % PD scaled by M(qd) + gravity
        else, tau = M(q)*(Kp*e + Kd*de) + C(q,dq) + G(q); end
        ddq = M(q) \ (tau - C(q,dq) - G(q));
        dq = dq + dt*ddq; q = q + dt*dq; Q(:,k) = q;
    end
    fprintf('%-16s: final error = [%.4f %.4f] deg\n', names{law}, rad2deg(qd - q));
    subplot(1,2,law); plot(T, rad2deg(Q), 'LineWidth', 1.4); hold on;
    plot(T, rad2deg(qd)*ones(size(T)), 'k--'); grid on; title(names{law});
    xlabel('t (s)'); ylabel('joint angle (deg)');
end
```

3.18.7 Simulink models

S3.1 Forward kinematics block. A `MATLAB Function` block implementing (3.11) driven by two `Sine Wave` joint inputs; plot the tool path with an `XY Graph`.

S3.2 Trajectory generator. Implement the cubic (3.22) with `Clock`, `Fcn` and `Derivative` blocks, or use the `Polynomial Trajectory` / `Trapezoidal Velocity Profile Trajectory` blocks of the Robotics System Toolbox.

S3.3 Robot dynamics and control. Build (3.28) for the two-link arm in a `MATLAB Function` block computing $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\boldsymbol{\tau} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G})$, followed by two `Integrator` blocks; close the loop with PD + gravity compensation and with computed torque; compare tracking of the cubic trajectory.

S3.4 Simscape Multibody. Import or build the two-link arm with revolute joints and torque actuation to verify the hand-derived Lagrangian model.

3.19 Unit III Summary and Review

3.19.1 Unit summary

1. Frames describe pose; ${}^{A}\mathbf{p} = {}^{A}_{B}\mathbf{R}\,{}^{B}\mathbf{p} + {}^{A}\mathbf{p}_{B,\mathrm{org}}$. Rotation matrices are orthonormal ($\mathbf{R}^{-1} = \mathbf{R}^{\mathsf{T}}$, $\det = 1$); their columns are the rotated axes; composition is non-commutative.
2. Homogeneous 4×4 matrices combine rotation and translation, chain by multiplication and invert cheaply as $[\mathbf{R}^{\mathsf{T}}, -\mathbf{R}^{\mathsf{T}}\mathbf{t}]$.
3. The DH convention describes each link with $(\theta_i, d_i, a_i, \alpha_i)$; ${}^{i-1}\mathbf{A}_i = \mathrm{Rot}_z\mathrm{Trans}_z\mathrm{Trans}_x\mathrm{Rot}_x$; forward kinematics is the product of the link matrices.
4. Inverse kinematics may have no, several or infinitely many solutions; the 2R solution uses the law of cosines and atan2; spherical wrists decouple position and orientation; numerical IK uses the Jacobian.

5. The Jacobian maps joint to Cartesian velocities ($\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$) and Cartesian forces to joint torques ($\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}\mathbf{F}$); singularities ($\det \mathbf{J} = 0$) lose DOF and demand infinite joint rates.
6. Workspaces are reachable or dexterous; boundaries are singular configurations.
7. Joint trajectories use cubic, quintic or LSPB timing laws; Cartesian trajectories interpolate the path in task space and solve IK at each step.
8. Dynamics take the form $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} = \boldsymbol{\tau}$, derived by the Lagrangian $\tau_i = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}$.
9. Joint-space control ranges from independent PID through PD + gravity compensation to computed torque; task-space control uses the Jacobian transpose (impedance) or resolved rates.

3.19.2 Formula sheet

Table 3.6: Unit III formula sheet.

Quantity	Formula	Ref.
General transformation	${}^{A}\mathbf{p} = {}^{A}_{B}\mathbf{R}\,{}^{B}\mathbf{p} + {}^{A}\mathbf{p}_{B,\text{org}}$	(3.2)
Elementary rotations	$\mathbf{R}_x, \mathbf{R}_y, \mathbf{R}_z$	(3.4)
RPY	$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_x(\gamma)$	(3.5)
Homogeneous transform	$\mathbf{T} = [\mathbf{R}\ \mathbf{t};\ \mathbf{0}\ 1]$	(3.6)
Inverse transform	$[\mathbf{R}^{\mathsf{T}}\ \ -\mathbf{R}^{\mathsf{T}}\mathbf{t};\ \mathbf{0}\ 1]$	(3.8)
DH link matrix	$\text{Rot}_z(\theta)\text{Trans}_z(d)\text{Trans}_x(a)\text{Rot}_x(\alpha)$	(3.9)
2R forward kinematics	$x = l_1c_1 + l_2c_{12},\ y = l_1s_1 + l_2s_{12}$	(3.11)
Articulated FK	$x = c_1(a_2c_2 + a_3c_{23}),$ $z = d_1 + a_2s_2 + a_3s_{23}$	(3.13)
2R inverse kinematics	$c_2 = (x^2 + y^2 - l_1^2 - l_2^2)/(2l_1l_2)$	(3.14), (3.15)
Jacobian	$\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; 2R: $\det \mathbf{J} = l_1l_2s_2$	(3.17), (3.18)
Geometric Jacobian column	$[\hat{\mathbf{z}}_{i-1} \times (\mathbf{o}_n - \mathbf{o}_{i-1}); \hat{\mathbf{z}}_{i-1}]$	(3.19)
Dexterous workspace (3R)	$\lvert L_1 - L_2\rvert + L_3 \le r \le L_1 + L_2 - L_3$	(3.21)
Cubic coefficients	$a_2 = 3\Delta/t_f^2,\ a_3 = -2\Delta/t_f^3$	(3.22)
Quintic	$\theta_0 + \Delta(10\tau^3 - 15\tau^4 + 6\tau^5)$	(3.23)
LSPB	$t_b = (\theta_0 - \theta_f + Vt_f)/V$	(3.24)
Minimum time (trapezoid)	$t_f = \Delta/V_{\max} + V_{\max}/a_{\max}$	(3.26)
Statics	$\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}\mathbf{F}$	(3.27)
Equation of motion	$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} = \boldsymbol{\tau}$	(3.28)
Lagrange	$\tau_i = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}$	(3.29)
Single link	$\tau = I_O\ddot{\theta} + mgl_c\cos\theta$	(3.30)

Quantity	Formula	Ref.
PD joint loop	$\omega_n = \sqrt{K_p/J},\ \zeta = (B + K_d)/(2\sqrt{K_p J})$	(3.32)
Computed torque	$\boldsymbol{\tau} = \mathbf{M}(\ddot{\mathbf{q}}_d + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e}) + \mathbf{C}\dot{\mathbf{q}} + \mathbf{G}$	(3.34)
Jacobian-transpose control	$\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}(\mathbf{K}_p\mathbf{e}_x - \mathbf{K}_d\dot{\mathbf{x}}) + \mathbf{G}$	(3.35)

3.20 Unit III Question Bank

Exactly **30 questions of 10 marks each** (300 marks). Distribution: concepts Q1–Q5; theory Q6–Q12; derivations Q13–Q17; numerical problems Q18–Q23; diagrams and working principles Q24–Q26; applications Q27–Q30.

Question-to-notes reference index

For every question the table gives the topic (section) number, the page of the question and the page on which the topic begins in these notes. The **Refer to** line under each question also lists the specific figures, tables, equations and worked examples to study.

Table 3.7: Unit III question-to-notes reference index (automatic page references).

Q	Topic	Q. page	Notes page	Q	Topic	Q. page	Notes page
Q1	3.1	155	120	Q16	3.13	157	141
Q2	3.2	155	121	Q17	3.15	157	143
Q3	3.5	155	128	Q18	3.3	157	124
Q4	3.8	155	135	Q19	3.5	157	128
Q5	3.10	155	136	Q20	3.6	157	130
Q6	3.2	155	121	Q21	3.7	157	132
Q7	3.4	155	126	Q22	3.10	157	136
Q8	3.7	156	132	Q23	3.15	157	143
Q9	3.11	156	139	Q24	3.8	158	135
Q10	3.12	156	139	Q25	3.10	158	136
Q11	3.14	156	142	Q26	3.16	158	145
Q12	3.17	156	146	Q27	3.3	158	124
Q13	3.3	156	124	Q28	3.6	158	130
Q14	3.4	156	126	Q29	3.12	158	139
Q15	3.6	156	130	Q30	3.17	158	146

A. Fundamental concepts and definitions

Q1 **Topic 3.1** – Coordinate Frames and Transformations **[10 marks]**

Define a coordinate frame and explain the need for coordinate transformations in robotics with examples (base, tool, user and camera frames). Distinguish pure translation, pure rotation and general transformation, writing the relation for each.

☞ **Refer to:** Section 3.1 (p. 120); Figure 3.1 (p. 121)

Q2 **Topic 3.2** – Rotation Matrices **[10 marks]**

Define a rotation matrix as a direction-cosine matrix and state and explain its five properties. Write the elementary rotation matrices about x, y and z and explain the difference between fixed-axis and current-axis composition.

☞ **Refer to:** Section 3.2 (p. 121); Figure 3.3 (p. 123); Equation (3.4) (p. 123); Example 3.1 (p. 123); Example 3.2 (p. 124)

Q3 **Topic 3.5** – Forward and Inverse Kinematics **[10 marks]**

Define forward and inverse kinematics. Explain why inverse kinematics is more difficult, discussing existence, multiplicity of solutions, closed-form versus numerical solutions and Pieper's condition.

☞ **Refer to:** Section 3.5 (p. 128); Example 3.4 (p. 129); Example 3.5 (p. 130); Algorithm 3.2 (p. 132)

Q4 **Topic 3.8** – Singularities and Workspace **[10 marks]**

Define a kinematic singularity and explain its consequences. Distinguish boundary and interior singularities, define manipulability, and define reachable and dexterous workspaces.

☞ **Refer to:** Section 3.8 (p. 135)

Q5 **Topic 3.10** – Path and Trajectory **[10 marks]**

Distinguish path and trajectory. State the requirements of a good trajectory (continuity of position, velocity, acceleration; limits; synchronisation) and explain via points and the meaning of jerk.

☞ **Refer to:** Section 3.10 (p. 136); Example 3.8 (p. 137); Example 3.9 (p. 138)

B. Theoretical explanations

Q6 **Topic 3.2** – Orientation Representations **[10 marks]**

Explain roll–pitch–yaw angles, Z–Y–Z Euler angles, axis–angle and quaternion representations of orientation. Write the RPY rotation matrix, explain how the angles are recovered from a rotation matrix and discuss representation singularities. Compare the representations in a table.

☞ **Refer to:** Section 3.2 (p. 121); Table 3.2 (p. 125); Equation (3.5) (p. 124); Example 3.1 (p. 123); Example 3.2 (p. 124)

Q7 **Topic 3.4** – Denavit–Hartenberg Convention **[10 marks]**

State the rules for assigning DH frames and define the four DH parameters with a neat sketch. Obtain the DH tables of a planar 2R arm, a SCARA robot and a three-DOF articulated arm.

☞ **Refer to:** Section 3.4 (p. 126); Figure 3.5 (p. 127); Table 3.3 (p. 128)

Q8 Topic 3.7 – Velocity Kinematics **[10 marks]**

Explain the geometric Jacobian of a serial manipulator, giving the column expressions for revolute and prismatic joints and their physical meaning. Explain resolved-rate motion control and the pseudo-inverse for redundant robots.

☞ **Refer to:** Section 3.7 (p. 132); Example 3.7 (p. 134); Algorithm 3.3 (p. 134)

Q9 Topic 3.11 – Cartesian Trajectory Planning **[10 marks]**

Explain trajectory planning in Cartesian space for straight-line and circular paths. Compare joint-space and Cartesian-space planning in a table, explaining with a sketch why joint interpolation does not give a straight tool path.

☞ **Refer to:** Section 3.11 (p. 139); Figure 3.13 (p. 140); Table 3.4 (p. 140); Algorithm 3.4 (p. 141)

Q10 Topic 3.12 – Velocity and Acceleration Profiles **[10 marks]**

Explain trapezoidal, triangular, S-curve and polynomial velocity profiles with sketches of velocity and acceleration. Derive the minimum time of a trapezoidal move and the condition for a triangular profile.

☞ **Refer to:** Section 3.12 (p. 139); Figure 3.12 (p. 138); Figure 3.11 (p. 138)

Q11 Topic 3.14 – Robot Dynamics **[10 marks]**

Explain the purpose of robot dynamic models. Write the general equation of motion of an n-link robot, explaining the inertia, Coriolis, centrifugal, gravity and friction terms and the important properties of the model. Compare the Newton–Euler and Lagrangian approaches.

☞ **Refer to:** Section 3.14 (p. 142)

Q12 Topic 3.17 – Task-Space Control **[10 marks]**

Explain task-space control of a robot by the Jacobian-transpose (impedance) method and by resolved-rate control, with block diagrams. Compare joint-space and task-space control schemes.

☞ **Refer to:** Section 3.17 (p. 146); Figure 3.17 (p. 147); Table 3.5 (p. 147)

C. Mathematical derivations

Q13 Topic 3.3 – Homogeneous Transformation **[10 marks]**

Derive the homogeneous transformation matrix from the general coordinate transformation and derive its inverse. Show how transformations are composed and explain the double interpretation of ${}^{A}_{B}\mathbf{T}$.

☞ **Refer to:** Section 3.3 (p. 124); Example 3.3 (p. 126)

Q14 Topic 3.4 – DH Link Transformation **[10 marks]**

Derive the DH link transformation matrix ${}^{i-1}\mathbf{A}_i$ as a product of four elementary transformations, and hence derive the forward kinematics of a planar 2R manipulator.

☞ **Refer to:** Section 3.4 (p. 126); Equation (3.11) (p. 129)

Q15 Topic 3.6 – Inverse Kinematics of a 2R Arm **[10 marks]**

Derive the closed-form inverse kinematics of a planar 2R manipulator, showing the two solution branches. Extend the method to a planar 3R manipulator with specified orientation.

☞ **Refer to:** Section 3.6 (p. 130); Figure 3.7 (p. 131); Example 3.6 (p. 131)

Q16 **Topic 3.13** – Jacobian and Statics **[10 marks]**

Derive the Jacobian of a planar 2R manipulator and show that $\det \mathbf{J} = l_1 l_2 \sin\theta_2$. Using the principle of virtual work, derive $\boldsymbol{\tau} = \mathbf{J}^{\mathsf{T}}\mathbf{F}$ and interpret it at a singularity.

☞ **Refer to:** Section 3.13 (p. 141); Example 3.10 (p. 141)

Q17 **Topic 3.15** – Lagrangian Formulation **[10 marks]**

State Lagrange's equation. Derive the equation of motion of a single link rotating in a vertical plane, and the inertia matrix and gravity vector of a two-link planar arm with point masses at the link ends.

☞ **Refer to:** Section 3.15 (p. 143); Equation (3.31) (p. 144); Example 3.11 (p. 144); Example 3.12 (p. 144)

D. Numerical problems

Q18 **Topic 3.3** – Transformation Calculations **[10 marks]**

Frame $\{B\}$ is obtained by rotating $\{A\}$ by 90° about $\hat{\mathbf{x}}_A$ and then translating its origin to $(1, 2, 0.5)$ m in $\{A\}$. (a) Write ${}^A_B\mathbf{T}$. (b) Find ${}^A\mathbf{p}$ for ${}^B\mathbf{p} = (0.2, 0, 0.1)$ m. (c) Find ${}^B_A\mathbf{T}$ and verify the result.

☞ **Refer to:** Section 3.3 (p. 124); Example 3.3 (p. 126)

Q19 **Topic 3.5** – Forward Kinematics Calculation **[10 marks]**

For the articulated arm of Table 3.3 with $d_1 = 0.35$ m, $a_2 = 0.4$ m, $a_3 = 0.35$ m, find the tool position for $\boldsymbol{\theta} = (60^\circ, 30^\circ, -45^\circ)$.

☞ **Refer to:** Section 3.5 (p. 128); Example 3.4 (p. 129); Example 3.5 (p. 130)

Q20 **Topic 3.6** – Inverse Kinematics Calculation **[10 marks]**

A planar 2R arm has $l_1 = l_2 = 0.4$ m. Find both inverse-kinematics solutions for the target $(0.5, 0.3)$ m and verify one by forward kinematics. Is the target $(0.9, 0.2)$ m reachable?

☞ **Refer to:** Section 3.6 (p. 130); Example 3.6 (p. 131)

Q21 **Topic 3.7** – Jacobian Calculation **[10 marks]**

For a 2R arm with $l_1 = 0.4$ m, $l_2 = 0.3$ m at $\boldsymbol{\theta} = (45^\circ, 60^\circ)$, find $\mathbf{J}$ and $\det\mathbf{J}$; the tool velocity for $\dot{\boldsymbol{\theta}} = (0.2, 0.3)$ rad/s; the joint rates for $\dot{\mathbf{x}} = (0, 0.1)$ m/s; and the joint torques to exert $\mathbf{F} = (10, 0)$ N. State the singular configurations.

☞ **Refer to:** Section 3.7 (p. 132); Example 3.7 (p. 134)

Q22 **Topic 3.10** – Trajectory Calculation **[10 marks]**

A joint moves from -20° to 40° in 2 s, at rest at both ends. (a) Find the cubic coefficients, the position at 0.5 s, and the peak velocity and acceleration. (b) Find the peak velocity of the quintic. (c) Design an LSPB with $V = 40^\circ$/s.

☞ **Refer to:** Section 3.10 (p. 136); Example 3.8 (p. 137); Example 3.9 (p. 138)

Q23 **Topic 3.15** – Dynamics Calculation **[10 marks]**

A uniform link of mass 3 kg and length 0.6 m rotates in a vertical plane about one end. Find the torque required at $\theta = 30^\circ$ above the horizontal with an angular acceleration of 4 rad/s^2, and the holding torque at $\theta = 0$. Compute also the inertia matrix of the two-link arm of Example 3.12 at $\theta_2 = 0$ and comment.

☞ **Refer to:** Section 3.15 (p. 143); Example 3.11 (p. 144); Example 3.12 (p. 144)

E. Diagrams and working principles

Q24 **Topic 3.8** – Singularity Configurations **[10 marks]**

With neat sketches, explain the boundary singularities of a 2R arm and the wrist, elbow and shoulder singularities of a six-axis robot. Explain how manipulability ellipses illustrate singularity and how controllers handle singularities.

☞ **Refer to:** Section 3.8 (p. 135); Figure 3.9 (p. 136); Figure 3.8 (p. 133)

Q25 **Topic 3.10** – Joint-Space Trajectories **[10 marks]**

With neat graphs of position, velocity and acceleration, explain cubic, quintic and LSPB trajectories and compare them. Explain multi-axis synchronisation.

☞ **Refer to:** Section 3.10 (p. 136); Figure 3.11 (p. 138); Figure 3.12 (p. 138); Example 3.8 (p. 137); Example 3.9 (p. 138)

Q26 **Topic 3.16** – Joint-Space Control **[10 marks]**

With block diagrams, explain independent joint PID control, PD control with gravity compensation and computed-torque control. Derive the closed-loop error equation of computed-torque control and select gains for $\omega_n = 12\,\text{rad/s}$, $\zeta = 1$.

☞ **Refer to:** Section 3.16 (p. 145); Figure 3.17 (p. 147); Example 3.13 (p. 145); Algorithm 3.5 (p. 146)

F. Applications and analytical questions

Q27 **Topic 3.3** – Vision-to-Robot Transformation **[10 marks]**

A camera mounted above a conveyor detects a part. Explain, using a chain of homogeneous transformations, how the part position in the camera frame is converted into a target pose for the robot tool, including the tool and user frames. What calibrations are needed?

☞ **Refer to:** Section 3.3 (p. 124); Example 3.3 (p. 126)

Q28 **Topic 3.6** – Multiple IK Solutions in Practice **[10 marks]**

A six-axis robot can reach a welding point in up to eight configurations. Explain how these arise and the criteria by which a controller or programmer selects one. What happens if a linear move requires a change of configuration?

☞ **Refer to:** Section 3.6 (p. 130); Example 3.6 (p. 131)

Q29 **Topic 3.12** – Cycle-Time Planning **[10 marks]**

A palletising joint must rotate 120° with $V_{\max} = 150^\circ/\text{s}$ and $a_{\max} = 500^\circ/\text{s}^2$. Determine whether the trapezoidal profile has a cruise phase and find the minimum move time. Repeat for 30°. Discuss how jerk limits would affect the result.

☞ **Refer to:** Section 3.12 (p. 139)

Q30 **Topic 3.17** – Controller Selection **[10 marks]**

Recommend and justify a control scheme for: (a) a heavy palletising robot doing point-to-point moves; (b) a high-speed direct-drive SCARA tracking fast trajectories; (c) a collaborative robot polishing a surface with controlled contact force; (d) a teleoperated robot driven by a joystick in Cartesian velocity.

☞ **Refer to:** Section 3.17 (p. 146)

Unit IV: Industrial Automation, Embedded Systems and Machine Vision

Unit overview

The previous units treated the robot as an isolated machine. In practice a robot is one element of an automated production system: it is sequenced by a PLC, fed by conveyors, guided by cameras, protected by safety devices, programmed through a teach pendant and connected to plant networks. This unit covers industrial automation and robotic workcells (Sections 4.1 and 4.2), PLCs (Section 4.3), industrial robots and their programming (Sections 4.4 to 4.6), the embedded systems that implement controllers (Sections 4.7 to 4.12), machine vision (Sections 4.13 to 4.20) and the principal applications (Sections 4.21 to 4.25).

Table 4.1: Unit IV syllabus items and the sections that cover them.

No.	Syllabus item	Section	No.	Syllabus item	Section
1	Industrial automation	4.1	14	Image acquisition	4.14
2	Robotic workcells	4.2	15	Cameras	4.15
3	PLCs	4.3	16	Lighting techniques	4.16
4	Industrial robots	4.4	17	Image processing	4.17
5	Robot programming	4.5	18	Feature extraction	4.18
6	Teach pendant concepts	4.6	19	Object detection	4.19
7	Embedded systems	4.7	20	Inspection techniques	4.20
8	Microcontrollers	4.8	21	Smart manufacturing	4.21
9	Sensor interfacing	4.9	22	Pick-and-place	4.22
10	Actuator interfacing	4.10	23	Assembly	4.23
11	PWM generation and control	4.11	24	Material handling	4.24
12	Communication interfaces	4.12	25	Autonomous robotics	4.25
13	Machine vision	4.13			

4.1 Industrial Automation

4.1.1 Definition and objectives

Industrial automation is the use of control systems – PLCs, robots, CNC machines, computers and networks – to operate industrial processes with minimal human intervention. Its objectives are higher productivity, consistent quality, lower cost, improved safety (removing people from hazardous tasks), flexibility and data for traceability [7]. The three types of automation (fixed, programmable, flexible) were introduced in Unit I (Table 1.2).

4.1.2 The automation hierarchy

Automation systems are organised in levels, formalised in the ISA-95/IEC 62264 model and drawn as the *automation pyramid* (Fig. 4.1):

- **Level 0 – field**: sensors and actuators that interact with the physical process.
- **Level 1 – control**: PLCs, robot controllers, CNCs and drives executing real-time control in milliseconds.
- **Level 2 – supervisory**: SCADA and HMI systems that monitor and command one or more cells.
- **Level 3 – manufacturing operations (MES)**: scheduling, quality management, traceability.
- **Level 4 – enterprise (ERP)**: orders, planning and logistics.

Moving up the pyramid, time scales lengthen from microseconds to days and data become more aggregated. Industry 4.0 flattens the pyramid: devices publish data directly to higher levels through industrial Ethernet, OPC UA and cloud services (Section 4.21).

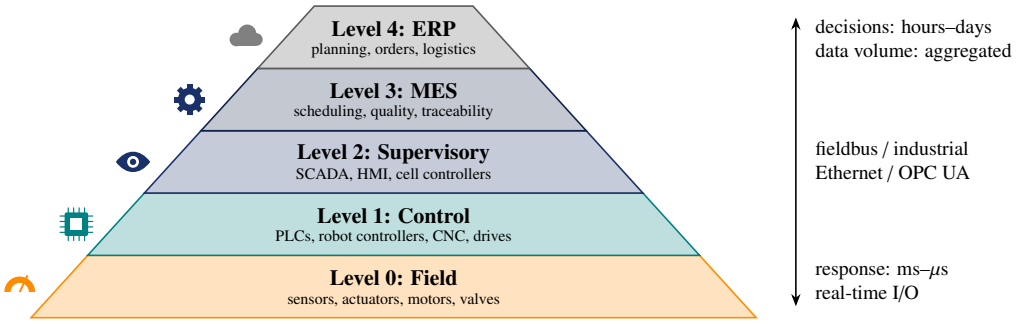


Figure 4.1: The automation pyramid (ISA-95 levels) with typical response times and communication.

4.1.3 Elements of an automated system

Every automated system contains (i) a *source of power* (electric, pneumatic, hydraulic); (ii) a *program of instructions* (PLC logic, robot program, CNC code); and (iii) a *control system* that executes the program and uses sensor feedback (open or closed loop). Control may be *continuous* (regulating analogue variables such as temperature or speed) or *discrete-event* (sequencing on/off actions – the domain of PLCs).

4.2 Robotic Workcells

4.2.1 Definition and components

A *robotic workcell* is a production unit in which one or more robots, together with machines, part-feeding devices, sensors, safety equipment and a cell controller, perform a defined task such as machine tending, welding or palletising (Fig. 4.2). Typical components:

- robot(s) with controller, end-effectors and tool changers;
- part presentation: conveyors, fixtures, pallets, bowl feeders, buffers;
- process equipment: CNC machines, presses, welding sources, test stations;
- sensors: part-presence, vision cameras, force sensors;
- cell controller (PLC) coordinating the robot and equipment by digital I/O or fieldbus, and an HMI;
- safety system: fences, interlocked doors, light curtains, area scanners, emergency stops, safety PLC.

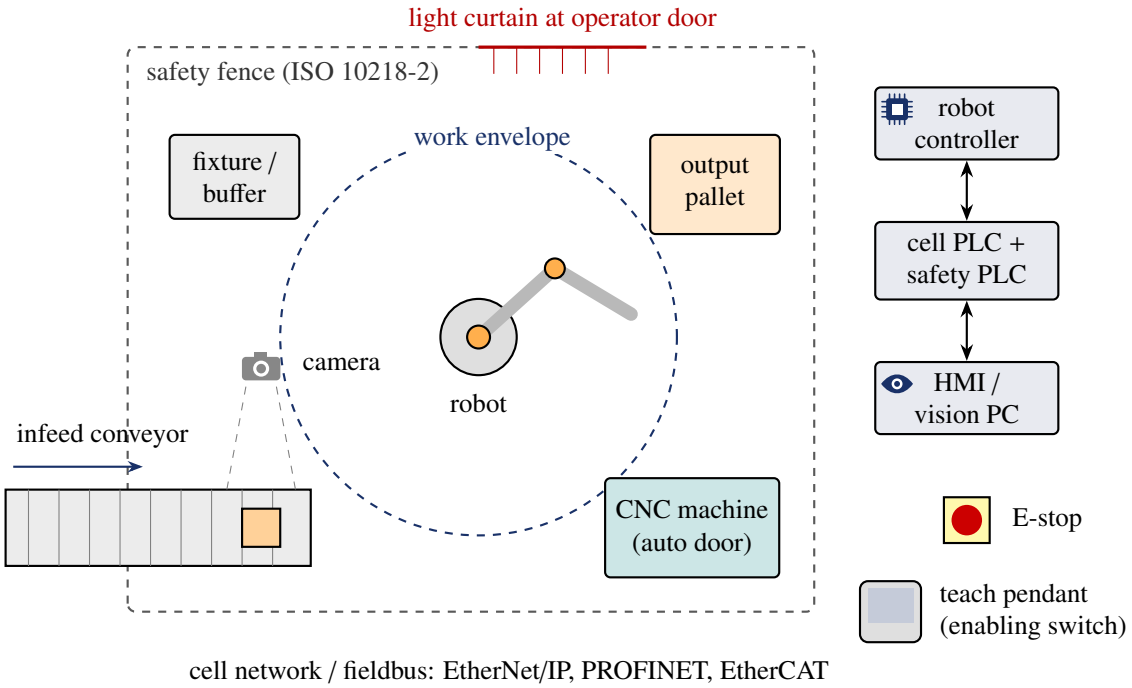


Figure 4.2: Layout of a robotic machine-tending workcell with infeed conveyor, vision camera, CNC machine, output pallet, safeguarding and controllers.

4.2.2 Layouts

- **Robot-centred cell**: machines arranged around a stationary robot within its work envelope (machine tending, Fig. 4.2).
- **In-line cell**: robots along a moving or indexing line (automotive body welding); robots must work on a moving part (tracking) or during dwell.
- **Mobile-robot cell**: robot on a track (seventh axis) or on an AMR serving several machines.

Layout is optimised by reach studies and cycle-time simulation in offline programming software.

4.2.3 Interlocks and safety

The robot and the equipment exchange *handshake* signals (e.g. “machine door open”, “part clamped”, “robot clear of machine”) so that neither moves into the other’s space; these *interlocks* are implemented in the PLC. Personnel safety follows ISO 10218-1/-2 and risk assessment to ISO 12100: perimeter guarding with interlocked gates; presence-sensing devices (light curtains, laser area scanners) whose response time determines the minimum safety distance ($S = K\,T + C$, ISO 13855); emergency stops; safety-rated monitored speed and separation for collaborative operation (ISO/TS 15066).

4.3 Programmable Logic Controllers

4.3.1 Definition and history

A PLC is a ruggedised industrial computer that controls machines and processes by executing a user program that reads inputs and sets outputs cyclically. It was introduced in 1968–69 (Modicon 084, for General Motors) to replace relay-logic panels, which were bulky and had to be rewired whenever the logic changed [21]. PLCs are designed for the factory floor: wide temperature range, noise immunity, isolated I/O, modular expansion and deterministic real-time behaviour.

4.3.2 Architecture

Figure 4.3 shows the main parts:

- **CPU** – executes the program, manages I/O and communication, runs diagnostics.
- **Memory** – program memory (non-volatile) and data memory including the input and output *image tables*, timers, counters and variables.
- **Input modules** – digital (24 V DC, opto-isolated) and analogue (4 mA to 20 mA, 0 V to 10 V, thermocouple, RTD), high-speed counters for encoders.
- **Output modules** – relay (AC/DC, slow, isolated), transistor (fast DC), triac (AC), analogue outputs.
- **Power supply** and **communication** ports (Ethernet, PROFINET, EtherNet/IP, Modbus) and a programming device.

4.3.3 The scan cycle

A PLC executes its program in a continuous loop called the *scan* (Fig. 4.4): (1) read all inputs into the input image table; (2) execute the program from first to last rung using the image values; (3) write the output image table to the physical outputs; (4) handle communications and diagnostics, and reset the watchdog timer. Because inputs are read once per scan and outputs written once per scan, the program sees a consistent snapshot of the process.

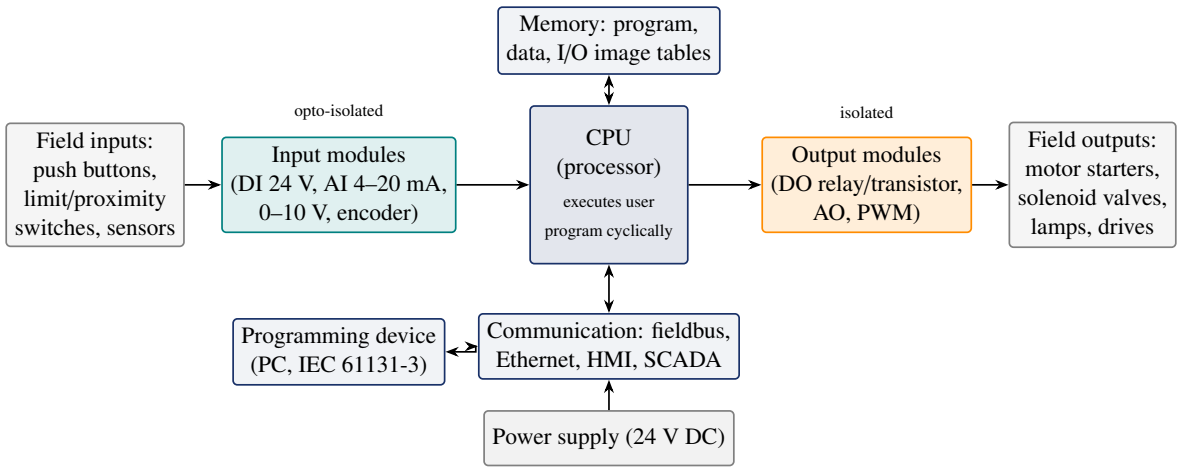


Figure 4.3: PLC architecture with field devices.

Scan time and response time. The scan time is

$$T_{\text{scan}} = T_{\text{in}} + T_{\text{prog}} + T_{\text{out}} + T_{\text{comm}}, \tag{4.1}$$

typically 1 ms to 20 ms. An input change occurring just after the inputs were read is not seen until the next scan and its effect appears at the end of that scan, so the worst-case response time is

$$T_{\text{resp,max}} \approx T_{\text{filter}} + 2\,T_{\text{scan}} + T_{\text{out,delay}}. \tag{4.2}$$

An input pulse shorter than about one scan (plus filter time) may be missed entirely; fast events are captured by interrupt inputs or high-speed counters.

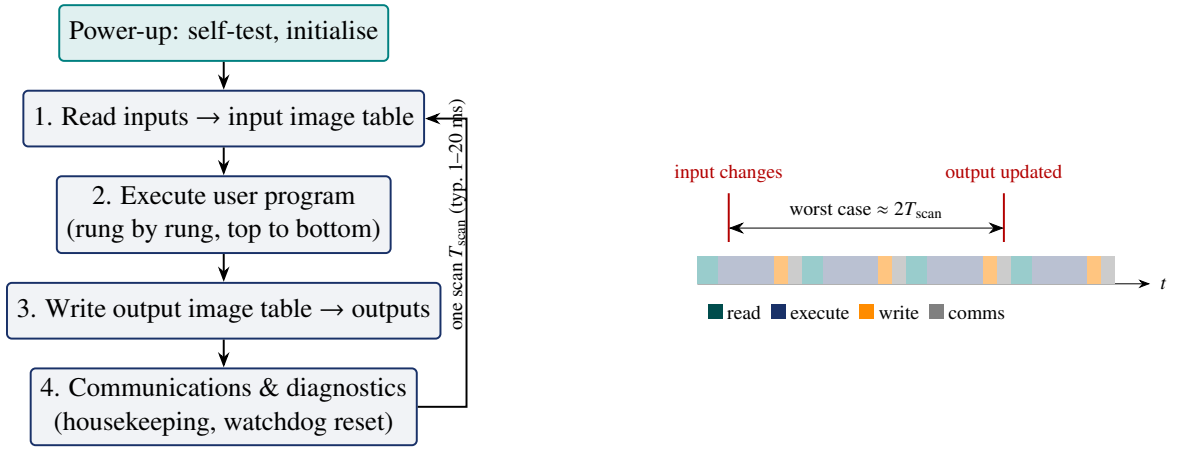


Figure 4.4: PLC scan cycle flowchart and timing diagram showing the worst-case response of about two scans.

Example 4.1: PLC scan and response time (Topic 4.3)

Given: Program of 4000 instructions at 0.05 µs each; input/output update 0.5 ms; communication overhead 0.3 ms; input filter 3 ms; relay output delay 10 ms.

Solution: $T_{\text{prog}} = 4000 \times 0.05 = 0.2$ ms; $T_{\text{scan}} = 0.2 + 0.5 + 0.3 = 1.0$ ms. Worst-case response

$= 3 + 2(1.0) + 10 = 15$ ms.

Interpretation: The relay output dominates; a transistor output (below 0.1 ms) would reduce the response to about 5 ms. A sensor pulse shorter than about 4 ms could be missed.

4.3.4 Programming languages and ladder logic

IEC 61131-3 defines five languages: *Ladder Diagram* (LD), *Function Block Diagram* (FBD), *Structured Text* (ST), *Instruction List* (IL, deprecated) and *Sequential Function Chart* (SFC). Ladder logic, derived from relay diagrams, remains the most widely used (Fig. 4.5):

- *Normally open (NO) contact* $-|\,|-$: true when the input bit is 1.
- *Normally closed (NC) contact* $-|/|-$: true when the bit is 0. (Stop buttons are wired NC and examined so that a broken wire stops the machine – fail-safe.)
- *Coil* $-(\,)-$: sets an output or internal bit when the rung is true. Series contacts form AND, parallel branches form OR.
- *Seal-in (latching) circuit*: the output's own contact in parallel with the START button keeps the motor running after START is released, until STOP opens the rung (rung 1). Boolean form: $\text{MOTOR} = (\text{START} + \text{MOTOR}) \cdot \overline{\text{STOP}} \cdot \overline{\text{OL}}$.
- *Timers*: TON (on-delay) sets its output when its input has been true for the preset time PT (rung 2); TOF (off-delay); TP (pulse). A timer with time base 0.1 s and preset 5 s counts 50 ticks.
- *Counters*: CTU counts rising edges up to the preset PV (rung 3, e.g. 12 parts per pallet); CTD counts down.

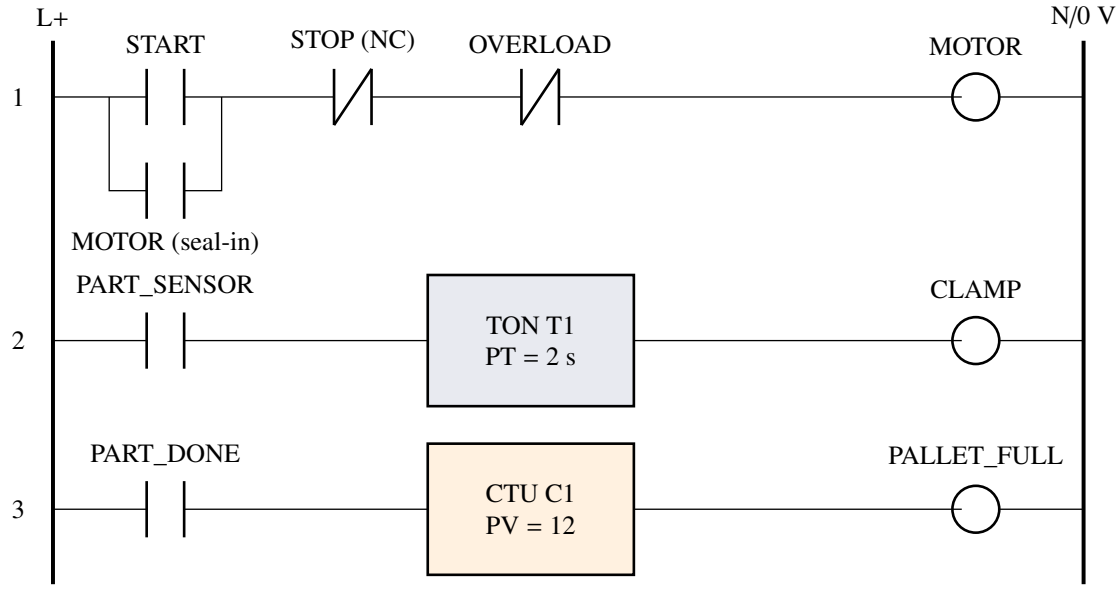


Figure 4.5: Ladder logic: (1) start/stop seal-in circuit with overload, (2) on-delay timer, (3) up-counter.

Algorithm 4.1 PLC-based sequence control of a robot machine-tending cell

Require: Inputs: PartPresent, RobotHome, RobotDone, DoorOpen, Cycle_Start, E_Stop_OK
Ensure: Outputs: RobotStart(job), DoorOpenCmd, ClampCmd, MachineStart

```
state ← IDLE
loop                                                    ▷ executed every scan
    if not E_Stop_OK then
        all outputs OFF; state ← FAULT
    end if
    if state = IDLE and Cycle_Start and PartPresent and RobotHome then
        DoorOpenCmd ← 1; state ← OPEN
    end if
    if state = OPEN and DoorOpen then
        RobotStart(LOAD); state ← LOADING
    end if
    if state = LOADING and RobotDone and RobotHome then
        ClampCmd ← 1; DoorOpenCmd ← 0; state ← MACHINING
    end if
    if state = MACHINING and door closed then
        MachineStart ← 1 (pulse); start TON(cycle time)
    end if
    if state = MACHINING and machining complete then
        ClampCmd ← 0; DoorOpenCmd ← 1; RobotStart(UNLOAD); state ← UNLOADING
    end if
    if state = UNLOADING and RobotDone then
        count part (CTU); state ← IDLE
    end if
end loop
```

The state-machine structure of Algorithm 4.1 is the standard way of writing sequences in a PLC (in ST, or graphically in SFC): each state waits for its transition condition, which always includes the relevant interlock.

Table 4.2: PLC versus microcontroller versus industrial PC.

	PLC	**Microcontroller**	**Industrial PC**
Environment	Rugged, certified, DIN rail	Board level; needs enclosure	Rugged PC hardware
I/O	Isolated 24 V/4–20 mA modules	Logic-level pins (3.3/5 V)	Via fieldbus or cards
Programming	IEC 61131-3 (ladder, ST)	C/C++, assembly	Any language, PLC runtime
Determinism	Guaranteed scan	Excellent (bare metal/RTOS)	Needs real-time OS
Cost per unit	High	Very low	High
Typical use	Machine and cell control	Embedded products, drives, sensors	HMI, vision, complex control

4.4 Industrial Robots

An industrial robot system (Unit I) comprises the manipulator, controller, teach pendant and end-effector. From the automation viewpoint the important features are:

- **Specifications**: payload, reach, number of axes, repeatability (ISO 9283), maximum speeds, protection class (IP rating), mounting (floor, wall, ceiling).
- **Controller functions**: trajectory generation (Unit III), servo control, program storage and interpretation, I/O and fieldbus interface to the PLC, safety functions (safe zones, safe speed, safe torque off), and options such as conveyor tracking, vision interface and force control.
- **Types by application**: articulated six-axis (welding, handling), SCARA (assembly), Delta (picking), Cartesian/gantry (large areas), palletising four-axis, *collaborative robots (cobots)* with power and force limiting that may work beside people after risk assessment.

Table 4.3: Industrial versus collaborative robots.

	Conventional industrial robot	**Collaborative robot (cobot)**
Safeguarding	Fences, interlocks, light curtains	Power and force limiting, speed and separation monitoring
Payload / speed	High	Low to medium
Programming	Pendant, offline	Hand guiding, graphical apps, pendant
Sensing	Encoders; optional external sensors	Joint torque sensors or current monitoring, collision detection
Typical use	High-volume welding, painting, handling	Small-batch assembly, machine tending, inspection

4.5 Robot Programming

4.5.1 Programming methods

- **Lead-through (walk-through)**: the operator physically guides the arm (or a lightweight replica) while the controller records the path at a fixed rate; used for spray painting and with cobots (hand guiding).
- **Teach-pendant (online) programming**: the robot is jogged to each required position, which is recorded, and instructions are entered on the pendant. Simple and accurate (uses repeatability) but occupies the robot and the production line.
- **Offline programming (OLP)**: programs are written and simulated on a computer using CAD models of the robot and cell, then downloaded. The line keeps running; paths can be generated from CAD geometry; reach, collisions and cycle time are verified virtually. It relies on robot absolute accuracy and cell calibration, so key points are usually touched up on the real robot.

- **Programming by demonstration and AI**: the robot learns from human demonstration or from data; an active research area entering industry.

4.5.2 Elements of a robot language

Robot languages (RAPID, KRL, KAREL/TP, INFORM, URScript) share common elements:

- **Motion instructions**: joint interpolation (MOVJ/PTP/MoveJ) – fastest, path not controlled; linear interpolation (MOVL/LIN/MoveL) – straight TCP path at programmed speed; circular (MOVC/CIRC/MoveC) through three points. Each specifies target, speed and a *zone*/blending radius (fine stop or fly-by), as derived in Unit III (Table 3.4).
- **Position data**: points (pose + configuration) taught or computed, in a selected tool and user frame; offsets for patterns (e.g. palletising grids).
- **I/O and synchronisation**: set/reset digital outputs (gripper), wait for inputs (part present), handshakes with the PLC.
- **Program flow**: IF, WHILE, FOR, CALL subroutines, registers, error handling and interrupts.

Example program (generic syntax) for a pick-and-place:

```
TOOL = GRIPPER ; UFRAME = CONVEYOR
MOVJ HOME V=50%
LOOP:
  WAIT DI[1]=ON ! part present at pick station
  MOVJ P_PICK + OFFSET(0,0,100) V=80% Z=50 ! approach 100 mm above
  MOVL P_PICK V=300mm/s FINE
  SET DO[1] ! close gripper
  WAIT DI[5]=ON TIMEOUT 1.0s ! gripper closed sensor
  MOVL P_PICK + OFFSET(0,0,100) V=500mm/s Z=20
  MOVJ P_PLACE + OFFSET(0,0,100) V=100% Z=50
  MOVL P_PLACE V=300mm/s FINE
  RESET DO[1] ! open gripper
  MOVL P_PLACE + OFFSET(0,0,100) V=500mm/s Z=20
  PULSE DO[3] 0.2s ! report "part placed" to PLC
  GOTO LOOP
```

Approach and depart points 100 mm above each target use fast joint or fly-by motions in free space, while the final approach is a slow linear move with a fine stop – the standard structure of every pick-and-place program.

Figure 4.6 summarises the development workflow, from task analysis and reach study through frame definition, teaching and testing at reduced speed to automatic operation.

4.6 Teach Pendant Concepts

The teach pendant is the hand-held operator interface of the robot (Fig. 4.7). Its essential elements are:

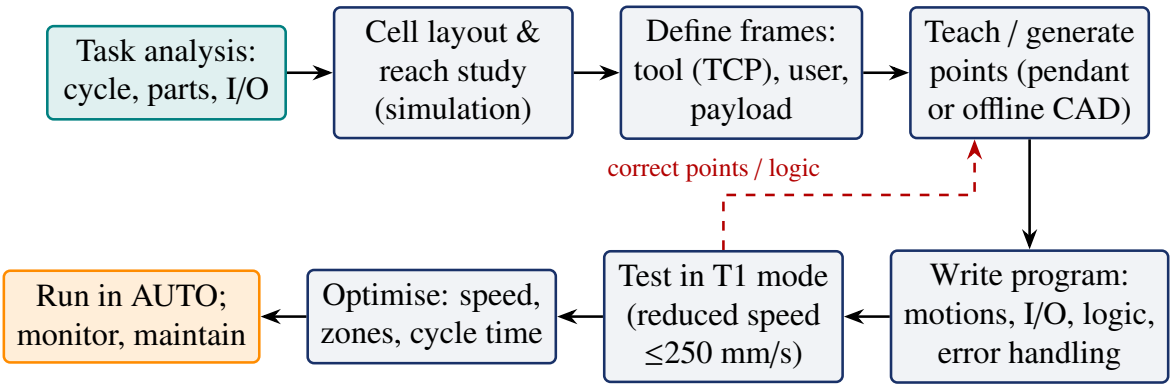


Figure 4.6: Workflow for developing an industrial robot program.

- **Display/touch screen** for programs, positions, I/O status, alarms and settings.
- **Jog keys** (or a 6-D joystick) to move the robot in a selected *coordinate system*: joint, world/base, tool or user frame (Unit I, Fig. 1.8), with selectable jog speed and incremental (step) mode.
- **Three-position enabling switch**: motion is possible only while the switch is held in the middle position; releasing it *or* squeezing it fully (a panic reaction) stops the robot. This is a key safety device required by ISO 10218-1.
- **Emergency stop** button (category 0 or 1 stop).
- **Mode selector** (key switch): *T1* – manual reduced speed, TCP speed limited to 250 mm/s, used for teaching inside the cell; *T2* – manual full speed for program verification; *AUTO* – automatic production with all safeguards active and nobody inside.
- **Program keys** to record positions (touch-up), insert instructions, step forward/backward through a program and override speed.

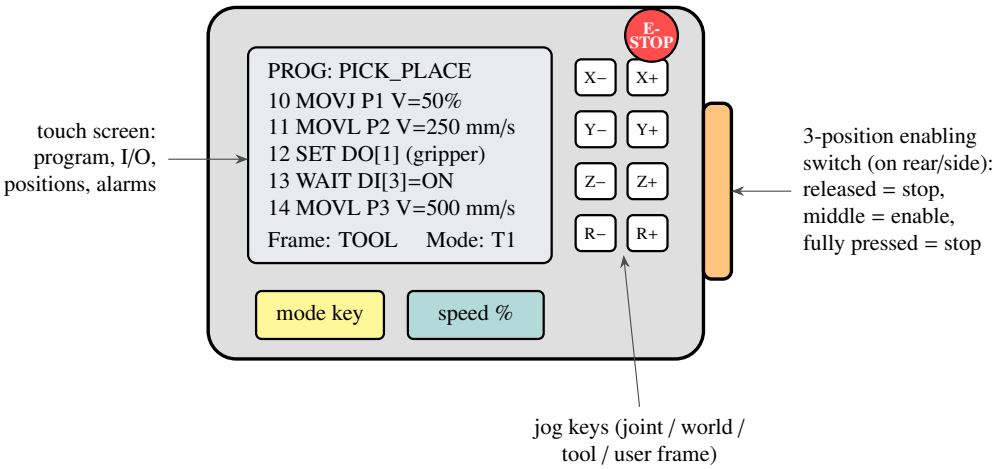


Figure 4.7: Teach pendant: display, jog keys, emergency stop, mode key and three-position enabling switch.

Teaching procedure. Select T1 mode and the tool/user frames; hold the enabling switch; jog to the point (joint mode for coarse moves, Cartesian tool mode for fine approach); record the

point; insert the motion instruction with speed and zone; repeat; step through the program at low speed to verify; then test at higher speed and in AUTO.

4.7 Embedded Systems

4.7.1 Definition and characteristics

An *embedded system* is a computer system designed to perform a dedicated function within a larger mechanical or electrical system, often under real-time constraints. Robots are full of them: every servo drive, encoder interface, gripper controller, force sensor, vision camera and mobile-robot base contains one or more embedded processors. Characteristic features are: dedicated function; real-time operation (correct results *on time*); tight constraints on cost, size and power; high reliability and long unattended operation (watchdog timers); and direct interaction with sensors and actuators.

4.7.2 Architecture

Figure 4.8 shows the general architecture. A microcontroller or processor runs the firmware; sensors are connected through signal conditioning to inputs (digital pins, ADC, counters); actuators are driven through power stages; memory holds code and data; communication interfaces link the system to other devices; a power supply with regulators and supervisory circuits keeps it running safely. Software may be a simple *super-loop* with interrupts, or a *real-time operating system* (RTOS) scheduling prioritised tasks with guaranteed response times – for example a 10 kHz current-control task, a 1 kHz trajectory task and a low-priority communication task.

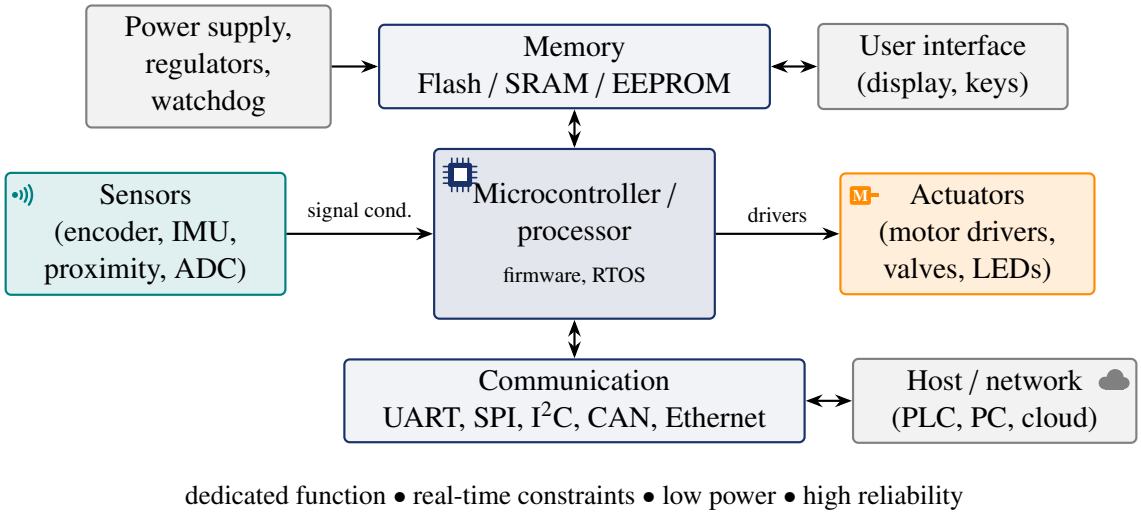


Figure 4.8: Architecture of an embedded system for robotic sensing and actuation.

Real-time classes. *Hard* real-time: a missed deadline is a failure (motor current loop, safety stop). *Soft* real-time: occasional misses degrade performance (display update). The worst-case execution time of each task must be known and the processor load kept below a safe margin.

4.8 Microcontrollers

4.8.1 Microcontroller versus microprocessor

A *microprocessor* is a CPU on a chip; memory and peripherals are external (PCs, high-end controllers). A *microcontroller* (MCU) integrates CPU, program memory (Flash), data memory (SRAM), clock, timers, ADC, communication interfaces and I/O ports on one chip – a complete computer for control tasks at low cost and power [14].

4.8.2 Internal architecture

Figure 4.9 shows a typical 32-bit MCU (ARM Cortex-M class):

- **CPU core** – registers, ALU, often with hardware multiply/divide and floating point; Harvard architecture (separate instruction and data buses) for speed.
- **Memory** – Flash (program, non-volatile), SRAM (variables, stack), EEPROM (parameters).
- **Clock system** – crystal oscillator, PLL multiplying to tens or hundreds of MHz; reset and brown-out circuits; *watchdog timer* that resets the MCU if firmware stops responding.
- **Interrupt controller** (NVIC) and **DMA** – respond to events within microseconds and move data without CPU load.
- **Peripherals** – GPIO; timers/counters (PWM, input capture, quadrature-encoder mode); ADC (10–16 bit) and DAC; UART, SPI, I^2C, CAN, USB, Ethernet.

Common families: 8-bit AVR (Arduino Uno, ATmega328P), PIC; 32-bit ARM Cortex-M (STM32, used widely in drives and robots), ESP32 (with Wi-Fi), and DSP-oriented motor-control MCUs.

4.8.3 Programming model

Firmware is written in C/C++ (or MicroPython), compiled and flashed. Peripherals are controlled by writing *configuration registers*; events are handled by *interrupt service routines*. A typical control firmware initialises clocks and peripherals, then runs a timer interrupt at the control rate which reads sensors, computes the control law and updates the PWM – the structure of Algorithm 2.1.

4.9 Sensor Interfacing

4.9.1 Digital inputs

Switches, proximity sensors and encoders produce logic-level signals. Points to observe (Fig. 4.10(a)):

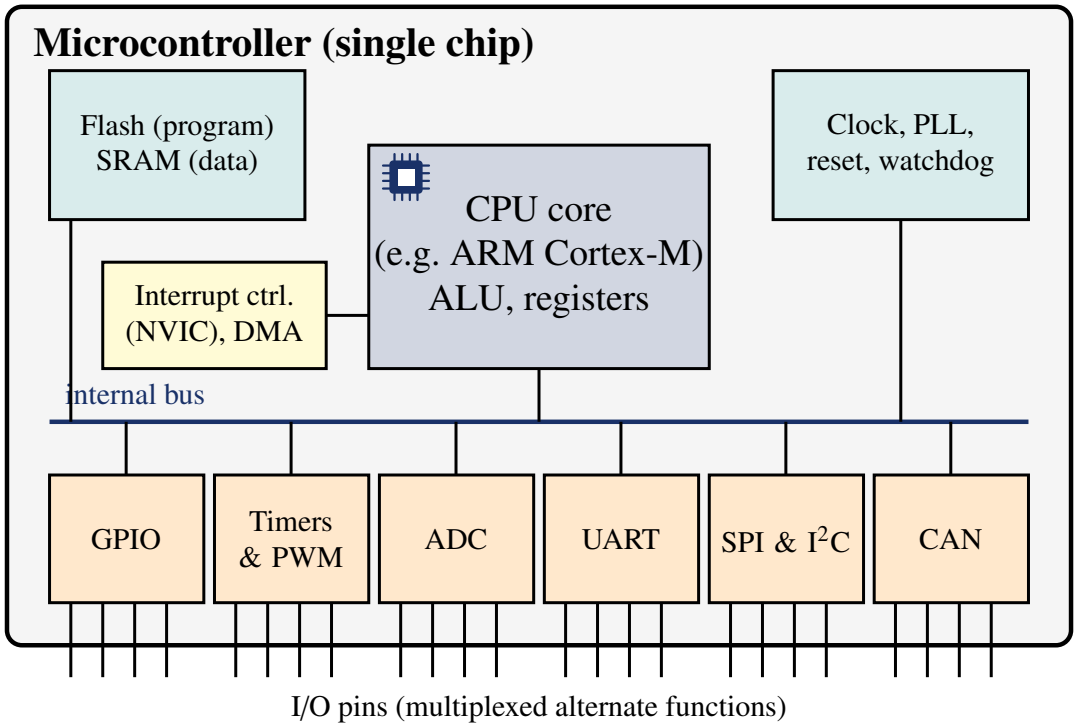


Figure 4.9: Internal architecture of a microcontroller.

- **Pull-up/pull-down resistors** define the input level when a switch or open-collector sensor is open.
- **Debouncing**: mechanical contacts bounce for 1–10 ms; an RC filter ($\tau = R_f C_f$) followed by a Schmitt-trigger input, or a software debounce (accept a change only after it is stable for several samples), removes false transitions.
- **Level compatibility**: 24 V industrial sensors (PNP/NPN) must be converted to 3.3/5 V logic with optocouplers or dividers; optocouplers also provide isolation.
- **Fast signals** (encoders) use timer input-capture or quadrature-encoder peripherals rather than polling.

4.9.2 Analogue inputs

Analogue sensors are connected to the ADC through the conditioning of Unit II. For a V_{ref}, N-bit ADC the code for input v is

$$D = \text{round}\left(\frac{v}{V_{\text{ref}}}\,(2^N - 1)\right), \qquad v = \frac{D}{2^N - 1} V_{\text{ref}} \tag{4.3}$$

(conventions differ slightly between 2^N and $2^N - 1$). Signals larger than the ADC range are scaled by a divider $v_o = v_i R_2/(R_1 + R_2)$ (Fig. 4.10(b)); the divider's Thévenin resistance must be low enough for the ADC's sampling capacitor to charge within the sampling time.

Example 4.2: Temperature sensor interfacing (Topic 4.9)

Given: An LM35-type sensor (10 mV/°C) feeds a 12-bit ADC with $V_{\text{ref}} = 3.3$ V. Temperature 37.5 °C. A second, 0–5 V sensor must be read by the same MCU.

Solution: $v = 0.375$ V; $D = \text{round}(0.375/3.3 \times 4095) = 465$. Resolution $= 3.3/4095/0.01 = 0.081$ °C per LSB. For the 5 V sensor use $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$: $v_o = 5 \times 20/30 = 3.33$ V at full scale – just at the limit; choose $R_2 = 18\,\text{k}\Omega$ (3.21 V) for margin.

Digital sensors (IMUs, digital temperature sensors, absolute encoders) communicate over I^2C, SPI or serial protocols (Section 4.12), delivering calibrated values directly.

4.10 Actuator Interfacing

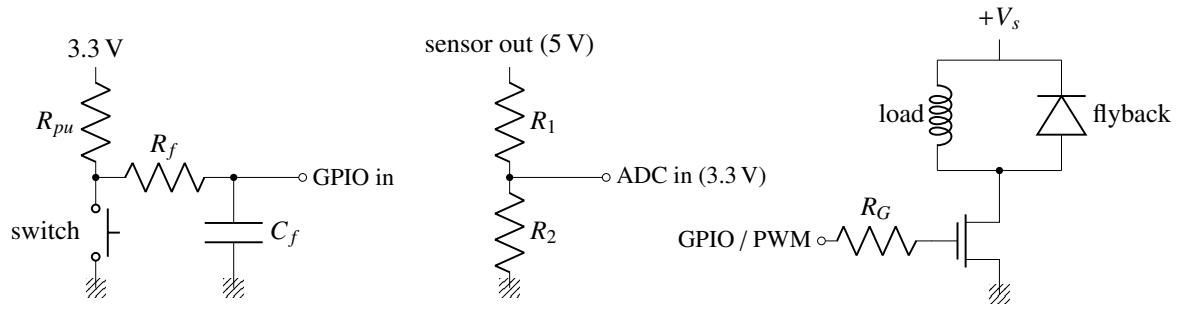
MCU pins supply only a few milliamperes at logic voltage, so actuators require driver stages:

- **Transistor/MOSFET switch** (Fig. 4.10(c)): a logic-level N-MOSFET as a low-side switch drives solenoids, relays, LEDs and small DC motors. For a bipolar transistor the base resistor is chosen for saturation: $R_B = (V_{\text{OH}} - V_{BE})/I_B$ with $I_B \approx (2\text{–}5)\, I_C/h_{FE}$.
- **Flyback diode**: an inductive load produces a large voltage spike $L\,di/dt$ when switched off; a diode across the load provides a path for the current and protects the transistor.
- **Relays and contactors** switch mains-powered loads with isolation; solid-state relays for frequent switching.
- **H-bridge drivers** (four switches) drive DC motors in both directions with PWM speed control (Unit II); integrated drivers (e.g. dual H-bridges) include current sensing and protection.
- **Stepper drivers** generate microstepping currents from STEP/DIR pulses; **servo drives** accept PWM/analogue/fieldbus commands; **RC servos** take a 1–2 ms pulse.
- **Isolation** with optocouplers or digital isolators separates the noisy power side from the logic side.

Example 4.3: Relay driver design (Topic 4.10)

Given: A 5 V MCU output drives a relay coil of 60 mA through an NPN transistor with $h_{FE} = 100$, $V_{BE} = 0.7$ V; overdrive factor 5.

Solution: $I_B = 5 \times 60/100 = 3$ mA; $R_B = (5 - 0.7)/0.003 = 1.43\,\text{k}\Omega$ (use 1.2 kΩ–1.5 kΩ). A 1N4148 flyback diode across the coil is mandatory.



(a) Pull-up + RC debounce $\tau = R_f C_f$

(b) Level shifting by divider $v_o = v_i\, R_2/(R_1 + R_2)$

(c) Low-side MOSFET driver with flyback diode

Figure 4.10: Interfacing circuits: (a) switch with pull-up and RC debounce; (b) level shifting with a voltage divider; (c) low-side MOSFET driver with flyback diode.

4.11 PWM Generation and Control

4.11.1 Principle

Pulse-width modulation switches a supply fully on and off at a fixed frequency $f_{\mathrm{PWM}} = 1/T$; the *duty cycle* $D = t_{\mathrm{on}}/T$ determines the average voltage delivered to a load whose time constant is much longer than T:

$$V_{\mathrm{avg}} = \frac{1}{T}\int_0^{T} v(t)\,dt = D\,V_{\mathrm{dc}}. \tag{4.4}$$

Because the switch is either fully on (small voltage drop) or fully off (no current), losses are very small – the reason PWM is used in every motor drive, LED dimmer and switching regulator.

4.11.2 Generation by a timer

A hardware timer counts clock pulses from 0 to an auto-reload value ARR and restarts (sawtooth, Fig. 4.11); the output is high while the counter is below the compare value CCR. Hence

$$f_{\mathrm{PWM}} = \frac{f_{\mathrm{clk}}}{(\mathrm{PSC}+1)(\mathrm{ARR}+1)}, \qquad D = \frac{\mathrm{CCR}}{\mathrm{ARR}+1}, \qquad \text{resolution} = \log_2(\mathrm{ARR}+1)\ \text{bits}, \tag{4.5}$$

where PSC is the prescaler. There is a trade-off: higher PWM frequency means fewer counts per period and lower duty-cycle resolution.

Example 4.4: PWM timer configuration (Topic 4.11)

Given: $f_{\mathrm{clk}} = 72\,\mathrm{MHz}$. (a) 1 kHz PWM at 30% duty. (b) 20 kHz motor PWM. (c) Average voltage at 30% from 12 V.

Solution: (a) PSC = 71 gives a 1 MHz timer clock; ARR = 999 gives 1 kHz; CCR = 300; resolution $\log_2 1000 \approx 10$ bits. (b) PSC = 0: ARR = $72\times10^6/20\times10^3 - 1 = 3599$; resolution

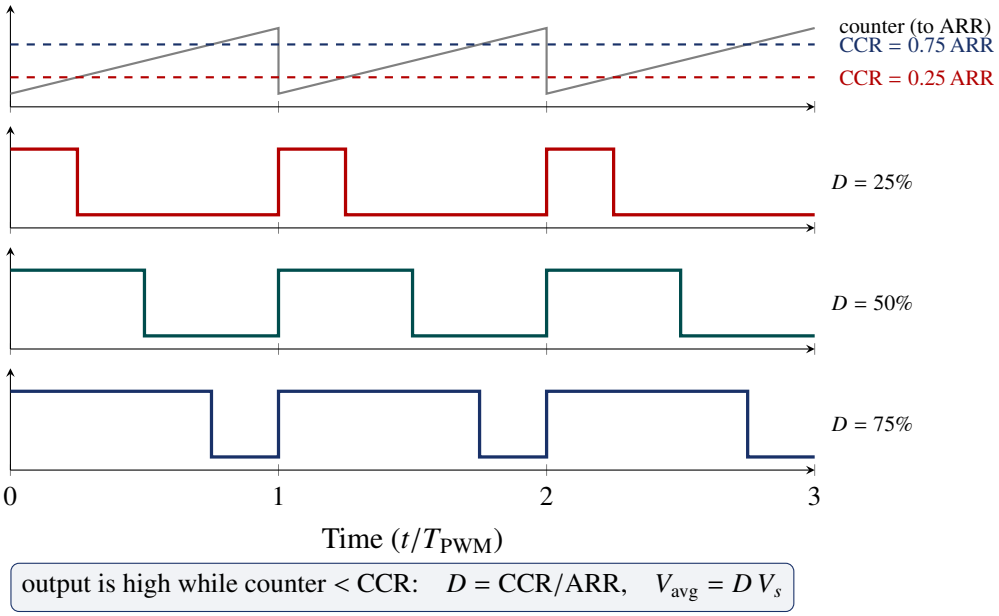


Figure 4.11: PWM generation: timer counter compared with two compare levels (top) and output waveforms at 25%, 50% and 75% duty cycle.

≈ 11.8 bits; inaudible frequency for motors. (c) $V_{\text{avg}} = 0.3 \times 12 = 3.6\,\text{V}$.

4.11.3 PWM applications and choice of frequency

- **DC motor speed control** via an H-bridge: f_{PWM} 10 kHz to 40 kHz (above audibility, well above $1/\tau_e$ so current ripple is small).
- **RC servo position**: 50 Hz frame, 1–2 ms pulse (Eq. (2.12)).
- **Three-phase inverters** for BLDC/PMSM: centre-aligned PWM with dead time between complementary switches to avoid shoot-through.
- **Solenoid valves and heaters**: PWM for proportional force or power; **LEDs**: brightness.
- **DAC substitute**: PWM followed by an RC low-pass gives an analogue voltage.

Algorithm 4.2 PWM generation and closed-loop motor speed control on a microcontroller

Require: f_{clk}, desired f_{PWM}; speed reference ω_r; PI gains K_p, K_i; control period T_c
Ensure: Duty cycle updated every T_c

choose PSC; ARR $\leftarrow f_{\text{clk}}/((\text{PSC}+1)f_{\text{PWM}}) - 1$; enable timer in PWM mode
configure encoder timer in quadrature mode; start control interrupt at $1/T_c$
loop ▷ interrupt every T_c
    $\omega \leftarrow \Delta\text{count} \cdot 2\pi/(4NT_c)$ ▷ M-method speed
    $e \leftarrow \omega_r - \omega$; $I \leftarrow I + K_i T_c e$; $u \leftarrow K_p e + I$ (with anti-windup)
    $D \leftarrow \text{clip}(u/V_{\text{dc}}, -1, 1)$; set direction pins from sign of D
    CCR $\leftarrow |D|\,(\text{ARR}+1)$
end loop

4.12 Communication Interfaces

4.12.1 Serial communication fundamentals

Robotic and embedded devices exchange data serially – one bit after another – to minimise wiring. Interfaces are classified as *synchronous* (a clock line shared: SPI, I^{2}C) or *asynchronous* (each side times bits from its own clock: UART, CAN); *point-to-point* or *bus*; *single-ended* or *differential* (noise-immune, long distance) (Fig. 4.12).

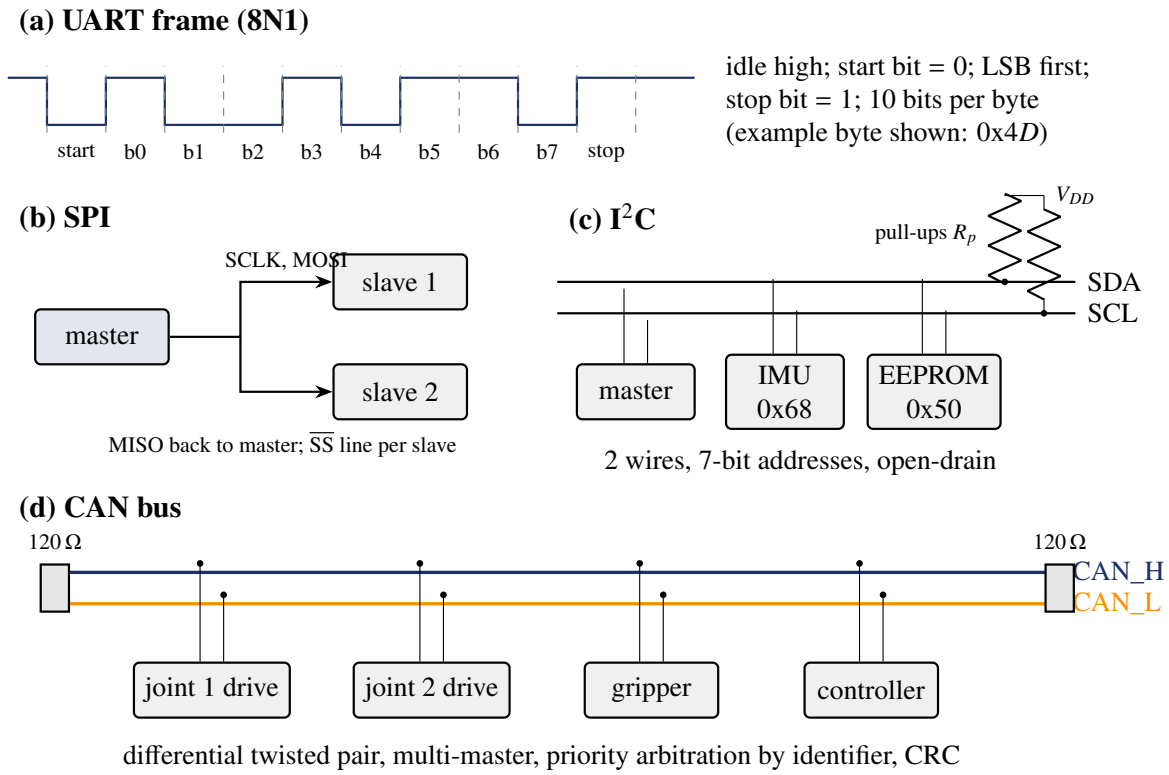


Figure 4.12: Communication interfaces: (a) UART frame, (b) SPI with chip selects, (c) I^{2}C two-wire bus, (d) CAN bus with termination.

4.12.2 UART

Universal Asynchronous Receiver/Transmitter: two data lines (TX, RX) and ground. A frame (Fig. 4.12(a)) has a start bit (0), 5–9 data bits (LSB first), optional parity and stop bit(s) (1); both sides must use the same *baud rate* (bits/s). With the common 8N1 format, 10 bits carry one byte:

$$\text{throughput} = \frac{\text{baud rate}}{\text{bits per frame}} \ \text{bytes/s}. \tag{4.6}$$

At 115 200 baud, 11 520 bytes/s; a 100-byte message takes 8.68 ms. Physical layers: TTL (short), RS-232 (±12 V, up to ~15 m), RS-485 (differential, multi-drop, up to 1200 m; used by Modbus RTU).

4.12.3 SPI

Serial Peripheral Interface: full-duplex synchronous bus with four lines – SCLK, MOSI (master out), MISO (master in) and a separate slave-select $\overline{\text{SS}}$ per device (Fig. 4.12(b)). Clock rates of 1–50 MHz make it ideal for fast peripherals on a board: ADCs, DACs, absolute encoders, displays, flash memory. A 16-bit transfer at 10 MHz takes 1.6 µs. Disadvantages: more wires per device, no acknowledgement, short distances.

4.12.4 I^2C

Inter-Integrated Circuit: two open-drain lines, SDA (data) and SCL (clock), with pull-up resistors; up to 127 devices with 7-bit addresses (Fig. 4.12(c)). Each byte is followed by an acknowledge bit (9 clock cycles per byte); transactions begin with START and address. Standard 100 kHz, fast 400 kHz, fast-plus 1 MHz. Reading 6 bytes of IMU data (address, register, repeated start, address, 6 data bytes ≈ 9 bytes × 9 bits + start/stop) at 400 kHz takes about 0.21 ms. Used for IMUs, temperature sensors, EEPROMs and port expanders on a board.

4.12.5 CAN

Controller Area Network (ISO 11898): a differential two-wire bus (CAN_H, CAN_L) terminated by 120 Ω at both ends (Fig. 4.12(d)). Messages carry an identifier rather than an address; when several nodes transmit, non-destructive bitwise *arbitration* lets the lowest identifier (highest priority) win. CRC, acknowledgement and automatic retransmission give high reliability. Classic CAN runs at up to 1 Mbit/s with 8 data bytes per frame (CAN FD: up to 64 bytes and faster data phase). A standard frame with 8 bytes is about 111 bits (before bit stuffing), i.e. 111 µs at 1 Mbit/s – about 9000 frames/s. CANopen over CAN is widely used for robot joint drives, mobile robots and grippers.

4.12.6 Industrial networks

- **Fieldbuses**: PROFIBUS, DeviceNet, Modbus RTU (RS-485).
- **Industrial Ethernet**: EtherNet/IP, PROFINET, Modbus TCP, and **EtherCAT**, which processes frames "on the fly" in each slave and achieves cycle times below 100 µs with distributed clocks – the dominant choice for multi-axis motion and robot joints.
- **IO-Link**: point-to-point digital link to smart sensors.
- **OPC UA, MQTT**: vendor-independent information exchange with MES and cloud (Industry 4.0).

4.13 Machine Vision

Table 4.4: Comparison of communication interfaces.

Interface	Wires	Speed	Distance	Typical robotic use
UART/RS-232	2 + GND	to 115 kbit/s+	~15 m	debugging, GPS, simple devices
RS-485	2 (diff.)	to 10 Mbit/s	1200 m	Modbus RTU, drives
SPI	3 + 1/device	1–50 MHz	on board	encoders, ADC, displays
I^2C	2	0.1–1 MHz	on board	IMU, sensors, EEPROM
CAN / CAN FD	2 (diff.)	1 / 5+ Mbit/s	40 m at 1 Mbit/s	joint drives, mobile robots, grippers
EtherCAT	Ethernet	100 Mbit/s	100 m/segment	multi-axis robot motion
EtherNet/IP, PROFINET	Ethernet	100 Mbit/s–1 Gbit/s	100 m/segment	PLC–robot–cell I/O

4.13.1 Definition and role

Machine vision is the acquisition and automatic analysis of images to extract information for controlling a process – locating parts for robots, inspecting quality, measuring dimensions, reading codes. It differs from general computer vision in its emphasis on controlled conditions (fixed lighting, known parts) and on speed, robustness and integration with automation [22,23]. Vision turns a robot from a playback machine into a sensor-guided one: parts need not be precisely fixtured, and every part can be inspected.

4.13.2 System architecture

A machine-vision system (Fig. 4.13) comprises:

1. **Illumination** designed to make the features of interest high in contrast (Section 4.16);
2. **Optics** (lens) forming an image of the scene on the sensor (Section 4.14);
3. **Camera** with image sensor converting light into digital pixel values (Section 4.15);
4. **Triggering** – a part-present sensor or encoder triggers acquisition at the right instant;
5. **Interface** (GigE Vision, USB3 Vision, Camera Link) or a smart camera with built-in processor;
6. **Processing software** executing the image-processing pipeline (Sections 4.17 to 4.19);
7. **Communication of results** – pass/fail to a PLC, coordinates (x, y, θ) to a robot, data to MES.

4.14 Image Acquisition

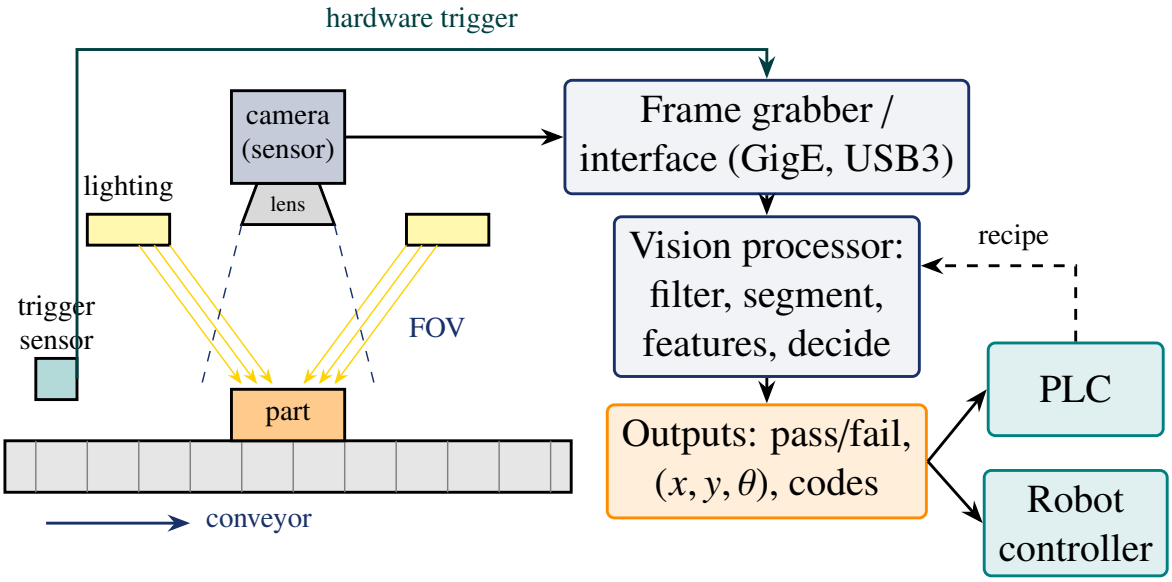


Figure 4.13: Machine-vision system architecture: lighting, lens and camera over a conveyor, triggered acquisition, processing and outputs to PLC and robot.

4.14.1 Image formation and the digital image

A lens projects the scene onto a 2-D array of photosites (pixels). Each pixel integrates the light falling on it during the exposure time and the resulting charge is digitised, typically to 8–12 bits. A digital grey-scale image is therefore a matrix $f(u, v)$ of M rows and N columns with integer values 0–255 (8 bit); a colour image has three such planes (R, G, B).

4.14.2 Optics – lens equation and field of view

For a thin lens of focal length f, object distance u and image distance v (Fig. 4.14):

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}, \qquad m = \frac{v}{u} = \frac{w_s}{W} \qquad \Rightarrow \qquad u = f\,\frac{1+m}{m}, \tag{4.7}$$

where m is the magnification, w_s the sensor width and W the field-of-view (FOV) width. The third relation follows by substituting $v = mu$ into the lens equation. The *spatial resolution* (object size per pixel) is

$$r_s = \frac{W}{N_{\text{px}}} \quad \text{(mm/pixel)}, \tag{4.8}$$

and a feature should span at least 2–3 pixels to be detected reliably (and gauging tolerances should be at least about 10 times r_s unless sub-pixel methods are used). *Depth of field* increases with larger f-number (smaller aperture) at the cost of light; telecentric lenses eliminate perspective error for precise gauging.

Example 4.5: Selecting camera and lens (Topic 4.14)

Given: FOV width 256 mm; smallest defect 0.5 mm; candidate camera 1280 × 1024 pixels,

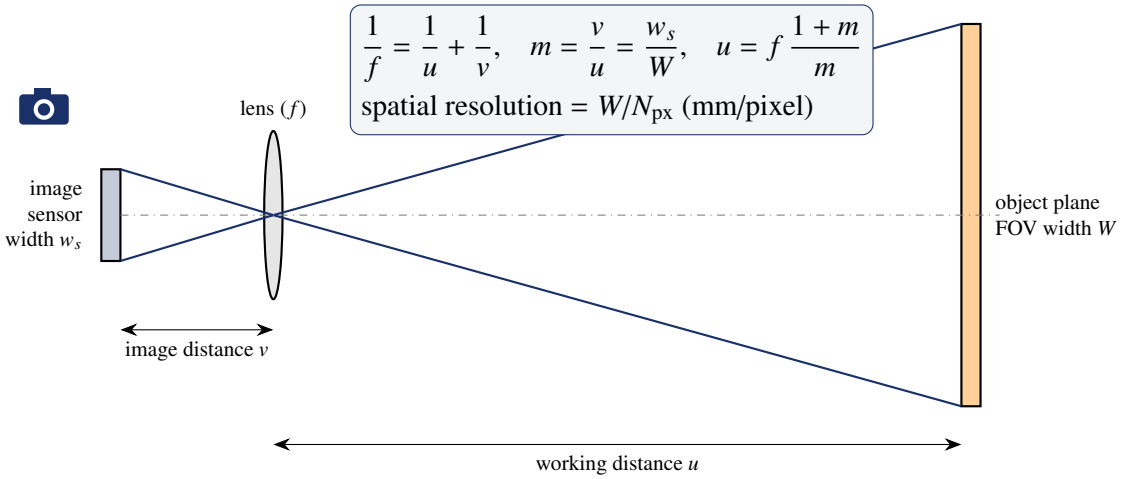


Figure 4.14: Camera–lens geometry: sensor, lens, working distance and field of view.

pixel size 5 µm; lens $f = 16$ mm; 30 frames/s, 8-bit.

Solution: Resolution $r_s = 256/1280 = 0.2$ mm/pixel; defect spans $0.5/0.2 = 2.5$ pixels – just adequate. Sensor width $= 1280 \times 5\,\mu\text{m} = 6.4$ mm; $m = 6.4/256 = 0.025$; working distance $u = 16(1.025)/0.025 = 656$ mm. Raw data rate $= 1280 \times 1024 \times 8 \times 30 = 315$ Mbit/s – within GigE Vision (1 Gbit/s).

Interpretation: For more margin on the defect, use a 5-megapixel camera or reduce the FOV.

4.14.3 Acquisition parameters

Exposure time must be short enough to freeze motion: blur in pixels $= v\,t_{\text{exp}}/r_s$ (e.g. at 0.5 m/s and 0.2 mm/pixel a 100 µs exposure gives 0.25 pixel blur); strobed lighting achieves very short effective exposures. *Gain* amplifies the signal (and noise); *triggering* synchronises the image with the part position; *region of interest* (ROI) readout increases frame rate.

4.15 Cameras

- **Sensor technology**: *CCD* – charge transferred to a single output amplifier; high uniformity, now largely obsolete; *CMOS* – amplifier and ADC per pixel/column; fast, low power, low cost; the standard today.
- **Shutter**: *global shutter* exposes all pixels simultaneously – essential for moving parts; *rolling shutter* exposes row by row – cheaper but distorts moving objects.
- **Area-scan vs line-scan**: area cameras capture 2-D frames; line-scan cameras capture one line at a time while the object moves past (web, cylinders, long parts), synchronised by an encoder.
- **Monochrome vs colour**: mono offers higher resolution and sensitivity; colour (Bayer filter) is needed for colour sorting and verification.

- **3-D cameras**: *stereo* (two cameras, depth from disparity $Z = fB/d$), *structured light* (projected patterns), *laser triangulation* (line profile), *time-of-flight* (phase of modulated light) – used for bin picking, depalletising and volume measurement.
- **Smart cameras** integrate sensor, processor and I/O; **PC-based systems** offer more power and flexibility.

Table 4.5: Camera technologies for robotic vision.

Type	Strengths	Typical use
Area-scan, global shutter	Freezes motion, simple	Pick-and-place, inspection
Line-scan	Very high resolution along motion	Web, sheet, cylindrical parts
Colour (Bayer)	Colour information	Sorting, label/colour checks
Stereo 3-D	Passive, outdoor	Mobile robots, depalletising
Structured light / laser	Accurate dense depth	Bin picking, weld-seam tracking
Time-of-flight	Fast, compact depth	Logistics, volume, obstacle detection
Smart camera	Compact, integrated I/O	Simple inspection stations

4.16 Lighting Techniques

Lighting is usually the most important factor in a vision application: good lighting maximises contrast of the features of interest and suppresses everything else, making software simpler and more robust. The main techniques (Fig. 4.15):

(a) **Backlighting**: light behind the object; the camera sees a high-contrast silhouette – ideal for dimensional gauging, hole presence and part location of flat parts.

(b) **Diffuse (dome/cloudy-day) lighting**: uniform light from all directions eliminates glare and shadows on shiny, curved surfaces.

(c) **Ring light / direct front lighting**: general-purpose illumination mounted around the lens.

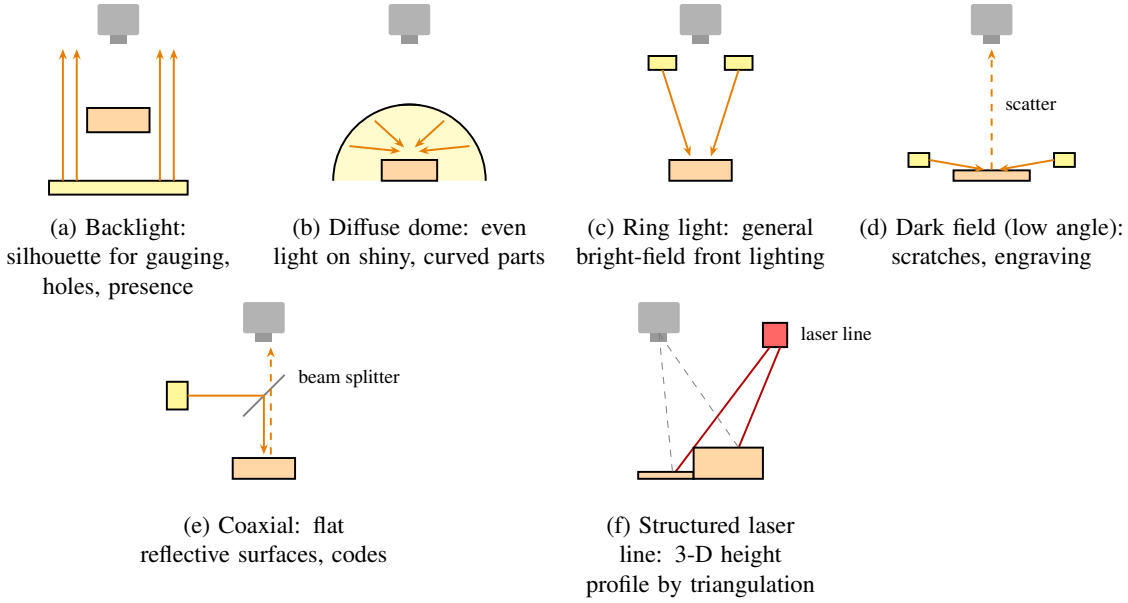
(d) **Dark-field (low-angle) lighting**: light grazes the surface; flat areas reflect away from the camera and appear dark, while scratches, edges, engraving and dust scatter light into the camera and appear bright.

(e) **Coaxial (on-axis) lighting**: light introduced along the optical axis through a beam splitter; flat reflective surfaces appear bright and defects or marks dark – used for codes on metal and wafers.

(f) **Structured light**: a laser line or pattern projected at an angle; its displacement in the image encodes height (triangulation).

Wavelength matters too: red LEDs suit monochrome sensors and many parts; infrared penetrates some plastics; blue/UV enhance fluorescence. Polarisers suppress specular reflections,

and band-pass filters reject ambient light.



(a) Backlight: silhouette for gauging, holes, presence

(b) Diffuse dome: even light on shiny, curved parts

(c) Ring light: general bright-field front lighting

(d) Dark field (low angle): scratches, engraving

(e) Coaxial: flat reflective surfaces, codes

(f) Structured laser line: 3-D height profile by triangulation

Figure 4.15: Lighting techniques for machine vision.

4.17 Image Processing

4.17.1 The processing pipeline

Figure 4.16 shows the standard pipeline: acquisition, pre-processing (noise removal, enhancement, ROI), segmentation (separating objects from background), feature extraction, recognition/decision and output [22].

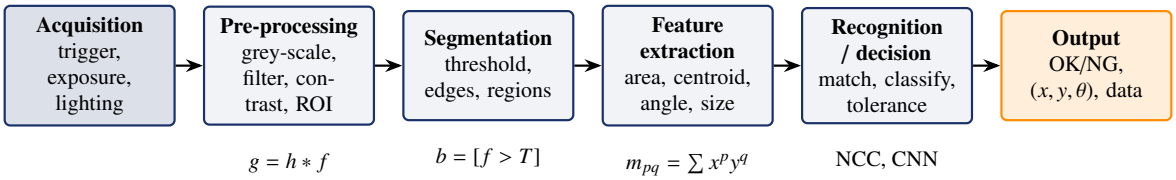


Figure 4.16: Image-processing pipeline of an industrial vision system.

4.17.2 Point operations and histogram

Point operations transform each pixel independently: brightness/contrast adjustment $g = af + b$, negative $g = 255 - f$, gamma correction, and *histogram equalisation*, which spreads grey levels to enhance contrast. The *histogram* $h(k)$ (number of pixels with grey level k) is the primary tool for choosing thresholds and checking exposure.

4.17.3 Spatial filtering (convolution)

Neighbourhood operations replace each pixel by a combination of its neighbours. Linear filtering is a 2-D *convolution* with a kernel (mask) h:

$$g(x,y) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} h(i,j)\, f(x-i,\, y-j). \tag{4.9}$$

- **Mean (box) filter** $h = \frac{1}{9}\mathbf{1}_{3\times 3}$ – smooths noise but blurs edges.
- **Gaussian filter** – weights fall off with distance; smooths with less ringing; used before edge detection.
- **Median filter** (non-linear) – replaces the pixel by the median of the window; removes impulse (salt-and-pepper) noise while preserving edges.

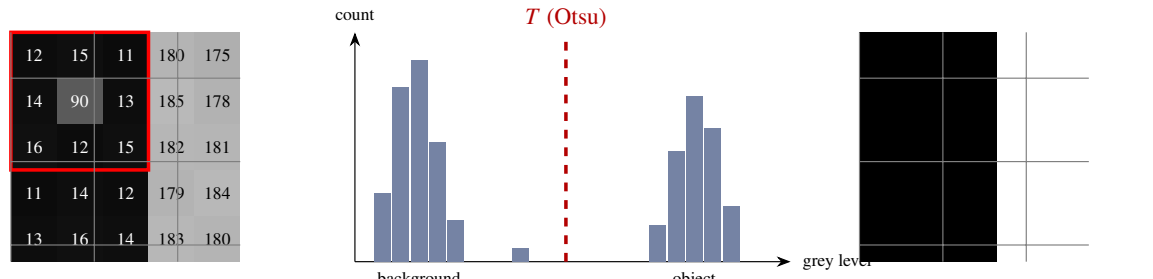
Example (Fig. 4.17(a)): the 3×3 window $\{12, 15, 11, 14, 90, 13, 16, 12, 15\}$ around the outlier 90 has mean 22 but median 14; the median filter removes the outlier completely, while the mean filter only spreads it.

4.17.4 Thresholding and segmentation

Global thresholding converts a grey image into a binary image:

$$b(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & \text{otherwise} \end{cases} \tag{4.10}$$

For a bimodal histogram (Fig. 4.17(b)) the threshold lies in the valley. **Otsu's method** chooses T automatically to maximise the between-class variance $\sigma_B^2(T) = \omega_0\omega_1(\mu_0 - \mu_1)^2$, where $\omega_{0,1}$ are the class probabilities and $\mu_{0,1}$ the class means. *Adaptive (local) thresholding* handles uneven illumination. Other segmentation methods are region growing, edge-based segmentation and colour segmentation.



(a) grey levels; outlier 90 (red: 3×3 window)

(b) bimodal histogram

(c) median filter + threshold $T = 100$: binary image (object = 1, white)

Figure 4.17: Image filtering and thresholding: (a) grey-level image with an impulse-noise pixel and a 3×3 window; (b) bimodal histogram and threshold; (c) binary image after median filtering and thresholding.

Morphological operations on binary images use a structuring element: *erosion* shrinks objects and removes small specks; *dilation* grows objects and fills small gaps; *opening* (erosion then

dilation) removes noise; *closing* (dilation then erosion) fills holes.

Algorithm 4.3 Image thresholding (Otsu) and binary segmentation

Require: Grey image f (L grey levels)
Ensure: Binary image b and threshold T^*
1: compute normalised histogram p_k, $k = 0..L-1$; global mean $\mu_T = \sum k p_k$
2: **for** $T = 0$ **to** $L-1$ **do**
3: $\omega_0 \leftarrow \sum_{k\le T} p_k$; $\mu(T) \leftarrow \sum_{k\le T} k p_k$
4: $\sigma_B^2(T) \leftarrow (\mu_T\omega_0 - \mu(T))^2/(\omega_0(1-\omega_0))$ ▷ between-class variance
5: **end for**
6: $T^* \leftarrow \arg\max_T \sigma_B^2(T)$
7: $b(x,y) \leftarrow [f(x,y) > T^*]$; apply opening to remove noise
8: **return** b, T^*

4.18 Feature Extraction

4.18.1 Edge detection

Edges – abrupt changes in intensity – carry most of an object's shape information. They are found from the image gradient $\nabla f = (\partial f/\partial x, \partial f/\partial y)$, approximated by convolution with derivative kernels such as **Sobel** (Fig. 4.18(a)):

$$g_x = G_x * f, \qquad g_y = G_y * f, \qquad |\nabla f| = \sqrt{g_x^2 + g_y^2}, \qquad \phi = \operatorname{atan2}(g_y, g_x). \tag{4.11}$$

The Sobel kernel combines differentiation in one direction with smoothing (weights 1–2–1) in the other, reducing noise sensitivity. For the vertical step edge $\begin{bmatrix} 10 & 10 & 200 \\ 10 & 10 & 200 \\ 10 & 10 & 200 \end{bmatrix}$, $g_x = (-10+200) + 2(-10+200) + (-10+200) = 760$ and $g_y = 0$: a strong vertical edge with gradient direction $0°$ (pointing across the edge). The **Canny** detector adds Gaussian smoothing, non-maximum suppression (thin edges) and hysteresis thresholding (connected edges). Figure 4.18(b) shows a 1-D intensity profile and its gradient peaking at the edge.

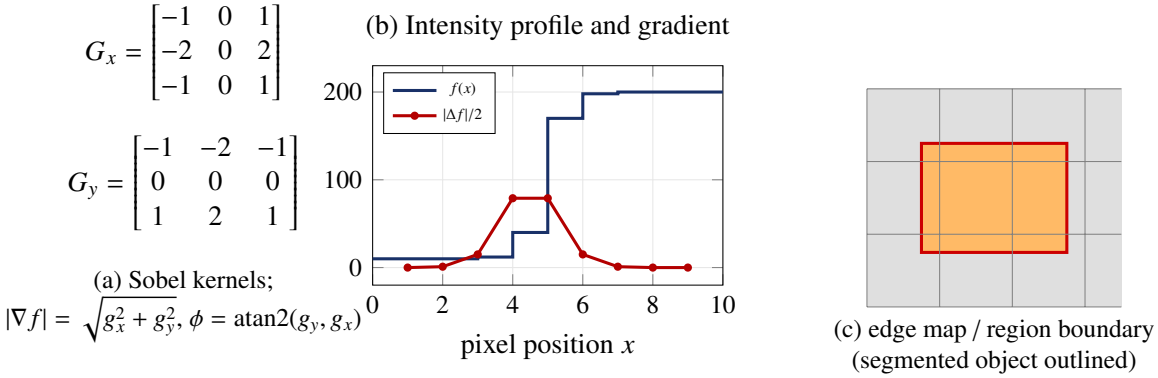


Figure 4.18: Edge detection: (a) Sobel kernels; (b) intensity profile across an edge and its gradient; (c) segmented object with its boundary.

4.18.2 Region (blob) features

After segmentation, connected-component labelling groups touching foreground pixels into *blobs*. Each blob is described by features computed from its *moments*

$$m_{pq} = \sum_x \sum_y x^p y^q \, b(x,y), \qquad \mu_{pq} = \sum_x \sum_y (x-\bar{x})^p (y-\bar{y})^q \, b(x,y). \tag{4.12}$$

- **Area** $A = m_{00}$ (pixels); **centroid** $\bar{x} = m_{10}/m_{00}$, $\bar{y} = m_{01}/m_{00}$ – the pick point for a robot.
- **Orientation** of the principal axis

$$\theta = \tfrac{1}{2} \operatorname{atan2}\left(2\mu_{11},\ \mu_{20} - \mu_{02}\right) \tag{4.13}$$

  – the gripper rotation.
- **Perimeter** P, **circularity** $4\pi A/P^2$ (1 for a circle, $\pi/4 = 0.785$ for a square), **bounding box**, aspect ratio, number of holes (Euler number).

Moment-based invariants (Hu moments) are unchanged by translation, rotation and scale and are used for shape recognition.

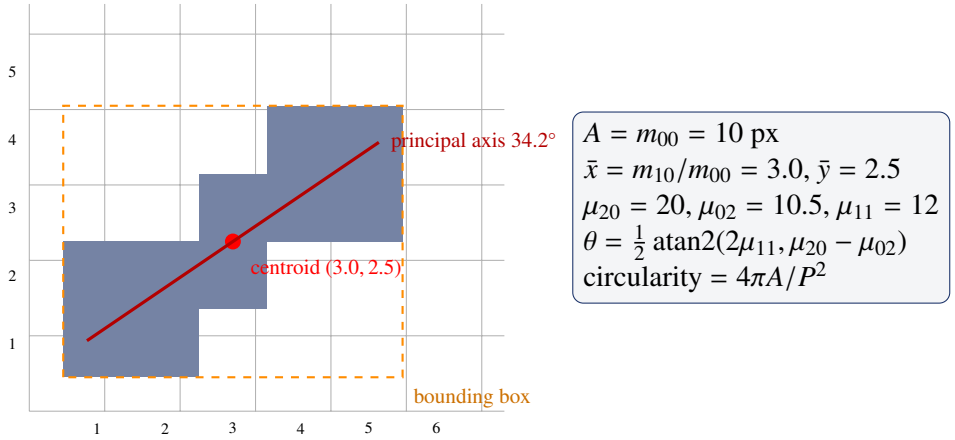


Figure 4.19: Feature extraction for a blob: area, centroid, principal axis and bounding box.

Example 4.6: Centroid and orientation of a blob (Topic 4.18)

Given: The blob of Fig. 4.19 has pixels at $(x,y) = (1,1), (2,1), (2,2), (3,2), (3,3), (4,3), (4,4), (5,4), (1,2), (5,3)$ (y upwards).

Solution: $A = 10$; $\sum x = 30 \Rightarrow \bar{x} = 3.0$; $\sum y = 25 \Rightarrow \bar{y} = 2.5$. Central moments: $\mu_{20} = \sum (x-3)^2 = 20$, $\mu_{02} = \sum (y-2.5)^2 = 10.5$, $\mu_{11} = \sum (x-3)(y-2.5) = 12$. $\theta = \frac{1}{2}\operatorname{atan2}(24,\ 9.5) = \frac{1}{2}(68.4^\circ) = 34.2^\circ$.

Interpretation: The robot should grasp at the centroid with the gripper rotated 34.2° (in image axes, then transformed to robot axes).

Algorithm 4.4 Object feature extraction (blob analysis)

Require: Binary image b; area limits $A_{\min}, A_{\max}$
Ensure: List of objects with $(A, \bar{x}, \bar{y}, \theta)$
label connected components of b (4- or 8-connectivity)
for each label k **do**
  $A \leftarrow m_{00}$; $(\bar{x}, \bar{y}) \leftarrow (m_{10}, m_{01})/m_{00}$
  compute $\mu_{20}, \mu_{02}, \mu_{11}$; $\theta \leftarrow \frac{1}{2}\operatorname{atan2}(2\mu_{11}, \mu_{20} - \mu_{02})$
  if $A_{\min} \le A \le A_{\max}$ **then**
    append $(A, \bar{x}, \bar{y}, \theta)$
  end if ▷ reject noise and touching parts
end for

4.19 Object Detection

Object detection decides *whether* and *where* objects of interest appear in an image. Industrial methods include:

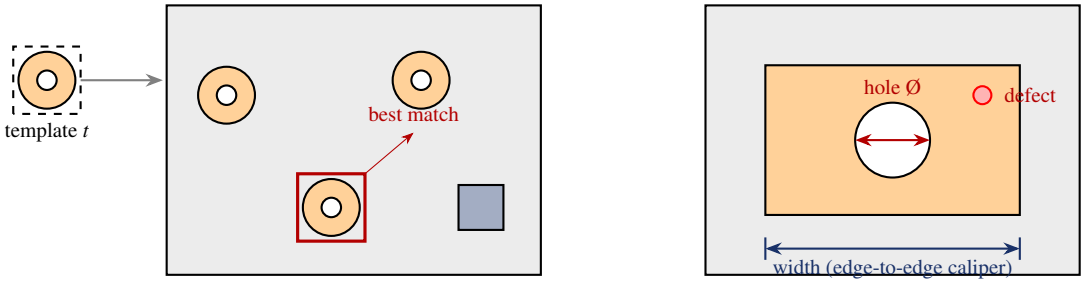
- **Blob analysis** – segmentation followed by feature filtering (Algorithm 4.4); fast and robust when contrast is good.
- **Template matching** (Fig. 4.20(a)) – a stored template t is compared with every position of the image by the *normalised cross-correlation*

$$\mathrm{NCC}(u,v) = \frac{\sum_{x,y}\left[f(x+u,y+v) - \bar{f}_{u,v}\right]\left[t(x,y) - \bar{t}\right]}{\sqrt{\sum\left[f(x+u,y+v) - \bar{f}_{u,v}\right]^2 \sum\left[t(x,y) - \bar{t}\right]^2}} \in [-1,1], \tag{4.14}$$

  which is insensitive to uniform brightness and contrast changes; peaks above a threshold (e.g. 0.8) indicate matches.
- **Geometric (edge-based) pattern matching** – matches edge contours, robust to rotation, scale, partial occlusion and lighting; the standard tool in commercial vision libraries for robot guidance.
- **Machine learning / deep learning** – convolutional neural networks (e.g. YOLO-type detectors) trained on labelled images detect and classify variable objects (food, logistics parcels, defects) where rules fail.

Algorithm 4.5 Object detection by normalised cross-correlation template matching

Require: Image f; template t ($m \times n$); threshold ρ
Ensure: Detected positions (u, v) and scores
for each position (u, v) (optionally on an image pyramid for speed) **do**
  extract window $W = f(u : u+m-1,\ v : v+n-1)$
  $s(u,v) \leftarrow \mathrm{NCC}(W, t)$ ▷ Eq. (4.14)
end for
find local maxima of s with $s > \rho$; refine to sub-pixel; return list



(a) Template matching: slide t over image f, NCC(u, v) peaks at objects

(b) Inspection: gauging, presence/absence, surface defects, tolerance check

Figure 4.20: (a) Object detection by template matching; (b) inspection by gauging, presence checks and defect detection.

Table 4.6: Comparison of object-detection methods.

Method	Principle	Strengths	Limitations
Blob analysis	Threshold + connected regions	Very fast; gives centroid and angle	Needs good contrast; touching parts merge
Template (NCC)	Grey-level correlation	Simple; brightness-invariant	Sensitive to rotation/scale; slow without pyramids
Geometric matching	Edge-contour matching	Rotation/scale/occlusion robust	Licensed libraries; needs clear edges
Deep learning (CNN)	Learned features	Handles natural variation	Needs data, GPU; less explainable

4.20 Inspection Techniques

Automated visual inspection checks every part at production speed (Fig. 4.20(b)):

- **Presence/absence and completeness** – are all components, screws, labels present? (blob count, intensity in ROI).
- **Dimensional gauging** – distances, diameters and angles measured by edge-to-edge *calipers* along scan lines, with sub-pixel edge interpolation; converted to millimetres by calibration (Eq. (4.8)); verified by gauge R&R studies.
- **Surface defect detection** – scratches, dents, contamination using dark-field lighting, texture analysis or deep learning.
- **Code reading and OCR/OCV** – 1-D barcodes, Data Matrix/QR codes, printed text, for traceability.
- **Colour and assembly verification** – correct part variant, wire colours, label position.

Inspection results drive reject mechanisms and are logged for statistical process control.

4.21 Smart Manufacturing Applications

Smart manufacturing (Industry 4.0) integrates automation with digital technologies (Fig. 4.21): **IIoT** connectivity of machines and sensors; **cyber-physical systems** in which physical processes are monitored and controlled by networked computation; **digital twins** – simulation models synchronised with the real cell for planning, virtual commissioning and predictive maintenance; **edge/cloud analytics** and AI for quality prediction and optimisation; **flexible, reconfigurable lines** with mobile and collaborative robots for lot-size-one production; and **traceability** through MES. A central performance measure is the *overall equipment effectiveness*

$$\text{OEE} = \text{availability} \times \text{performance} \times \text{quality}. \tag{4.15}$$

Example: availability 0.90, performance 0.95, quality 0.98 give OEE = 0.838 (83.8%), slightly below the commonly quoted world-class benchmark of about 85%; availability is usually the factor with most scope for improvement.

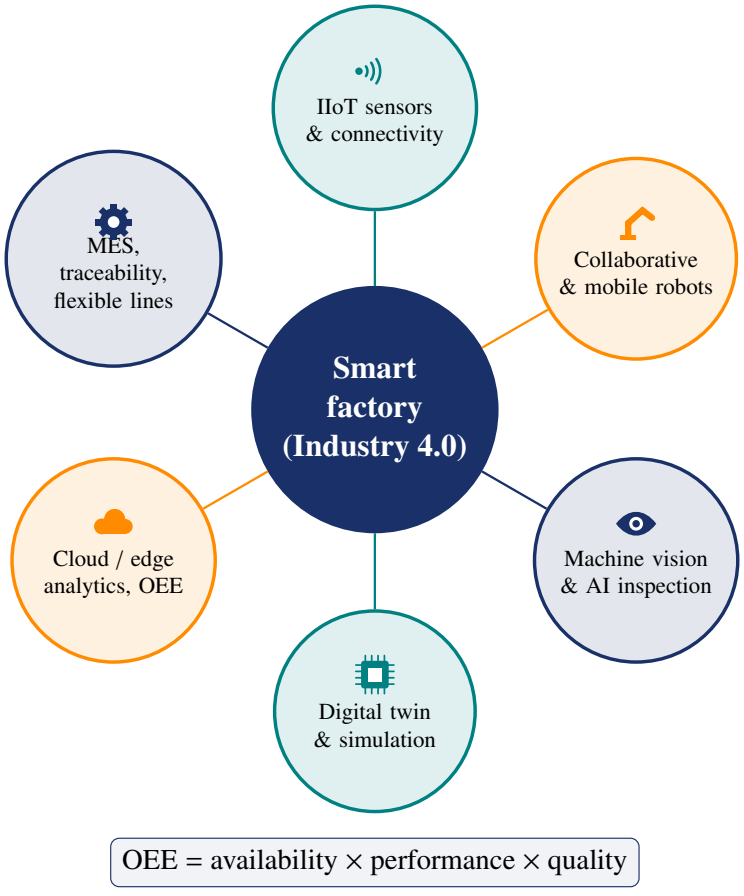


Figure 4.21: Elements of a smart factory (Industry 4.0).

4.22 Pick-and-Place Applications

Pick-and-place – transferring parts from one location to another – is the most common robot task. A *vision-guided* pick-and-place system (Fig. 4.22) locates randomly positioned parts and computes the grasp for each.

Pixel-to-robot calibration. For a fixed camera looking perpendicularly at a plane, a pixel (u, v) maps to camera-plane coordinates by the scale s (mm/pixel) and principal point (u_0, v_0), and then to the robot base frame by the homogeneous transform ${}^{B}_{C}\mathbf{T}$ (Unit III):

$$ {}^{C}\mathbf{p} = \begin{bmatrix} s\,(u - u_0) \\ -s\,(v - v_0) \\ 0 \end{bmatrix}, \qquad {}^{B}\tilde{\mathbf{p}} = {}^{B}_{C}\mathbf{T}\,{}^{C}\tilde{\mathbf{p}} \tag{4.16} $$

(the sign on v reflects image rows increasing downwards). ${}^{B}_{C}\mathbf{T}$ is found by *hand–eye calibration*: the robot touches or presents several known points that the camera also measures, and a least-squares fit gives the transform. General calibration uses the pinhole model with intrinsic matrix and lens distortion.

Example 4.7: Pixel to robot coordinates (Topic 4.22)

Given: $s = 0.2$ mm/px, principal point $(640, 512)$; camera frame axes parallel to the robot's with its origin at $(500, 0, 0)$ mm in the base frame. A part centroid is detected at pixel $(820, 412)$ with orientation $25°$. The conveyor moves at 0.3 m/s in $+x$ and the total latency from image to grasp is 0.2 s.

Solution: ${}^{C}\mathbf{p} = (0.2(180),\ -0.2(-100)) = (36,\ 20)$ mm. ${}^{B}\mathbf{p} = (500 + 36,\ 0 + 20) = (536,\ 20)$ mm. Conveyor shift $\Delta x = v_c \Delta t = 300 \times 0.2 = 60$ mm; grasp target $(596,\ 20)$ mm with gripper yaw $25°$.

Interpretation: Without conveyor tracking the robot would miss by 60 mm; real systems read the conveyor encoder at the instant of image capture and track continuously.

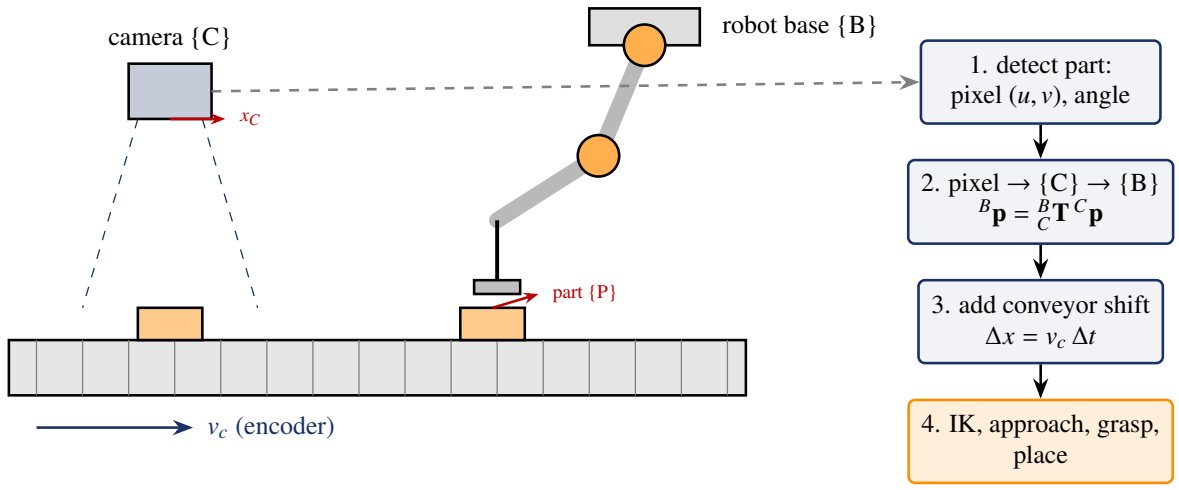


Figure 4.22: Vision-guided pick-and-place: camera and robot frames, conveyor tracking and processing steps.

Cycle time and throughput. Throughput $= 60/t_{\text{cycle}}$ picks per minute; Delta robots achieve

Algorithm 4.6 Vision-guided robotic pick-and-place

Require: Calibration ${}^{B}_{C}\mathbf{T}$, s, (u_0, v_0); tool frame; grasp offset; conveyor speed v_c
Ensure: Parts transferred to place positions

loop
  wait for trigger; capture image and conveyor encoder count n_0
  detect parts (blob / pattern matching); for each: centroid (u, v), angle θ
  ${}^{B}\mathbf{p} \leftarrow {}^{B}_{C}\mathbf{T}\,[s(u-u_0), -s(v-v_0), 0, 1]^{\mathsf{T}}$ ▷ Eq. (4.16)
  add conveyor displacement $(n - n_0)\cdot$(mm/count) at pick time
  move to approach pose above the part (joint move), descend linearly tracking the conveyor
  close gripper; verify grasp (sensor/vacuum); depart; move to place pose; release
end loop

100–200 picks/min, SCARAs 60–120. Cycle time is reduced by fly-by zones, overlapping vision processing with motion, and grasping several parts per trip.

4.23 Assembly Applications

Robotic assembly joins parts by insertion, screwing, press fitting, snapping, gluing and welding. The canonical problem is **peg-in-hole insertion** (Fig. 4.23): position errors of the robot, part and fixture (typically 0.05–0.5 mm) exceed the clearance (often < 0.02 mm), so pure position control would jam. Solutions:

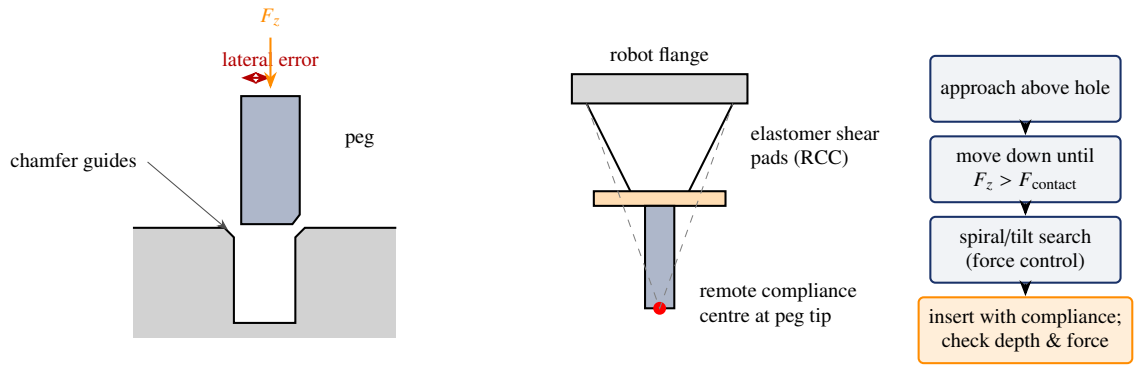
- **Part design for assembly**: chamfers and lead-ins that guide the peg; self-locating features.
- **Passive compliance**: the *remote centre compliance* (RCC) device places a compliance centre near the peg tip, so lateral contact forces translate the peg and contact moments rotate it about the tip – correcting errors without jamming or wedging.
- **Active compliance**: force/torque sensing and impedance or hybrid force/position control (Unit III, Section 3.17) with search strategies (spiral, tilt) to find the hole.
- **Vision** for fine alignment (visual servoing) and verification.

SCARA robots dominate small-part assembly because of their vertical rigidity and horizontal compliance; cobots with joint torque sensing perform force-guided assembly in small batches.

4.24 Material Handling Applications

Material handling covers moving, storing and controlling materials:

- **Palletising and depalletising** – robots stack cartons or bags in programmed patterns; pattern positions are computed from a base point and offsets ($x_{ij} = x_0 + i\,\Delta x$, $y_{ij} = y_0 + j\,\Delta y$, $z_k = z_0 + k\,h$); 3-D vision enables mixed-case depalletising.
- **Machine tending** – loading/unloading CNC machines, presses and injection-moulding machines (Fig. 4.2).



(a) Peg-in-hole with chamfer (b) Remote-centre-compliance device (c) Force-guided insertion

Figure 4.23: Robotic assembly: (a) chamfered peg-in-hole; (b) remote-centre-compliance device; (c) force-guided insertion strategy.

- **Conveyor tracking and sorting** – robots pick from moving conveyors using encoder feedback and vision.
- **Bin picking** – 3-D vision locates randomly piled parts; path planning avoids the bin walls.
- **Intralogistics** – AGVs (following fixed paths: magnetic tape, wires) and AMRs (navigating freely with LiDAR and maps) move material between stations and warehouses; mobile manipulators combine an AMR with a cobot arm.

Gripper choice (Unit II), payload including the gripper, cycle time and safety determine robot selection.

4.25 Autonomous Robotics

4.25.1 The sense–plan–act paradigm

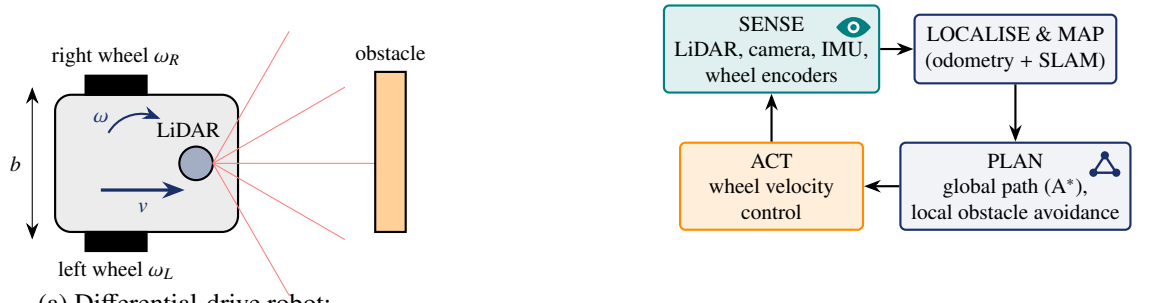
An autonomous robot operates without continuous human guidance in a partially unknown environment. It repeatedly *senses* (LiDAR, cameras, IMU, wheel encoders, GPS outdoors), *localises and maps* (estimates its pose and builds a map), *plans* (global path to the goal and local obstacle avoidance) and *acts* (wheel or joint velocity control) (Fig. 4.24(b)) [9, 24].

4.25.2 Differential-drive kinematics and odometry

For a robot with two driven wheels of radius r separated by track b (Fig. 4.24(a)), the forward and angular velocities follow from rigid-body kinematics:

$$v = \frac{r}{2}(\omega_R + \omega_L), \qquad \omega = \frac{r}{b}(\omega_R - \omega_L), \tag{4.17}$$

and the pose is updated by *odometry*: $x_{k+1} = x_k + v\,\Delta t \cos\phi_k$, $y_{k+1} = y_k + v\,\Delta t \sin\phi_k$, $\phi_{k+1} = \phi_k + \omega\,\Delta t$. Odometry errors accumulate (wheel slip, radius errors), so the pose is corrected by matching sensor data to a map. *Example:* $r = 0.05\,\text{m}$, $b = 0.4\,\text{m}$, $\omega_R = 10$, $\omega_L = 8\,\text{rad/s}$: $v = 0.025(18) = 0.45\,\text{m/s}$, $\omega = 0.125(2) = 0.25\,\text{rad/s}$ – a left turn of radius $v/\omega = 1.8\,\text{m}$.



(a) Differential-drive robot: $v = \frac{r}{2}(\omega_R + \omega_L)$, $\omega = \frac{r}{b}(\omega_R - \omega_L)$

(b) Sense–plan–act loop of an autonomous robot

Figure 4.24: Autonomous robotics: (a) differential-drive mobile robot with LiDAR; (b) sense–plan–act loop.

4.25.3 Localisation, mapping and planning

- **Localisation** estimates the pose from noisy sensors using probabilistic filters: Kalman filters (EKF) fuse odometry and IMU with landmark measurements; particle filters (Monte Carlo localisation) handle global localisation [9].
- **SLAM** (simultaneous localisation and mapping) builds an occupancy-grid or feature map while localising in it – the basis of AMRs that are taught a new site simply by driving them around.
- **Path planning**: global planners such as Dijkstra and A* search the map for the shortest collision-free path (A* uses cost $f = g + h$ with an admissible heuristic h); local planners (dynamic window approach, potential fields) avoid moving obstacles in real time.
- **Safety**: safety-rated laser scanners define protective and warning fields that slow or stop the robot near people.

Applications include warehouse AMRs, hospital delivery, agricultural robots, inspection drones, autonomous vehicles and mobile manipulators that combine navigation with the manipulation techniques of Units II and III.

4.26 MATLAB and Simulink Implementation

The programs below use only core MATLAB (tested in GNU Octave) so that every step of PLC logic, interfacing, PWM control and image processing is visible; equivalent toolbox functions are noted.

4.26.1 Program 4.1 – PLC sequence logic

Objective. Emulate the PLC scan cycle (read inputs, execute rungs, write outputs) for the seal-in rung of Fig. 4.5 and the state-machine sequence of Algorithm 4.1, with timers implemented by accumulating the scan time. **Expected output:** the motor latches at 1 s, the sequence passes CLAMP → MACHINE → UNLOAD and completes one part, starts a second, and everything returns to IDLE when STOP is pressed at 10 s. (Small timing offsets of one scan illustrate scan-time quantisation.)

Program 4.1: PLC scan-cycle simulation of a seal-in circuit and a machine-tending sequence.

```
% Program 4.1: PLC scan-cycle simulation of a start/stop circuit and a machine-tending sequence
clear; clc;
Tscan = 0.01; t = 0:Tscan:12; N = numel(t);
START = (t >= 1 & t < 1.2); STOP = (t >= 10 & t < 10.1); % push-button inputs
PART = (t >= 2); % part-present sensor
MOTOR = false; state = 'IDLE'; prev = ''; tmr = 0; count = 0;
for k = 1:N
    % --- 1. read inputs (input image) ---
    in_start = START(k); in_stop = STOP(k); in_part = PART(k);
    % --- 2. execute program ---
    MOTOR = (in_start || MOTOR) && ~in_stop; % rung 1: seal-in
    switch state % sequence (state machine)
        case 'IDLE'
            if MOTOR && in_part, state = 'CLAMP'; tmr = 0; end
        case 'CLAMP' % TON: clamp for 2 s
            tmr = tmr + Tscan;
            if tmr >= 2, state = 'MACHINE'; tmr = 0; end
        case 'MACHINE' % machining 3 s
            tmr = tmr + Tscan;
            if tmr >= 3, state = 'UNLOAD'; tmr = 0; end
        case 'UNLOAD'
            tmr = tmr + Tscan;
            if tmr >= 1, count = count + 1; state = 'IDLE'; end
    end
    if ~MOTOR, state = 'IDLE'; end
    % --- 3. write outputs / log transitions ---
    if k == 1 || ~strcmp(state, prev), fprintf('t = %5.2f s MOTOR = %d state -> %s\n', t(k), MOT
    prev = state;
end
fprintf('Parts completed = %d\n', count);
```

4.26.2 Program 4.2 – Microcontroller sensor interfacing

Objective. Reproduce Example 4.2, filter noisy ADC samples with a moving average, and debounce a bouncing switch in software. **Expected output:** code 465, 0.081 °C/LSB, read-back 37.47 °C; divider output 3.21 V; eight raw transitions reduced to two debounced transitions.

Program 4.2: Microcontroller sensor interfacing: ADC scaling, divider, filtering and debouncing.

```
% Program 4.2: Microcontroller sensor interfacing -- ADC scaling, divider, filtering, debouncing
clear; clc; close all;
Vref = 3.3; N = 12; full = 2^N - 1;
T = 37.5; v = 0.010*T; % LM35: 10 mV/degC
D = round(v/Vref*full);
fprintf('LM35 at %.1f degC -> %.3f V -> ADC code %d (%.3f degC/LSB)\n', T, v, D, Vref/full/0.010
```

```
fprintf('Reading back: %.2f degC\n', D/full*Vref/0.010);
R1 = 10e3; R2 = 18e3; fprintf('Divider 5 V -> %.2f V\n', 5*R2/(R1+R2));
% ---- moving-average filtering of noisy ADC samples
fs = 100; t = 0:1/fs:5; vtrue = 0.010*(30 + 5*(t > 2));
codes = round((vtrue + 0.004*randn(size(t)))/Vref*full);
M = 8; filt = filter(ones(1,M)/M, 1, codes);
plot(t, codes/full*Vref/0.010, '.', t, filt/full*Vref/0.010, 'LineWidth', 1.5); grid on;
xlabel('Time (s)'); ylabel('Temperature (degC)'); legend('raw ADC', '8-sample moving average');
% ---- software debounce: accept a change only after 5 identical samples (1 kHz)
raw = [zeros(1,20) 1 0 1 1 0 1 ones(1,30) 0 1 0 zeros(1,30)];
state = 0; cnt = 0; deb = zeros(size(raw));
for k = 1:numel(raw)
    if raw(k) ~= state, cnt = cnt + 1; else, cnt = 0; end
    if cnt >= 5, state = raw(k); cnt = 0; end
    deb(k) = state;
end
fprintf('Raw transitions = %d, debounced transitions = %d\n', sum(abs(diff(raw))), sum(abs(diff(
```

4.26.3 Program 4.3 – PWM-based actuator control

Objective. Compute the timer registers of Example 4.4(b) and simulate Algorithm 4.2: a 1 kHz PI loop that sets the PWM duty cycle of the DC motor of Unit II (average-voltage model). **Expected output:** ARR = 3599 (11.8-bit resolution); speed settles at 300 rad/s with duty cycle ≈ 0.63 (CCR ≈ 2261).

Program 4.3: PWM timer configuration and PI speed control of a DC motor.

```
% Program 4.3: PWM timer configuration and PI speed control of a DC motor
clear; clc; close all;
fclk = 72e6; fpwm = 20e3; PSC = 0; ARR = fclk/((PSC+1)*fpwm) - 1;
fprintf('ARR = %d, resolution = %.1f bits\n', ARR, log2(ARR+1));
Ra = 1.2; K = 0.05; J = 2e-5; b = 1e-5; Vdc = 24; % motor of Unit II
Tc = 1e-3; t = 0:Tc:0.5; wref = 300*(t > 0.05); % speed step to 300 rad/s
Kp = 0.08; Ki = 4.0; I = 0; w = 0; dt = 1e-5;
W = zeros(size(t)); Dlog = W;
for k = 1:numel(t)
    e = wref(k) - w; I = I + Ki*Tc*e; u = Kp*e + I; % PI in volts
    D = min(max(u/Vdc, 0), 1); % duty cycle (one direction)
    if u/Vdc > 1 || u < 0, I = I - Ki*Tc*e; end % anti-windup
    CCR = round(D*(ARR+1)); % value written to compare register
    for j = 1:round(Tc/dt) % plant between control updates
        ia = (D*Vdc - K*w)/Ra; % average-voltage model
        w = w + dt*(K*ia - b*w)/J;
    end
    W(k) = w; Dlog(k) = D;
end
fprintf('Final speed = %.1f rad/s, final duty = %.3f, CCR = %d\n', W(end), Dlog(end), CCR);
```

```
subplot(2,1,1); plot(t, W, t, wref, '--'); grid on; ylabel('\omega (rad/s)'); title('PWM speed c
subplot(2,1,2); plot(t, Dlog); grid on; xlabel('t (s)'); ylabel('duty cycle');
```

4.26.4 Program 4.4 – Image acquisition and filtering

Objective. Without any toolbox, synthesise a camera image of two parts under uneven lighting with Gaussian and impulse noise; apply mean (convolution, Eq. (4.9)) and median filters; compute the histogram and Otsu threshold (Algorithm 4.3). **Expected output:** Otsu threshold $T \approx 88$ and about 9.5% foreground pixels; the median-filtered image is free of salt-and-pepper noise while edges remain sharp. (With the Image Processing Toolbox the same steps are `medfilt2`, `imhist` and `graythresh`.) The binary image is saved for Program 4.5.

Program 4.4: Synthetic image acquisition, mean/median filtering, histogram and Otsu thresholding.

```
% Program 4.4: Image acquisition (synthetic), filtering, histogram and Otsu thresholding
clear; clc; close all;
rng(1);
[X, Y] = meshgrid(1:200, 1:150);
img = 40*ones(150, 200); % dark background
img((X-60).^2 + (Y-70).^2 < 25^2) = 200; % disc
img(abs(X-140) < 20 & abs(Y-60) < 12) = 190; % rectangle
img = img + 0.2*(X - 100); % uneven lighting
img = img + 10*randn(size(img)); % sensor noise
sp = rand(size(img)); img(sp < 0.01) = 255; img(sp > 0.99) = 0; % salt & pepper
img = min(max(round(img), 0), 255);
% ---- mean filter by convolution and 3x3 median filter
mean3 = conv2(img, ones(3)/9, 'same');
med3 = img;
for r = 2:size(img,1)-1
    for c = 2:size(img,2)-1
        w = img(r-1:r+1, c-1:c+1); med3(r,c) = median(w(:));
    end
end
% ---- Otsu threshold
h = histc(med3(:), 0:255)'; p = h/sum(h); k = 0:255;
w0 = cumsum(p); mu = cumsum(k.*p); muT = mu(end);
sB = (muT*w0 - mu).^2 ./ (w0.*(1 - w0) + eps);
[~, iT] = max(sB); T = iT - 1;
bw = med3 > T;
fprintf('Otsu threshold T = %d; foreground fraction = %.3f\n', T, mean(bw(:)));
subplot(2,2,1); imagesc(img); colormap(gray); axis image; title('acquired (noisy)');
subplot(2,2,2); imagesc(med3); axis image; title('3x3 median');
subplot(2,2,3); bar(0:255, h); xlim([0 255]); hold on; plot([T T], ylim, 'r'); title('histogram
subplot(2,2,4); imagesc(bw); axis image; title('binary');
save('bw.mat', 'bw', 'med3');
```

4.26.5 Program 4.5 – Feature extraction and object detection

Objective. Verify the Sobel and moment calculations of Section 4.18, label the connected components of the binary image from Program 4.4 by flood fill (Algorithm 4.4), and locate the disc by NCC template matching (Algorithm 4.5). **Expected output:** $g_x = 760$, $g_y = 0$; blob centroid $(3.00, 2.50)$, $\theta = 34.2°$; two objects (after rejecting noise blobs) with centroids $(60, 70)$ and $(140, 60)$ pixels – the true centres of the synthetic disc and rectangle; best NCC = 1.000 at $(60, 70)$.

Program 4.5: Sobel edges, connected-component blob features and NCC template matching.

```
% Program 4.5: Edge detection, blob features and NCC template matching (no toolboxes)
clear; clc; close all;
P = [10 10 200; 10 10 200; 10 10 200]; Gx = [-1 0 1; -2 0 2; -1 0 1]; Gy = Gx';
fprintf('Sobel on step patch: gx = %d, gy = %d\n', sum(sum(P.*Gx)), sum(sum(P.*Gy)));
% ---- blob of the worked example (x,y with y upwards)
xy = [1 1;2 1;2 2;3 2;3 3;4 3;4 4;5 4;1 2;5 3];
A = size(xy,1); c = mean(xy,1); d = xy - c;
mu20 = sum(d(:,1).^2); mu02 = sum(d(:,2).^2); mu11 = sum(d(:,1).*d(:,2));
fprintf('Blob: A = %d, centroid = (%.2f, %.2f), theta = %.1f deg\n', A, c, ...
        0.5*rad2deg(atan2(2*mu11, mu20 - mu02)));
% ---- connected-component labelling and features on the binary image of Program 4.4
S = load('bw.mat'); bw = S.bw; [R, C] = size(bw); L = zeros(R, C); n = 0;
for r = 1:R
  for col = 1:C
    if bw(r,col) && L(r,col) == 0
      n = n + 1; stack = [r col]; L(r,col) = n;
      while ~isempty(stack)
        p = stack(end,:); stack(end,:) = [];
        for dr = -1:1, for dc = -1:1
          rr = p(1)+dr; cc = p(2)+dc;
          if rr>=1 && rr<=R && cc>=1 && cc<=C && bw(rr,cc) && L(rr,cc)==0
            L(rr,cc) = n; stack(end+1,:) = [rr cc];
          end
        end, end
      end
    end
  end
end
for k = 1:n
  [rr, cc] = find(L == k);
  if numel(rr) < 50, continue; end % reject noise blobs
  fprintf('Object %d: area = %5d px, centroid (x,y) = (%6.1f, %6.1f)\n', k, numel(rr), mean(cc),
end
% ---- Sobel gradient magnitude of the image
g = sqrt(conv2(double(S.med3), Gx, 'same').^2 + conv2(double(S.med3), Gy, 'same').^2);
% ---- NCC template matching: template = 21x21 patch centred on the disc
tpl = S.med3(60:80, 50:70); tz = tpl - mean(tpl(:)); best = -1;
for r = 1:R-20
```

```
  for col = 1:C-20
    w = S.med3(r:r+20, col:col+20); wz = w - mean(w(:));
    s = sum(wz(:).*tz(:))/sqrt(sum(wz(:).^2)*sum(tz(:).^2) + eps);
    if s > best, best = s; loc = [col+10, r+10]; end
  end
end
fprintf('Best NCC = %.3f at (x,y) = (%d, %d)\n', best, loc);
imagesc(g); colormap(gray); axis image; title('Sobel gradient magnitude');
```

4.26.6 Program 4.6 – Vision-guided robotic pick-and-place

Objective. Implement Algorithm 4.6 for Example 4.7: convert pixels to the camera plane and robot base with (4.16), add the conveyor shift, solve SCARA inverse kinematics (Unit III) and plan a cubic joint move. **Expected output:** camera $(36, 20)$ mm, base $(536, 20)$ mm, pick $(596, 20)$ mm; joint targets $\theta_1 = -24.93°$, $\theta_2 = 63.89°$, $\theta_4 = -13.96°$; mid-trajectory joints $(-12.47°, 76.95°)$.

Program 4.6: Vision-guided pick-and-place: pixel-to-robot transformation, conveyor compensation and inverse kinematics.

```
% Program 4.6: Vision-guided pick-and-place -- pixel to robot coordinates, conveyor shift, IK
clear; clc;
s = 0.2; u0 = 640; v0 = 512; % mm/pixel, principal point
T_BC = [eye(3) [500; 0; 0]; 0 0 0 1]; % camera frame in robot base (mm)
uv = [820 412]; theta = 25; % detected centroid and angle (deg)
pC = [s*(uv(1)-u0); -s*(uv(2)-v0); 0; 1];
pB = T_BC*pC;
vc = 300; latency = 0.2; % conveyor mm/s, s
pick = pB(1:2) + [vc*latency; 0];
fprintf('Camera frame: (%.1f, %.1f) mm -> base: (%.1f, %.1f) mm -> pick at (%.1f, %.1f) mm, yaw
        pC(1:2), pB(1:2), pick, theta);
% ---- SCARA-type inverse kinematics (l1 = 400 mm, l2 = 300 mm), elbow branch +
l1 = 400; l2 = 300; x = pick(1); y = pick(2);
Dk = (x^2 + y^2 - l1^2 - l2^2)/(2*l1*l2);
if abs(Dk) > 1, error('pick point out of reach'); end
t2 = atan2(sqrt(1 - Dk^2), Dk); t1 = atan2(y, x) - atan2(l2*sin(t2), l1 + l2*cos(t2));
t4 = deg2rad(theta) - t1 - t2; % tool yaw for SCARA
fprintf('Joint targets: theta1 = %.2f, theta2 = %.2f, theta4 = %.2f deg\n', rad2deg([t1 t2 t4]))
% ---- cubic joint trajectory from home (0,90 deg) in 0.6 s, checked at mid-time
q0 = deg2rad([0 90]); qf = [t1 t2]; tf = 0.6; tm = tf/2;
qm = q0 + (qf - q0)*(3*(tm/tf)^2 - 2*(tm/tf)^3);
fprintf('Mid-trajectory joints = [%.2f %.2f] deg\n', rad2deg(qm));
```

4.26.7 Simulink and hardware exercises

S4.1 Stateflow sequence. Model the machine-tending sequence of Algorithm 4.1 as a Stateflow chart with timers and interlocks; drive it with `Signal Builder` inputs.

S4.2 PWM motor drive. Use `PWM Generator` and an H-bridge (Simscape Electrical) feeding a DC motor model; compare current ripple at 2 kHz and 20 kHz.

S4.3 Embedded deployment. With the Simulink Support Package for Arduino/STM32, read an analogue sensor and generate PWM to a motor driver; implement the PI loop of Program 4.3 on the hardware.

S4.4 Vision pipeline. Using the Computer Vision Toolbox blocks (or a `MATLAB Function` block with Program 4.5), process camera frames from a webcam (`webcam` support package) and output the centroid of the largest blob.

4.27 Unit IV Summary and Review

4.27.1 Unit summary

1. Industrial automation is organised in the ISA-95 pyramid from field devices to ERP; robots are integrated into workcells with conveyors, fixtures, machines, vision and safety systems.
2. PLCs execute ladder/ST programs in a scan cycle (read–execute–write); worst-case response is about two scans plus I/O delays; sequences are written as state machines with interlocks.
3. Robots are programmed by lead-through, teach pendant or offline simulation, using joint, linear and circular motion instructions with frames, speeds and zones; the pendant's enabling switch and T1/T2/AUTO modes ensure safe teaching.
4. Embedded systems with microcontrollers implement real-time sensing and control; sensors are interfaced via pull-ups, debouncing, dividers and ADCs; actuators via MOSFETs, flyback diodes, relays and H-bridges.
5. PWM gives $V_{\text{avg}} = DV_{\text{dc}}$, generated by timers with $f = f_{\text{clk}}/((\text{PSC}+1)(\text{ARR}+1))$.
6. UART, SPI, I^2C, CAN and industrial Ethernet link devices; choice depends on distance, speed, wiring and determinism.
7. Machine vision combines lighting, optics, cameras and processing; resolution W/N_{px}, lens $1/f = 1/u + 1/v$.
8. Image processing: filtering (convolution, median), thresholding (Otsu), morphology; features: edges (Sobel), moments, centroid, orientation; detection: blob analysis, NCC, geometric matching, CNNs; inspection: gauging, presence, defects, codes.
9. Applications: smart manufacturing (IIoT, digital twin, OEE), vision-guided pick-and-place with hand–eye calibration and conveyor tracking, compliant assembly (RCC, force control), material handling (palletising, AGV/AMR), and autonomous robots using sense–plan–act, odometry, SLAM and A^*.

4.27.2 Formula sheet

Table 4.7: Unit IV formula sheet.

Quantity	Formula	Ref.
PLC scan time	$T_{\text{scan}} = T_{\text{in}} + T_{\text{prog}} + T_{\text{out}} + T_{\text{comm}}$	(4.1)
Worst-case response	$T_{\text{filter}} + 2T_{\text{scan}} + T_{\text{out}}$	(4.2)
ADC code	$D = \text{round}(v(2^N - 1)/V_{\text{ref}})$	(4.3)
PWM average	$V_{\text{avg}} = DV_{\text{dc}}$	(4.4)
Timer PWM	$f = f_{\text{clk}}/((\text{PSC} + 1)(\text{ARR} + 1))$; $D = \text{CCR}/(\text{ARR} + 1)$	(4.5)
UART throughput	baud / bits per frame	(4.6)
Lens	$1/f = 1/u + 1/v$; $m = w_s/W$; $u = f(1 + m)/m$	(4.7)
Spatial resolution	W/N_{px}	(4.8)
Convolution	$g = \sum\sum h(i, j)f(x - i, y - j)$	(4.9)
Threshold	$b = [f > T]$; Otsu maximises σ_B^2	(4.10)
Sobel gradient	$\lvert\nabla f\rvert = \sqrt{g_x^2 + g_y^2}$	(4.11)
Moments, centroid	$m_{pq} = \sum x^p y^q b$; $\bar{x} = m_{10}/m_{00}$	(4.12)
Orientation	$\theta = \frac{1}{2}\,\text{atan2}(2\mu_{11}, \mu_{20} - \mu_{02})$	(4.13)
NCC	normalised correlation $\in [-1, 1]$	(4.14)
Pixel to robot	${}^B\tilde{\mathbf{p}} = {}^B_C\mathbf{T}[s(u - u_0), -s(v - v_0), 0, 1]^\top$	(4.16)
OEE	availability $\times$ performance $\times$ quality	(4.15)
Differential drive	$v = \frac{r}{2}(\omega_R + \omega_L)$, $\omega = \frac{r}{b}(\omega_R - \omega_L)$	(4.17)

4.28 Unit IV Question Bank

Exactly **30 questions of 10 marks each** (300 marks). Distribution: concepts Q1–Q5; theory Q6–Q12; derivations Q13–Q17; numerical problems Q18–Q23; diagrams and working principles Q24–Q26; applications Q27–Q30.

Question-to-notes reference index

For every question the table gives the topic (section) number, the page of the question and the page on which the topic begins in these notes. The **Refer to** line under each question also lists the specific figures, tables, equations and worked examples to study.

Table 4.8: Unit IV question-to-notes reference index (automatic page references).

Q	Topic	Q. page	Notes page	Q	Topic	Q. page	Notes page
Q1	4.1	199	159	Q16	4.18	201	183
Q2	4.7	199	169	Q17	4.25	201	190
Q3	4.13	199	176	Q18	4.3	201	162
Q4	4.6	200	167	Q19	4.9	202	170
Q5	4.21	200	187	Q20	4.11	202	173
Q6	4.2	200	160	Q21	4.12	202	175
Q7	4.3	200	162	Q22	4.14	202	177
Q8	4.5	200	166	Q23	4.22	202	187
Q9	4.12	200	175	Q24	4.3	202	162
Q10	4.15	200	179	Q25	4.8	202	170
Q11	4.16	200	180	Q26	4.17	203	181
Q12	4.19	201	185	Q27	4.22	203	187
Q13	4.3	201	162	Q28	4.23	203	189
Q14	4.11	201	173	Q29	4.24	203	189
Q15	4.14	201	177	Q30	4.25	203	190

A. Fundamental concepts and definitions

Q1 **Topic 4.1** – Industrial Automation **[10 marks]**

Define industrial automation and state its objectives. Explain the five levels of the automation pyramid with examples of devices and response times at each level, and explain how Industry 4.0 changes this hierarchy.

☞ **Refer to:** Section 4.1 (p. 159); Figure 4.1 (p. 160)

Q2 **Topic 4.7** – Embedded Systems **[10 marks]**

Define an embedded system and explain its characteristics. Distinguish hard and soft real-time systems, super-loop and RTOS software structures, and microprocessor and microcontroller, with examples from robotics.

☞ **Refer to:** Section 4.7 (p. 169); Figure 4.8 (p. 169)

Q3 **Topic 4.13** – Machine Vision **[10 marks]**

Define machine vision and distinguish it from general computer vision. List the elements of a machine-vision system and the main applications of vision in robotic automation.

☞ **Refer to:** Section 4.13 (p. 176); Figure 4.13 (p. 178)

Q4 **Topic 4.6** – Teach Pendant Concepts **[10 marks]**

Describe the functions and controls of a teach pendant. Explain the three-position enabling switch, the T1, T2 and AUTO operating modes, and the coordinate systems available for jogging.

☞ **Refer to:** Section 4.6 (p. 167); Figure 4.7 (p. 168)

Q5 **Topic 4.21** – Smart Manufacturing **[10 marks]**

Explain the concept of smart manufacturing (Industry 4.0) and its key technologies – IIoT, cyber-physical systems, digital twin, cloud/edge analytics and collaborative/mobile robots. Define OEE.

☞ **Refer to:** Section 4.21 (p. 187); Figure 4.21 (p. 187)

B. Theoretical explanations

Q6 **Topic 4.2** – Robotic Workcells **[10 marks]**

Explain the components and layouts of a robotic workcell. Explain interlocks and handshake signals between a robot and a machine, and describe the safeguarding of a robotic cell according to ISO 10218.

☞ **Refer to:** Section 4.2 (p. 160); Figure 4.2 (p. 161); Algorithm 4.1 (p. 165)

Q7 **Topic 4.3** – PLC Architecture **[10 marks]**

With a neat block diagram, explain the architecture of a PLC and the function of each part. List the IEC 61131-3 languages and compare PLCs with microcontrollers and industrial PCs.

☞ **Refer to:** Section 4.3 (p. 162); Figure 4.3 (p. 163); Table 4.2 (p. 165); Example 4.1 (p. 163)

Q8 **Topic 4.5** – Robot Programming **[10 marks]**

Explain the lead-through, teach-pendant and offline methods of robot programming with their merits and limitations. Explain the elements of a robot programming language, including joint, linear and circular motion instructions, zones and I/O synchronisation.

☞ **Refer to:** Section 4.5 (p. 166)

Q9 **Topic 4.12** – Communication Interfaces **[10 marks]**

Explain the UART, SPI, I^2C and CAN interfaces with neat diagrams, describing their signals, framing or addressing, speed and typical use. Compare them in a table and mention industrial Ethernet protocols used in robotics.

☞ **Refer to:** Section 4.12 (p. 175); Figure 4.12 (p. 175); Table 4.4 (p. 177)

Q10 **Topic 4.15** – Cameras and Image Acquisition **[10 marks]**

Explain CCD and CMOS sensors, global and rolling shutters, area-scan and line-scan cameras, and 3-D cameras (stereo, structured light, time of flight). Explain the acquisition parameters exposure, gain and triggering.

☞ **Refer to:** Section 4.15 (p. 179)

Q11 **Topic 4.16** – Lighting Techniques **[10 marks]**

Explain with sketches backlighting, diffuse dome, ring, dark-field, coaxial and structured lighting, giving an application of each. Why is lighting called the most critical part of a vision system?

☞ **Refer to:** Section 4.16 (p. 180); Figure 4.15 (p. 181)

Q12 **Topic 4.19** – Object Detection and Inspection **[10 marks]**

Explain blob analysis, template matching, geometric pattern matching and deep-learning-based object detection, comparing them. Explain the main automated inspection techniques used in industry.

☞ **Refer to:** Section 4.19 (p. 185); Table 4.6 (p. 186); Equation (4.14) (p. 185)

C. Mathematical derivations

Q13 **Topic 4.3** – PLC Scan Cycle and Response Time **[10 marks]**

Explain the PLC scan cycle with a flowchart and derive expressions for the scan time and the worst-case input-to-output response time. Explain why a short input pulse may be missed and how this is prevented. Write the Boolean expression of a start/stop seal-in rung.

☞ **Refer to:** Section 4.3 (p. 162); Figure 4.4 (p. 163); Example 4.1 (p. 163)

Q14 **Topic 4.11** – PWM **[10 marks]**

Derive the average output voltage of a PWM waveform. Derive the timer relations between clock frequency, prescaler, auto-reload value, compare value, PWM frequency, duty cycle and resolution, and discuss the frequency–resolution trade-off.

☞ **Refer to:** Section 4.11 (p. 173); Example 4.4 (p. 173)

Q15 **Topic 4.14** – Optics of Image Acquisition **[10 marks]**

Using the thin-lens equation, derive the working distance in terms of focal length and magnification. Derive the spatial resolution and motion blur of a vision system and state the rules for choosing camera resolution.

☞ **Refer to:** Section 4.14 (p. 177); Example 4.5 (p. 178)

Q16 **Topic 4.18** – Image Moments and Features **[10 marks]**

Define raw and central image moments and derive the area, centroid and principal-axis orientation of a binary object. Define circularity and show its value for a circle and a square. Explain Sobel edge detection mathematically.

☞ **Refer to:** Section 4.18 (p. 183); Example 4.6 (p. 184)

Q17 **Topic 4.25** – Mobile Robot Kinematics **[10 marks]**

Derive the forward kinematics of a differential-drive mobile robot and the odometry update equations. Explain why odometry drifts and how localisation corrects it.

☞ **Refer to:** Section 4.25 (p. 190)

D. Numerical problems

Q18 **Topic 4.3** – PLC Timing Calculations **[10 marks]**

A PLC program of 6000 instructions executes at 0.08 µs per instruction; I/O update takes 0.6 ms and communication 0.42 ms. The input filter is 5 ms and the transistor output delay 0.1 ms. Find the scan time, worst-case response time and minimum reliably detected pulse width. Find the preset of a TON timer of 3 s with a 10 ms time base.

☞ **Refer to:** Section 4.3 (p. 162); Example 4.1 (p. 163)

Q19 Topic 4.9 – Interfacing Calculations **[10 marks]**

(a) A 10-bit, 5 V ADC reads a 270° potentiometer on a joint; code 600 is read. Find the angle and the angular resolution. (b) Design a divider to read a 0 V to 24 V signal with a 3.3 V ADC. (c) A MOSFET with $R_{DS(on)} = 50\,\text{m}\Omega$ switches a 1.5 A solenoid; find its conduction loss and state the protection needed.

☞ **Refer to:** Section 4.9 (p. 170); Example 4.2 (p. 171)

Q20 Topic 4.11 – PWM Calculations **[10 marks]**

A 16 MHz MCU timer with prescaler 8 generates the 50 Hz signal for an RC servo. Find ARR, the compare value for a 1.5 ms pulse, and the angular resolution (servo range 180° over 1–2 ms). Find the average voltage applied to a 12 V motor at 75% duty.

☞ **Refer to:** Section 4.11 (p. 173); Example 4.4 (p. 173)

Q21 Topic 4.12 – Communication Calculations **[10 marks]**

(a) Find the time to send 50 bytes by UART (8N1) at 9600 and 115200 baud. (b) A six-axis robot needs two CAN frames (8 data bytes, ≈111 bits) per joint every 1 ms. Is this possible at 500 kbit/s or 1 Mbit/s? (c) Estimate the time to read 10 bytes from an I^2C sensor at 100 kHz.

☞ **Refer to:** Section 4.12 (p. 175)

Q22 Topic 4.14 – Vision System Design **[10 marks]**

A part 120×90 mm must be inspected for defects of 0.3 mm while moving at 0.2 m/s. Select the camera resolution, compute the mm/pixel, the lens focal length for a working distance of about 300 mm (pixel size 4.8 µm), the actual working distance with a standard 16 mm lens, and the motion blur for a 200 µs exposure.

☞ **Refer to:** Section 4.14 (p. 177); Example 4.5 (p. 178)

Q23 Topic 4.22 – Image Processing and Vision-Guidance Calculations **[10 marks]**

(a) Compute the mean, median and Sobel gradient of the patch $\begin{bmatrix} 50 & 50 & 50 \\ 50 & 50 & 50 \\ 200 & 200 & 200 \end{bmatrix}$ at its centre. (b) A camera with 0.25 mm/px and principal point $(320, 240)$ is mounted with its x-axis along the robot y-axis (rotation 90° about z) and origin at $(300, -100)$ mm. A part is detected at pixel $(400, 180)$. Find its robot coordinates.

☞ **Refer to:** Section 4.22 (p. 187); Example 4.7 (p. 188)

E. Diagrams and working principles

Q24 Topic 4.3 – Ladder Logic **[10 marks]**

Draw and explain ladder diagrams for (a) a start/stop motor circuit with overload and seal-in, (b) a conveyor that runs for 5 s after a part is detected (timer), (c) a counter that signals a full pallet after 12 parts. Explain why stop buttons are wired normally closed.

☞ **Refer to:** Section 4.3 (p. 162); Figure 4.5 (p. 164); Example 4.1 (p. 163)

Q25 Topic 4.8 – Microcontroller and Interfacing **[10 marks]**

With a neat diagram explain the internal architecture of a microcontroller. Draw and explain circuits for interfacing a push button (with debouncing), an analogue sensor and an inductive load to a microcontroller.

☞ **Refer to:** Section 4.8 (p. 170); Figure 4.9 (p. 171); Figure 4.10 (p. 173)

Q26 **Topic 4.17** – Image Processing Pipeline **[10 marks]**

With a block diagram explain the image-processing pipeline. Explain filtering (mean, Gaussian, median), histogram-based thresholding with Otsu's method, and morphological operations, illustrating each.

☞ **Refer to:** Section 4.17 (p. 181); Figure 4.16 (p. 181); Figure 4.17 (p. 182); Algorithm 4.3 (p. 183)

F. Applications and analytical questions

Q27 **Topic 4.22** – Vision-Guided Pick-and-Place **[10 marks]**

Design a vision-guided pick-and-place system that transfers randomly oriented parts from a moving conveyor to trays. Describe the hardware, calibration (pixel-to-robot), conveyor tracking, software steps and cycle-time considerations.

☞ **Refer to:** Section 4.22 (p. 187); Figure 4.22 (p. 188); Equation (4.16) (p. 188); Example 4.7 (p. 188); Algorithm 4.6 (p. 189)

Q28 **Topic 4.23** – Robotic Assembly **[10 marks]**

Explain the difficulties of robotic peg-in-hole assembly and the solutions: part design, remote centre compliance, force-controlled search and vision. Explain why SCARA robots are preferred for small-part assembly.

☞ **Refer to:** Section 4.23 (p. 189); Figure 4.23 (p. 190)

Q29 **Topic 4.24** – Material Handling **[10 marks]**

Explain robotic palletising, machine tending, bin picking and intralogistics with AGVs and AMRs. Compute the positions of a 3×4 pallet pattern with pitch 200 mm by 150 mm from a base point $(800, -300, 100)$ mm and layer height 120 mm, for the item in row 2, column 3, layer 2 (counting from 0).

☞ **Refer to:** Section 4.24 (p. 189); Figure 4.2 (p. 161)

Q30 **Topic 4.25** – Autonomous Mobile Robots **[10 marks]**

Explain the sense–plan–act architecture, localisation (EKF, particle filter), SLAM and A* path planning of an autonomous mobile robot. A differential-drive robot has $r = 0.075\,\text{m}$, $b = 0.5\,\text{m}$, $\omega_R = 6$ and $\omega_L = 4\,\text{rad/s}$; find v, ω and the turning radius.

☞ **Refer to:** Section 4.25 (p. 190); Figure 4.24 (p. 191)

Glossary of Important Technical Terms

Table 4.9: Glossary of important technical terms (Unit I).

Term	Definition
Accuracy	Closeness of agreement between a measured value and the true value.
Absolute encoder	Encoder giving a unique digital code for every angular position.
Calibration	Comparison of instrument output with a standard to establish the input–output relation.
Cold-junction compensation	Correction of a thermocouple reading for the reference-junction temperature.
Dead band	Range of input producing no change of output.
Degree of freedom	Independent coordinate needed to specify configuration.
Dexterous workspace	Points reachable with every orientation of the end-effector.
Drift	Slow change of zero (bias) or sensitivity (span) with time or temperature.
End-effector	Gripper or tool attached to the robot wrist flange.
Gauge factor	Fractional resistance change per unit strain.
Gray code	Binary code in which adjacent values differ in one bit.
Hysteresis	Difference between outputs for increasing and decreasing inputs.
Loading effect	Disturbance of the measured system by the measuring instrument.
Manipulator	Chain of links and joints that moves the end-effector.
Payload	Maximum load a robot can carry at rated performance.
Precision	Closeness of agreement between repeated measurements.
Quadrature	Two signals 90° apart, allowing direction detection.
Repeatability	Precision under identical conditions; for robots, closeness of repeated poses.
Resolution	Smallest detectable change of input.
Seebeck effect	EMF generated in a circuit of two metals with junctions at different temperatures.
Sensitivity	Change of output per unit change of input.

Term	Definition
Settling time	Time for a response to enter and remain within a tolerance band.
Tool centre point	Reference point of the end-effector used in programming.
Transducer	Device converting one form of energy or signal into another.
Uncertainty	Parameter characterising the dispersion of values attributable to the measurand.
Work envelope	Region of space reachable by the robot’s wrist or TCP.

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